SECTION **6.0**

CARDIOVASCULAR SURGERY

CHAPTER **6.1**

Cardiac Surgery

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CARDIOPULMONARY BYPASS

SURGICAL CONSIDERATIONS

The development of cardiopulmonary bypass (CPB) technology has allowed the repair of many congenital and acquired lesions of the heart and great vessels. Designed to replace cardiac and pulmonary functions, full CPB requires a blood pump and oxygenator. The pump may be of the roller-head or centrifugal variety with the latter producing less trauma to formed blood elements. The oxygenator allows O_2 and CO_2 to diffuse through a thin membrane into the surrounding blood (membrane oxygenator), though older forms of oxygenators may bubble gases through a blood-filled reservoir (bubble oxygenator). Utilization of any blood pump requires at least partial heparinization (ACT > 180 sec), and introduction of an oxygenator mandates full heparinization (ACT > 400 sec). Heparin-coated components, which partially eliminate the necessity for heparin, are available.

Full CPB typically drains systemic venous return via the right atrium into a venous reservoir, from which the blood is pumped through an oxygenator and then returned to the ascending aorta, subclavian artery, innominate artery, or femoral artery, completely bypassing the heart and lungs (Figs. 6.1-1 and 6.1-2). **Partial CPB** usually supports only a portion of the body—typically the infradiaphragmatic portion—and requires maintenance of native cardiac output and ventilation in addition to an oxygenator-equipped CPB circuit in order to supply oxygenated blood to both the supradiaphragmatic and infradiaphragmatic portions of the body. **Left heart bypass** relies solely on the patient’s lungs as the oxygenator and diverts a portion of oxygenated blood from the left atrium to the infradiaphragmatic portion of the body, while supradiaphragmatic perfusion is supplied by native cardiac output. Full CPB is utilized during a sternotomy for work on the heart, ascending aorta, and transverse arch, while partial CPB is normally used for work on the descending or thoracoabdominal aorta.

After exposure of the relevant organs (heart or descending thoracic aorta), and after heparinization, venous and arterial cannulae must be placed intraluminally. **Cannulation of the heart** usually involves venous drainage from the right atrium, with either two cannulae inserted through the atrium into the SVC and IVC (bicaval) or via a larger dual-stage cannula draining the right atria and IVC. Bicaval cannulation reduces venous return (and rewarming) to the heart and allows caval snares to be placed so that the right atrium can be opened without introducing air into the venous return. Occasionally, atrial manipulation for cannulation can depress cardiac output with resultant hypotension. This usually can be reversed with volume replacement. Aortic cannulation usually is not associated with any physiologic perturbation, although hypertension must be avoided to minimize aortic complications. After the cannulae are in place and connections are made to the bypass circuit, CPB may be instituted electively. Depending on surgical procedure and surgeon preference, cardiac operations may be conducted under mild (32–35°C), moderate (28–32°C), or deep (<28°C) hypothermia. Deep hypothermia is utilized to facilitate aortic arch repair in the setting of profound hypothermic circulatory arrest (PHCA). However, recent advancements in cerebral protective strategies, such as selective antegrade cerebral perfusion (SACP) and retrograde cerebral perfusion (RCP), have allowed for aortic arch repair to be performed under more moderate hypothermia. For operations on the descending thoracic aorta, normothermia is maintained unless surgical access of the aortic arch is necessary, in which case PHCA may be utilized.

With initiation of CPB, many physiologic variables are now controlled by the anesthesiologist, perfusionist, and surgeon, including systemic flow rate and perfusion pressure, arterial O_2 and CO_2, temperature, and hematocrit. Other physiologic parameters follow either directly or indirectly. Thus, physiologic monitoring for the anesthesiologist and perfusionist includes, at a minimum, arterial pressure, central venous pressure, arterial blood gas determination (preferably online during CPB), cardiac output, urine output, and cardiac electrical activity. Constant communication among surgeon, perfusionist, and anesthesiologist is essential for a smooth operation. TEE is used to verify cannula positioning, characterize primary pathology and identify previously undiagnosed pathology, evaluate effectiveness of surgical repair, and provide information about overall cardiac function and volume status. As such, the use of TEE has become standard practice for cardiac surgery.

The **physiologic response to CPB** is complex and is associated with a massive catecholamine release that resolves after bypass. Subsequent changes include abnormal bleeding, increased capillary permeability, leukocytosis, renal dysfunction, and impairment of the immune response. Hemodilution, nonpulsatile flow, hypothermia, exposure of formed elements to nonendothelial surfaces, complement activation, protein denaturation, cascading effects within the coagulation and fibrinolytic system, and activation of the kallikrein-bradykinin cascade all contribute to this physiologic derangement and account for much of the morbidity and mortality after CPB.

For periods of **cardiac arrest**, during which the heart is deprived of its arterial blood supply, the metabolic demands of the myocardium must be minimized. This usually is accomplished by achieving electrical and mechanical arrest with a hyperkalemic cardioplegic solution, by lowering myocardial temperature to <15°C and by ensuring the ventricles are adequately decompressed to minimize myocardial wall tension. Frequent reinfusions of cardioplegia maintain hypothermia and electrical silence, prevent lactic acid accumulation, and deliver some minimally available dissolved O_2.

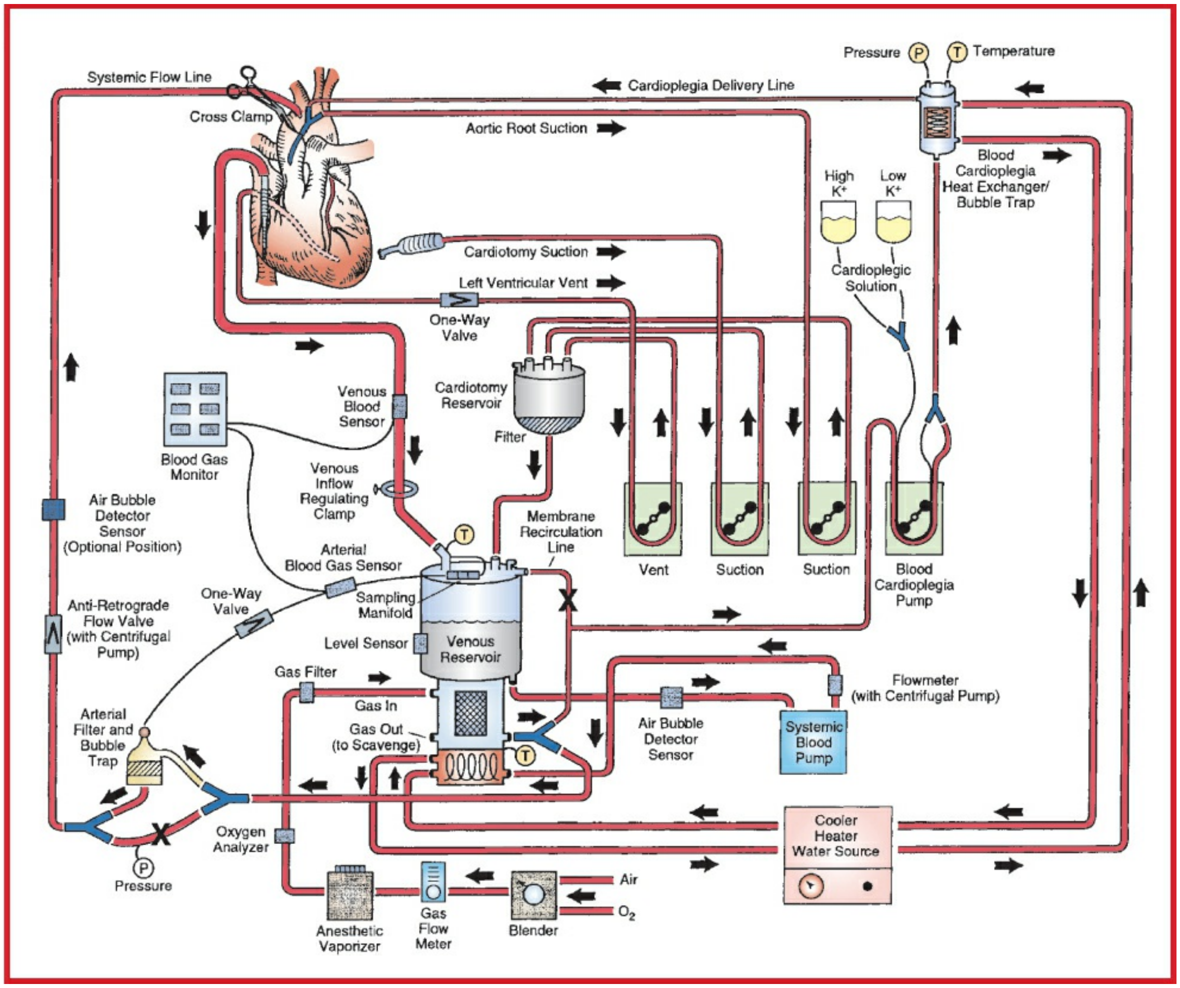

Figure 6.1-1 Detailed schematic diagram of the arrangement of a typical cardiopulmonary bypass circuit using a membrane oxygenator with integral hard-shell venous reservoir (lower center) and external cardiotomy reservoir. Venous cannulation is by a cavoatrial cannula, and arterial cannulation is in the ascending aorta. Some circuits do not incorporate a membrane recirculation line; in these cases, the cardioplegia blood source is a separate outlet connector built into the oxygenator near the arterial outlet. The systemic blood pump may be either a roller or centrifugal type. The cardioplegia delivery system (right) is a one-pass combination blood/crystalloid type. The cooler-heater water source may be operated to supply water to both the oxygenator heat exchanger and cardioplegia delivery system. The air bubble detector sensor may be placed on the line between the venous reservoir and systemic pump, between the pump and membrane oxygenator inlet, or between the oxygenator outlet and arterial filter (neither shown) or on the line after the arterial filter (optional position on drawing). One-way valves prevent retrograde flow (some circuits with a centrifugal pump also incorporate a one-way valve after the pump and within the systemic flow line). Other safety devices include an oxygen analyzer placed between the anesthetic vaporizer (if used) and the oxygenator gas inlet and a reservoir level sensor attached to the housing of the hard-shell venous reservoir (on the left center). Arrows, directions of flow; X, placement of tubing clamps; P and T (within circles), pressure and temperature sensors. Hemoconcentrator (described in text) not shown.

Cessation of CPB is accomplished by gradually decreasing pump flows, allowing for right heart filling, and gradually replenishing the circulating blood volume. Pulmonary and coronary vasodilators may be required during this phase, as there may be heightened vasoreactivity after periods of ischemia and hypothermia.

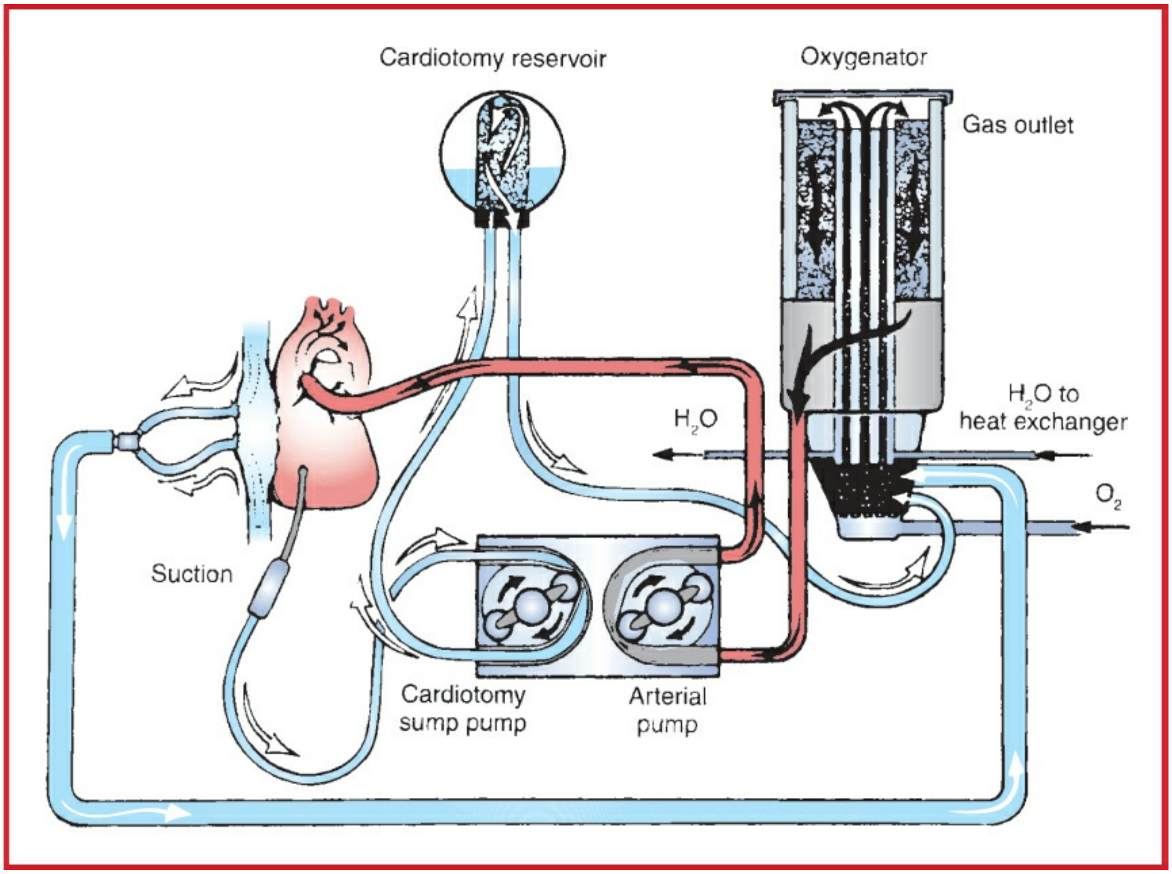

Figure 6.1-2 Schematic representation of the CPB circuit. (Reproduced with permission from Hardy JD: Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)

Secondary effects of CPB demand special considerations during the final stages of the procedure and chest closure. Adverse effects on coagulation have already been mentioned, and vigorous attention to maintenance and replacement of coagulation factors is essential. The capillary leak phenomenon results in interstitial myocardial and pulmonary edema. Decreased myocardial performance and compliance mandate an increase in preload and inotropic support, especially during the physical act of chest closure, where a transient rise in intramediastinal pressure may depress systemic venous return. Similarly, decreased pulmonary compliance and gas exchange mandate control of inspiratory pressures and lung volumes during chest closure because mediastinal volume is physically decreased.

ANESTHETIC CONSIDERATIONS FOR CARDIOPULMONARY BYPASS

This segment is not meant to be a definitive text on CPB, but rather a guide to the anesthetic management of bypass. Communication among surgeons, anesthesiologists, and pump technicians is of vital importance in carrying out this procedure.

■ PREPARATION FOR BYPASS

Prebypass/anticoagulation	✓ baseline ACT (normal = 90–130 sec). Heparin (3 mg [~300 U]/kg) is administered via a central vein (✓ back-bleeding to verify intravascular position) or by the surgeon directly into the atrium (preferred). ✓ ACT 3 min after heparin. It is essential to ensure adequate anticoagulation (ACT = 400–500 sec, depending on whether CPB circuit is heparin-coated).
Aortic cannulation	Control MAP to ~70 mm Hg or SBP < 100 mm Hg for cannulation to ensure that the aortotomy does not extend. If needed, vasodilators may be used. The arterial line should be inspected for air bubbles.
Venous cannulation	Venous cannulation may be associated with atrial dysrhythmias. Blood loss can be excessive. Be prepared to infuse volume boluses, if necessary.
Pupils	Assess pupil symmetry for later comparison.

■ TRANSITION ONTO CPB

Stop ventilation	Commencing bypass is a dangerous period for the patient as a result of the many hemodynamic changes that occur. D/C ventilation once there is no pulmonary blood flow (as determined by nonpulsatile arterial line or pulmonary arterial catheter tracing).
Withdraw PA catheter	Withdraw PA catheter 4–5 cm to minimize pulmonary artery injury.
Stop infusions	Stop unnecessary iv infusions (typically inotropic agents) and reduce iv fluid infusions to KVO.
Anticoagulation	ACT should be rechecked after 5 min on CPB to ensure anticoagulation (ACT > 400 sec); this is typically performed by the perfusionist.

Oxygenation	Verify oxygenation by checking arterial inflow color and inline sensors and by ABG within 5 min of beginning CPB.
Flow	✓ for adequate venous drainage (CVP falls to a low level). Ensure adequate arterial inflow. Initial pressures may be very low due to acute hemodilution and release of inflammatory mediators, but usually will increase. View the descending thoracic aorta on TEE to verify flow (no aortic dissection).
Anesthesia	Anesthetics (e.g., fentanyl and midazolam) may be needed. A repeat dose (e.g., 50 mg of rocuronium) should be given to prevent movement or shivering (increases O_2 requirements).
Pupils	Assess pupils. Unilateral dilation may indicate arterial inflow into the innominate artery (unilateral carotid perfusion).

BYPASS PERIOD

Anticoagulation	ACT levels should be checked regularly (q20–30min) and kept at >400. Add heparin (5000–10,000 U), if needed. This is typically performed by the perfusionist.
Pressure/flow	There is controversy about safe flows and pressures. Generally, flows of 1.2–3 L/m^2/min are used, with pressures of 30–80 mm Hg. A MAP of 50–60 mm Hg is generally accepted for adequate cerebral perfusion while preventing excessive noncoronary blood flow.
Acid-base status	α-stat (ABG measured and interpreted at 37°, regardless of actual patient temperature) regulation of acid-base status is preferred because of maintenance of normal cerebral flow and autoregulation on CPB (see Table 6.1-1).
Hct/electrolytes	Hematocrit generally decreases with CPB initiation and maintenance and is typically acceptable in most patients unless Hct < 25. Hyperkalemia is common and should be corrected prior to weaning off CPB. A

	K^+ ~4.5 mEq/L is desirable. Blood glucose levels should be monitored q1h. Treat blood glucose >160 mg/dL with iv regular insulin (see Table 6.1-2).
UO	Keep UO > 1 mL/kg/h. If needed, mannitol and/or furosemide should be given (assuming pump flow is adequate).
Temperature	During bypass, temperature is usually maintained between 28 and 34°C per surgeon preference and surgical procedure, unless PCHA is employed.

Table 6.1-1 Difference Between α-Stat and pH-Stat Management During CPB

α-Stat (Temperature-Uncorrected)	pH Stat (Temperature-Corrected)
Plasma pH maintained at 7.4 (when measured at 37°C)	Plasma pH maintained at 7.4 at actual patient temperature
Arterial PCO_2 maintained at 40 mm Hg (when measured at 37°C)	Arterial PCO_2 maintained at 40 mm Hg at actual patient temperature. Requires addition of CO_2 to inspired gases
Relative respiratory alkalosis during hypothermia	Relative normocapnia during hypothermia
Cerebral autoregulation preserved—CBF maintained	Results in loss of cerebral autoregulation

Table 6.1-2 Continuous Insulin Infusion in Intraoperative Adult Cardiac Surgery Patients

1. Bolus and start infusion pump as follows:

Blood Glucose	Insulin Bolus (U)	Insulin Drip (U/h)
<150	0	0

150–200	3	2
200–250	5	4
>250	BG/50	5
2. Frequency of blood glucose determination: every 60 min intraoperatively		
3. Insulin titration as follows:		

Blood Glucose	**Action**
<120	Stop insulin; recheck blood glucose in 30 min. When blood glucose >150, restart with rate 50% of previous rate. If blood glucose <70 mg/dL, give D50W 25 g and recheck in 30 min
120–150	No action
150–175	Bolus dose 2 U, infusion 2 U/h
175–200	Bolus dose 3 U, infusion 3 U/h
200–225	Bolus dose 4 U, infusion 4 U/h
225–250	Bolus dose 5 U, infusion 5 U/h
>250	Bolus dose (BG/50), infusion 6 U/h

TERMINATION OF BYPASS

Rewarming	Prior to discontinuing CPB, the patient should have a core temperature of at least 36°C.
Anesthesia/relaxation	Patient awareness may be a problem during rewarming. Volatile agents ± benzodiazepine (e.g., midazolam 3–5 mg) ± a narcotic to prevent awareness. In addition, a muscle relaxant also should be given.
Acid-base/electrolytes/Hct	✓ electrolytes, acid-base status, and Hct. Correct acidosis. K^+: 4.5–5.5, normal ionized Ca^{++}, and Hct ≥ 24 should be ensured.
Air maneuvers	Air maneuvers (to remove intracardiac and intra-aortic air) are carried out when the heart is opened.

	Ventilation is commenced after there is pulmonary blood flow and will aid in the evacuation of air. Pleural fluid should be removed.

WEANING FROM BYPASS

Prior to weaning	Prior to weaning from CPB, the aortic cross-clamp should have been off to allow rewarming and reperfusion of the heart. Reperfusion duration is typically determined by baseline cardiac function, surgical procedure, and duration of cross-clamp time. Defibrillation is often needed, and epicardial pacing may be required. NSR or AV pacing is preferred. Vasoactive drugs should be available. After normothermia, adequate ventilation, stable cardiac rhythm and function, and reperfusion are established, bypass may be terminated. The heart is gradually volume-loaded (transfused from oxygenator) to adequate filling pressures and bypass flow slowly decreased over 15–45 sec until it is off. Return to bypass may be needed in the event of progressive cardiac distension or dysfunction. Assess CO and BP and adjust vascular resistance as necessary. Inotropic agents often are required at this stage.
Reversal of anticoagulation	After the patient is off CPB, anticoagulation must be reversed with protamine (1–1.3 mg/100 U heparin), first given as a 10-mg test dose and then administered slowly over 10–20 min because rapid administration can be associated with ↓ BP (treat with volume, calcium, α-agonists as necessary). Other less common reactions include pulmonary HTN and true allergic reactions. ✓ ACT to ensure that it has returned to control (90–130 sec).
BP management	Vasoactive support often is needed in the postbypass period; the need varies with surgical procedure, disease process, and underlying cardiac function. In general, SBP should be limited to 120 mm Hg to avoid stress on the aortotomy site and to minimize surgical

	bleeding.

COAGULATION AND CPB

General	Bleeding is common post-CPB and may be considerable. Both preop and intraop factors contribute to this. Knowledge of these factors and the tests involved will aid with the management of these patients (Table 6.1-3).
Preop factors	History of previous bleeding during surgery is important. Many drugs may contribute to bleeding: aspirin/NSAIDs/clopidogrel, anticoagulants (heparin, Coumadin), and fibrinolytic agents. These should be stopped preop, if possible, or their action reversed. Other pathologic processes (e.g., liver failure/congestion, renal failure, or hemophilia) also play a role. Preop testing is important and should include PT, PTT, platelet count, and bleeding time, as a minimum.
Intraop factors	CPB is associated with ↓ platelet count and function. Circulating clotting factors are decreased, and fibrinolysis occurs. Many surgical teams routinely use either aminocaproic acid (Amicar, 5- to10-g loading dose; 1 g/h, maintenance) or tranexamic acid (TXA, 10–30 mg/kg loading dose; 1–10 mg/kg/h maintenance) prophylactically during cardiac surgery to inhibit the fibrinolytic process.
Post-CPB bleeding	The common causes of post-CPB bleeding are surgical causes, inadequate heparin reversal, decreased platelet count and function, ↓ clotting factors, fibrinolysis, DIC, excessive blood pressure, and hypothermia. Surgical causes and inadequate heparin reversal (✓ ACT) should be ruled out. If no surgical cause is found and ACT is normal, platelets (1–2 plateletpheresis units) should be infused. PT, PTT, Plt count, and fibrinogen should be checked. TEG may prove very useful. (See Fig. 7.12-9 and Table 7.12-3 in liver/kidney transplantation.)

Table 6.1-3 A Treatment Plan for Excessive Bleeding After Cardiac Surgery		
Action	**Dosage**	**Indication**
Rule out surgical cause	—	Oozing at puncture sites and incision
More protamine	0.5–1 mg/kg	ACT > 150 sec or aPTT > 1.5 times control
Warm the patient	—	"Core" temperature <35°C
Apply positive end-expiratory pressure (PEEP)* Desmopressin	5–10 cm H_2O 0.3 mcg/kg iv	Prolonged bleeding time
Aminocaproic acid	50 mg/kg, then 25 mg/kg/h	Elevated D-dimer or teardrop-shaped TEG tracing
Tranexamic acid	10 mg/kg, then 1 mg/kg/h	Elevated D-dimer or teardrop-shaped TEG tracing
Platelet transfusion	1 U/10 kg	Platelet count <100,000/mm^3
Fresh frozen plasma	15 mL/kg	PT or aPTT > 1.5 times control
Cryoprecipitate	1 U/4 kg	Fibrinogen <1 g/L or 100 mg/dL

*PEEP is contraindicated in hypovolemia.

Suggested Readings

1. Bergeron EJ, Mosca MS, Aftab M, et al: Neuroprotection strategies in aortic surgery Cardiol Clin 2017; 35(3):453–65.
2. Conlon N, Grocott HP, Mackensen GB: Neuroprotection during cardiac surgery Expert Rev Cardiovasc Ther 2008; 6(4):503–20.
3. De Somer F: What is optimal flow and how to validate this? J Extra Corpor Technol 2007; 39(4):278–80.
4. Ferrando C, Soro M, Belda FJ: Protection strategies during cardiopulmonary bypass ventilation, anesthetics and oxygen. Curr Opin Anaesthesiol 2015; 28(1):73–80.
5. Kok WF, van Harten AE, Koene BM, et al: A pilot study of cerebral tissue

oxygenation and postoperative cognitive dysfunction among patients undergoing coronary artery bypass grafting randomised to surgery with or without cardiopulmonary bypass. Anaesthesia 2014; 69(6):613–22.
6. Serraino GF, Murphy GJ: Effects of cerebral near-infrared spectroscopy on the outcome of patients undergoing cardiac surgery: a systematic review of randomized trials. BMJ Open 2017; 7(9):e016613. [no clinical benefit]
7. Shaefi S, Mittel A, Klick J, et al: Vasoplegia after cardiovascular procedures-pathophysiology and targeted therapy. J Cardiothorac Vasc Anesth 2018; 32(2):1013–22.

CORONARY ARTERY BYPASS GRAFT SURGERY

SURGICAL CONSIDERATIONS

Description

Coronary artery bypass grafting (CABG) is the most frequently performed cardiac operation. Since the discovery that coronary thrombosis is the causative event for MI, many schemes have been devised to augment the restricted coronary blood flow, including collateral pericardial blood flow to epicardial arteries and implantation of the internal mammary artery (IMA) with unligated side branches into the LV muscle. With the discovery by Favalaro that saphenous veins can be anastomosed to the epicardial coronary arteries, a new era of myocardial revascularization began. Basically, the technique involves bypass to a narrowed or occluded epicardial coronary >1 mm in diameter with a small-diameter conduit (usually reversed saphenous vein or IMA) distal to the narrowed segment, with the proximal arterial inflow source being the ascending aorta. The IMA may be mobilized from the chest wall, leaving its proximal origin with the subclavian artery intact (pedicled graft), or the IMA may be transected and its proximal end anastomosed to the aorta or saphenous vein as a "free mammary graft." An "**all-arterial revascularization,**" using the left and right IMA and nondominant radial artery, is being utilized with increasing frequency, while other arterial conduits including the left gastroepiploic artery and superficial epigastric artery are being studied.

The heart is approached through a median sternotomy, with the patient typically supported on full **CPB** (see separate section on CPB). Although other operative strategies may be used—including full CPB support without cardiac arrest (on-pump beating heart) and no CPB support (off pump)—the most common regimen is for all distal (epicardial) anastomoses to be performed during a single period of aortic cross-clamping and cardiac arrest. During that period of induced asystole, myocardial protection is achieved by hypothermia and occasional reperfusion via antegrade or retrograde cardioplegia. Cardiac standstill and a bloodless field are necessary when small-diameter anastomoses are to be constructed, with no obstruction to flow, in a minimal amount of time. The cross-clamp is then removed and the heart allowed to resume beating. A partially occluding aortic cross-clamp can then be applied to allow construction of the proximal aortic anastomoses. After a sufficient period of resuscitation, the patient is weaned from CPB, and decannulation, heparin reversal, and chest closure are allowed to proceed as previously noted.

The choice of conduit depends on availability and durability. Historically, the saphenous vein was the first small vessel conduit with acceptable patency, but with prolonged experience it appears that 50% of vein grafts will be significantly diseased or occluded at 10 yr. The IMA appears to have superior long-term performance with ~90% 10-yr patency rates. Other arterial conduits, such as the left gastroepiploic artery, superficial epigastric artery, and the radial artery, are being studied for long- and short-term durability. Typical target arteries include the distal right coronary and its major terminal branch and the posterior descending artery. From the left circulation, the left anterior descending (LAD), with its diagonal and septal branches, is the most important, having been estimated to supply blood to 60% of the left ventricle.

The left circumflex coronary artery courses in the posterior atrioventricular groove and is not easily accessible for bypass, which is usually performed to its obtuse marginal or posterolateral branches.

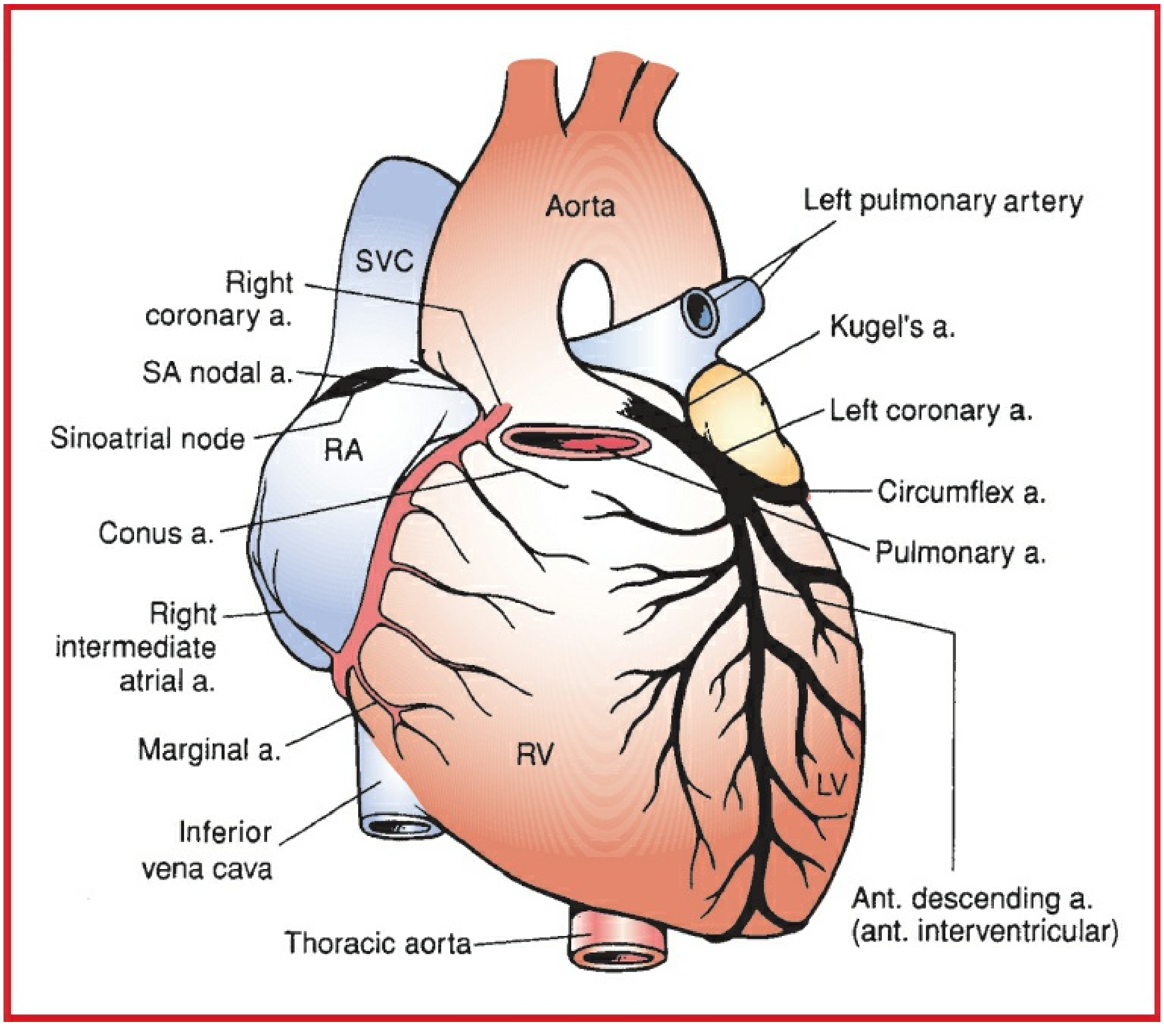

Figure 6.1-3 Coronary artery circulation—anterior view. (Reproduced with permission from Edwards EA, Malone PD, Collins JJ Jr: Operative Anatomy of the Thorax. Lea & Febiger, Philadelphia: 1972.)

In a randomized study on coronary artery surgery (CASS), coronary bypass was notec to be superior to medical management for relief of angina and to prolong life in patients with left main CAD and in those with three-vessel disease and impaired LV function. Other patients may receive bypass for intractable angina refractory to medical management.

Variant procedure or approaches

Port-access coronary artery revascularization; **off-pump,** on-pump without cardiac arrest; **and minimally invasive coronary artery bypass**

Usual preop diagnosis

CAD with class 3 or 4 angina (angina with minimal exertion or at rest)

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist**

should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. **See Appendix K: Enhanced Recovery After Surgery, p. K-1.**

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Median sternotomy with legs prepped for saphenous vein harvest
Special instrumentation	Complete hemodynamic monitoring; TEE
Unique considerations	CPB
Antibiotics	Cefazolin 2 g iv prior to incision and then 2 g q4h (3 g if BW >120 kg)
Surgical time	3.5–4.5 h
Closing considerations	Prevention of coagulopathy; maintenance of preload
EBL	500–600 mL
Postop care	ICU: 1–3 d, intubated 6–24 h
Mortality	2–4%
Morbidity	Cognitive decline in patients >60 yr, 26%; MI, 3–6%; pneumonia, 5%; CVA/stroke, 2–4% ↓ renal function
Pain score	7–8

■ PATIENT POPULATION CHARACTERISTICS	
Age range	60–80 yr (mean = 74 yr)
Male:Female	2:1
Incidence	Common
Etiology	Coronary atherosclerosis
Associated	LV/RV failure; pulmonary HTN; ischemic mitral

conditions	regurgitation; diabetes mellitus; obstructive pulmonary disease

ANESTHETIC CONSIDERATIONS

(Procedures covered: CABG; LV aneurysmectomy)

PREOPERATIVE

H&P and tests will divide these patients broadly into two groups: (a) high-risk, characterized by poor LV function (cardiac failure; EF < 40%; LVEDP > 18 mm Hg; C < 2.0 L/min/m^2; ventricular dyskinesia; three-vessel disease; occlusion of left main or left main equivalent; valvular disease; recent MI; ventricular aneurysm; VSD; MI in progress; and old age), and (b) low-risk, characterized by good LV function. The type of monitoring chosen will depend on the patient's group.

Respiratory	Hx of smoking or COPD—patient should be encouraged to stop smoking at least 2 wk before surgery. Treat COPD and optimize therapy before surgery. **Tests:** CXR; PFTs; as indicated by H&P
Cardiovascular	In the preop assessment, the following factors will affect patient management and surgical outcome: • Hx of angina (stable, unstable at rest, and precipitating factors) • Patient's exercise tolerance will provide a clue to LV function and surgical outcome. • The presence of CHF (Sx: SOB, PND, orthopnea, DOE, pulmonary edema, JVD, third-heart sound). • Recent (<6 mo) MI, dysrhythmias, HTN, and vascular disease (particularly carotid stenosis, aortic disease) • Valvular disease (particularly MR or AS) or the presence of a VSD or LV aneurysm may imply increased risk of periop complications. **Tests:** 12-lead ECG: ✓ ischemia (area involved), LVH, previous MI, dysrhythmias. Exercise stress testing: ✓ exercise tolerance, area of ischemia, maximal HR and BP before ischemia occurs,

	dysrhythmias. Thallium scan: ✓ component of reversible ischemia. ECHO (may be combined with stress ECHO): ✓ LV function, wall motion abnormalities, valvular disease, VSD. Cardiac catheterization: ✓ extent and location of disease, LV function, valvular pathology, VSD, LVEDP, LV aneurysm
Postinfarction VSD	The development of a postinfarction VSD is associated with high operative morbidity and mortality because of the difficulty in repairing the lesion due to friable tissue, difficulty in obtaining hemostasis, emergent nature of the condition, and possible pulmonary edema. These patients effectively have poor LV function and should be considered as high-risk patients. They often require support, including IABP, during induction, prebypass, and postbypass.
LV aneurysm	LV aneurysm is usually a late complication of infarction; however, it can occur early, when it usually is associated with cardiac rupture (and high mortality). These patients usually have poor myocardial function and should be anesthetized with full monitoring, including PA catheterization. Postbypass, the LV cavity is reduced in size and compliance. To ensure an adequate CO, maintain adequate preload, a higher than normal HR (A-V pacing if needed), and sinus rhythm, and consider the use of inotropes. LV aneurysms are often associated with dysrhythmias and may require cardiac mapping. Hemostasis is often difficult to obtain, and adequate iv access is a necessity. Treatment usually entails the use of blood products.
Emergency revascularization	Emergency revascularization occurs in the setting of acute MI, often with acute LV failure or after failed PTCA, where the patient may be stable, suffering from acute ischemia and hemodynamically unstable, or even in full cardiac arrest. Factors to consider in

	these cases are full stomach (and the need for rapid-sequence induction in the face of ischemia), prior use of fibrinolytic agents (with increased risk of hemorrhage), need for inotropes, antianginals, IABP, and dysrhythmias. These patients have a higher morbidity and mortality. In patients who have received fibrinolytic agents, consider postbypass use of antifibrinolytic agents (e.g., aminocaproic acid and tranexamic acid).
Neurologic	Previous stroke or Hx/Sx of carotid artery disease should be documented and evaluated.
Endocrine	Diabetes is common and periop control of blood glucose is important. **Tests:** Blood glucose
Renal	✓ baseline renal function, as CPB places these patients at risk for renal failure. **Tests:** Cr; BUN; electrolytes (particularly K^+)
Hematologic	Patients are often on aspirin or other antiplatelet therapy, which may lead to increased intraop hemorrhage. These agents should be stopped 7–10 d before surgery, if possible. Some patients may be on anticoagulants (usually heparin in the immediate preop period). Heparin should be stopped 6–8 h preop; however, in some patients, heparin infusion is continued into the OR. Other patients may have received thrombolytic agents that put them at increased risk for intraop hemorrhage. **Tests:** Consider Plt count, Plt function mapping, and PTT, if indicated.
Laboratory	Hb/Hct; other tests as indicated from H&P. T&C 2–4 U PRBCs
Premedication	Patients should be instructed to continue all medications (e.g., nitrates, β-blockers; Ca^{++} antagonists, antidysrhythmics, and antihypertensives with the exception of ACEi, ARBs, and diuretics on the day of surgery. Allaying anxiety may decrease

	the incidence of periop ischemia and help with the preinduction placement of lines. If necessary, preop sedation may include diazepam (10 mg po) or lorazepam (1–2 mg po) the night before and again 1–2 h before arrival in the OR with the addition of morphine (0.1 mg/kg im) and scopolamine (0.3 mg im). Most often, midazolam (1–5 mg iv) immediately prior to OR is used. Severely compromised patients will require less premedication.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: An arterial line should be inserted, using liberal amounts of local anesthetic, before induction. The presence of real-time BP monitoring can be critical in the care of these patients, especially during induction. Although it is helpful to have a CVP line before induction for preload monitoring and drug infusion, it is not essential. This line usually is inserted after the patient is intubated. If infused drugs are necessary before the CVP catheter is in place, they can be administered through a separate peripheral iv.

Induction	Generally, a moderate- to high-dose narcotic technique (e.g., fentanyl 100–500 mcg or sufentanil 10–50 mcg), supplemented by propofol (0.5–1.5 mg/kg), etomidate (0.1–0.3 mg/kg), or midazolam (50–350 mcg/kg), is appropriate. As with all cardiac cases, the speed of induction and total drug dose depend on the patient’s cardiac function and pathology. Muscle relaxation may be obtained using

	rocuronium (0.6–1 mg/kg) or vecuronium (possibility of bradycardia, especially if the patient is β-blocked). It is important to avoid the sympathetic response to laryngoscopy. The use of high-dose narcotics (see above), esmolol (100–500 mcg/kg over 1 min, followed by 40–100 mcg/kg/min infusion), SNP (0.5–3 mcg/kg/min), lidocaine (1–2 mg/kg), or a combination of these agents may decrease or ablate this response. NTG (0.5–2 mcg/kg/min) also may be used during induction if evidence of ischemia occurs.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.	
Maintenance	Usually, narcotic (total: fentanyl 10–20 mcg/kg or sufentanil 2–5 mcg/kg) with midazolam (2–5 mg) for amnesia. Patients with good LV function may benefit from the decreased myocardial O_2 demand associated with the use of volatile agents (2° ↓ contractility). N_2O is generally avoided. Propofol infusion may be used while rewarming and postbypass. Recommend the use of antifibrinolytics for all cases requiring CPB.	
Emergence	Transported to ICU, sedated, intubated, and ventilated. Extubate when able—often <6 h if lower-dose narcotic technique used (fast-track). Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 14 ga × 1–2 BSS at 6–8 mL/kg/h	See Goal-Directed Fluid Management, Appendix

	UO 0.5–1 mL/kg/h Warm all fluids. Humidify gases.	K, p. K-4.
Monitoring	Standard monitors	Standard monitors and an A-line are placed before induction.
	Arterial line	✓ BP in both arms. Ensure placement in contralateral arm if radial artery harvest to be performed for grafting.
	CVP or PA catheter	CVP (and/or PA line, if indicated) usually is placed after intubation. In the low-risk/good LV function group, CVP is adequate; in high-risk/poor LV function, a PA catheter is useful for hemodynamic monitoring, weaning from bypass, and vasoactive therapy. Some groups use routine PA catheterization for all patients undergoing CABG surgery.
	ECG	Five-lead monitoring using leads II and V_5 (or area most at risk for ischemia)
	TEE	Will reflect regional wall motion abnormalities, papillary muscle dysfunction, and MR
Myocardial O_2 balance	Supply—coronary blood flow: Perfusion pressure	The balance of myocardial O_2 supply vs. demand is important in

	(DBP-LVEDP) Diastolic filling time (HR) blood viscosity (optimal Hct = 30) Coronary vasoconstriction: Spasm $PaCO_2$ (hypocapnia →constriction) α-sympathetic activity Supply—O_2 delivery: O_2 sat Hct Oxyhemoglobin dissociation curve Demand—O_2 consumption: BP (afterload) Ventricular volume (preload) Wall thickness (↓ subendocardial perfusion) HR Contractility	the management of these patients. The goal of anesthesia is to ensure that this balance remains in equilibrium and that no ischemia occurs, or, if it does, that it is treated promptly. Those patients with poor LV function or complicated disease will benefit from maintenance of contractility (avoid volatile agents) and a high FiO_2, whereas those with good LV function may benefit from mild cardiac depression (↓ demand associated with the addition of low-dose volatile agents). Certain events are associated with **increased risk of intraop ischemia:** intubation, incision, sternotomy, cannulation, tachycardia, ↑ BP or ↓ BP, ventricular fibrillation or distension, inadequate cardioplegia, emboli, spasm, or inadequate revascularization. Care should be taken to avoid these complications and to ablate responses to stimuli.
Detection of ischemia	ECG	ST segment depression or elevation or a new T-wave alteration may

		suggest ischemia. Monitoring two leads—one lateral (e.g., V_5) and one inferior (e.g., II)—gives the best detection rate.
	PA catheter	Elevations of PCWP may be indicative of ischemia. A new V-wave on the PCWP trace is a better sign of possible ischemia (papillary muscle dysfunction).
	TEE	The appearance of a new regional wall motion abnormality is the most sensitive indicator of ischemia, but it requires constant monitoring. The transgastric LV midpapillary short-axis view allows for monitoring of walls supplied by all coronary arteries.
Treatment of ischemia	Caused by tachycardia: Esmolol (100–500 mcg/kg) ↑ anesthesia Verapamil (2.5–10 mg iv) Caused by ↑ BP: NTG (0.5–4 mcg/kg/min); ↑ anesthesia Caused by ↓ BP: Phenylephrine (0.2–	Although avoidance of ischemia is the goal, when it does occur, it should be treated aggressively. Treatment may include inotropic support (e.g., dopamine 1–5 mcg/kg/min, epinephrine 25–150 ng/kg/min, or milrinone 0.375–0.75 mcg/kg/min). If LV failure persists

	0.75 mcg/kg/min) ↑ preload Caused by ↓↓ HR: A-V pacing	despite other therapy, an IABP or VAD may be inserted. ECMO initiation may also be considered.
Positioning	✓ and pad pressure points ✓ eyes	

POSTOPERATIVE

Complications	Infarction Ischemia Tamponade Dysrhythmias Cardiac failure Coagulopathy Hemorrhage	Postop control of ischemia is important because hemodynamic instability may be associated with inadequate pain relief, awakening, and ventilation.
Multimodal pain management	Limit opioids	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast track extubation, early enteral nutrition, Foley catheter removal, and mobilization.
Tests	ECG CPK CXR	

	Electrolytes ABG Coag profiles

Suggested Readings

1. Barry AE, Chaney MA, London MJ. Anesthetic management during cardiopulmonar bypass: a systematic review. Anesth Analg. 2015 Apr;120(4):749–69.
2. Crumley S, Schraag S: The role of local anaesthetic techniques in ERAS protocol for thoracic surgery. J Thorac Dis 2018; 10(3):1998–2004.
3. Devgun JK, Gul S, Mohananey D, et al: Cerebrovascular events after cardiovascula procedures: risk factors, recognition, and prevention strategies. J Am Coll Cardiol 2018; 71(17):1910–20.
4. Munsch C: What cardiology trainees should know about coronary artery surgery—and coronary artery surgeons: ischaemic heart disease? Heart 2008; 94(2):230–6.
5. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery.J Cardiothorac Vasc Anesth 2018; 32(6):2760–70.
6. Rogers WJ, Coggin CJ, Green B, et al: 10-year follow-up of quality of life in patient randomized to receive medical treatment or coronary artery bypass graft surgery. Circulation 1990; 82(5):1647–58.
7. Shaefi S, Mittel A, Loberman D, et al: Off-pump versus on-pump coronary arter bypass grafting-a systematic review and analysis of clinical outcomes. J Cardiothorac Vasc Anesth 2019; 33(1):232–44.
8. Wong WT, Lai VK, Chee YE, et al: Fast-track cardiac care for adult cardiac surgica patients. Cochrane Database Syst Rev 2016; 9:CD003587.

LEFT VENTRICULAR ANEURYSMECTOMY

SURGICAL CONSIDERATIONS

Description

Extensive MI may → large areas of myocardial necrosis, with subsequent aneurysm formation. LV failure may ensue as a result of continuous LV dilation or mitral insufficiency 2° annular dilation or involvement of papillary muscles. Indications for this surgery include worsening CHF and increased dysrhythmias.

Typically, apical dilation with maintenance of basilar myocardial contractility allows for aneurysm resection and preservation of both myocardial contractility and chamber size to produce an adequate CO. Operation is commenced in the usual manner—by establishing CPB, cross-clamping the aorta, establishing myocardial protection, and then assessing the left ventricle. A thinned, dilated ventricular segment with full-thickness scar formation can be resected and ventricular continuity restored with improvement of ventricular geometry and myocardial energy demands. Coronary bypass can be performed during this same period. Then, air is removed from the left side of the heart, the cross-clamp is removed, and coronary perfusion is reestablished. After a sufficient period of resuscitation and return of vigorous contractility, bypass is D/C'd, not infrequently with the assistance of intra-aortic balloon pump (IABP) to augment forward output. Decannulation, protamine administration, and closure proceed as described in CPB.

Usual preop diagnosis

LV aneurysm with CHF

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. See Appendix K, p. K-1.**

SUMMARY OF PROCEDURE	
Position	Supine
Incision	Median sternotomy, ± leg incision for saphenous vein harvest if coronary bypass is planned
Unique considerations	Preparations (ECG leads) should be made for pre- or postop IABP or LV assist device (LVAD).
Antibiotics	Cefazolin 2 g iv prior to incision and then 2 g q4h (3 g if BW > 120 kg)
Surgical time	Aortic cross-clamp: 40–100 min CPB: 70–130 min Total: 3–4 h
EBL	300–400 mL
Postop care	ICU × 1–3 d, intubated 6–24 h; usually requires inotropic support ± mechanical support (LVAD, IABP)

Mortality	5–7%
Morbidity	Overall: 8–10% Requirement for IABP: 10% Respiratory insufficiency: 5% CVA: 2–3%
Pain score	7–10

■ PATIENT POPULATION CHARACTERISTICS	
Age range	50–70 yr
Male:Female	3:1
Incidence	Uncommon
Etiology	Usually the end result of MI 2° CAD
Associated conditions	Mitral insufficiency; pulmonary HTN; CAD

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations Following Coronary Artery Bypass Graft Surgery.

Suggested Reading

1. Ruzza A, Czer LSC, Arabia F, et al: Left ventricular reconstruction for postinfarction left ventricular aneurysm: review of surgical techniques. Tex Heart Inst J 2017; 44(5):326–35.

AORTIC VALVE REPLACEMENT

SURGICAL CONSIDERATIONS

Description

Disease of the aortic valve may present as valvular stenosis, insufficiency, or a combination of the two. Valvular disease may occur as a result of rheumatic disease, but also may occur 2° to calcific degeneration (aortic sclerosis) in the elderly. Congenitally bicuspid valves and endocarditis account for most of the remainder. Repair of the aortic valve is occasionally possible, but most conditions require valve replacement. The three most commonly used **prostheses** are porcine bioprosthesis, especially in the older patient; mechanical prostheses with the necessity for lifelong anticoagulation; and cryopreserved homografts, which, unfortunately, are expensive and in short supply. The operation, on full CPB, usually is performed through a median sternotomy. After routine venous and aortic cannulation, the patient is taken onto full CPB. Left heart drainage may be accomplished via a vent placed in the pulmonary artery, left atrium, or left ventricle. Because LV hypertrophy is a common sequela of aortic stenosis, myocardial protection is of utmost importance. Most centers favor hyperkalemic, hypothermic cardioplegic arrest, augmented by topical cooling, using either a continuous infusion of cold saline into the pericardial well or a cooling jacket. Cardioplegic administration can be achieved either antegrade into the coronary ostia or retrograde via the coronary sinus. Myocardial temperature is monitored continuously.

After the heart is arrested, the aorta is opened to expose the aortic valve. Continuous insufflation of the operative field with CO_2 reduces the amount of dislodged or trapped air. The rheumatic, stenotic valve, including aortic annulus, is frequently heavily calcified, and all Ca^{++} must be debrided to allow the prosthetic valve to be securely seated. This is frequently a tedious and time-consuming procedure, but one that then allows the remainder of the procedure to proceed in a timely fashion. After excision of the valve leaflets and debridement of the annulus, ensuring that no particulate debris embolizes into the ventricle or coronary arteries, the annulus is measured to ensure a proper match between prosthetic valve and annulus, and an appropriate valve prosthesis is selected. Interrupted sutures are placed through the annulus for its entire circumference and then passed through the sewing ring of the prosthesis. The prosthesis is lowered into the annulus and securely tied in place. Proper sizing and positioning are mandatory to prevent paravalvular leaks or impingement on the coronary ostia. Systemic rewarming is initiated during the final stages of the valve implantation, and the LV is allowed to fill during aortic closure. With the patient in the head-down position, all remaining air is vented from the left heart and aorta, and the cross-clamp is removed to allow resumption of myocardial perfusion. The heart is allowed to recover from this period of ischemia, and after sufficient resuscitation, with continuous venting of air from the aorta, the patient is weaned from CPB. Vasodilators are frequently utilized, as there may be excessive vasospasm present in both the coronary and pulmonary circulations after hypothermia. Decannulation and heparin reversal with protamine are then accomplished in the routine manner.

Alternative procedures: Percutaneous, transcatheter procedures for aortic valve repair/replacement are discussed below and on p. 403.

Usual preop diagnosis

Severe AS with syncope, chest pain, or CHF; aortic insufficiency with CHF

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. See** **Appendix K, p. K-1.**

SUMMARY OF PROCEDURES

	Valve Replacement with Prosthesis	Homograft Valve Replacement
Position	Supine	⇐
Incision	Median sternotomy	⇐
Special instrumentation	TEE; hemodynamic monitoring; CPB	TEE or surface ECHO
Unique considerations	ECHO assessment of valve function and regional wall motion intraop	ECHO assessment of annular size and intraop evaluation of valve function after implantation
Antibiotics	Cefazolin 2 g iv prior to incision and 2 g q4h	⇐
Surgical time	Aortic cross-clamp: 45 min CPB: 90 min Total: 3 h	60 min 105 min ⇐
EBL	300–400 mL	⇐
Postop care	ICU: 1–3 d, intubated 6–24 h; hypertrophied, noncompliant LV requiring high preload	⇐
Mortality	5–8%	⇐
Morbidity	Pneumonia: 5–10% Neurologic sequela:	⇐

	Transient: 3–7% CVA: 1–2% Permanent: 1–2% Infection: 1%	⇐
Pain score	7–10	7–10

PATIENT POPULATION CHARACTERISTICS

Age range	Bicuspid valves, 50–60 yr; rheumatic, 55–80 yr (mean = 58 ± 13 yr)
Male:Female	3:1
Incidence	70–100 cases/yr in tertiary care center
Etiology	Postrheumatic (majority of patients worldwide); aortic sclerosis (most common in the United States); progressive stenosis of a bicuspid aortic valve; endocarditis
Associated conditions	Poststenotic dilation of the ascending aorta (may require separate surgical attention); rheumatic mitral valvular involvement; CAD; CHF

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Respiratory	Respiratory compromise may occur 2° pulmonary congestion (LV failure) and pleural effusion. An effusion, if significant, should be drained prior to surgery, as it may impair oxygenation and, with IPPV, may impair venous return →↓ CO + ↓ myocardial perfusion. **Test:** CXR
Cardiovascular	**Aortic stenosis (AS):** Sx are those of angina pectoris (if at rest may indicate concurrent CAD), syncope, and CHF (indicates severe disease with 2-yr life expectancy). The ejection murmur of AS is best heard at the 2nd right interspace. ECG shows LVH. Important points in the preop investigations include:

- Aortic orifice size: severe AS = 0.7–0.9 cm^2; critical AS ≤ 0.5 cm^2 (normal = 2.6–3.5 cm^2)
- Peak aortic valvular gradient: severe ≥ 70 mm Hg
- Ejection fraction (EF): ↓ EF indicates evidence of LV failure (normal ≥ 0.6).
- Coronary angiography: associated CAD often demonstrated. AS → ↑ LV work (↑ pressure load) → LV concentric hypertrophy → ↑ LV diastolic dysfunction + ↑ risk for ischemia (MVO_2 + O_2 supply).
- Reliance on atrial "kick:" important to maintain NSR. • Sensitivity to changes in SVR: ↓ SVR → ↓↓ BP → ↓ myocardial perfusion + ↓ CO → ↓↓↓ BP.
- Sensitivity to volume changes: hypovolemia → ↓ preload → ↓ ↓ CO.
- Sensitivity to rate changes: tachycardia → ↓ ejection time → ↓ myocardial perfusion.
- LV wall tension + ↑ duration of systole → ↑ MVO_2.
- ↑ LVEDP + ↑ wall thickness and tension + ↓ diastolic aortic pressure → ↓ O_2 supply.

Aortic regurgitation (AR): Sx include DOE, orthopnea, PND, palpitations, and (less frequently) angina. Exercise tolerance may remain reasonably good, even with severe chronic AR. Acute AR is very poorly tolerated. The pandiastolic murmur of AR is loudest over the sternum and left lower sternal border. AR → chronic LV volume overload → LV eccentric hypertrophy → massive cardiomegaly → LV failure (CHF) → ↑ LVEDP → ↑ PA pressure, and pulmonary congestion.

- Possibility of ischemia: ↑ MVO_2 and ↓ supply (↓ diastolic pressure, ↑ HR).
- Sensitivity to rate changes: ↓ HR → ↑ AR + ↓ CO.
- Sensitivity to changes in SVR: ↑ SVR → ↑ regurgitation + ↓ CO.

Tests: ECG: ✓ hypertrophy, LV strain, ischemia, rhythm. ECHO: ✓ LV function, valve area, regurgitant fraction. Angiography: ✓ LV function, right heart pressure, CAD, valve area, regurgitant

	fraction
Hepatic	CHF may result in passive liver congestion with ↓ liver function and possible coagulopathy. **Tests:** LFTs; PT; PTT
Neurologic	Syncopal episodes may have resulted in neurologic deficits. These should be well documented.
Renal	Prerenal failure often is associated with AR 2° ↓ CO. **Tests:** BUN; Cr, creatinine clearance, if indicated; electrolytes
Hematologic	**Tests:** Hb/Hct; clotting profile to investigate abnormalities. T&C for 4 U PRBCs
Laboratory	✓ digitalis level and electrolytes; other tests as indicated from H&P
Premedication	Beware of oversedation in patients with AS where ↓ BP could be detrimental. Light premedication with an anxiolytic is usually sufficient. Digitalis and diuretics should be continued.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: A preinduction arterial line should be inserted, using liberal amounts of local anesthetic. The presence of real-time BP monitoring can be critical in the care of these patients, especially during induction. Although it is helpful to have a CVP line before induction for preload monitoring and drug infusion, it is not essential. This line usually is inserted after the patient is intubated. If infused drugs are necessary before the CVP catheter is in place, they can be administered through a separate peripheral iv.

Induction	**AS:** Typically, O_2 with moderate- to high-dose narcotic (e.g., fentanyl 5–10 mcg/kg). Avoid sufentanil in AS because of ↓ BP. Etomidate (0.1–0.3 mg/kg) and midazolam (50–350 mcg/kg) may be used to supplement the above. Paralysis with vecuronium or rocuronium. Induction is a critical period. CPB and surgeons should be available and ready to proceed. Danger is hypotension with a cycle of ischemia, further hypotension, and more ischemia. Hypotension should be treated aggressively with fluid and α-adrenergic agonists (phenylephrine 50–100 mcg iv bolus, infusion 0.1–0.75 mcg/kg/min). Critical AS patients may benefit from the use of phenylephrine as an infusion during induction and prebypass. The avoidance of hypotension is even more critical in the presence of CAD. Drugs causing tachycardia should be avoided. Atrial fibrillation (AF) or SVT should be treated with cardioversion. β-Blockers and other negative inotropes are generally contraindicated. Ventricular irritability should be treated early as ventricular fibrillation may be refractory to defibrillation. Beware of vasodilators, including NTG. **AR:** Again, O_2 with moderate- to high-dose narcotic (e.g., fentanyl 5–10 mcg/kg or sufentanil 2.5–10 mcg/kg). Etomidate (0.1–0.3 mg/kg) or benzodiazepines (e.g., midazolam 50–350 mcg/kg) may be used to supplement the opiates. Muscle relaxation with rocuronium (0.6 mg/kg) or vecuronium (0.1 mg/kg). These patients benefit from fluid augmentation, high normal HR (90 bpm) with afterload reduction (e.g., SNP 0.25–2 mcg/kg/min) to improve forward flow.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.
Maintenance	Narcotic (e.g., total fentanyl 10–20 mcg/kg). Low-dose

<table>
<tr><td></td><td colspan="2">volatile agent/air/O_2. Relaxant. (See Anesthetic Considerations for CPB.) Recommend use of antifibrinolytics for all cases requiring CPB. The following table summarizes the goals of intraop management:
<table>
<tr><th>Parameter</th><th>AS</th><th>AR</th></tr>
<tr><td>LV preload</td><td>↑</td><td>Normal-to-↑</td></tr>
<tr><td>HR</td><td>Normal-to-slow ↓</td><td>Modest ↑</td></tr>
<tr><td>Rhythm</td><td>NSR</td><td>NSR</td></tr>
<tr><td>Contractility</td><td>Maintain</td><td>Maintain</td></tr>
<tr><td>SVR</td><td>Modest ↑</td><td>↓</td></tr>
<tr><td>PVR</td><td>Maintain</td><td>Maintain</td></tr>
</table></td></tr>
<tr><td>Postbypass</td><td colspan="2">AS: Postbypass, patients may be hyperdynamic and require vasodilators for HTN, although inotropes also may be needed. Because of the hypertrophied, noncompliant ventricle, filling pressure may be higher than normally required.
AR: In the immediate postbypass period, patients with AR may require inotropic support (e.g., dopamine 3–10 mcg/kg/min, dobutamine 5–10 mcg/kg/min, epinephrine 25–100 ng/kg/min). Maintain LV filling. Other supportive measures (e.g., IABP) may be necessary.</td></tr>
<tr><td>Emergence</td><td colspan="2">Transport to ICU sedated, intubated (6–24 h), and ventilated. Early extubation may be possible if a lower-dose narcotic technique is used (fast-track). Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.</td></tr>
<tr><td>Blood and fluid requirements</td><td>IV: 14 ga × 1–2
BSS at 6–8 mL/kg/h
UO 0.5–1 mL/kg/h
Warm all fluids.
Humidify gases.
T&C 2–4 U blood</td><td>See Goal-Directed Fluid Management, Appendix K, p. K-4.</td></tr>
<tr><td>Monitoring</td><td>Standard monitors</td><td>Standard monitors and arterial line</td></tr>
</table>

	Arterial line CVP line PA catheter ECG TEE Urinary catheter	should be placed before induction. CVP (and/or PA line, if indicated) usually is placed after intubation. CVP may underestimate left-side pressures. In acute AR, PCWP may underestimate true LVEDP. V_5 lead should be monitored for ischemia. TEE may be useful to estimate LV filling and regional wall motion abnormalities.
Positioning	✓ and pad pressure points ✓ eyes	

POSTOPERATIVE

Complications	Hemorrhage Tamponade Cardiac failure Dysrhythmias Ischemia	Inotropic and vasodilator therapy usually is continued into the postop period and then weaned.
Multimodal pain management	Limit opioids	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.

Tests	ECG CXR Electrolytes Coag profile Hct

MINIMALLY INVASIVE AORTIC VALVE PROCEDURES (ALSO SEE P. 440)

SURGICAL CONSIDERATIONS

Description

Perhaps, the most significant advance in valve replacement surgery was the development of transcatheter aortic valve replacement/transcatheter aortic valve implantation (TAVR/TAVI), in which a biologic valve is attached to a metallic stent. The replacement valve can then be delivered transarterially into the aortic root, where the stenotic aortic valve can be balloon dilated, and the TAVR valve positioned carefully within the aortic annulus. The stent is then expanded, deploying the valve in proper position within the stenotic native valve. Initially utilized for the elderly, frail, and high surgical risk patients, TAVR/TAVI is being widely adopted for lower-risk patients, as the avoidance of a median sternotomy is attractive to both cardiologists and patients.

These procedures are often performed in hybrid surgical suites suitable for performing an open surgical aortic valve replacement (SAVR), which may become necessary for the TAVR/TAVI gone awry. Initially, TAVR procedures were performed under GETA, with full cardiac monitoring including arterial line, PA catheter and TEE monitoring. Presently, many groups perform TAVR/TAVI procedures under conscious sedation, with percutaneous femoral access. Arterial access is usually achieved via a femoral cutdown, although other access sites have included iliac, subclavian, and carotid arteries, as well as the ascending aorta and the left ventricular apex. Adequate vascular access is ascertained by CT angiography.

For procedural and anesthetic considerations, see p. 444.

Suggested Readings

1. Balkhy HH, Lewis CTP, Kitahara H: Robot-assisted aortic valve surgery: state of th

art and challenges for the future. Int J Med Robot 2018; 14(4):e1913.
2. Bashir M, Harky A, Bleetman D, et al: Aortic valve replacement: are we spoiled for choice? Semin Thorac Cardiovasc Surg 2017; 29(3):265–72.
3. Kanwar A, Thaden JJ, Nkomo VT: Management of patients with aortic valve stenosis. Mayo Clin Proc 2018; 93(4):488–508.
4. Murtuza B, Pepper JR, Stanbridge RD, et al: Minimal access aortic valve replacement: is it worth it? Ann Thorac Surg 2008; 85(3):1121–31.
5. Neuburger PJ, Saric M, Huang C, et al: A practical approach to managing transcatheter aortic valve replacement with sedation. Semin Cardiothorac Vasc Anesth 2016; 20(2):147–57.
6. Szabo TA, Toole JM, Payne KJ, et al: Management of aortic valve bypass surgery. Semin Cardiothorac Vasc Anesth 2012; 16(1):52–8.
7. Tjang YS, van Hees Y, Körfer R, et al: Predictors of mortality after aortic valve replacement. Eur J Cardiothorac Surg 2007; 32(3):469–74.

MITRAL VALVE REPAIR OR REPLACEMENT

SURGICAL CONSIDERATIONS

Description

Mitral valve repair or replacement is utilized typically for the correction of postrheumatic mitral valvular stenosis or insufficiency, as well as mitral valve prolapse, degenerative mitral insufficiency, or repair after endocarditis. For mitral regurgitation (MR) 2° posterior leaflet abnormalities (myxomatous degeneration, torn chordae) or pure annular dilation, most valves can be repaired. For severe rheumatic calcific mitral stenosis (MS), mitral valve replacement with preservation of subannular structures may be necessary.

The technique of mitral valve repair or replacement is similar regardless of the mitral valve pathology. After aortic and bicaval venous cannulation, CPB is established, and the left heart is vented through the PA. After cross-clamping of the aorta, diastolic arrest is accomplished with cardioplegia administered via the aortic root, augmented with topical cooling with either continuous pericardial saline infusion or a cooling jacket. Exposure is accomplished via a vertical incision in the left atrium just posterior to atrial septum. If the left atrium is not large enough, access may be gained via an incision in the right atrium and then incising the atrial septum, with caval snares in place to prevent air entry into the venous cannulae.

After suitable exposure, the atrium, atrial appendage, and mitral valve are carefully inspected, and a decision is made to repair or replace the valve. Repair of regurgitant valves caused primarily by posterior leaflet problems is usually possible. Similarly, annular dilation 2° LV enlargement is also usually possible by means of a ring annuloplasty. Regurgitation 2° anterior leaflet abnormalities is more problematic, requiring significant expertise, and may be less durable. Preop and postop evaluations are greatly facilitated by the use of TEE. Valve replacement can be performed after excising the valve leaflets, or the leaflets may be preserved in an attempt to maintain the benefits of the subvalvular apparatus to global ventricular performance. After appropriate excision and debridement, the annulus is rimmed with interrupted sutures passed through the sewing ring of the valve prosthesis. The prosthesis is carefully positioned, and the sutures are tied. After filling both the LV and atrium with blood, the atrium is closed, air is evacuated from the left heart, and the cross-clamp is removed to allow coronary perfusion. After a satisfactory period of resuscitation, and after all air has been evacuated from the circulation, CPB may be D/C'd, usually with the assistance of vasodilator agents. TEE may be utilized at this stage to assess adequacy of mitral valve repair.

Variant procedure or approaches

Mitral valve repair may also be accomplished via a right anterior thoracotomy approach, with the patient positioned right side up in slight lateral decubitus position. Peripheral (femoral artery-femoral vein +/− internal jugular vein) or central (femoral vein-aorta) bypass is used, and operation may be performed during ventricular fibrillation or without aortic clamping and cardioplegia. Single lung anesthesia is necessary, either with a double-lumen endotracheal tube or with a right-sided bronchial blocker. Cannulation of the coronary sinus may be requested, and the operative field will be flooded with CO_2. TEE is extensively used for preop diagnosis, and for postop monitoring of the surgical result. **Mitral commissurotomy** may be done as an endovascular procedure (e.g., during pregnancy).

Percutaneous repair of the regurgitant mitral valve (MitraClip™) has been approved as an alternative to open surgery (see p. 403).

Usual preop diagnosis

Class 3 or 4 CHF 2° mitral insufficiency or mitral stenosis

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of**

nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. **See Appendix K, p. K-1.**

SUMMARY OF PROCEDURES

	Mitral Valve Replacement/Repair	Closed Commissurotomy
Position	Supine	Right lateral decubitus
Incision	Median sternotomy	Left lateral thoracotomy
Special instrumentation	TEE and epicardial ECHO to assess valve function and regional wall motion; CPB	Performed without CPB
Unique considerations	Special loading conditions may be used to estimate the amount of MR, which is frequently afterload-dependent.	⇐
Antibiotics	Cefazolin 2 g iv prior to incision and then 2 g q4h	⇐
Surgical time	Aortic cross-clamp: 45–150 min CPB: 90–200 min	Total: 2 h Total: 3–4 h
EBL	300–400 mL	200–400 mL
Postop care	ICU × 1–3 d, intubated 6–24 h; vasodilator therapy for reversal of pulmonary HTN and afterload reduction	⇐
Mortality	5–8%	<5%
Morbidity	Pneumonia: 10–15% CNS complications: Transient neurologic dysfunction: 5–10%	⇐ ⇐ ⇐

	CVA: 1–2% Infection: <1%	⇐
Pain score	7–10	7–10

PATIENT POPULATION CHARACTERISTICS

Age range	40–75 yr	20–40 yr
Male:Female	1:1	Almost exclusively pregnant females
Etiology	Postrheumatic: majority of cases; increased aging; myxomatous degeneration; ischemic etiologies (increasing)	⇐
Associated conditions	Aortic valvular involvement with rheumatic disease, CAD, pulmonary HTN, and tricuspid regurgitation	⇐

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Respiratory	Pulmonary congestion, edema, and pleural effusions may be present, with an overall restrictive lung pattern. Pleural effusions should be drained before surgery if significant (↓ oxygenation, ↓ venous return with IPPV). ↑ LA volume may compress the left recurrent laryngeal nerve → left vocal cord paralysis (Ortner syndrome). **Tests:** CXR; PFTs, if indicated
Cardiovascular	**Mitral stenosis (MS):** Sx include exertional dyspnea and fatigue, progressing to pulmonary edema, atrial fibrillation (AF), and hemoptysis. Embolic events occur in 15% of patients. The opening snap and diastolic murmur of MS are best heard between the apex and left sternal border. The

ECG may show AF or wide-notched P-waves. It is important to grade the severity of MS by symptomatology and valve area: mild, 1.5–2 cm^2; moderate, 1–1.5 cm^2; and severe, <1.0 cm^2 (nl = 4–6 cm^2). AF is often the factor precipitating deterioration in these patients. Because of the decreased LV filling, these patients are sensitive to:

- Loss of atrial "kick:" maintain NSR.
- Volume changes: keep full.
- Rate changes: avoid ↑ HR → ↓diastolic filling time → ↓ CO.

Pathophysiology:

- MS → ↓ LVH filling → ↓ CO.
- MS → ↑ LA pressure →pulmonary edema + ↑ PA pressure → ↑ PVR → RV failure + TR + left shift of iv septum → ↓ CO → ↑ LA pressure → ↑ LA volume → AF and thrombi.

Mitral regurgitation (MR): May be acute (MI/endocarditis) or chronic (often associated with MS). Chronic Sx include palpitations, DOE, PND, fatigue, and orthopnea. Acute MR may → sudden ↓↓ CO and pulmonary edema. MR can be classified as mild (<30% RF), moderate (30–60% RF), or severe (>60% RF). The pansystolic murmur is loudest at the apex. The ECG may show LA and LV overload. MR patients are sensitive to:

- Changes in SVR: ↑ SVR → ↑ RF → ↓ CO + ↓ BP.
  ↓ SVR → ↓ RF → ↑ CO + ↑ BP.
- Change in HR: ↓ HR →acute LA volume overload.
  Mild ↑ HR→ ↑ CO.

Pathophysiology:

- Acute MR → ↑ ↑ LA pressure → ↓ CO + ↓ BP → ↑ HR → ↑ contractility → ↑ O_2 demand.
  → ↑ LV diastolic volume → ↑ LVEDP →↑ O_2 demand. → Pulmonary edema.
- Chronic MR → ↑ LA volume + AF → pulmonary edema → RV failure.

	→ LV volume overload → LVH → ↓ CO + LV failure. **Tests:** ECG: ✓ LVH, left and right atrial enlargement, rhythm. ECHO: ✓ RF, valve pressure gradients, LV function, valve pressure gradient and area.
Neurologic	The large left atrium and AF may result in thrombus formation, with the possibility of embolism. Neurologic deficits should be documented.
Gastrointestinal	Hepatic congestion may → decreased function, which may be reflected in coagulation problems. **Tests:** Consider LFTs; PT; PTT.
Renal	↓ CO may lead to renal failure. **Tests:** BUN; Cr; electrolytes
Hematologic	Because of the potential for thromboembolism, these patients may be on anticoagulants, which may be given up to the day before surgery. In the case of Coumadin, anticoagulant effects can be reversed by the use of FFP and vitamin K. **Tests:** Consider PT; PTT.
Laboratory	T&C for 2–4 U PRBCs. Hb/Hct; coagulation profile; other tests as indicated from H&P
Premedication	Premedication with an anxiolytic (e.g., lorazepam 1–2 mg po, midazolam 0.05–0.2 mg/kg im) or an analgesic (e.g., morphine 0.1–0.2 mg/kg) or a combination of these agents may be used, depending on patient status.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: An arterial line should be inserted, using liberal amounts of local anesthetic, before induction. The presence of real-time BP monitoring can be critical in the care of these patients, especially during induction. Although it is helpful to have a CVP line before induction for preload monitoring and drug infusion, it is not essential. This line usually is inserted after the patient is intubated. If infused drugs are necessary before the CVP catheter is in place, they can be administered through a separate peripheral iv.

Induction	Typically, moderate- to high-dose narcotic (fentanyl 5–10 mcg/kg or sufentanil 2.5–20 mcg/kg), supplemental midazolam (50–350 mcg/kg), or etomidate (0.1–0.3 mg/kg). Use rocuronium or vecuronium **MS:** ↓ BP should be treated with fluid, but beware of precipitating pulmonary edema. Occasionally, phenylephrine may be needed to maintain SVR. Tachycardia should be avoided; if it occurs, it should be treated (↑ anesthesia, esmolol if ventricular function is preserved). Sinus rhythm should be maintained. New-onset atrial flutter/AF should be treated with defibrillation, ideally after LA thrombus is ruled out by TEE. Avoid factors that may increase PVR (N_2O, acidosis, hypoxia, hypercarbia). **MR:** Maintain or augment preload, depending on response to fluid load. Inotropic agents may be useful to maintain contractility. Afterload reduction will improve forward flow. Generally, SNP (0.5–4 mcg/kg/min) is used, but NTG (0.5–4 mcg/kg/min) may be more appropriate in patients with ischemia-induced regurgitation. Avoid ↑ PVR caused by N_2O, acidosis, hypoxia, or hypercarbia. IABP may be useful periop in patients with acute MR 2° MI (↓ afterload, ↑ coronary perfusion pressure).
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine

	(0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.
Maintenance	Narcotic (total = fentanyl 10–20 mcg/kg, or sufentanil 5–20 mcg/kg) with benzodiazepine (e.g., midazolam 50–350 mcg/kg) for amnesia; low-dose isoflurane; O_2/air. Recommend use of antifibrinolytics for all cases requiring CPB. The following table summarizes the goals of intraop management:

Parameter	MS	MR
LV preload	Normal-to-↑	Normal-to-↑
HR	↓	↑
Rhythm	NSR	NSR
Contractility	Maintain	Maintain
SVR	Normal	↓
PVR	Avoid ↑	Avoid ↑

Postbypass	Although patients with MS generally do well after valve replacement, inotropes (e.g., dopamine, dobutamine) occasionally are necessary. This is especially true late in the disease when cardiomyopathy may be present.	
Emergence	Transport to ICU, intubated and ventilated 6–24 h. Early extubation is possible if lower-dose narcotic technique is used (fast-track). Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 14 ga × 1–2 BSS at 6–8 mL/kg/h Maintain UO 0.5–1 mL/kg/h. Warm all fluids. Humidify gases.	See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Arterial line	Standard monitors and an A-line are placed before induction. CVP (and/or

	CVP line PA catheter Urinary catheter	PA line, if indicated) usually is placed after intubation. Care should be exercised with PA catheter insertion because the dilated PA may be susceptible to rupture with the catheter. In MS, PCWP may overestimate the LV filling pressure due to stenosis.
	TEE	TEE may help in volume management, regional wall motion abnormalities (postacute MI), and assessing the success of mitral valve repair. Occasionally, the use of a valve ring during repair may cause systolic anterior motion of the valve leaflet with resultant regurgitation, which may be seen on TEE.
Positioning	✓ and pad pressure points ✓ eyes	

POSTOPERATIVE

Complications	Hemorrhage Tamponade Cardiac failure Dysrhythmias Conduction defects Atrioventricular disruption	Inotropic support or vasodilator therapy may be needed. IABP for mitral incompetence, especially in the presence of acute MR 2° infarction. Systolic anterior motion (SAM) of the mitral valve after MV repair should be excluded by TEE prior to completion of the case.
Multimodal pain management	Limit opioids	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with

		regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.
Tests	ECG CXR ABG Coag profile Electrolytes	

Suggested Readings

1. Barker CM: Percutaneous repair of mitral regurgitation. Methodist Debakey Cardiovasc J 2017; 13(3):114–9.
2. Calvinho P, Antunes M: Current surgical management of mitral regurgitation. Expert Rev Cardiovasc Ther 2008; 6(4):481–90.
3. Crumley S, Schraag S: The role of local anaesthetic techniques in ERAS protocol for thoracic surgery. J Thorac Dis 2018; 10(3):1998–2004.
4. Dunning J, Versteegh M, Fabbri A, et al: Guideline on antiplatelet and anticoagulation management in cardiac surgery. Eur J Cardiothorac Surg 2008; 34(1):73–92.
5. Greco E, Zaballos JM, Alvarez L, et al: Video-assisted mitral surgery through a micro-access: a safe and reliable reality in the current era. J Heart Valve Dis 2008; 17(1):48–53.
6. Nagendran J, Catrip J, Losenno KL, et al: Minimally invasive mitral repair surgery why does controversy still persist? Expert Rev Cardiovasc Ther 2017; 15(1):15–24.
7. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery. J Cardiothorac Vasc Anesth 2018; 32(6):2760–70.
8. Sharma A, Agrawal S, Goel S, et al: Surgical treatment of ischemic mitral regurgitation: valve repair versus replacement. Curr Cardiol Rep 2017; 19(1):3.
9. Telila T, Mohamed E, Jacobson KM: Endovascular therapy for rheumatic mitral and

aortic valve disease: review article. Curr Treat Options Cardiovasc Med 2018; 20(7):59.
). Tinsley MM, Martin DE: Anesthetic management for the surgical treatment o valvular heart disease. In: Hensley FA, Martin DE, Gravlee GP, eds.A Practical Approach to Cardiac Anesthesia, 5th edition. Lippincott Williams & Wilkins, Philadelphia: 2013, 319–58.
l. van der Merwe J, Casselman F: Mitral valve replacement-current and future perspectives. Open J Cardiovasc Surg 2017; 9:1179065217719023.

TRICUSPID VALVE REPAIR

SURGICAL CONSIDERATIONS

Description

Insufficiency of the tricuspid valve is almost always 2° left-sided valvular disease, with tricuspid annular dilation and resultant valvular insufficiency usually 2° pulmonary HTN. Some congenital conditions (e.g., Ebstein deformity) may persist into early adulthood, when replacement is usually necessary. Tricuspid repair is normally possible in the absence of primary involvement of tricuspid leaflets. The procedure is usually accomplished on CPB, either with the heart fibrillating or during a brief period of aortic cross-clamping and diastolic arrest. After instituting CPB and snaring the venous cannulae to prevent air entry into the pump circuit, the tricuspid valve is exposed through an incision in the right atrium. In the absence of leaflet involvement by the rheumatic process, repair usually can be accomplished by a simple **annuloplasty.** After atrial closure, evacuation of air, and myocardial resuscitation, the patient is weaned from CPB with vasodilator agents. Temporary pacing wires usually are inserted and passed to the anesthesiologist in case inadvertent injury to the AV node or His bundle causes complete heart block. Maintenance of RV systolic function after repair of tricuspid regurgitation is of utmost importance, as the RV may be exposed to increased afterload with the elimination of a regurgitant tricuspid valve (loss of "pop-off valve"). This is frequently accomplished by the initiation of pulmonary vasodilators (i.e., epoprostenol) and inotropic support.

Variant procedure or approaches

Transcatheter tricuspid valve repair/replacement. Transcatheter procedures have a high success rate with an average 6% device migration, 2% conversion to open surgery, and 5% 30-d mortality in early trials. Residual TR occurs more frequently following transcatheter interventions compared to open surgery, and further studies will be necessary for the postop implications (see p. 412).

Tricuspid Valve Replacement

Usual preop diagnosis

Tricuspid regurgitation; 2° annular dilation 2° pulmonary HTN and left-side failure

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. See Appendix K, p. K-1.

SUMMARY OF PROCEDURE	
Position	Supine
Incision	Median sternotomy
Special instrumentation	Intraop ECHO; hemodynamic monitoring; CPB
Antibiotics	Cefazolin 2 g iv prior to incision and then 2 g q4h
Surgical time	Aortic cross-clamp: 30–40 min CPB: 70–80 min Total: 3–4 h
EBL	400–800 mL (Long-standing tricuspid regurgitation may → impaired hepatic production of coagulation factors.)
Postop care	ICU × 1–3 d, intubated 6–24 h; vasodilator therapy for pulmonary HTN
Mortality	2–3%
Morbidity	Third-degree heart block: 10% RV failure: 2–3%
Pain score	7–10

PATIENT POPULATION CHARACTERISTICS

Age range	50–75 yr
Male:Female	1:1
Incidence	Rare
Etiology	Rheumatic; congenital (Ebstein anomaly)
Associated conditions	Rheumatic involvement of left-side valves with pulmonary HTN

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Cardiovascular	**Tricuspid regurgitation (TR)** usually is well tolerated, and Sx (↓ CO) may go unnoticed, masked by Sx associated with left-side valvular disease and pulmonary HTN. Occasionally, TR is 2° endocarditis and raises the suspicion of iv drug abuse. TR may be caused by pulmonary HTN → ↑ RV afterload → RV dilation, ↑ RV wall tension → dilation of tricuspid valve annulus, and TR. TR may also occur in heart transplant patients due to endomyocardial biopsies and in patients with PM and ICDs, particularly after lead removal. Pathophysiology: 1. TR → ↓ CO 2° ↑ RV size → shift of the intraventricular septum to the left → ↓ LV size, ↓ LV compliance → LV underloading due to ↓ LV size and ↓ RV stroke volume. 2. TR → atrial fibrillation (AF) 2° ↑ right atrial size. In isolated insufficiency, the ↑ in right atrial pressure may result in shunting across a patent foramen ovale (PFO), leading to paradoxical embolization with potentially disastrous consequences. **Tests:** ECG: ✓ AF. ECHO: ✓ relative chamber size, contractility, PFO, valve lesions. Cardiac catheterization: ✓ contractility, CO, pulmonary pressures and their response to vasodilators, other valvular pathology.

Respiratory	Pulmonary HTN (>25 mm Hg mean pressure), pulmonary edema, and effusions may be present. **Tests:** CXR: ✓ pulmonary edema, pleural effusion
Renal	Chronic venous congestion may → prerenal failure. **Tests:** BUN; Cr; electrolytes
Hepatic	Hepatic congestion may → impaired synthetic function, particularly coag factors. **Tests:** Consider PT; PTT
Hematologic	**Tests:** Hb/Hct; other tests as indicated from H&P and Sx. Consider viral testing (HIV, hepatitis in isolated tricuspid endocarditis or in iv drug abusers). T&C 2–4 U PRBCs
Laboratory	Other tests as indicated from H&P
Premedication	Standard premedication is usually appropriate.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: An arterial line should be inserted, using liberal amounts of local anesthetic, before induction. The presence of real-time BP monitoring can be critical in the care of these patients, especially during induction. Although it is helpful to have a CVP line before induction for preload monitoring and drug infusion, it is not essential. This line usually is inserted after the patient is intubated. If infused drugs are necessary before the CVP catheter is in place, they can be administered through a separate peripheral iv.

Induction	Typically, O_2 and premedicate with midazolam (2–5 mg). Induce with moderate- to high-dose narcotic (fentanyl 5–10 mcg/kg or sufentanil 2.5–20 mcg/kg) with etomidate (0.1–0.3 mg/kg) or propofol (0.5–1

	mg/kg). Muscle relaxation is obtained with a nondepolarizing agent (e.g., rocuronium 0.6–1.0 mg/kg, vecuronium 0.1 mg/kg).
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.
Maintenance	The choice of narcotic/benzodiazepine/O_2 or volatile agent and O_2 will be determined by the underlying lesion and ventricular function. Avoid N_2O (pulmonary HTN). Low-dose isoflurane may be used. Benzodiazepines (e.g., midazolam 50–200 mcg/kg) may be used for amnesia. Exact choice of agents for maintenance depends on underlying lesion, ventricular function, and coexisting disease. Management goals of TR are often those of coexisting valvular problems. The general principles, however, are (a) adequate preload (↓ preload →↓ RV stroke volume); (b) HR, normal-to-increased; (c) contractility, normal (may require inotropic support for ↑ PVR, anesthetic myocardial depression, or IPPV); (d) PVR, normal or decreased (high PVR →↓ RV stroke volume); and (e) SVR, little effect unless it affects left-side pathology. Recommend use of antifibrinolytics for all cases requiring CPB.
Emergence	To ICU intubated and ventilated × 6–24 h. Early extubation may be possible if a lower-dose narcotic technique is used (fast-track). Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Blood and fluid requirements	IV: 14 ga × 1–2 BSS at 6–8 mL/kg/h UO 0.5–1 mL/kg/h Warm all fluids. Humidify gases. T&C 2–4 U PRBC	Because of the possibility of PFO and R→L shunt, ensure that all venous lines are clear of air. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Arterial line CVP or PA catheter Urinary catheter	Standard monitors and an A-line are placed before induction. CVP (and/or PA line, if indicated) usually is placed after intubation. CVP may be a poor indicator of RV or LV filling. The PA catheter may be helpful in the management of fluid balance, CO, and pulmonary HTN but may be difficult to place and may require removal while valve or annular ring is placed.
	TEE	Because RV distension may affect LV filling and stroke volume, TEE may be useful in judging filling and relative LV size.
Pulmonary HTN	↑ PVR can result from: ↓ PaO_2 ↑ $PaCO_2$ ↓ pH N_2O α-Agonists Inadequate anesthesia	In TR, control of PVR is important because ↑ PVR may result in ↓ CO. Hypoxia, hypercarbia, acidosis, N_2O, or α-agonists may ↑ PVR. PVR may be reduced with hypocarbia, inotropic

		support with dopamine (5–10 mcg/kg/min), epinephrine (25–100 ng/kg/min), or milrinone (0.375–0.75 mg/kg/min), and by using pulmonary vasodilators—NTG (1–4 mcg/kg/min), SNP (0.5–2 mcg/kg/min), prostaglandin E_1 (0.05–0.4 mcg/kg/min), or inhaled NO (10–100 ppm).
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Coagulopathy Hemorrhage RV failure	Coagulopathy 2° prolonged bypass time for multiple valve replacements. RV failure 2° ↑ PVR (previously, RV decompressed by tricuspid incompetence). Rx includes inotropes, dopamine (5–10 mcg/kg/min), milrinone (infusion 0.3–0.75 mcg/kg/min), and epinephrine (25–100 ng/kg/min), avoidance of factors causing ↑ PVR, and the use of pulmonary vasodilators.

POSTOPERATIVE

Complications	Hemorrhage	Inotropic support and

	Coagulopathy Dysrhythmias RV failure Infection Renal impairment Methemoglobinemia	pulmonary vasodilators will need to be continued. Avoid those factors that ↑ PVR. Measure metHb levels during NO therapy.
Multimodal pain management	Limit opioids	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.
Tests	ECG Coag profile Renal panel: BUN, Cr ABG Electrolytes	

Suggested Readings

1. Asmarats L, Puri R, Latib A, et al: Transcatheter tricuspid valve interventions landscape, challenges, and future directions. J Am Coll Cardiol 2018; 71(25):2935–56.
2. Cevasco M, Shekar PS: Surgical management of tricuspid stenosis.Ann Cardiothorac Surg 2017; 6(3):275–82.

3. Crumley S, Schraag S: The role of local anaesthetic techniques in ERAS protocol for thoracic surgery. J Thorac Dis 2018; 10(3):1998–2004.
4. Fender EA, Zack CJ, Nishimura RA: Isolated tricuspid regurgitation: outcomes an therapeutic interventions. Heart 2018; 104(10):798–806.
5. Isomura T, Fukada Y, Miyazaki T, et al: Tricuspid surgery for tricuspid regurgitation: review. Gen Thorac Cardiovasc Surg 2016; 64(10):577–83.
6. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery.J Cardiothorac Vasc Anesth 2018; 32(6):2760–70.

SEPTAL MYECTOMY/MYOTOMY

SURGICAL CONSIDERATIONS

Description

Patients with asymmetric septal hypertrophy usually present with symptoms 2° LV outflow tract obstruction (LVOTO), increased diastolic dysfunction and stiffness, or a combination of the two, manifested as syncope, CHF, or severe chest pain. The systolic hemodynamic abnormality is caused by the anterior leaflet of the mitral valve being drawn into the LV outflow tract (LVOT) and abutting the asymmetrically hypertrophied intraventricular septum, narrowing the LVOT, and producing a large dynamic intracavitary gradient. In addition, severe mitral insufficiency may result. The most common surgical procedure for asymmetric septal hypertrophy is **septal myectomy/myotomy.** After institution of CPB, aortic cross-clamping, and cardioplegic arrest, the aorta is opened, and visualization of the subvalvar ventricular septum is attained. Manual palpation, as well as TEE visualization, can localize the asymmetric hypertrophy. Using the right coronary orifice as a landmark, the ventricular septum is longitudinally incised with two parallel incisions ~1 cm apart, with care being taken to avoid injury of the papillary muscle or mitral valve chordae. A trough of the hypertrophied septum is then excised, alleviating the LVOTO. Removal of a portion of the asymmetrically hypertrophied myopathic septum also usually reduces the systolic anterior motion (SAM) of the anterior mitral leaflet and reduces the intracavitary gradient and MR. Infrequently, mitral valve replacement may be necessary for persistent MR.

Usual preop diagnosis

Asymmetric septal hypertrophy with CHF; chest pain; and/or syncope

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. See Appendix K, p. K-1.

SUMMARY OF PROCEDURE

Position	Supine
Incision	Median sternotomy
Special instrumentation	TEE; hemodynamic monitoring; CPB
Unique considerations	Because of LV hypertrophy, maintenance of adequate preload is essential. Afterload also must be maintained to prevent LVOTO. Temporary pacing wires usually are placed at the start of the procedure.
Antibiotics	Cefazolin 2 g iv prior to incision and then 2 g q4h
Surgical time	2.5 h
EBL	150 mL
Postop care	ICU, intubated on ventilator 6–24 h
Mortality	3–4%
Morbidity	Complete heart block: 5% Persistent MR: 5% New aortic insufficiency: 2–5% CVA: <1% VSD: <1%
Pain score	6–10

PATIENT POPULATION CHARACTERISTICS

Age range	20–80 yr (mean = 45 yr)
Male:Female	1:1
Incidence	30/yr at Stanford University Medical Center

Etiology	Asymmetric septal hypertrophy

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Respiratory	Affected only 2° to cardiac failure.
Cardiovascular	Patients with idiopathic hypertrophic obstructive cardiomyopathy (HOCM) often present with Sx of syncope, angina pectoris (CAD), CHF, and palpitations. The main feature of HOCM is dynamic LVOTO 2° to septal hypertrophy and a possible Venturi effect that draws the anterior mitral valve leaflet into the outflow tract. LVOTO →↑ LV hypertrophy.
	→ ↓ compliance and ↓ diastolic function → ↑ LVEDP, making diastolic filling dependent on preload and atrial contraction. Obstruction is worsened by decreased preload or afterload, increased contractility, or HR. HOCM also results in ↑ MVO_2 and ↓ coronary perfusion, especially to the septum and subendocardium, which increases the risk of ischemia. Hypertrophy occurs throughout the whole myocardium and may lead to further dysfunction. MR is often present and is exacerbated by the same conditions that increase LVOTO. Patients are often on β-blockers or Ca^{++} antagonists. These should be continued to the day of surgery. **Tests:** ECG: ✓ Q-waves (indicative of septal hypertrophy); short PR interval with slurred QRS complex; supraventricular tachycardia; LV hypertrophy. ECHO: ✓ septal hypertrophy; MR; LV hypertrophy, myocardial dysfunction. Cardiac catheterization: ✓ LVOT pressure gradient, which may increase with provocation (e.g., Valsalva maneuver); obliteration of the LV cavity; MR; CAD.

Neurologic	Document syncope and any neurologic deficits.
Laboratory	Hb/Hct; electrolytes; other tests as indicated from H&P.
Premedication	Avoid activation of the sympathetic nervous system because this will cause ↑ HR and ↓ inotropy →↓ CO + ↓ BP. Thus, adequate anxiolysis is essential and can be obtained by premedication with a benzodiazepine (e.g., midazolam 0.05–0.2 mg/kg im, lorazepam 1–2 mg po, or diazepam 5–10 mg po). Avoid ↓ SVR, and maintain β-blockade and Ca^{++} channel blocker therapy.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: An arterial line should be inserted, using liberal amounts of local anesthetic, before induction. The presence of real-time BP monitoring is critical in the care of these patients, especially during induction. Although it is helpful to have a CVP line before induction for preload monitoring and drug infusion, it is not essential. This line usually is inserted after the patient is intubated. If infused drugs are necessary before the CVP catheter is in place, they can be administered through a separate peripheral iv.

Induction	Typically, moderate-dose narcotic (fentanyl 5–10 mcg/kg; avoid sufentanil due to vasodilation). Supplement with etomidate (0.1–0.3 mg/kg), midazolam (50–350 mcg/kg), or propofol (0.5–1 mg/kg). Ketamine should be avoided due to the activation of the sympathetic nervous system. Introduction of a volatile agent prior to intubation may be helpful. Vecuronium (0.1 mg/kg) or

	rocuronium (0.6–1.0 mg/kg) is used for muscle relaxation. Avoid inotropic agents.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.	
Maintenance	O_2 and narcotic (total fentanyl 10–20 mcg/kg) + volatile agent. Midazolam (50–350 mcg/kg) can be used for amnesia. The cornerstone of anesthesia for HOCM is to avoid factors that will increase LVOTO: (a) ↓ preload, (b) ↓ afterload, (c) ↑ contractility, (d) loss of NSR, and (e) ↑ HR. Recommend use of antifibrinolytics for all cases requiring CPB.	
Emergence	To ICU intubated and ventilated × 6–24 h. Early extubation is possible if a lower-dose narcotic technique is used (fast-track). Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 14 ga × 1–2 BSS at 6–8 mL/kg/h Maintain UO. 0.5–1 mL/h Warm fluids. Humidify gases.	See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Arterial line PA catheter Urinary catheter	Standard monitors and an A-line are placed before induction. CVP (and/or PA line, if indicated) usually is placed after intubation. A PA catheter is useful for assessment of LV filling

		(keep high normal PCWP) and for monitoring SVR (normal-to-high). A pacing or pace port PA catheter may be useful for conduction problems occurring postop.
	TEE	TEE is useful to judge LV filling, LV contractility, LVOTO, MR, and VSD (post myectomy).
Intraoperative problems	↓ Preload ↓ Afterload ↑ Contractility ↑ HR Loss of NSR Complete heart block	Rx: volume, phenylephrine Rx: phenylephrine Rx: esmolol Rx: esmolol Rx: cardioversion, verapamil. Should have reliable means of pacing available, including temporary ventricular leads
	VSD	Septum may be damaged, → VSD (diagnosed by TEE). If present, it should be repaired.
Position	✓ and pad pressure points ✓ eyes	

POSTOPERATIVE

Complications	Hemorrhage	Dx: from chest tube drainage. Rx with blood and factors as needed.

		✓ coag status. May require reexploration
	Complete heart block	Pacing should be available. Occasionally, inotropes of vasodilators are required. β-Blockers and Ca^{++} antagonists are usually D/C'd.
	Late VSD	Late VSD, indicated by development of a murmur, will require repair.
	Tamponade	Dx: by raising filling pressure, equalization of CVP, and PADP, ↓ CO, ↓ UO, and by ECHO. Rx: re-exploration and drainage
Multimodal pain management	Limit opioids	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.
Tests	ECG: ✓ conduction problems ABG ECHO: ✓ LVOTO; VSD	

	Coag profile Electrolytes

Suggested Readings

1. Crumley S, Schraag S: The role of local anaesthetic techniques in ERAS protocol for thoracic surgery. J Thorac Dis 2018; 10(3):1998–2004.
2. Geske JB, Ommen SR, Gersh BJ: Hypertrophic cardiomyopathy: clinical update JACC Heart Fail 2018; 6(5):364–75.
3. Maron BJ, Maron MS: Hypertrophic cardiomyopathy.Lancet 2013; 381(9862):242–55.
4. Nishimura RA, Seggewiss H, Schaff HV: Hypertrophic obstructive cardiomyopathy surgical myectomy and septal ablation. Circ Res 2017; 121(7):771–83.
5. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery.J Cardiothorac Vasc Anesth 2018; 32(6):2760–70.
6. Togni M, Billinger M, Cook S, et al: Septal myectomy: cut, coil, or boil?Eur Heart J 2008; 29(3):296–8.
7. Vives M, Roscoe A: Hypertrophic cardiomyopathy: implications for anesthesia. Minerva Anestesiol 2014; 80(12):1310–9.

PACEMAKER INSERTION

SURGICAL CONSIDERATIONS

Description

Pacemaker insertion may be required for relief of abnormalities of the conduction system. A transvenous pacemaker lead may be placed via the subclavian vein, through the tricuspid valve into the RV (single-chamber pacing), or two leads may be placed, one into the right atrium and the other into the RV (dual-chamber pacing). Typically, second- or third-degree heart block is the diagnostic indication, although sick sinus syndrome (SSS) and other abnormalities also may be found. Access to the subclavian veins usually is attained percutaneously, although a cutdown may be used to expose the cephalic vein in the deltopectoral groove. Passage of a guide wire into the RV may cause frequent premature ventricular beats, which usually subside spontaneously with repositioning of the guide wire or lead. After ventricular and/or atrial lead placement, the pacing lead will have to be tested for sensing threshold, pacing threshold, depolarization amplitude, and lead resistance. After satisfactory placement of the pacing leads, the actual pacemaker generator unit is connected and then placed in a subcutaneous pocket at the site of percutaneous lead placement.

Pacemakers are frequently placed in the cath lab (see p. 418).

Usual preop diagnosis

Abnormalities of S-A nodal function (SSS) or A-V nodal function (heart block)

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. See Appendix K, p. K-1.**

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Left subclavicular; mostly percutaneous
Special instrumentation	Fluoroscopy with fluoro-table
Unique considerations	Patient usually awake with mild sedation
Antibiotics	Cefazolin 2 g iv prior to incision
Surgical time	1 h
EBL	Minimal
Postop care	PACU → CCU; ECG monitoring × 2–4 h; chest radiograph to document lead configuration
Mortality	0–1%
Morbidity	Lead displacement, 1–2%; pneumothorax, 1–2%
Pain score	2

■ PATIENT POPULATION CHARACTERISTICS	
Age range	50–80 yr
Male:Female	2:1
Incidence	Infrequent
Etiology	CAD; cardiac valve repair/replacement

Associated conditions	Complete heart block with syncope; SSS; paroxysmal tachycardia; SVT

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

The indications for permanent pacemaker insertion are usually bradydysrhythmias (e.g., third-degree heart block, sinus node dysfunction) or, less commonly, tachycardia (e.g., atrial flutter not responsive to medical therapy). There are many different types of pacemakers, which are classified according to the chamber paced, chamber sensed, response to sensing, programmability, and antitachyarrhythmia functions. The anesthesiologist should be aware of the type of pacemaker to be implanted and the means for external control (Table 6.1-4).

Table 6.1-4 Five-Position Pacemaker Code (ICHD)

I	II	III	IV	V
Chamber(s) Paced	**Chamber(s) Sensed**	**Response(s) to Sensing**	**Programmability, Rate Modulation**	**Special Tachyarrhythm Functions**
V—Ventricle	V—Ventricle	T—Triggered	P—Simple programmable (rate and/or output)	B—Bursts
A—Atrium	A—Atrium	I—Inhibited	M—Multiprogrammable	N—Normal rate competition
D—Dual	D—Dual	D—Dual (T + I)	C—Communicating	S—Scanning
O—None	O—None	O—None	R—Rate modulation	
			O—None	E—External

Cardiovascular	Evaluate patient for associated disease, including CAD (~50%), HTN (~20%), cardiomyopathy, CHF, and valvular defect, and for any symptoms or recent

	changes related to the conduction problems (syncope, CHF). It is important to know the reason for pacemaker implantation and the patient's escape rhythm. These patients are often on antidysrhythmics, diuretics, cardiac glycosides, and a variety of other cardiovascular agents, which should be continued up to the day of surgery. **Tests:** Exercise tolerance by Hx. ECG: ✓ rate, ischemic changes, rhythm. Other tests as indicated by H&P. ✓ serum digoxin and other antidysrhythmic levels.
Pacemaker	If a permanent pacemaker is already in place, its type and present functional state should be assessed. (Sx of problems include chest pain, palpitations, syncope, and weakness.) If no permanent pacemaker, a temporary transvenous pacemaker should be placed before surgery, except in unusual circumstances. **Tests:** ECG: ✓ Valsalva maneuver or carotid sinus massage—may slow the heart sufficiently to allow the permanent pacemaker to fire and allow assessment of function. A rate lower than the set rate may indicate battery failure. CXR: ✓ lead continuity.
Hematologic	Hct or Hb
Laboratory	✓ K^+ level, which can affect pacing threshold, → loss of pacemaker capture, if low, or ventricular tachycardia, if high. Other tests as indicated from H&P.
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water

	for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Permanent pacemakers commonly are placed via the transvenous route, requiring only local anesthesia and MAC with sedation. For epicardial pacemaker placement, GETA is required. These patients should be monitored during transport to OR. Care is taken to avoid dislodging temporary pacing electrodes. The function of the temporary pacemaker should be checked prior to induction.

General anesthesia

Induction	Usually, anesthesia can be induced with propofol (1–2 mg/kg) or etomidate (0.2–0.3 mg/kg) in combination with an opiate (e.g., fentanyl 1–2 mcg/kg) to attenuate the response to intubation. Vecuronium (0.1 mg/kg) or rocuronium (1 mg/kg) will provide adequate muscle relaxation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.
Maintenance	Standard maintenance. Avoid excessive hyperventilation (→↓ K^+), and use volatile agents with caution because they may increase A-V conduction time. Recommend use of antifibrinolytics for all cases requiring CPB.
Emergence	Generally, patient is extubated at the end of the case. Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Blood and fluid requirements	Minimal fluid requirements IV: 18–16 ga × 1 BSS at 4–6 mL/kg/h	See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Urinary catheter	Rarely, arterial line if patient disease indicates.
Electrocautery interference	Minimize by: 1. Using bipolar cautery, if possible 2. Grounding pad on the leg 3. Limiting cautery output 4. Limiting cautery use	The use of electrocautery may interfere with pacemaker function and → dysrhythmias or failure of the pacemaker. Monitor ECG and pulse (because electrical activity does not mean CO) for interference. Have a magnet available to convert the pacemaker to asynchronous mode (check that this is possible and will not change the programming of the pacemaker).
Complications	Dysrhythmias Air embolism Cardiac tamponade Hemo/pneumothorax Hemorrhage	Availability of temporary pacing is important because complete heart block or dislodgement of leads may occur. In addition, pharmacologic agents (atropine, 0.5–2 mg; isoproterenol 10–100 ng/kg/min) to increase HR should be available.

POSTOPERATIVE

Complications	Dislodging of electrodes

	Dysrhythmias Pneumothorax Tamponade	
Multimodal pain management	Limit opioids	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.
Tests	ECG CXR Electrolytes	

Suggested Readings

1. Arora L, Inampudi C: Perioperative management of cardiac rhythm assist devices in ambulatory surgery and nonoperating room anesthesia. Curr Opin Anaesthesiol 2017; 30(6):676–81.
2. Baranchuk A, Refaat MM, Patton KK, et al: Cybersecurity for cardiac implantable electronic devices: what should you know? J Am Coll Cardiol 2018; 71(11):1284–8.
3. Crumley S, Schraag S: The role of local anaesthetic techniques in ERAS protocols for thoracic surgery. J Thorac Dis 2018; 10(3):1998–2004.
4. Ellison K, Sharma PS, Trohman R: Advances in cardiac pacing and defibrillation. Expert Rev Cardiovasc Ther 2017; 15(6):429–40.
5. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery. J Cardiothorac Vasc Anesth 2018; 32(6):2760–70.
6. Reynolds DW, Murray CM: New concepts in physiologic cardiac pacing. Curr Cardiol Rep 2007; 9(5):351–7.
7. Rozner MA: The patient with a cardiac pacemaker or implanted defibrillator and

management during anaesthesia. Curr Opin Anaesthesiol 2007; 20(3):261–8.

PERICARDIECTOMY

SURGICAL CONSIDERATIONS

Description

Constrictive pericarditis, either acute or chronic, interferes with ventricular filling, reducing stroke volume and depressing the cardiac index (CI). Although there are many possible etiologies (infectious, nephrogenic, postradiation), the cause remains unknown for a majority of patients. Typically, patients present with a progressive Hx of breathlessness, fatigability, or peripheral or abdominal swelling, often months to years after the inciting event. The Dx may be confirmed by cardiac catheterization, with equalization of end diastolic pressures, although volume loading may be necessary to demonstrate this in the patient under medical management. The differentiation between constrictive pericardial disease and restrictive myocardial disease may be difficult, if not impossible, and may coexist in a single patient. After this Dx has been confirmed, surgical **pericardiectomy** should be undertaken because the outlook without surgical relief is one of gradual but persistent deterioration. Although surgical mortality remains in the 10–15% range, long-term relief for survivors is good. Because these patients are usually significantly compromised hemodynamically, intensive monitoring is indicated. Approach may be through a **median sternotomy** or left **anterolateral thoracotomy.** Removal of both visceral and parietal pericardia is essential for relief, but dense adhesions of these layers to underlying muscle may make this dissection very difficult, tedious, and bloody, especially if the visceral pericardium and epicardium are involved in the constrictive process. CPB may be utilized for hemodynamic instability, but it obviously increases bleeding complications. Complete excision from both ventricular surfaces is mandatory. Most periop difficulties evolve from cardiac failure.

Variant procedure or approaches

A limited **pericardial window,** draining fluid into the left hemithorax, may relieve tamponade, but will be of no benefit for a true constrictive process.

Usual preop diagnosis

Constrictive pericarditis

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist**

should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. See Appendix K, p. K-1.

SUMMARY OF PROCEDURES

	Median Sternotomy	Anterolateral Thoracotomy
Position	Supine	Supine, with elevation of left hemithorax
Incision	Midline	5th interspace
Special instrumentation	TEE; full hemodynamic monitoring; CPB standby	⇐
Antibiotics	Cefazolin 2 g iv prior to incision and then 2 g q4h	⇐
Surgical time	2–5 h, depending on tenacity of visceral peel	⇐
Closing considerations	Avoid volume overload and cardiac distension. Chronically depressed hearts may require inotropic or mechanical support (i.e., IABP) postop	⇐
EBL	100–500 mL	⇐
Postop care	ICU; intubated ×4–6 h. Hemodynamic monitoring. Low CO state may persist postop	⇐
Mortality	5–15%, predominantly from cardiac failure	⇐
Morbidity	Persistent CHF: 5%	⇐

	Transient phrenic nerve dysfunction: <1%	
Pain score	7–10	7–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	10–80 yr (median = 45 yr)
Male:Female	2:1
Incidence	3–4 patients/yr at tertiary referral center
Etiology	Majority unknown. Infectious, radiation, prior cardiac operation, rheumatic pericarditis, and amyloid deposition
Associated conditions	Restrictive myocardial diseases

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Pericardiectomy is performed most commonly for patients with constrictive pericarditis, whereas pericardial window procedures are used for patients with cardiac tamponade.

Respiratory	Restrictive disease may be present 2° fibrosis (e.g., post-TB) or pleural effusions. This may impair oxygenation and, if the effusions are significant, may ↓ venous return on institution of IPPV → rapid decompensation (↓ CO). Drain prior to surgery. **Tests:** CXR: ✓ active disease, fibrosis, effusions, pericardial calcification. Consider ABG, PFT, as indicated by CXR and Sx, and if time permits.
Cardiovascular	Of importance is the presence or absence of a pericardial effusion. A large effusion that develops slowly (chronic pericarditis) may cause minimal Sx. Conversely, a small and rapidly forming effusion may → cardiac tamponade. Although the cardiovascular signs for both tamponade and constriction are similar

	(pulsus paradoxus, venous HTN, exaggerated venous pulsations, ↓ BP, tachycardia), it is important to differentiate between them because it may affect intraop management. Constrictive pericarditis can be differentiated from cardiac tamponade by ECHO, by pulsus paradoxus (frequent, with tamponade; rare, with constrictive pericarditis). Kussmaul sign (distension of the jugular veins on inspiration) is rare with tamponade and common with constrictive pericarditis. Electrical alternans is present with tamponade and absent in constrictive pericarditis. Examination of the RV pressure wave form is unchanged in tamponade, but shows a dip and prominent Y descent in constrictive pericarditis. Anesthetic management is influenced by the planned procedure, the underlying process (constriction or tamponade), and its severity. Clues to severity are the physical symptoms and the degree of tachycardia, ↓ BP, and the filling pressure. (Although it is not possible to give exact figures, a HR of >100 bpm, a systolic BP < 100 mm Hg, and a filling pressure >15 mm Hg are probably significant.) In addition, it is important to assess concurrent cardiac problems: cardiomyopathy, CAD, or valvular disease (especially in constrictive disease associated with TB and radiation therapy or in rheumatoid diseases, such as lupus). **Tests:** ECG: ✓ low-voltage complexes and electrical alternans. ECHO: ✓ pericardial effusion, calcification of pericardium, valvular lesions, myocardial function.
Gastrointestinal	Chronic hepatic congestion may → ↓ synthetic function (↓ procoagulants). Development of ascites may → ↑ intra-abdominal pressure. Because of this and the fact that these are sometimes emergency procedures, consider possible full stomach. **Tests:** Consider LFTs; PT; PTT.
Renal	Renal failure may cause pericarditis, and conversely, pericarditis may cause renal failure (2° prerenal

	factors: ↑ venous pressure, ↓ perfusion pressure). This may affect the choice of drugs used for anesthesia that depend on renal clearance (particularly muscle relaxants). **Tests:** BUN; Cr; consider creatinine clearance; electrolytes.
Hematologic	Some renal or hepatic conditions may be associated with coagulation disorders. These include both procoagulant and Plt problems. If possible, any coagulopathy should be corrected before surgery with FFP, Plt, or both. Consult with a hematologist, if necessary. **Tests:** Hb/Hct; PT; PTT; Plt count
Laboratory	Other tests as indicated from H&P
Premedication	Little or no premedication may be indicated; otherwise, a benzodiazepine (e.g., midazolam 0.05–0.2 mg/kg im) may be used. Consider full-stomach precautions: H_2 antagonists (e.g., ranitidine 50 mg iv), metoclopramide (10 mg iv), and antacids (e.g., Na citrate 0.3 M 30 mL po).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: An arterial line should be inserted, using liberal amounts of local anesthetic, before induction. The presence of real-time BP monitoring can be critical in the care of these patients, especially during induction. Although it is helpful to have a CVP line before induction for preload monitoring and drug infusion, it is not essential. This line usually is inserted after the patient is intubated. If infused drugs are necessary before the CVP catheter is in place, they can be administered through a separate peripheral iv. Consider pericardiocentesis or pericardial window under local anesthesia prior to induction, as drainage of even a small amount of fluid may improve the patient's status dramatically. The considerable manipulation of the heart, extensive dissection, blood loss, dysrhythmias, and unrelieved tamponade make pericardiectomy cases a challenge.

Induction	Typically, ketamine (1–2 mg/kg) or etomidate (0.2–0.3 mg/kg) ± narcotic (fentanyl 5–10 mcg/kg), depending on patient status. Consider maintaining spontaneous ventilation in tamponade patients until drained, as institution of IPPV may result in rapid decompensation and cardiac arrest due to ↓↓ venous return. Otherwise, succinylcholine (1 mg/kg) with cricoid pressure (full stomach) or vecuronium (0.1 mg/kg) for muscle relaxation. For unstable patients, may prep and drape and have surgeons standing by before induction of anesthesia.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.
Maintenance	Narcotic (total fentanyl 10–20 mcg/kg), low-dose volatile agent, midazolam (50–350 mcg/kg), or a combination of these agents in 100% O_2. CO is dependent on maintaining a high preload to ensure adequate cardiac filling, avoiding and treating bradycardia, and preserving myocardial contractility. The use of inotropes (dopamine, isoproterenol, or epinephrine) may be necessary; α-agonists should be avoided but, on occasion, may be needed to increase coronary perfusion. These anesthetic considerations apply to both tamponade and constrictive disease. In patients with acute tamponade, dramatic increases in BP may occur when the pericardium is opened.

	Aggressive use of vasodilators and additional anesthetic agents may be necessary. Recommend use of antifibrinolytics for all cases requiring CPB.	
Emergence	In general, plan for extubation in the OR in the case of pericardial window and transport to ICU for postop ventilation 4–24 h following pericardiectomy. Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 14 ga (or 7 Fr) × 1–2; BSS UO: 0.5–1 mL/kg/h Warm fluids and humidify gases. T&C 2–4 U for pericardiectomy	During pericardiectomy, anticipate rapid blood loss (a major cause of mortality). For this reason, CPB should be available on standby for all pericardiectomy procedures. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Arterial line CVP line PA catheter TEE Urinary catheter	Standard monitors and an A-line are placed before induction. CVP (and/or PA line, if indicated) usually is placed after intubation. PA catheters aid the management of filling pressures, CO, and afterload. TEE may be useful to gauge filling volume and degree of relief of pericardial constriction.
Complications	Cardiac tamponade	Intrathoracic pressure

	Dysrhythmias	associated with IPPV may produce a ↓↓ CO in these patients (because of ↓ venous return). Spontaneous ventilation, therefore, is preferred until the tamponade is drained.
	Hemorrhage Heart failure	Hemorrhage is not usually a problem unless penetrating trauma is the cause of the tamponade. Once constriction is relieved, myocardial function does not return to normal quickly. Inotropes are often needed.
	Coagulopathy	Due to extensive dissection and hemorrhage, coagulopathies may develop and should be treated aggressively.
Positioning	✓ and pad pressure points ✓ eyes	

POSTOPERATIVE

Complications	Hemorrhage Coagulopathy Ventricular hypofunction Dysrhythmias Ischemia	Following pericardial window surgery, patients improve with the relief of the tamponade. Postpericardiotomy patients, however, may have a continuation of intraop dysrhythmias and myocardial depression.

		Inotropes (e.g., dopamine 5–10 mcg/kg/min) may be needed for 24–48 h. Normal cardiac function can take 4–6 wk to return.
Multimodal pain management	Limit opioids	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.
Tests	ECG CXR ABG Coag profile Electrolytes	

Suggested Readings

1. Crumley S, Schraag S: The role of local anaesthetic techniques in ERAS protocol for thoracic surgery. J Thorac Dis 2018; 10(3):1998–2004.
2. Grocott HP, Gulati H, Srinathan S, et al: Anesthesia and the patient with pericardia disease. Can J Anaesth 2011; 58(10):952–66.
3. Hemmati P, Greason KL, Schaff HV: Contemporary techniques of pericardiectom for pericardial disease. Cardiol Clin 2017; 35(4):559–66.
4. Hoit BD: Pericardial disease and pericardial tamponade. Crit Care Med 2007; 35(8 Suppl):S355–64.

5. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery.J Cardiothorac Vasc Anesth 2018; 32(6):2760–70.
6. Tuck BC, Townsley MM. Clinical update in pericardial diseases.J Cardiothorac Vasc Anesth 2019; 33(1):184–99.

Suggested Viewing

Links are available online to the following videos:

Bypass Surgery on a Beating Heart. Arie Blitz, MD, FACC: https://www.youtube.com/watch?v=MGzyhuCs43o

Heart Transplantation: Surgical Technique. Arie Blitz, MD: https://www.youtube.com/watch?v=lB_2Xvmqxjo

Surgical Repair of a Ruptured Type A Aortic Dissection: Arie Blitz, MD. https://www.youtube.com/watch?v=O398CeblR3U

Mitral Valve Replacement for Rheumatic Heart Disease: https://www.youtube.com/watch?v=W2FBtSysaPU

Repair of Ascending Aortic Aneurysm: Arie Blitz, MD: https://www.youtube.com/watch?v=h_u6SfPhHyc

Minimally Invasive Mitral Valve Surgery—Right Thoracotomy Approach: https://www.youtube.com/watch?v=EnJQh_W3r3A

NEJM Procedure: Deployment of an Endovascular Graft in an Abdominal Aortic Aneurysm https://www.youtube.com/watch?v=qUpXJBoAoWI

CHAPTER 6.2

Minimally Invasive Cardiac Surgery

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OFF-PUMP AND MINIMALLY INVASIVE CORONARY ARTERY BYPASS GRAFTING

SURGICAL CONSIDERATIONS

Description

Although coronary artery bypass grafting (CABG) without cardiopulmonary bypass (CPB) was originally proposed more than six decades ago, development and advances in CPB resulted in the abandonment of off-pump cardiac surgery by most cardiac centers until the 1990s. The re-emergence of off-pump coronary revascularization was due in part to reported adverse effects of CPB (e.g., cerebral and systemic inflammatory response), increased comorbidities in an aging population, medical economics, and the development of reliable mechanical stabilization devices (Fig. 6.2-1). More recently, some of the advantages of off-pump CABG have been questioned. Off-pump surgery is technically more challenging and has been associated with a higher risk of incomplete revascularization. Results from a 5-yr randomized outcome study comparing off-pump versus on-pump CABG demonstrated the inferiority of off-pump surgery. Other studies have suggested that off-pump surgery may be beneficial in select patient populations (e.g., octogenarians and patients with chronic kidney disease).

From the standpoint of terminology, off-pump coronary artery bypass (OPCAB) grafting applies to all cases of coronary revascularization not using the CPB circuit, and although it does not refer directly to the surgical approach, most are via conventional median sternotomy. Minimally invasive direct coronary artery bypass (MIDCAB/MICAB) grafting refers to off-pump coronary revascularization via a sma anterolateral thoracotomy incision, while totally endoscopic coronary artery bypass grafting (TECAB) refers to robotic off-pump coronary revascularization accomplishe entirely endoscopically without thoracotomy.

Challenges of off-pump coronary revascularization include accurate vascular anastomosis while minimizing hemodynamic perturbations during the procedure. Interrupting flow to the target artery can → regional ischemia, arrhythmias, and hemodynamic instability; displacing the heart to expose lateral or posterior arteries may → ventricular compression and profound hemodynamic compromise. Pharmacologic interventions to decrease HR and ischemic preconditioning may facilitate the anastomosis. Although not fully defined, ischemic preconditioning results from exposure to transient myocardial ischemia and is an endogenous adaptation that may mitigate the effects of subsequent prolonged myocardial ischemia. Thus, mechanically occluding the coronary artery for a brief period may confer some protection from ischemic injury associated with coronary occlusion during the anastomosis. Important preop considerations include the number and suitability of distal-target coronary arteries, cardiac function and pulmonary status, and other medical comorbidities. The presence of cardiomegaly may limit the degree of intraop cardiac manipulation. The perfusionist is present during the procedure so that rapid conversion to conventional on-pump CABG can be achieved if there are hemodynamic compromise, unsuitable distal target inability to expose lateral or posterior target vessels, or regional myocardial ischemia. Occasionally, placement of intra-aortic balloon pump intraop may facilitate the off-pump approach in a patient with ischemic cardiomyopathy.

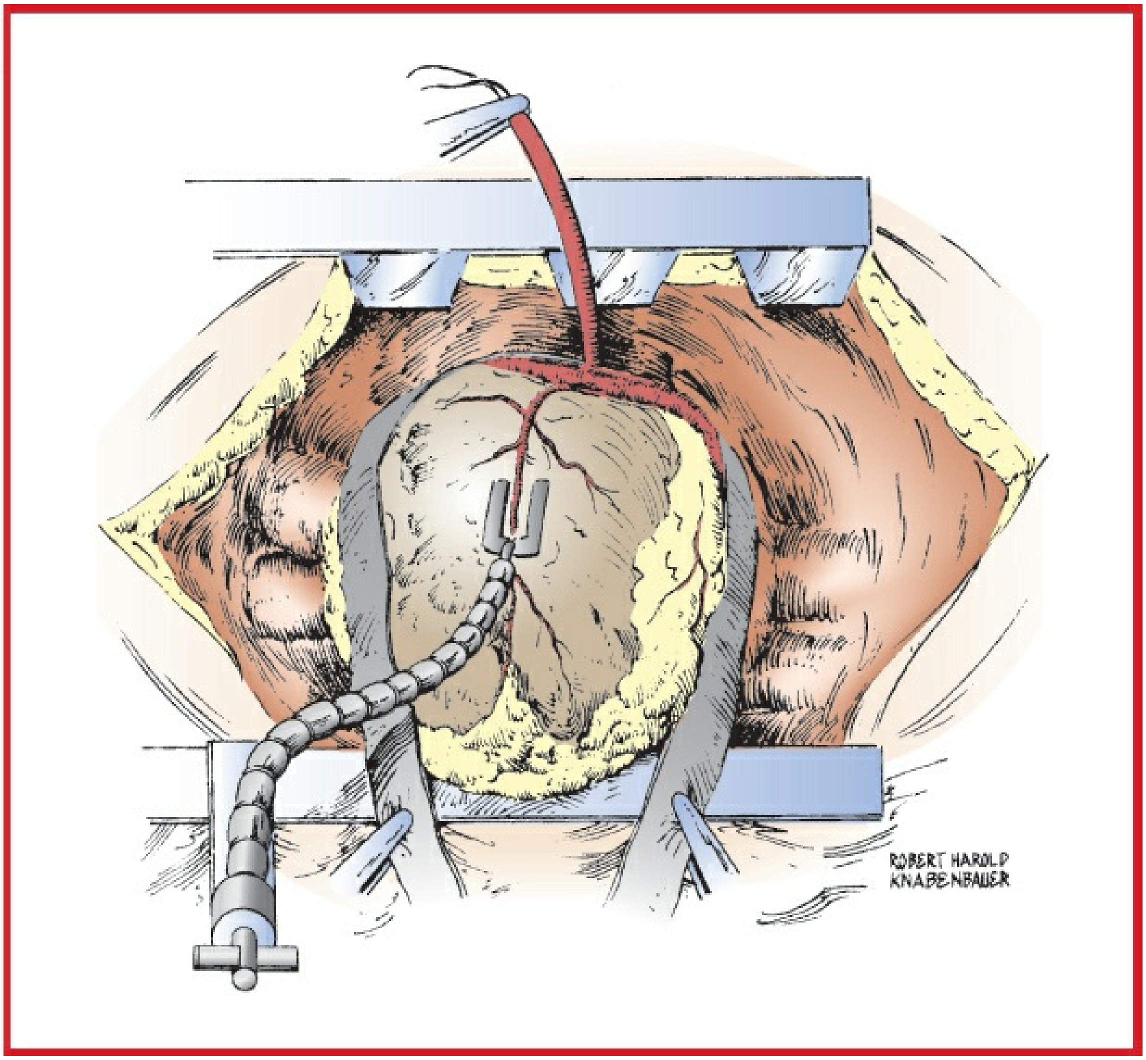

Figure 6.2-1 Manipulation of the heart and placement of a pressure-plate mechanical stabilizer is performed in preparation for OPCAB. (Reproduced with permission from Estafanous FG, Barash PG, Reves JG:Cardiac Anesthesia: Principles and Clinical Practice, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

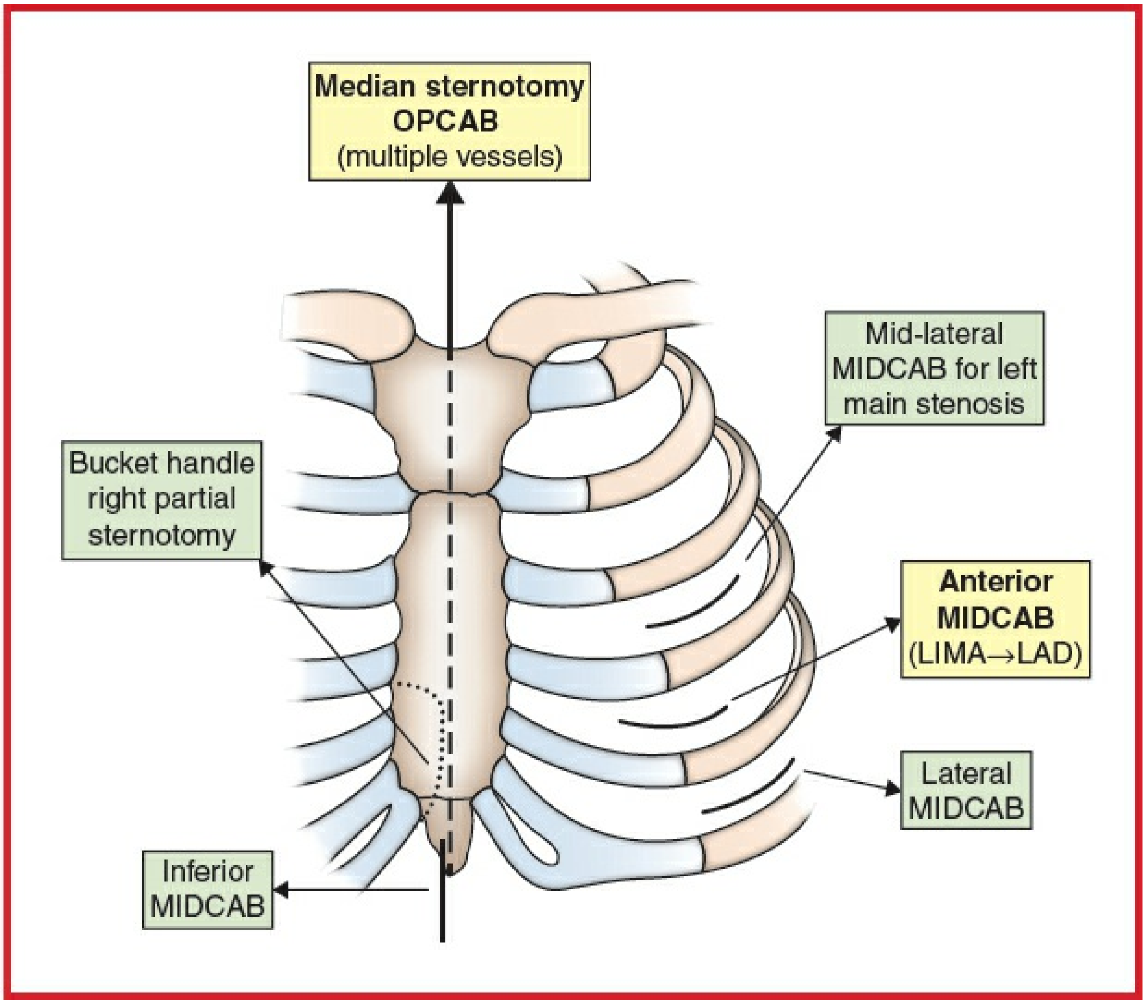

Figure 6.2-2 Incision sites for access to various target coronary arteries in MICAB. (Reproduced with permission from Estafanous FG, Barash PG, Reves JG. Cardiac Anesthesia: Principles and Clinical Practice, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

For conventional OPCAB, a standard median sternotomy is made (Fig. 6.2-2). If the internal mammary artery (IMA) is to be used as a graft, it is harvested in the usual fashion. The patient is partially heparinized, and an intravenous bolus of lidocaine is given. A sternal retractor with attachments for coronary stabilization is placed (Fig. 6.2-1). If vein grafts are used, the proximal anastomoses may be performed at this point or later, after the completion of the distal anastomoses, using a partial side-biting aortic cross-clamp or automated anastomotic device. During the period of partial aortic clamping, BP typically is lowered to reduce potential complications associated with the use of the clamp. The goal of the operation is to establish adequate perfusion of the most critical vascular bed first. Provided it is a target vessel, the left anterior descending (LAD) artery is often approached first because an IMA graft (or vein graft) can provide immediate perfusion to the anterior, septal, and anterolateral LV walls and requires minimal cardiac manipulation to construct the anastomosis. Mechanical coronary artery stabilizers, based on local myocardial compression (Fig. 6.2-1) or vacuum suction (Fig.

6.2-3) and attached to the retractor, provide stable local epicardial motion restraint, thereby facilitating the anastomosis. After stabilization, the artery is occluded (following a period of ischemic preconditioning), and an arteriotomy is made. Equipment that minimizes blood in the operative site includes standard suction attachments, temporary intracoronary shunts, vessel occluders, and “blower-misters” that displace blood by delivering a combination of gas (CO_2) and heparinized saline. The distal anastomosis is then performed. The patient is monitored closely at this point for any signs of myocardial ischemia and/or hemodynamic instability. To expose the lateral and posterior target vessels (most often branches of the circumflex and right coronary arteries), manipulation of the heart is necessary and may not be well tolerated. During lateral and posterior pericardial suture placement, the surgeon displaces and compresses the heart, resulting in temporary hemodynamic compromise due to decreased preload and ventricular distortion. Ventilation may need to be decreased temporarily to facilitate this maneuver. Additional exposure techniques include the use of an apical suction device to facilitate cardiac manipulation with potentially less hemodynamic compromise, Trendelenburg position and tilting the operating table to the right, release of right pericardial stay sutures, opening of the right pleura and incising the right pericardium, and placement of laparotomy sponges. The target vessel is stabilized again, ischemic preconditioning is carried out, and the artery is opened. After construction of the distal anastomosis, the graft is relieved of any residual air before securing the sutures and the proximal anastomosis performed if not already done so as noted earlier. Graft flow is assessed using a Doppler flow probe.

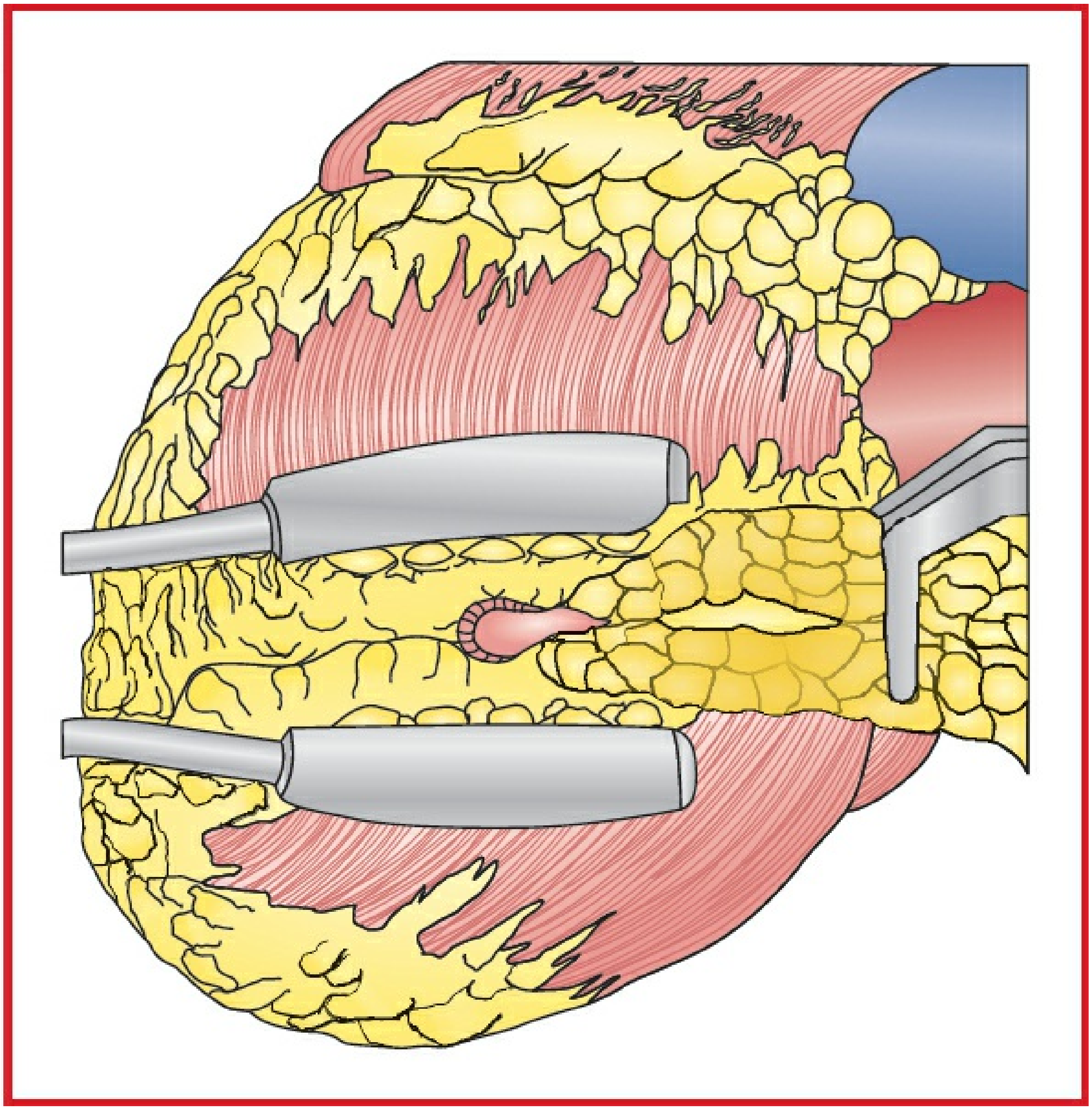

Figure 6.2-3 The Octopus stabilizer (Medtronic, Minneapolis, MN) uses a series of suction cups on two fixed arms that adhere to the epicardial surface and reduce myocardial motion at the anastomotic site during OPCAB. (Reproduced with permission from Estafanous FG, Barash PG, Reves JG: *Cardiac Anesthesia: Principles and Clinical Practice*, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. See Appendix K: Enhanced Recovery After Surgery, p. K-1.**

SUMMARY OF PROCEDURE

Position	Supine

Incision	Median sternotomy (OPCAB); alternatively, limited left anterolateral thoracotomy (MIDCAB; Fig. 6.2-2)
Antibiotics	Cefazolin 2 g iv
Surgical time	3–4 h
Closing considerations	No special considerations. Patients often can be extubated in OR
EBL	200–400 mL (consider using Cell-Saver)
Postop care	ICU for cardiac monitoring ± ventilator management, less volume required than with conventional CABG
Mortality	0–5% (higher with increasing age)
Morbidity	AF: 10–30% Pulmonary insufficiency: 3–5% Reoperation for bleeding: 1–4% Cardiac (e.g., myocardial infarction): 0–4% Mediastinal infection: 1–2% Neurologic (cerebrovascular accident): 1–2% Renal dysfunction: 1–2% IABP use: 0–2%
Pain score	4–6

PATIENT POPULATION CHARACTERISTICS

Age range	40 to >90 yr
Male:Female	3:2
Incidence	150,000/yr in the United States
Etiology	Multifactorial for atherosclerosis
Associated conditions	HTN, diabetes, PVD, COPD, cigarette smoking

ANESTHETIC CONSIDERATIONS FOR TECAB/MICAB/OPCAB

PREOPERATIVE

Preop assessment is similar to that used for standard CABG patients. Relative contraindications for MICAB include low EF, morbid obesity, previous cardiac surgery, AF, and COPD or other pulmonary pathology precluding the use of one-lung ventilation (OLV). It is important to know the coronary artery anatomy and the planned surgical procedure and sequence. For example, proximal surgical occlusion of a coronary artery with high-grade distal disease may be poorly tolerated as compared with severe proximal disease with collateralization.

Premedication	Adequate control of preop anxiety using midazolam (titrated to effect) may reduce periop tachycardia.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: For MICAB/MIDCAB, selective lung ventilation with DLT, Univent ETT, or bronchial blocker. Unlike CABG on CPB, the anesthetic demands for OPCABG are somewhat different, as CPB will not be providing hemodynamic support. Positioning of the heart for access to the target vessel and mechanical stabilization of the heart to immobilize the vessel for accurate anastomosis tend to produce hemodynamic compromise. In particular, elevation of the LV apex to allow access to lateral and posterior wall vessels can limit LV filling and obstruct RV outflow tract. In addition, ↓ TV may be necessary to prevent obstruction of visualization. Snares may be placed around the coronary target vessel to create a dry operative field; however, this may provoke regional ischemia. Many of these changes are assessed by intraop TEE, and they may require resuscitative measures with volume loading, vasoconstriction, and inotropic support (e.g., dopamine). Interventions on the RCA are particularly prone to arrhythmias, and appropriate antiarrhythmics and pacing should be available. For further discussion, see Intraoperative considerations for CABG.

Induction	As for CABG surgery. For MICAB, these patients will undergo a small anterior thoracotomy, and the aim is to extubate them at the end of the procedure or shortly thereafter. Thus, the dose of narcotic should be moderate (e.g., fentanyl 5–15 mcg/kg). Long-acting muscle relaxants should be avoided for the same reason. Consider regional anesthetic techniques (e.g., paravertebral or intercostal blocks) and local anesthetic infiltration.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.
Maintenance	Because early extubation is planned, volatile agents or propofol should be used and the overall dose of narcotic reduced. Remifentanil infusion may also be useful. Recommend the use of antifibrinolytics for all cases requiring CPB.
Emergence	For further discussion, see Postoperative Considerations for CABG. If the patient is not suitable for extubation in the OR, a DLT may be exchanged for a conventional ETT (over a tube changer) before transport to ICU. Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.
Blood and fluid requirements	See CABG surgery Without the use of CPB, hemodilution from the pump prime is avoided. See Goal-Directed Fluid Management, Appendix K, p. K-4.

Monitoring	See CABG surgery	Central venous access. PA catheters are useful for detecting ischemia in patients with poor LV function. If a continuous CO thermodilution catheter is used, consideration must be given to the delays in data display. ST segment analysis.
	TEE	Useful for detecting ischemia, LV dysfunction, and load assessment. Views may be limited by surgical positioning of the heart within the chest.
Special considerations	Anticoagulation	Heparin should be administered before surgical coronary artery occlusion.
	Ischemic preconditioning temporary occlusion → ischemia	Prior to occlusion, ischemic preconditioning may be accomplished with surgical manipulation and by pharmacologic means (e.g., isoflurane, sevoflurane). This is a great opportunity to consider how well transient ischemia is tolerated. During grafting, the vessel is temporarily occluded to avoid flooding the field with blood. The myocardium distal to this may become ischemic.

	✓ ECG, TEE	TEE imaging will be affected by surgical mechanical manipulation of the heart, and ECG voltage may be decreased.
	Cardiac positioning →↓ CO	Cardiac output may be markedly reduced by the combination of heart positioning and coronary artery occlusion. Vasoconstriction may be required to maintain adequate blood pressure. Patients poorly responsive to vasoconstrictors may have extremely depressed cardiac output and severe acidosis. Inotropic support or intra-aortic balloon pumping may be necessary.
	Coronary artery shunt	Surgical placement of a coronary artery shunt may be necessary. Treatment of rhythm disorders with pharmacologic agents, pacing, or defibrillation may be required. The capability to institute CPB rapidly should exist. A rest period with the heart returned to the normal position within the chest should be considered following each

		anastomosis to restore cardiac output and permit resolution of acidosis.
Positioning	✓ pad pressure points ✓ eyes	

POSTOPERATIVE

Complications	See CABG surgery Premature extubation	
Multimodal pain management	Same as CABG surgery	Preop placement of paravertebral catheters and postop intercostal blocks will decrease opioid use and aid in early extubation. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-1)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.
Tests	See CABG surgery	

Suggested Readings

1. Caputo M, Alwair H, Rogers CA, et al: Thoracic epidural anesthesia improves earl

outcomes in patients undergoing off-pump coronary artery bypass surgery: a prospective, randomized, controlled trial. Anesthesiology 2011; 114(2):380–90.

2. Crumley S, Schraag S: The role of local anaesthetic techniques in ERAS protocol for thoracic surgery. J Thorac Dis 2018; 10(3):1998–2004.

3. Dorsa AG, Rossi AI, Thierer J, et al: Immediate extubation after off-pump coronar artery bypass graft surgery in 1,196 consecutive patients: feasibility, safety and predictors of when not to attempt it. J Cardiothorac Vasc Anesth 2011; 25(3):431–6.

4. Hemmerling TM, Romano G, Terrasini N, Noiseux N: Anesthesia for off-pum coronary artery bypass surgery. Ann Card Anaesth 2013; 16(1):28–39.

5. Lamy A, Devereaux PJ, Prabhakaran D, et al: Effects of off-pump and on-pum coronary-artery bypass grafting at 1 year. N Engl J Med 2013; 368(13):1179–88.

6. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery.J Cardiothorac Vasc Anesth 2018; 32(6):2760–70.

7. Raja SG, Berg GA: Impact of off-pump coronary artery bypass surgery on systemi inflammation: current best available evidence. J Card Surg 2007; 22:445–55.

8. Roosens C, Heerman J, De Somer F, et al: Effects of off-pump coronary surgery o the mechanics of the respiratory system, lung, and chest wall: comparison with extracorporeal circulation. Crit Care Med 2002; 30:2430–7.

9. Shaefi S, Mittel A, Loberman D, et al: Off-pump versus on-pump coronary arter bypass grafting-a systematic review and analysis of clinical outcomes. J Cardiothorac Vasc Anesth 2018; 33(1):232–44.

0. Shroyer AL, Grover FL, Hattler B, et al: Veterans affairs randomized on/off bypas (ROOBY) study group. On-pump versus off-pump coronary-artery bypass surgery.N Engl J Med 2009; 361(19):1827–37.

1. Shroyer L, Hattler B, Wagner TH, et al: Five-year outcomes after on-pump and off pump coronary-artery bypass. N Engl J Med 2017; 377(7):623–32.

2. Takagi H, Mitta S, Ando T, et al: The lion and the unicorn were fighting for the crown: on-pump versus off-pump coronary-artery bypass grafting. J Thorac Di 2017; 9(12):4893–95.

PORT-ACCESS AND TECAB CORONARY REVASCULARIZATION

SURGICAL CONSIDERATIONS

Description

Advances in videoscopic technology have led to less-invasive approaches in the treatment of many general and thoracic surgical disorders. The development of less-invasive surgery has resulted in alternative and novel approaches to cardiac surgery (Fig. 6.2-4), including port-access cardiac surgery and off-pump coronary revascularization. The port-access approach was developed in the mid-1990s and is now infrequently used because of its complexity. With the development of robotic (or total endoscopic) techniques, the port-access technology is being employed as a means to achieve CPB and cardioplegic arrest. Using a port-access approach, the surgeon can perform cardiac operations (e.g., CABG) and valve surgery in a motionless, bloodless field through smaller chest incisions. This approach typically relies on peripheral CPB (femoral artery and femoral vein cannulation, Fig. 6.2-5). The femoral artery is cannulated with a 19- to 23-Fr Y-shaped cannula, which permits arterial inflow and insertion of the endoaortic clamp. Venous drainage is provided by the 22- to 25-Fr cannula, introduced through a femoral vein. Drainage may be augmented by 20–40%, using vacuum-assisted venous drainage or a centrifugal venous drainage pump placed between the venous cannula and the reservoir. The port-access system includes a 10.5-Fr endoaortic “clamp” (EAC), a triple-lumen catheter with an inflatable balloon at its distal end. This clamp is positioned in the ascending aorta using fluoroscopy and TEE guidance. The lumen used for balloon inflation is connected to a manometer to monitor balloon pressure. Cardioplegic solution is delivered through a central lumen, which also acts as an aortic root vent after cardioplegia delivery. A third lumen serves as an aortic root pressure monitor. In addition, a percutaneous PA-venting catheter, placed via the jugular approach, helps in ventricular decompression. The left IMA is harvested under direct vision (Fig. 6.2-6), with video-assisted thoracoscopy (VAT; Fig. 6.2-7) or robotic assistance. After peripheral CPB is achieved, the EAC provides aortic occlusion and cardioplegic arrest. Exposure of the lateral and posterior aspects of the heart is easily accomplished in the arrested heart, thereby permitting two- and three-vessel coronary revascularization. Notably, if the ascending aorta is accessible, a dual-armed, Y-shaped cannula can be placed directly into the ascending aorta with one arm of the cannula connected to the arterial inflow of the CPB circuit and the other arm as a conduit for introducing the EAC.

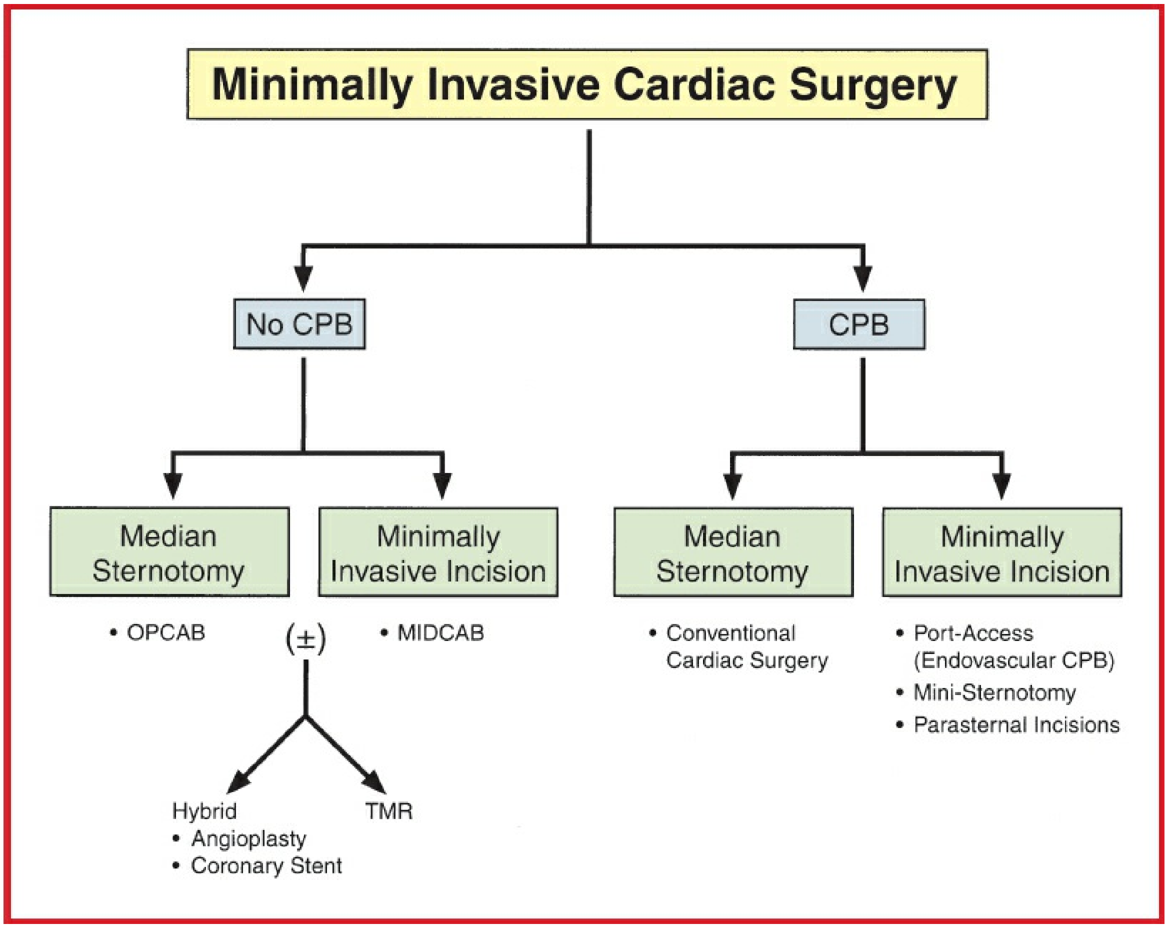

Figure 6.2-4 Variations in coronary artery bypass and valvular heart surgery, with minimally invasive techniques. CPB, cardiopulmonary bypass; OPCAB, off-pump coronary artery bypass; MIDCAB, minimally invasive direct coronary artery bypass; TMR, transmyocardial revascularization. (Redrawn with permission from Estafanous FG, Barash PG, Reves JGCardiac Anesthesia: Principles and Clinical Practice, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

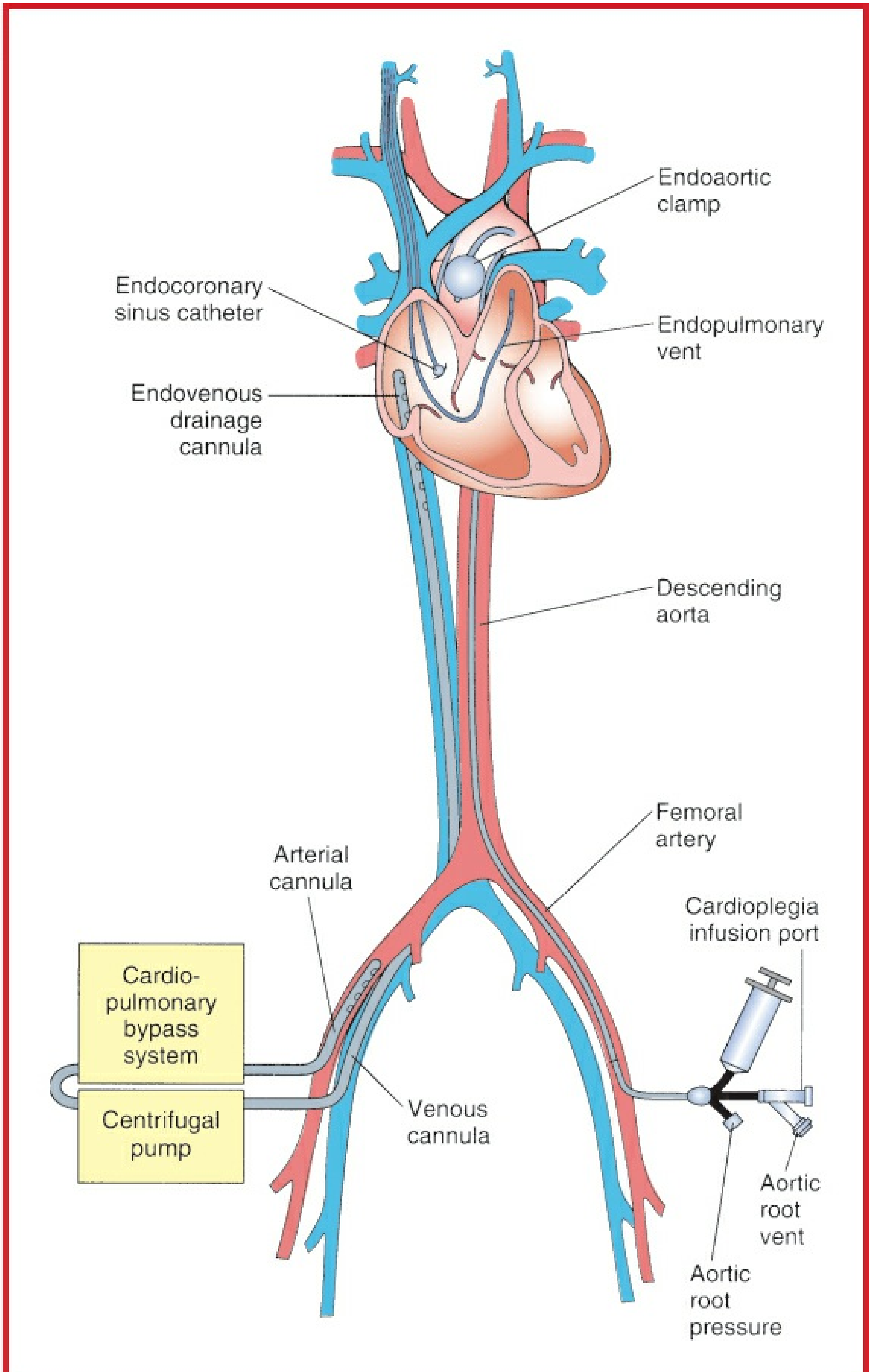

Figure 6.2-5 The port-access cardiopulmonary bypass (CPB) system. Femorofemoral (fem-fem) CPB

is utilized, and a centrifugal pump augments venous drainage. The endoaortic balloon occlusion catheter is inflated in the ascending aorta, and antegrade cardioplegia is delivered through the central lumen. The endopulmonary vent assists in ventricular decompression. (Redrawn from Fann JI, Pompili MF, Stevens JH, et al: Port-access cardiac surgery with cardioplegic arrest. Ann Thorac Surg 1997; 63:S35–9. Copyright © 1997 Elsevier. With permission.)

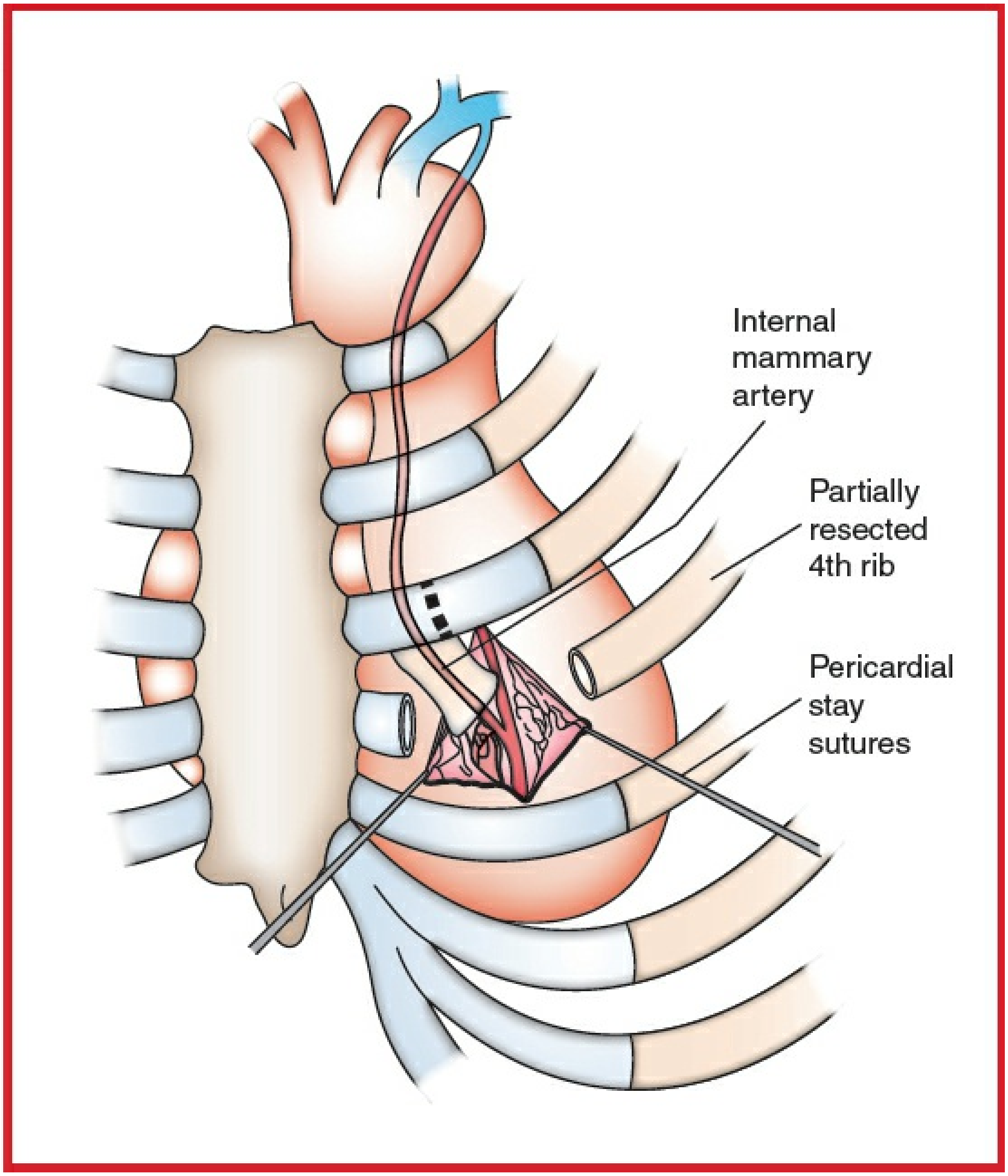

Figure 6.2-6 The left IMA-to-left anterior descending artery anastomosis is performed under direct vision. Stay sutures suspend the pericardium and bring the heart into view. (Redrawn from Acuff TE, Landreneau RJ, Griffith BP, et al: Minimally invasive coronary artery bypass grafting. Ann Thorac Surg 1996; 61(1):135–7. Copyright © 1996 Elsevier. With permission.)

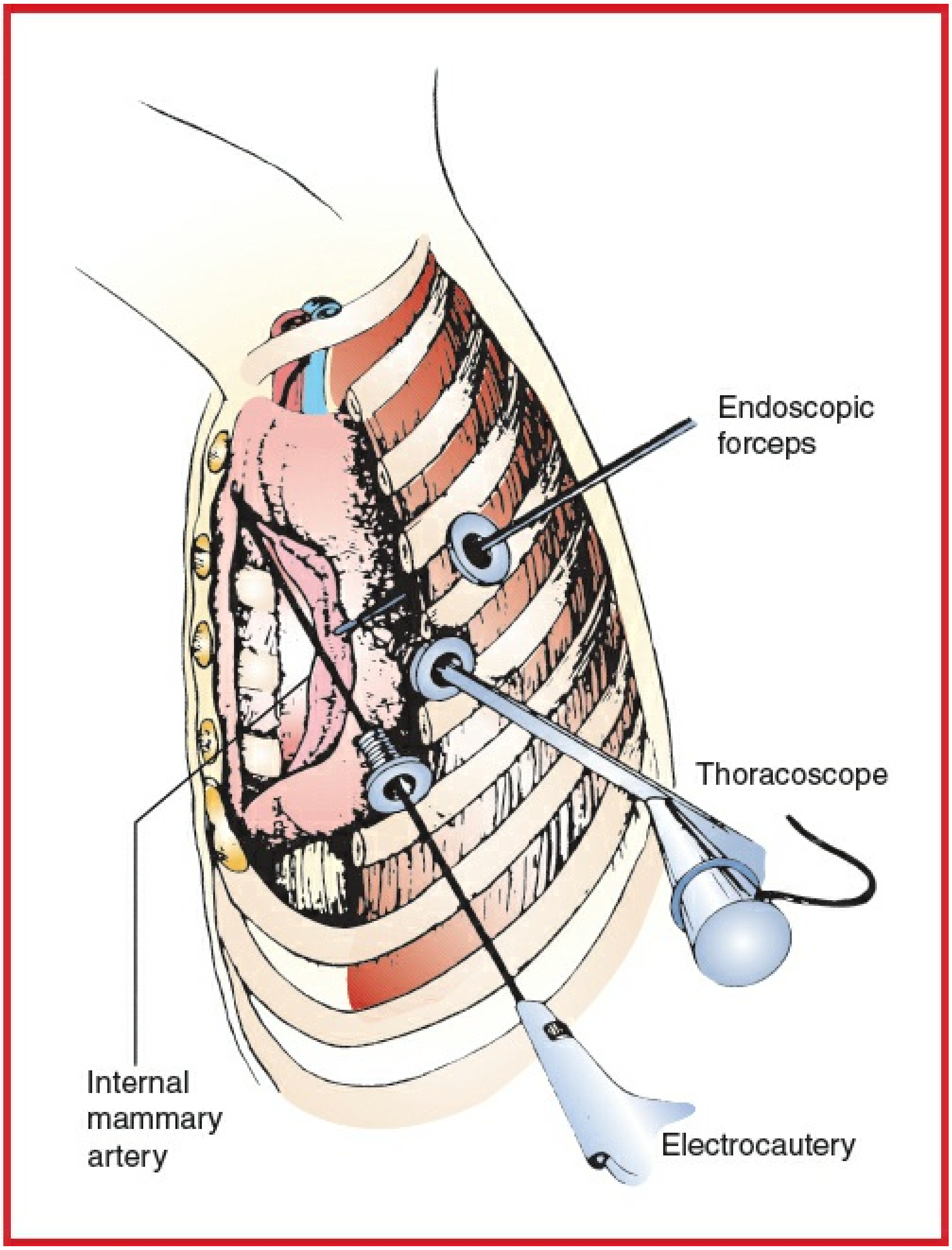

Figure 6.2-7 Dissection and harvesting of the left IMA using a thoracoscopic approach. (Redrawn from Acuff TE, Landreneau RJ, Griffith BP, et al: Minimally invasive coronary artery bypass grafting. Ann Thorac Surg 1996; 61(1):135–7. Copyright © 1996 Elsevier. With permission.)

Usual preop diagnosis

Coronary artery disease (CAD)

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and**

physiotherapy plan. **See Appendix K, p. K-1.**

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	4th interspace, left anterior thoracotomy for CABG; may need to resect the 4th rib; may need thoracoscopy for IMA harvest
Special instrumentation	TEE and fluoroscopy
Unique considerations	CPB used with peripheral femoral cannulation, percutaneous insertion of PA vent and retrograde cardioplegic catheter
Antibiotics	Cefazolin 2 g iv at induction
Surgical time	3–5 h
Closing considerations	Assess chest wall for hemorrhage, chest tube insertion. Bupivacaine for field block
EBL	250–500 mL
Postop care	ICU monitoring, early extubation, may require inotropic support
Mortality	1–3%, depending on comorbidities
Morbidity	Arrhythmias (including atrial fibrillation): 10% Conversion to sternotomy: 4% MI: 0–4% Reoperation for bleeding: 3% Stroke: 2% Thrombosis of graft (graft failure): rare Fem-fem bypass complications (e.g., arterial dissection): rare
Pain score	1–4

■ PATIENT POPULATION CHARACTERISTICS	
Age range	40–80 yr

Male:Female	3:2
Incidence	<50,000/yr in the United States
Etiology	Multifactorial for atherosclerosis
Associated conditions	LV dysfunction, COPD, PVD, HTN, diabetes mellitus, cigarette smoking

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Port-Access Procedures.

Suggested Readings

1. Dogan S, Graubitz K, Aybek T, et al: How safe is the port access technique in minimally invasive coronary artery bypass grafting? Ann Thorac Surg 2002; 74:1537–43.
2. Ishikawa N, Watanabe G: Robot-assisted cardiac surgery. Ann Thorac Cardiovasc Surg 2015; 21(4):322–8.
3. Kofler M, Schachner T, Reinstadler SJ, et al: Comparative analysis of perioperative and mid-term results of TECAB and MIDCAB for revascularization of anterior wall. Innovations (Phila) 2017; 12(3):207–13.
4. Langer NB, Argenziano M: Minimally invasive cardiovascular surgery: incisions and approaches. Methodist Debakey Cardiovasc J 2016; 12(1):4–9.
5. Maselli D, Pizio R, Borelli G, et al: Endovascular balloon versus transthoracic aortic clamping for minimally invasive mitral valve surgery: impact on cerebral microemboli. Interact Cardiovasc Thorac Surg 2006; 5:183–6.
6. See Suggested Readings for Port-Access Procedures.

LIMITED THORACOTOMY AND PORT-ACCESS APPROACHES TO VALVE SURGERY

SURGICAL CONSIDERATIONS

Description

Although the field of mitral valve surgery has seen marked advances in biomaterials and innovative repair techniques since the early 1960s, the **conventional median sternotomy approach** to access and expose the mitral valve has not changed substantially. Because of the progress in video-assisted surgery, a less-invasive approach to cardiac surgery has been developed, and various techniques of mitral valve surgery through **limited thoracotomy** or **upper sternotomy** incisions and a **port-access technique** to achieve cardioplegic arrest are now used in the clinical setting. The transcatheter approach to valve surgery (see p. 440) may eventually supplant open surgical valve repair/replacement techniques.

Limited thoracotomy

The right thoracotomy incision is a less-invasive approach (compared to median sternotomy) for mitral valve procedures (Fig. 6.2-8). Utilizing hypothermic fibrillatory or cardioplegic arrest, the mitral valve, annulus, and subvalvular apparatus can be visualized directly and the valve procedure carried out. The addition of VAT to mitral valve surgery or using robotic technology has resulted in even smaller incisions for these procedures.

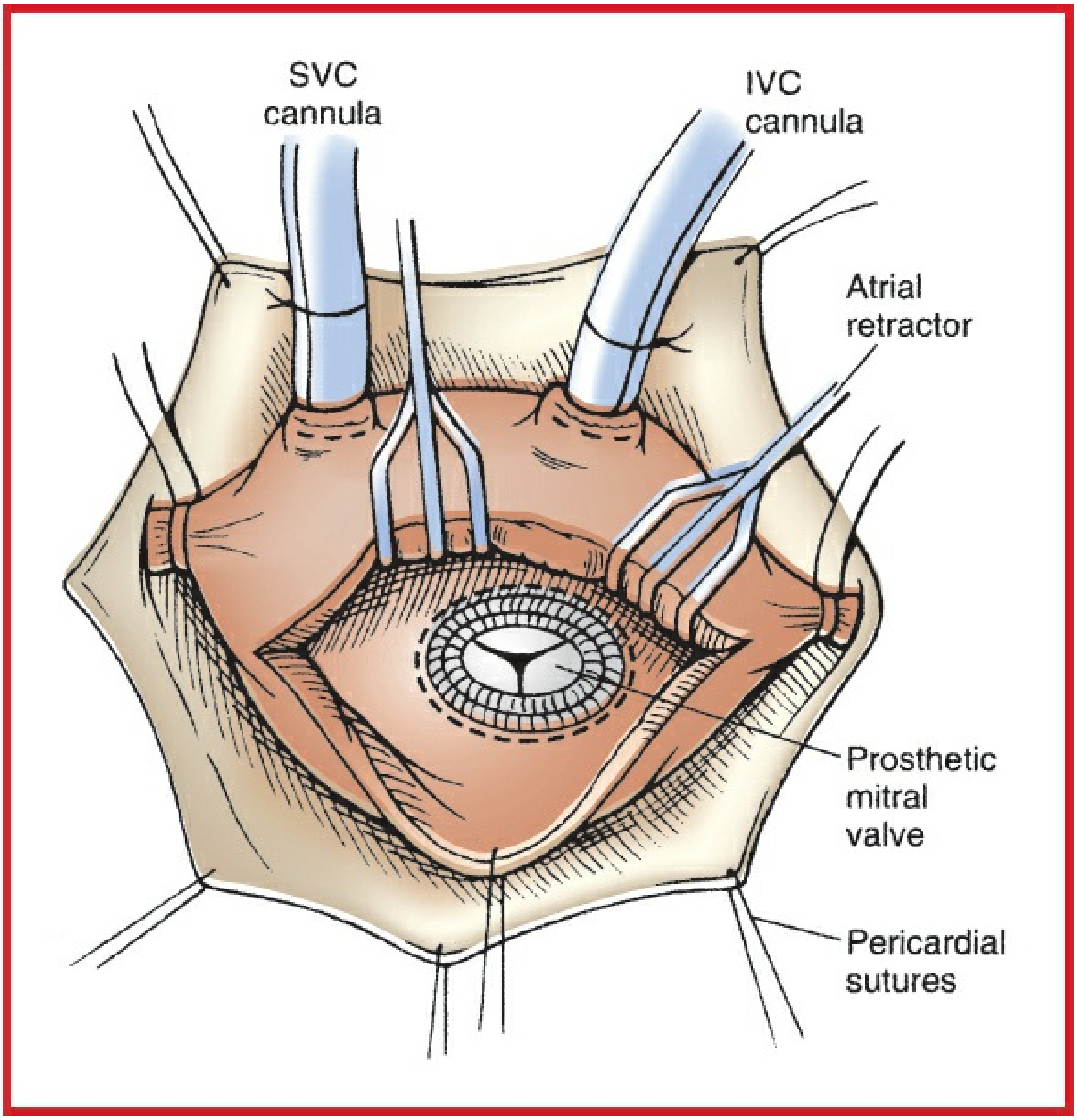

Figure 6.2-8 The right thoracotomy approach with left atriotomy and exposure of the mitral valve area

with prosthetic valve in place. (Reprinted from Tribble CG, Killinger WA Jr, Harman PK, et al: Anterolateral thoracotomy as an alternative to repeat median sternotomy for replacement of the mitral valve. Ann Thorac Surg 1987; 43:380–2. Copyright © 1987 The Society of Thoracic Surgeons. With permission.)

A "micromitral" approach to mitral valve replacement has been described, using VAT through a limited thoracotomy. Peripheral CPB (fem-fem) is used, and a catheter is placed in the coronary sinus for retrograde cardioplegia delivery. An external aortic cross-clamp is introduced through a separate incision in the chest. After achieving cardioplegic arrest, the mitral valve is replaced with thoracoscopic assistance. Proposed advantages of the micromitral approach include the avoidance of a sternotomy, with decreased chest wall trauma and patient discomfort.

An alternative **partial sternotomy approach** to mitral and aortic valve surgery has been described. The 7-cm partial sternotomy permits aortic and right atrial cannulation for CPB. The external aortic cross-clamp is positioned, and a left ventricular vent is placed through the right superior pulmonary vein. Mitral valve exposure is achieved through the dome of the left atrium.

Port-access mitral valve surgery

The port-access system has been used successfully in mitral valve surgery via limited thoracotomy incision using special instrumentation or even less-invasive robotic technology. To facilitate dissection and to provide adequate exposure of the left atrium, DLT intubation for OLV is used. Under fluoroscopic and TEE guidance, a retrograde cardioplegia catheter is directed into the coronary sinus via an introducer placed in the jugular vein; a PA-venting catheter is inserted through another jugular vein introducer. A limited right thoracotomy is made, with or without dividing the 4th rib, followed by the placement of a soft tissue retractor. A separate port is placed in the 6th interspace for introduction of a thoracoscope, if necessary. The pericardium is opened anterior to the phrenic nerve. After systemic heparinization, the femoral artery and vein are cannulated. The endoaortic clamp is introduced through the side limb of the femoral arterial cannula and its tip positioned in the ascending aorta. CPB is initiated and systemic hypothermia achieved. The balloon of the endoaortic clamp is inflated, achieving effective aortic occlusion. Cold blood cardioplegia is delivered using the distal port of the endoaortic clamp; retrograde cardioplegia is administered via the coronary sinus catheter. A left atriotomy is made, and an atrial retractor is placed through a separate port. Valve repair or replacement is carried out using specially designed instruments. Before completion of atriotomy closure, deairing maneuvers are accomplished. These include temporarily discontinuing pulmonary and aortic root venting, inflating the lungs to displace residual air, and increasing the patient's blood volume from the venous reservoir. Also, the patient is placed in a Trendelenburg and left lateral decubitus position for further deairing. The balloon of the endoaortic catheter is deflated, and the catheter is left in place for further deairing through the aortic vent lumen. Temporary ventricular pacing wires are placed. After being weaned from CPB, the patient is decannulated and the anticoagulation reversed.

Usual preop diagnosis

Mitral valve disease

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. **See Appendix K, p. K-1.**

SUMMARY OF PROCEDURE

	Port-Access/Limited Thoracotomy
Position	Supine, with right side slightly elevated
Incision	4th interspace limited right thoracotomy, ports for possible thoracoscopy, port for atrial retractor
Special instrumentation	Specially designed retractors; TEE and fluoroscopy for guiding placement of endopulmonary vent, retrograde cardioplegia catheter, and endoaortic clamp
Unique considerations	CPB is achieved with peripheral femoral cannulation, placement of endopulmonary vent, placement of retrograde cardioplegia catheter, may require conversion to open procedure
Antibiotics	Cefazolin 1–2 g iv at induction
Surgical time	3–5 h
Closing considerations	Assess chest wall for hemorrhage
EBL	250–500 mL
Postop care	ICU monitoring, early extubation, may require inotropic

	support
Mortality	1–6%, depending on comorbidities (lower for mitral valve repairs)
Morbidity	AF:10–40% Chest wall hemorrhage: 2–9% Stroke: 1–3% Conversion to sternotomy: 2% Vascular injury: rare Pneumonia Renal failure
Pain score	1–4

PATIENT POPULATION CHARACTERISTICS

Age range	40–70 yr
Male:Female	1:1
Incidence	<20,000/yr in the United States
Etiology	Myxomatous mitral valve (most common in developed countries), rheumatic valve disease (most common in developing countries), endocarditis, ischemic (papillary muscle dysfunction), mitral annular calcification
Associated conditions	Pulmonary HTN, atrial arrhythmias, CAD, aortic valve disease, tricuspid regurgitation

TRANSCATHETER VALVE REPLACEMENT PROCEDURES

SURGICAL CONSIDERATIONS

Description

Minimally invasive approaches to aortic valve replacement include transvenous (antegrade), transarterial (retrograde), and transapical (transthoracic). These minimally invasive valve replacement procedures are usually performed in the cath lab, may only require sedation (transvenous and transarterial approach), and can be accomplished without CPB. Transcatheter aortic valve replacement (TAVR) is becoming the preferred treatment for patients with severe aortic stenosis (AS) who are otherwise at high risk for conventional sternotomy; however, trials are underway to evaluate its use in lower-risk populations. An alternative treatment, percutaneous balloon aortic valvuloplasty (BAV), has a high rate of restenosis and is now infrequently performed except to provide temporary relief while patients await more definitive treatment and during TAVR in preparation for valve placement.

Transcatheter approaches to mitral valve replacement have been slower to develop, in part because of the irregular shape of the valve orifice and the absence of heavy calcification to help anchor the prosthetic valve.

Transvenous aortic or mitral valve replacement: Under sedation, a guide wire is inserted through a femoral vein sheath into the right atrium. The atrial septum is punctured allowing the guide wire to enter the left atrium to cross the mitral valve and aortic valve, where it is snared by a device in the aorta introduced via the femoral artery (Fig. 6.2-9). The atrial septal puncture and the diseased valve are dilated, and then the compressed prosthetic valve is positioned and deployed during rapid ventricular pacing (RVP), which is used to ↓ LV stroke volume and the risk of unwanted device movement. This approach has largely been replaced by transarterial and transapical approaches for aortic valve replacement.

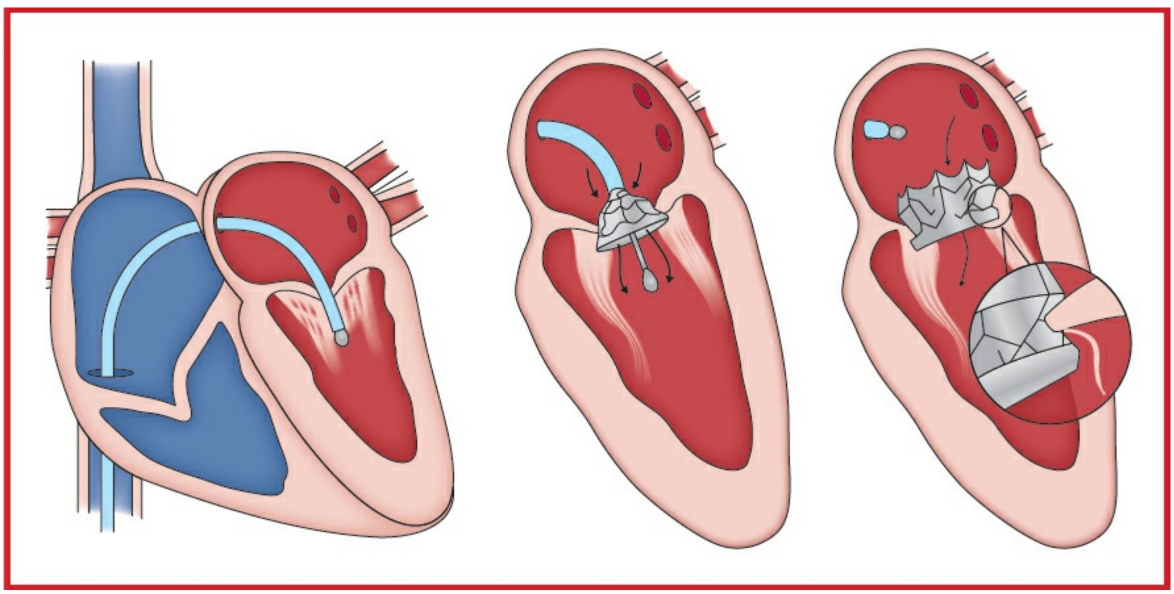

Figure 6.2-9 Transseptal (transvenous) approach to mitral valve (and aortic valve).

Transarterial aortic valve replacement: Under MAC or GETA, a guide wire is inserted from a femoral artery through the aorta (or directly into the aorta through the chest wall) and across the aortic valve. As in the transvenous approach, the diseased valve is dilated, and then, the compressed prosthetic valve is positioned and deployed (Fig. 6.2-

10) during RVP to ↓ LV stroke volume and the risk of unwanted device movement. A self-expanding aortic valve, which is deployed without the need for RVP, is available, although the incidence of postop heart block is high and necessitates placement of a temporary transvenous pacemaker. Although the direct arterial approach to the valve has some advantages, patients with small, tortuous, or diseased iliac and femoral arteries may not be suitable candidates for distal arterial access, and more proximal access to the aorta may be obtained through a minithoracotomy.

Transapical aortic valve replacement: Under GETA, a cardiothoracic surgeon exposes the LV apex through an anterolateral intercostal incision. Purse-string sutures are preplaced around the guide wire entry site at the apex, and the beating heart is punctured with insertion of a guide wire across the aortic valve. A sheath is placed, the diseased valve is dilated, and then, the compressed prosthetic valve is positioned and deployed (Fig. 6.2-11) during rapid epicardial pacing to ↓ LV stroke volume and the risk of unwanted device movement. The sheath is removed and the myocardium closed using the preplaced sutures.

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. See Appendix K, p. K-1.

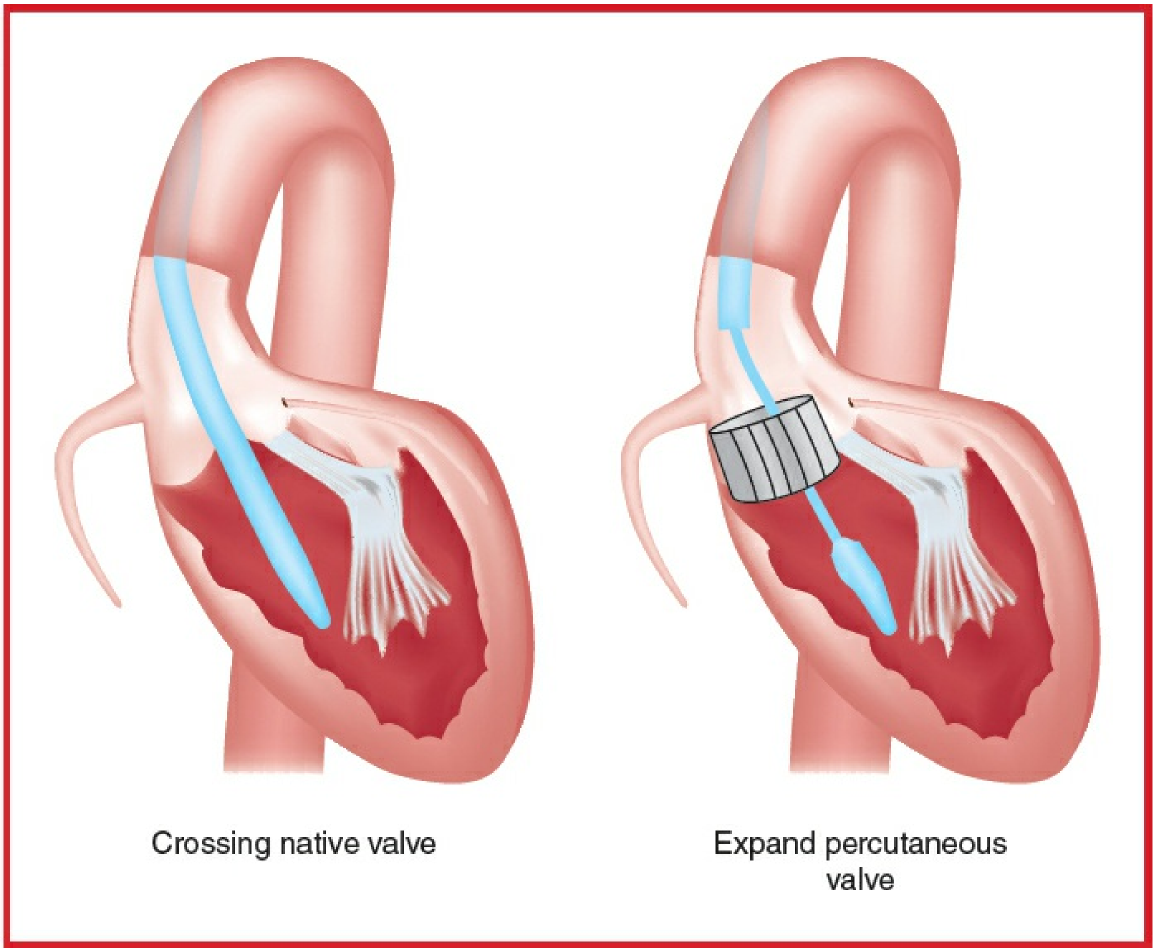

Figure 6.2-10 TAVR—Transcatheter aortic valve implantation: retrograde approach.

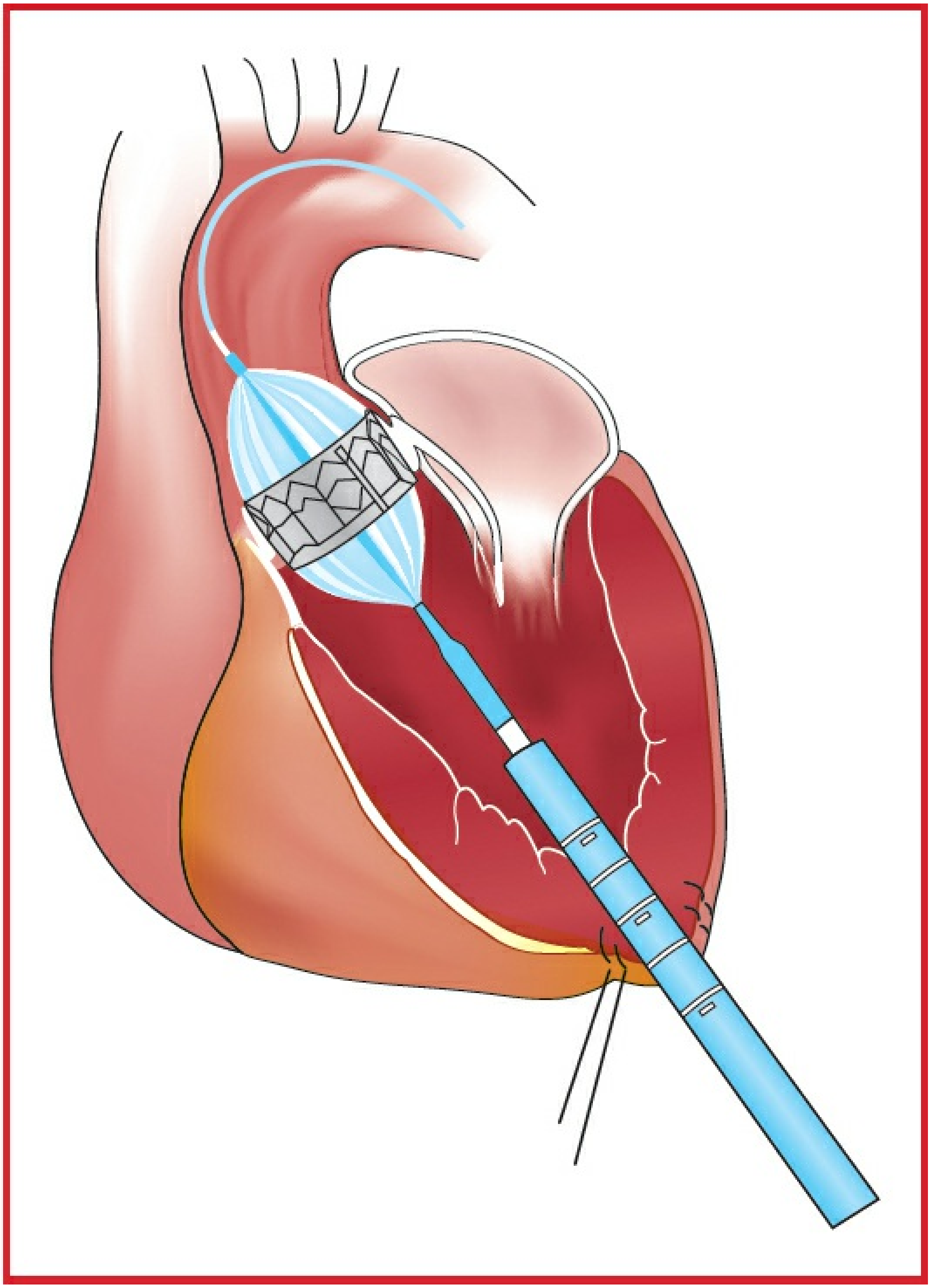

Figure 6.2-11 Transapical approach to aortic valve.

■ SUMMARY OF PROCEDURES

	Transvascular	Transapical
Position	Supine	⇐
Incision	Vascular sheaths	⇐ + Left

		minithoracotomy
Unique considerations	Anticoagulation (100 U heparin/kg) CPB standby (additional heparin will be necessary for CPB). Inotropic/vasopressor support may be necessary. TEE may be useful for valve sizing determination, positioning verification, and hemodynamic monitoring	⇐
Key procedural steps [with typical complications]	Retrograde placement of pigtail catheter in the noncoronary sinus (serves as landmark to depict the level of the aortic annulus) [intraventricular position with provocation of arrhythmias]	⇐
	—	Anterolateral minithoracotomy Pericardiotomy [pneumothorax, lung injury, pleural bleeding]
	Placement of temporary transvenous (transjugular or transfemoral) pacemaker in the right ventricle, testing of capture [ventricular perforation by pacemaker, pacer-induced hemodynamic deterioration]	Placement of epicardial pacemaker electrodes, testing of capture [bleeding, pacer-induced hemodynamic deterioration]
Key procedural steps [with typical complications] (cont'd)	—	Placement of purse-string sutures at the anterolateral left ventricular wall, myocardial puncture, placement of delivery

		sheath [bleeding, tamponade, air embolism, MV damage]
	Retrograde placement of a guide wire across the aortic valve [dysrhythmias]	Antegrade placement of a guide wire across the aortic valve [dysrhythmias]
	Start of rapid ventricular pacing (160–220 bpm) to ensure a stable balloon position during BAV [hemodynamic deterioration, shock, myocardial ischemia,↑ MR, VF]	⇐
	BAV [calcific embolism with myocardial or cerebral infarction, AR, rupture of aortic annulus/LVOT, aortic dissection, AV block]	⇐
	Recovery	Recovery
	Retrograde placement of transcatheter aortic delivery catheter [dysrhythmias]	Antegrade placement of transcatheter aortic delivery catheter [dysrhythmias]
	Start of rapid ventricular pacing (160–220 bpm) to ensure a stable device position during valve deployment [hemodynamic deterioration, shock, myocardial ischemia,↑ MR, VF]	⇐
	Valve positioning and deployment [calcific embolism with myocardial or cerebral infarction, coronary flow obstruction, valve	⇐

	embolization/malposition/failure, paravalvular or transvalvular regurgitation, MV damage, rupture of aortic annulus/LVOT, aortic dissection, AV block]	
	Recovery and assessment of regular valve position and function	⇐
	Removal of delivery sheath using a vascular closure device or surgical closure, crossover iliofemoral angiogram to assess hemostasis and patency [bleeding, dissection, ischemia, atrioventricular fistula, pseudoaneurysm]	Retrieval of delivery sheath, closure of apical puncture site and pericardium, closure of surgical incision [bleeding, dysrhythmias]
Antibiotics	Cefazolin 2 g iv	⇐
Surgical time	1–3 h	2–4 h
Closing considerations	Assess entry sites for hemorrhage	⇐
EBL	Usually <200 mL	⇐
Postop care	May be monitored on cardiac floor, ICU stay routinely not necessary	Short ICU stay and may remain intubated
Mortality	30 d, 5–11%; 1 yr, 18–27%	⇐
Morbidity	MI (<72 h): 0.2–2% Stroke (<30 d): 4–9% Renal injury: 16–26% Bleeding: 12–21% Vascular complications: 14–24% Moderate-severe AR: 5–10%	⇐
Pain score	2–3 (6–8 if transaortic	6–8

	approach)

PATIENT POPULATION CHARACTERISTICS

Age range	65–90
Male:Female	3:2
Incidence	1.3–4% of the population >65 yo
Etiology	Degenerative calcification most common in this age group
Associated conditions	CAD (50%); HTN (40%); CHF, AF. Aortic regurgitation, mitral valve disease, and von Willebrand disease with impaired platelet function are common with severe AS

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for transcatheter aortic valve replacement (TAVR) are typically too frail to successfully undergo open surgical valve replacement. In addition to the problems associated with critical AS, these patients often have severe comorbidities including pulmonary hypertension, decreased cardiac function, significant mitral regurgitation (MR) or tricuspid regurgitation (TR), COPD, and renal disease that may challenge anesthetic management. TAVR is not currently recommended in patients with active endocarditis, thrombi in the left ventricle or ascending aorta, high risk of intraprocedure coronary ostium occlusion, or a valve annulus outside the range of 18–29 mm. For a complete preop discussion of AS, see Anesthetic Considerations for Aortic Valve Replacement.

INTRAOPERATIVE

Anesthetic technique

GETA is used for a transapical approach and is often used for transvascular approaches, although sedation may be appropriate for some patients (↓ hemodynamic compromise, faster recovery). Although these cases were primarily managed by GETA when this procedure was first being developed, with improvements and familiarity with the procedure, more and more centers are performing this procedure under MAC/sedation. Considerations for MAC versus GA include patient preference cooperation, cardiopulmonary function, and need for intraop TEE guidance. CPB is no used in these patients; however, depending on the type of valve to be placed, there may be brief periods of RVP at 160–220 bpm that will produce a functional asystole (facilitates balloon and valve placement). For a complete discussion of intraop management in AS, see Anesthetic Considerations for Aortic Valve Replacement. A radial art line, external defibrillation pads, and standard monitors should be placed before induction. If TEE is utilized, it should be inserted postinduction. Maintain hemodynamic stability and coronary perfusion by guided volume repletion, avoiding hypotension or tachycardia, and preserving a sequential A-V rhythm. Before the start of RVP, ensure that MAP is >75 mm Hg (systolic BP > 120 mm Hg) using vasopressors as necessary. Hypertension after valve deployment increases the risks of ventricular rupture.

POSTOPERATIVE

After GETA, most patients can be extubated in the procedure room. For most patients hemodynamics are improved immediately following valve replacement and may require vasodilators to avoid HTN. However, diastolic dysfunction often persists and may worsen in the postop period, requiring continued hemodynamic support. Hypertension should be avoided to decrease the risk of ventricular rupture. In patients requiring a minithoracotomy, intercostal nerve blocks placed through the incision by the surgeons will provide good pain relief. Alternatively, paravertebral catheters can be used to infuse local anesthetic for continuous analgesia.

Suggested Reading

1. Gregory SH, Zoller JK, Shahanavaz S, et al: Anesthetic considerations fo transcatheter pulmonary valve replacement. J Cardiothorac Vasc Anesth 2018; 32(1):402–11.
2. Kanwar A, Thaden JJ, Nkomo VT: Management of patients with aortic valv stenosis. Mayo Clin Proc 2018; 93(4):488–508.
3. Mader T, Ngai J: "Transcatheter Valve Procedures and the Anesthesiologist", Int Anesthesiol Clin 2018 Fall;56(4):74–97.
4. Neuburger PJ, Saric M, Huang C, et al: A practical approach to managin transcatheter aortic valve replacement with sedation. Semin Cardiothorac Vasc Anesth 2016; 20(2):147–57.
5. Sharma A, Agrawal S, Goel S, et al: Surgical treatment of ischemic mitra regurgitation: valve repair versus replacement. Curr Cardiol Rep 2017; 19(1):3.
6. Telila T, Mohamed E, Jacobson KM: Endovascular therapy for rheumatic mitral anc aortic valve disease: review article. Curr Treat Options Cardiovasc Med 2018;

20(7):59.
7. van der Merwe J, Casselman F: Mitral valve replacement-current and future perspectives. Open J Cardiovasc Surg 2017;9:1179065217719023.

ANESTHETIC CONSIDERATIONS FOR PERCUTANEOUS AND PORT-ACCESS PROCEDURES

PREOPERATIVE

As with all anesthetic procedures, the use of the port-access system should not be attempted without appropriate training. Catheter placement and monitoring relies on TEE and/or fluoroscopy. The evaluation of the patient for port-access cardiac surgery should parallel that of patients having conventional cardiac surgery (for mitral valve surgery and for CABG surgery). There are, however, a number of specific conditions that preclude the use of the port-access system:

1. Aortic regurgitation (AR): Although mild-to-moderate AR is not a contraindication, it may make the delivery of antegrade cardioplegia via the endoaortic clamp catheter (EAC) problematic; therefore, insertion of the endocoronary sinus catheter (ESC) is very important. Severe AR is a contraindication.
2. Atherosclerotic disease of the aorta: Perfusion may be associated with embolization of atheromatous plaques (particularly pedunculated lesions).
3. PVD: Femoral bypass may be associated with retrograde arterial dissection. PVD and tortuous femoral and iliac vessels are contraindications to the use of femoral arterial cannulation.
4. Thoracic aortic aneurysm/Marfan syndrome: Port-access procedures require passage of the EAC into the thoracic aorta and inflation of the EAC balloon; therefore aneurysms or a weakened aortic wall is contraindication.
5. Scarring of the pleural cavity (e.g., chest trauma, previous thoracotomy) may make surgical access difficult although the surgical approach may be desirable in the patient with previous median sternotomy.
6. Inability to obtain TEE or fluoroscope imaging because placement of catheters relies on imaging.

Preop evaluation should focus on the underlying pathophysiology (valvular disease vs. CAD). In addition, the following aspects should be considered.

Respiratory	OLV is used to facilitate surgical exposure; thus, evaluation of patients with severe lung disease may include CXR, ABG, and PFTs (see Thoracic Surgery).

Cardiovascular	The vascular system should be evaluated with respect to insertion of the catheters and cannulae for endovascular CPB. Severity of arterial occlusive or atherosclerotic disease should be evaluated (embolization/dissection risk). The possible presence of a persistent left SVC should be considered. **Tests:** CXR; aortography; iliofemoral arteriography; vascular MRI/MRA; vascular ultrasound; TEE
Premedication	As for the underlying condition. Usually, a mild anxiolytic (e.g., midazolam 1–3 mg iv) is sufficient.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

Table 6.2-1 Common Abbreviations Used in Minimally Invasive Cardiac Surgery

Abbreviation	Definition
MICS	Minimally invasive cardiac surgery
MICAB	Minimally invasive coronary artery bypass (includes OPCAB, MIDCAB, port-access, and ministernotomy techniques)
OPCAB	Off-pump coronary artery bypass ("beating heart surgery": median sternotomy)
MIDCAB	Minimally invasive direct coronary artery bypass (direct vision, no sternotomy, CPB, or cardioplegia)
MIDCABG	Minimally invasive direct coronary artery bypass graft (same as MIDCAB)
MITACAB	Minimally invasive thoracoscopically assisted coronary artery bypass

VADCAB	Video-assisted direct coronary artery bypass
LAST	Left anterior small thoracotomy
PACAB	Port-access coronary artery bypass (also called Port-CAB, Port-CABG)
TECAB	Totally endoscopic coronary artery bypass grafting

INTRAOPERATIVE

Anesthetic technique

GETA (DLT or BB): Anesthetic technique depends on the underlying pathophysiology (e.g., valvular disease vs. CAD). Early extubation and postop pain relief can be facilitated by using intrathecal narcotics (e.g., Duramorph 10 mcg/kg intrathecal) injected before induction or paravertebral catheters placed preop.

Induction	Typically, induction can be accomplished with etomidate (0.1–0.3 mg/kg) or propofol (0.5–2 mg/kg), with fentanyl (5–20 mcg/kg) and rocuronium (0.6–1 mg/kg) or vecuronium (0.1 mg/kg). The overall length of port-access procedures is similar to or slightly longer than conventional approaches, so that a long-acting muscle relaxant can be used. Intubation is accomplished with a DLT, and its placement is checked by auscultation and/or bronchoscopy. A bronchial blocker or Univent tube may also be used to provide selective lung ventilation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.
Maintenance	With the goal of early extubation, it is important to avoid oversedation. The use of volatile agents or a propofol infusion (25–100 mcg/kg/min)

	before, during, and after CPB will facilitate early extubation. During dissection of the IMA or exposure of the left atrium, OLV is needed. Recommend the use of antifibrinolytics for all cases requiring CPB.
Emergence	At the end of the procedure, intercostal blocks with ropivacaine or bupivacaine may be placed, and infiltration of the skin incision also is helpful. Extubation in the OR may be appropriate for stable patients. Alternatively, the patient is transported to ICU and ventilated. The DLT may be exchanged for a single-lumen tube when postop mechanical ventilation is required. Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.
Special considerations: placement/monitoring of endocardiopulmonary system	The port-access system consists of CPB with venous drainage via an endovenous drain (EVD) and an aortic or femoral arterial return cannula, which is Y-ed to accept the EAC. These are placed by the surgeon with the aid of fluoroscopy and/or TEE. Additional drainage of the right side of the heart is accomplished via an endopulmonary vent (EPV) placed by the anesthesiologist. Immediately after intubation and before CVP insertion, a TEE exam should be performed to exclude any contraindication to endo-CPB (e.g., severe AR, aortic atheromatous disease, aortic aneurysm); the size of the ascending aorta should be measured (aids the surgeon in inflation of the aortic balloon); and the coronary sinus should be identified to aid placement of the ESC (coronary sinus catheter). The right IJ vein is then cannulated with two sheaths (9 Fr for EPV and 11 Fr for the ESC). Cardioplegia is delivered in an anterograde fashion via the EAC or retrograde via

an ESC, also placed by the anesthesiologist.

Placement of ESC	The ESC can be placed with fluoroscopy and/or TEE. The coronary sinus is identified on TEE (transverse view of the right atrium or longitudinal bicaval view). Prior to placement, the patient should be partially anticoagulated with 70 U/kg of heparin. The ESC is placed through the 11-Fr sheath. After the tip of the catheter engages the coronary sinus, the catheter is advanced either directly or over a guiding wire until the occlusion balloon is 2–4 cm inside the sinus. Position is confirmed by inflating the balloon and obtaining ventricularization of the pressure tracing. Careful note should be taken of the volume of fluid required to occlude the coronary sinus (1–2 mL) so that overinflation and possible coronary sinus trauma do not occur. Contrast injection will define the correct positioning of the ESC in the coronary sinus. Care should be taken to avoid injecting contrast too quickly, thus pushing the ESC out of the coronary sinus. A time limit (20–30 min) should be set to pass this catheter, as repeated attempts increase the risk of cardiac injury (e.g., cardiac perforation).
Placement of EPV	The EPV (pulmonary vent) is positioned by advancing the catheter, using balloon flotation with pressure monitoring (as with a Swan-Ganz catheter). Fluoroscopy and/or TEE also may be used.
Placement of EVD	The EVD (venous drain) cannula is placed via the femoral vein into the right atrium over a guide wire. Fluoroscopy or TEE is used to ensure that the wire enters the SVC before advancing the EVD into position. The tip of the EVD should be at the SVC/RA junction or just inside the SVC.
Placement of arterial cannula	The aorta may be cannulated with a "Y" cannula with an incising introducer. For femoral arterial

	cannulation, a guide wire is advanced under fluoroscopy, and no resistance to its passage should be felt. The guide wire is advanced into the descending aorta and its intraluminal position confirmed on TEE prior to advancing the aortic return cannulation into its final position. The perfusionist should confirm normal line pressures with a test bolus of fluid and ✓ that normal arterial pulsation is present. These precautions will decrease the likelihood of arterial dissection.
Placement of EAC	The EAC (endoaortic clamp) may be advanced via the "Y" arterial cannula. For a femoral arterial cannula, the EAC is advanced over a guide wire with imaging into the ascending aorta so that the tip of the catheter is just proximal to the sinotubular ridge (2–3 cm above the aortic valve). Position is confirmed by contrast injection and/or TEE. It is also useful to identify the takeoff of the innominate artery in relation to the balloon. Migration of the balloon proximally or distally can occur, and thus, its position needs to be monitored. Monitoring of the pressure in the aortic root via this catheter should show that the mean aortic root pressure is the same as mean radial artery pressure prior to the initiation of bypass.
Sequence of events during endo-CPB	After initiation of bypass, the anesthesiologist should observe the descending aorta to exclude aortic dissection and open the EPV to allow venting. EPV pressures during bypass should be negative (positive pressures may indicate inadequate decompression of the heart or kinking of the EPV). Once adequate CPB is established and the heart is drained, the EAC balloon can be inflated (balloon pressure = 250–350 mm Hg). Occlusion of the aorta is confirmed by a differential between the radial and aortic root pressures and by an aortic root contrast injection. Antegrade cardioplegia is delivered via the EAC

	with careful monitoring to ensure that the balloon is not displaced distally, thus occluding the innominate artery (see Monitoring of endo-CPB: regional perfusion, below). After cardioplegic arrest, retrograde cardioplegia may be delivered via the ESC, and the left side of the heart may be vented via the EAC. Avoid overventing, high systemic pressures, or high CPB flow, which could displace the EAC toward the aortic valve. Initiation of retrograde cardioplegia should begin with a low flow to avoid displacing the ESC, followed by inflation of the ESC balloon until the pressure in the coronary sinus just starts to rise, indicating coronary sinus occlusion. No further inflation of the balloon should take place. Normal coronary sinus perfusion pressure is <40 mm Hg. After retrograde cardioplegia is delivered, the ESC balloon is deflated. Once the surgery is completed, the EAC balloon is deflated (after deairing procedures, as needed); the heart is reperfused; and the EAC is removed. Prior to weaning from CPB, mobility of the ESC and EPV should be assessed in mitral procedures to ensure that they have not been incorporated in the atrial suture line. Weaning from bypass is routine. The ESC should be removed prior to heparin reversal, and the EPV can be replaced with a pulmonary artery catheter, if needed.	
Blood and fluid requirements	As for underlying condition: For mitral valve surgery For CABG	See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	As for underlying condition: For mitral valve surgery For CABG	Special monitoring considerations for port-access are discussed below. External defibrillator pads should be placed on all

		patients.
Positioning	Supine Roll under the left chest for CABG and under the right chest for mitral valve	
Monitoring of endo-CPB	Regional perfusion	Distal migration of the EAC (innominate artery occlusion → ↓ cerebral blood flow) is possible, especially while cardioplegia is being delivered when pressure in the aortic root exceeds that in the systemic arterial system. Monitoring for regional perfusion includes: 1. Right-sided arterial pressure—radial, brachial, or axillary. The pump-induced artifact may be of use if roller heads are being used. 2. Bilateral arterial lines—right-sided, plus a second line in the left radial or in the femoral vessels to allow comparison 3. TEE—EAC visible in the ascending aorta proximal to the innominate artery. 4. Transcranial or carotid artery Doppler 5. Fluoroscopy
	Proximal EAC balloon	Most likely to occur

	migration	during aortic root venting when the systemic pressure exceeds aortic root pressure. Monitored by TEE
	Cardiac decompression	TEE can be useful in monitoring for adequate decompression of the heart.
Monitoring of endo-CPB (cont'd)	EAC balloon pressure	This is usually monitored by the perfusionist and ranges from 250 to 550 mm Hg. Decreases in pressure may be 2° proximal migration of the balloon into a wider area of the ascending aorta, rupture of the balloon, or prolapse into the left ventricle. Loss of balloon occlusion is indicated by blood in the surgical field, return of cardiac activity, cardiac distension, increased root venting, and TEE evidence.
	Aortic root pressure	<80 mm Hg while giving cardioplegia, 0–10 mm Hg during venting. Overventing should be avoided, as should venting when the left side of the heart is open, to avoid drawing air into the aortic root.

	Intracardiac air	TEE is very useful for determining that adequate deairing of the heart has occurred.
Complications of endo-CPB	Retrograde aortic dissection Coronary sinus damage EAC balloon migration	
	EAC balloon rupture	Especially 2° sutures placed in the mitral valve annulus
	Retained EPV in atrial suture line Chest wall hemorrhage	It is important that the surgeon ✓ the chest wall before closure.
	Limb ischemia	May follow femoral cannulation

POSTOPERATIVE

Complications	As for the primary procedure Low CO Chest wall hemorrhage Problems associated with peripheral cannulation (e.g., dissection, embolization) Perivalvular leak Heart block	See mitral valve surgery or CABG.
Multimodal pain management	Parenteral narcotics NSAIDs	Pain control is important to facilitate early extubation. Intrathecal narcotics and/or intercostal blocks with wound infiltration may be useful. Emphasize multimodal approach combining nonopioid and

		bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-1)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.
Tests	ECG Electrolytes ABG	

Suggested Reading

1. Crumley S, Schraag S: The role of local anaesthetic techniques in ERAS protocol for thoracic surgery. J Thorac Dis 2018;10(3):1998–2004.
2. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery.J Cardiothorac Vasc Anesth 2018;32(6):2760–70.

CHAPTER **6.3**

Vascular Surgery

SURGEONS

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CAROTID ENDARTERECTOMY (VASCULAR)

SURGICAL CONSIDERATIONS

Description

Carotid endarterectomy (CEA) continues to be commonly performed open vasculai surgery procedures in the United States, even as carotid stenting is being employed as an alternative approach. Because of presumed microemboli from stenotic/ulcerated plaques at the carotid bifurcation, CEA is an effective procedure to reduce the risk oi subsequent stroke. Nearly 30 yr ago, the NASCET Collaborators determined, in ɛ prospective, randomized, blinded trial, that CEA is more effective than medical therapy for symptomatic patients with internal carotid artery narrowing between 60% and 90%. Symptoms are usually hemispheric (contralateral, upper or lower extremity paresis or numbness) or retinal (unilateral monocular blindness). Symptoms may be transient (transient ischemic attach [TIA] or reversible ischemic neurologic deficit [RIND]) o permanent (CVA). The Asymptomatic Carotid Atherosclerosis Study (ACAS) ha shown benefit from prophylactic CEA in asymptomatic patients with >60% stenosis oi the internal carotid artery. The Asymptomatic Carotid Surgery Trial 1 (ACST-1) is a randomized study of patients assigned to immediate CEA or deferral of any carotid artery procedure. Recruiting patients from 1993 to 2003 and following until 2006–2008, this study showed that successful CEA for asymptomatic patients younger than 75 yr of age reduces 10-yr stroke risks, particularly for disabling or fatal strokes. The ACST-2 is currently studying the outcome differences between CEA versus carotid stenting in asymptomatic individuals. Additionally, a recent Cochrane report found moderate-quality evidence for "some" benefit of CEA in patients with 50–69% symptomatic stenosis and "high" benefit in patients with 70–99% stenosis. The report also found high-quality evidence to suggest lack of benefit of CEA in patients with carotid near-occlusion.

Although some surgeons routinely prefer local anesthesia, most prefer general anesthesia (GA) with careful hemodynamic monitoring because of the frequent concomitant coronary artery disease (CAD). The carotid artery is approached through an oblique neck incision along the anterior border of the sternocleidomastoid muscle. After division of the common facial vein, the carotid sheath is opened and the carotid artery is exposed, avoiding injury to the phrenic, vagus, ansa hypoglossi, and hypoglossal nerves (Fig. 6.3-1). After controlling the internal, external, and common carotid arteries, heparin is administered, and the internal, external, and common carotid arteries are clamped sequentially. To maintain carotid perfusion, an indwelling shunt may be utilized (Fig. 6.3-2B), at the discretion of the surgeon. During clamping, mean arterial blood pressure is maintained at ~100 mm Hg with phenylephrine infusion or boluses. An endarterectomy plane is established proximally (Fig. 6.3-2C) and developed distally into both the external and internal branches, with establishment of a fine tapered end point. After removal of all thrombus, loose smooth muscle fibers and endothelium, the arteriotomy is closed, with or without a patch, the artery flushed, and flow restored. The incision is closed after meticulous hemostasis has been ensured.

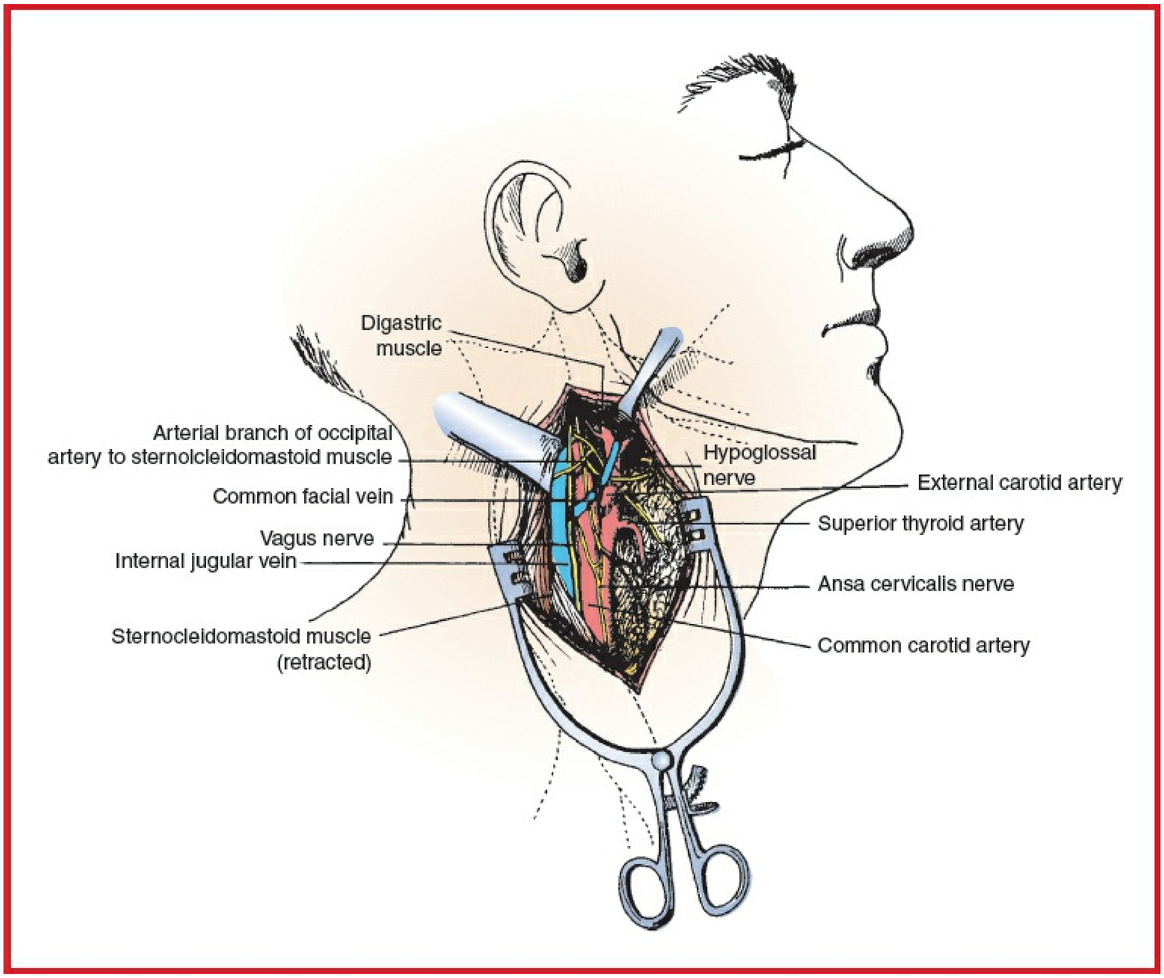

Figure 6.3-1 Exposure of the carotid bifurcation. (Reproduced with permission from Scott-Conner CEH, Dawson DL: Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

Usual preop diagnosis

Carotid artery disease

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See Appendix K: Enhanced Recovery After Surgery, p. K-1.**

SUMMARY OF PROCEDURE	
Position	Supine with neck extended and turned away from the side of the lesion
Incision	Anterior to the sternocleidomastoid from the earlobe to base of the neck or curvilinear in a skin crease over the carotid bifurcation

Special instrumentation	EEG monitor; ± shunt
Unique considerations	Capability to measure stump pressure may be needed. This can be accomplished by a high-pressure arterial line passed off the field to a pressure transducer. Avoid ↓ BP during carotid cross-clamping. An indwelling shunt may be utilized to restore carotid perfusion. Heparin 5000–10,000 U 5 min prior to cross-clamping, and protamine may be given after restoration of flow.
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg) after induction of anesthesia
Surgical time	Carotid cross-clamp: 30 min Total operating time: 90 min
Closing considerations	To assess the patient's neurologic status, it is best to be able to awaken and extubate the patient at the conclusion of the procedure. Avoidance of HTN is also critical during this period because the endarterectomy tissues are thin and friable.
EBL	100–200 mL
Postop care	ICU × 12–24 h; BP control; cardiac monitoring
Mortality	1%
Morbidity	MI: Major cause of postop mortality Cranial nerve injury (recurrent and superior laryngeal nerves): 39% Restenosis Asymptomatic: 9–12% Symptomatic: <3% Neurologic complications: <2% Hemorrhage: 1% False aneurysm: <0.5%
Pain score	3–5

■ PATIENT POPULATION CHARACTERISTICS	
Age range	55–80 yr
Male:Female	3:1
Incidence	Second most common vascular surgical procedure (after AAA repair)
Etiology	Arteriosclerosis; fibromuscular dysplasia
Associated conditions	Significant CAD coexists with carotid artery disease in at least 30% of patients, necessitating careful cardiac and hemodynamic monitoring.

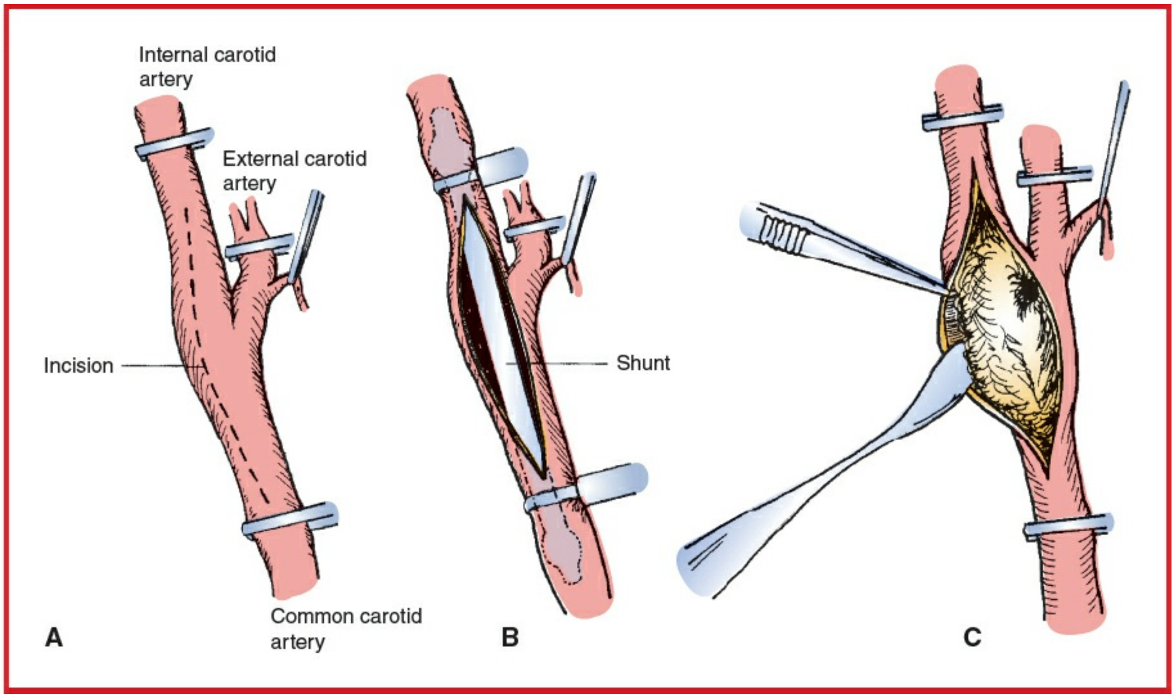

Figure 6.3-2 Carotid endarterectomy. **A:** Following occlusion of the superior thyroid and internal, external, and common carotid arteries, an arteriotomy is performed opposite the external carotid take off. **B:** A shunt may be placed to restore internal carotid flow. **C:** Plaque is separated from the artery wall. (Reproduced with permission from Scott-Conner CEH, Dawson DLOperative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Carotid Endarterectomy (Neurosurgical) i Extracranial Procedures.

Suggested Readings

1. Brott TG, Halperin JL, Abbara S, et al

ASA/ACCF/AHA/AANN/AANS/ACR/ASNR/CNS/SAIP/SCAI/SIR/SNIS/SVM/SVS guideline on the management of patients with extracranial carotid and vertebral artery disease: a report of the American College of Cardiology Foundation/American Heart Association Task Force. Circulation 2011; 124(4):e54–130.
2. Bulbulia R, Halliday A: The Asymptomatic Carotid Surgery Trial-2 (ACST-2): an ongoing randomised controlled trial comparing carotid endarterectomy with carotid artery stenting to prevent stroke. Health Technol Assess 2017; 21(57):1–40.
3. Doig D, Brown M: Carotid stenting versus endarterectomy. Annu Rev Med 2012; 63:259–76.
4. Halliday A, Harrison M, Hayter E, et al.; Asymptomatic Carotid Surgery Trial (ACST) Collaborative Group: 10-Year stroke prevention after successful carotid endarterectomy for asymptomatic stenosis (ACST-1): a multicentre randomized trial. Lancet 2010; 376:1074–84.
5. Howell SJ: Carotid endarterectomy. Br J Anaesth 2007; 99:119–31.
6. Lewis SC, Warlow CP, Bodenham AR, et al: General anaesthesia versus local anaesthesia for carotid surgery (GALA): a multicentre, randomised controlled trial. Lancet 2008; 372(9656):2132–42.
7. Mantese VA, Timaran CH, Chiu D, et al: Carotid Revascularization Endarterectomy versus Stenting Trial (CREST): stenting versus carotid endarterectomy for carotid disease. Stroke 2010; 41:S31–4.
8. Moritz S, Kasprzak P, Arlt M, et al: Accuracy of cerebral monitoring in detecting cerebral ischemia during carotid endarterectomy: a comparison of transcranial Doppler sonography, near-infrared spectroscopy, stump pressure, and somatosensory evoked potentials. Anesthesiology 2007; 107:563–9.
9. Orrapin S, Rerkasem K: Carotid endarterectomy for symptomatic carotid stenosis. Cochrane Database Syst Rev 2017; 6:CD001081.
10. Raman G, Moorthy D, Hadar N, et al: Management strategies for asymptomatic carotid stenosis: a systematic review and meta-analysis. Ann Intern Med 2013; 158:676–85.

REPAIR OF THORACIC AORTIC ANEURYSMS

SURGICAL CONSIDERATIONS

Description

Repairs of aneurysms of the ascending, transverse arch, and descending thoracic aorta are performed to repair expanding or leaking aneurysms or prophylactically to prevent rupture. Patients with Sx of rapid expansion or aneurysm leaking may require urgent repair. Each surgical type—ascending, arch, and descending—is considered separately, as follows. Aneurysms of the **ascending aorta** may arise 2° the degenerative changes of atherosclerosis (exacerbated by old age, hypertension [HTN], tobacco use), from inborn errors of metabolism (Marfan syndrome), or from poststenotic dilation and continued expansion of a chronic dissection. Diseases of the entire ascending aorta, including the sinuses of Valsalva, as in Marfan syndrome and annuloaortic ectasia, require replacement of the ascending aorta and aortic root with a composite valved conduit or a valve-sparing aortic root replacement (VSARR), whereas acquired diseases limited to the ascending aorta usually allow replacement of the aorta distal to the sinotubular ridge, avoiding the need to replace the aortic root and aortic valve. Repair of the ascending aorta is usually accomplished on full cardiopulmonary bypass (CPB) with an aortic cross-clamp placed just proximal to the innominate artery and arterial inflow through the femoral artery, distal to the aortic clamp on the ascending aorta, or via cannulation of the innominate or axillary arteries. The aneurysmal ascending aorta is replaced with a Dacron tube graft from the sinotubular ridge to the innominate artery or proximally if anatomically feasible (if nonaneurysmal tissue is present proximal to the innominate artery). Dilation of the sinuses of Valsalva mandates replacement with a composite valved conduit sewn proximally to the aortic annulus and distally to a normal-caliber ascending aorta, with coronary ostia reimplanted in the side of the tube graft, although in some cases with a normal-appearing aortic valve VSARR may be performed.

Aneurysms of the **aortic arch** are the least common of thoracic aortic aneurysms. Because of the need for concomitant replacement of the arch vessels, however, they are the most complex to repair. Total CPB is utilized, and cerebral protection is accomplished either by CPB perfusion of one or both carotid arteries, by retrograde cerebral perfusion (RCP), or by profound hypothermic circulatory arrest In the past profound hypothermia (cooling to 15–18°C) was required for the safe implementation of circulatory arrest and aortic arch repair. Now strategies including selective antegrade and RCP are employed to provide ongoing perfusion and cooling of the brain during circulatory arrest. As a result, most surgeons are cooling to more moderate levels of hypothermia (i.e., 24–28°). Repair can originate from the aortic annulus and extend distally to the middescending thoracic aorta at the level of the carina. Routine caval cannulation is accomplished via a median sternotomy, and arterial access is gained via the femoral artery or axillary artery. If circulatory arrest is to be used, the patient is cooled to 15–18°C; the heart is arrested; and, with no distal cross-clamp, distal anastomosis is accomplished, followed by implantation of the cerebral vessels attached to an island of aorta. If cerebral perfusion is augmented via RCP, air and debris may be eliminated from the cerebral circulation. However, RCP carries the risk of cerebral edema if venous pressure becomes elevated (goal central venous pressure [CVP] typically <20 mm Hg). Some surgeons, however, prefer antegrade cerebral perfusion during circulatory arrest. This can be achieved by directly cannulating the innominate and carotid arteries via a Y from the arterial CPB limb. This technique requires multiple cannulas in the surgical field and is a bit cumbersome. To avoid this some surgeons instead utilize unilateral or selective antegrade cerebral perfusion (SACP). This involves cannulating only the innominate artery or alternatively cannulating the axillary artery (which connects to the innominate but is accessed in a right chest incision remote from the surgical field). This unilateral perfusion technique of SACP allows for antegrade arterial flow to be delivered into the right carotid via retrograde flow in the axillary artery and antegrade flow in the innominate artery. SACP avoids the concerns of elevated cerebral venous pressures associated with RCP but requires good collateral connections to the left carotid system via the circle of Willis to ensure flow to the left cerebral hemisphere. Close monitoring of the left cerebral blood flow is generally utilized either using cerebral oximetry (artifact prone) or EEG monitoring. Evidence of inadequate flow to the L cerebral circulation requires the placement of a direct left carotid perfusion cannula. Perfusion is then reinstituted via full CPB, the graft clamped proximal to the innominate artery, and the proximal anastomosis performed while the patient is being rewarmed. Alternatively, if one elects to perfuse the cerebral vessels, the innominate and left carotid arteries can be individually cannulated and perfused via a "Y" connection from the femoral arterial perfusion line. If axillary cannulation is used alone, antegrade cerebral perfusion can be achieved by control of the proximal innominate artery. This is termed "unilateral cerebral perfusion" and requires close monitoring with cerebral oximetry to estimate the adequacy of contralateral (e.g., left-sided) perfusion. If a unilateral drop in cerebral oximetry occurs, the surgeon may convert to bilateral perfusion by placing a catheter into the left carotid artery from inside the arch. The necessity for profound hypothermic circulatory arrest is thus avoided. After completion of the distal aortic arch, vessel island, and proximal aortic anastomoses, weaning from CPB and subsequent steps proceed in a routine fashion.

Open repair of aneurysms of the **descending thoracic aorta** is usually performed for symptomatic and leaking aneurysms, enlarging aneurysms, and aneurysms of sufficient size to warrant prophylactic repair. Due to increased morbidity and mortality associated with open descending thoracic aortic repair, endovascular approaches are generally favored and considered prior to committing the patient to open repair. **Aneurysm repair** is accomplished through a left posterolateral thoracotomy on full or partial CPB. After entry into the left thorax, venous drainage for CPB may be obtained from the left atrium (partial bypass) or the femoral vein (partial or full bypass), and arterial return is via the femoral artery. Maintenance of ventilation is necessary for partial bypass. If partial bypass without an oxygenator is elected, thus minimizing the amount of heparin necessary, venous access can be gained via the pulmonary veins or left atrium and arterial return via the femoral artery or distal thoracic aorta. After institution of bypass, the aorta is cross-clamped above and below the aneurysm, the aorta is divided, a tube graft is interposed, and clamps are removed. The patient is weaned from bypass, and the operation is terminated in the routine fashion.

Usual preop diagnosis

Enlarging or symptomatic aortic aneurysm

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan.** **See Appendix K, p. K-1.**

■ SUMMARY OF PROCEDURES			
	Ascending Aorta	**Transverse Arch**	**Descending Aorta**
Position	Supine	⇐	Lateral decubitus with left side up
Incision	Median sternotomy	⇐	Left posterolateral thoracotomy with access to the femoral artery and vein
Special instrumentation	CPB, if used	Complete hemodynamic monitors; CPB	⇐ + DLT or BB; lower extremity BP monitor; epidural catheter; spinal catheter; ICP monitoring; full neuromonitoring (EEG, SSEP, MEP)
Unique	Routine CPB	If profound	OLV; partial CPB

considerations	hemodynamic monitoring	hypothermic arrest is utilized, neuroprotective adjuncts, including local hypothermia, barbiturates, and steroids, should be used.	
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg)	⇐	⇐
Surgical time	Aortic cross-clamp: 40–120 min CPB: 70–150 min Total: 2.5–5 h	Aortic cross-clamp: 75–120 min Circulatory arrest: 30–45 min CPB: 3–4.5 h Total: 4–6 h	Aortic cross-clamp: 25–45 min CPB: 30–60 min Total: 2.5–4.5 h
Closing considerations	Aggressive management of coagulopathy, if a long pump run is necessary	Aggressive management of coagulopathy	Replacement of DLT with single-lumen tube
EBL	300–400 mL	400–700 mL	200–300 mL
Postop care	ICU, intubated 5–20 h, depending on preop condition	ICU, intubated 6–24 h	ICU, intubated 5–24 h
Mortality	5–10%	10–15%	⇐
Morbidity	Renal failure: 5–10% CVA: 4–6% Respiratory failure: 3–5% MI: 2–5%	— 2–5% 10% ⇐	10–15% 2–4% 10–15% 2–4%
Pain score	7–10	7–10	9–10

PATIENT POPULATION CHARACTERISTICS			
Age range	23–80 yr (mean = 55 yr)	50–75 yr	34–79 yr (mean = 65 yr)
Male:Female	3:1	2:1	2.5:1
Incidence	10–30/yr at tertiary center	5–10/yr at tertiary center	10–30/yr at tertiary center
Etiology	Degenerative disease Atherosclerotic disease	⇐ ⇐ Chronic dissections	⇐ ⇐ ⇐
Associated conditions	CHF (50%); angina (30%); HTN (30%); COPD (15%)	Aortic valve disease (30%); COPD (20%); CAD (15%)	HTN (65%); CAD (50%); COPD (30%); CHF (10%)

ANESTHETIC CONSIDERATIONS

ERAS: Enhanced Recovery after Cardiac Surgery (ERACS) protocols are increasingl advocated within the relevant societies (Society of Cardiac Anesthesiologists, American Association of Thoracic Surgeons, et al.). Principles of ERACS are based o evidence-based care pathways targeting specific aspects of perioperative cardiac surgical patient care. These principles are broadly applicable to all patients coming for major cardiac and vascular procedures.

Early mobilization	**Prehabilitation** for patients with significant deconditioning prior to surgery **Fast-track extubation** as soon as extubation criteria, hemodynamics, and bleeding are satisfactory **Sitting up in chair** from postop day 1 onward
Analgesic management	**Opioid-minimizing strategies** including multimodal approach with NSAIDs, gabapentin or pregabalin, acetaminophen, and regional techniques where feasible **Scheduled antiemetics** (e.g., ondansetron) for the first 24–48 h

Goal-directed therapy (GDT)	**Fluid and vasoactive management** using GDT to reduce postop complications **Targeted euglycemia** with insulin infusion
Hemostasis	**Antifibrinolytics** for all cases requiring cardiopulmonary bypass **Active management of chest tube patency** to avoid retention of mediastinal blood
Delirium management	**Delirium screening** at regular intervals, for example, every nursing shift **ICU liberation strategies** including sedation vacations **Preop screening** for excessive alcohol use
Temperature management	**Avoidance of postop hypothermia (<35°C) and hyperthermia (>38°C)**

PREOPERATIVE

Typically, patients with thoracic aortic aneurysms have atherosclerosis and HTN. A subset of patients will have a connective tissue disorder (e.g., Marfan or Loeys-Dietz syndrome). In contrast with thoracic aortic dissections, thoracic aortic aneurysms may be of a more chronic and asymptomatic nature. A ruptured or leaking aneurysm, however, may have a more precipitous presentation.

Cardiovascular	**Arch and ascending aneurysms** (60–70% of aneurysms): commonly associated with HTN, cystic medial necrosis, connective tissue disorder (e.g., Marfan syndrome), atherosclerosis, or syphilis. CHF may occur 2° to dilation of the aortic valve annulus and aortic regurgitation (AR). Aneurysmal compression or intrinsic disease of the coronary arteries may result in myocardial ischemia. **Descending thoracic aneurysms (30%):** usually associated with HTN, cystic medial necrosis, Marfan syndrome, and atherosclerosis **Tests:** ECG: ✓ for LVH, ischemia. ECHO: ✓ for valvular disease, size and extent of aneurysm, LV function. Angiography: ✓ exact extent of aneurysm (allows planning of procedure and sites for arterial

	monitoring), coronary artery anatomy, and degree of occlusion
Respiratory	Recurrent laryngeal nerve palsy may → hoarseness (ascending/arch aneurysms). Tracheal deviation ± stridor or dyspnea may be present 2° to tracheal or bronchial compression (✓ CT scan). Hemoptysis or a hemothorax suggests aneurysmal leakage or rupture warranting immediate surgical consultation. The implications include the difficult intubation with DLT, risk of hypoxemia, massive hemorrhage on thoracotomy, decreased thoracic compliance, and consequent decreased venous return (especially when IPPV is instituted). **Tests:** CXR: ✓ for widened mediastinum, distortion of the trachea and left main bronchus (because it may affect the placement of DLT); MRI/CT with contrast: ✓ anatomic relation of aneurysm to surrounding structures (especially the trachea/bronchi for intubation considerations and the esophagus for TEE precautions); others as indicated from H&P
Neurologic	Full neurologic exam including strength in lower extremities particularly hip flexion. Any deficit should be well documented as neurologic sequelae (e.g., paraplegia/paraparesis) may occur after surgery secondary to spinal cord ischemia.
Renal	Renal problems may occur 2° to AR (↓CO) and heart failure, HTN, or involvement of renal arteries in the aneurysm. **Tests:** BUN; Cr; consider Cr clearance; electrolytes
Gastrointestinal	Descending aneurysms that involve the celiac or superior mesenteric arteries may → bowel ischemia. **Tests:** Consider ABG: ✓ persistent metabolic acidosis (2% bowel ischemia). If indicated by H&P, consider abdominal CT/x-ray: ✓ ileus
Hematologic	If time permits, consider autologous blood donation;

	preexisting coagulopathy increases risk of the procedure. **Tests:** PT; PTT; Hct/Hb
Musculoskeletal	✓ Marfanoid appearance; others as indicated from H&P
Laboratory	Lactic acid if concerned for organ hypoperfusion Others as indicated from H&P
Premedication	Pain and anxiety may significantly contribute to HTN and should be treated (e.g., morphine 0.1 mg/kg iv ± midazolam 0.025–0.1 mg/kg iv); but avoid obtundation. If patients present emergently with dissection or rupture, consider full-stomach precautions with rapid-sequence induction.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

The anesthetic management of patients with aortic dissections and aortic aneurysms are similar in many respects. For intraop and postop management of these conditions, see Anesthetic Considerations for Repair of Acute Aortic Dissections and Dissecting Aneurysms.

Suggested Readings

1. Bisschoff MS, Di Luozzo G, Griepp EB, et al: Spinal cord preservation in thoracoabdominal aneurysm repair. Perspect Vasc Surg Endovasc Ther 2011; 23:214–22.
2. Etz CD, Weigang E, Hartert M, et al: Contemporary spinal cord protection during thoracic and thoracoabdominal aortic surgery and endovascular aortic repair: a position paper of the vascular domain of the European association for cardio-thoracic surgery. Eur J Cardio-thoracic Surg 2014; 47(6):943–57.

3. Fischbein MP, Miller DC: Long-term durability of open thoracic and thoracoabdominal aneurysm repair. Semin Vasc Surg 2009; 22:74–80.
4. Goodwin MR, Blasius KR, Brand J, et al: One-lung ventilation for surgical repair of thoracic aortic aneurysm. Semin Cardiothorac Vasc Anesth 2013; 17:146–51.
5. Harrer M, Waldenberger FR, Weiss G, et al: Aortic arch surgery using bilateral antegrade selective cerebral perfusion in combination with near-infrared spectroscopy. Eur J Cardiothorac Surg 2010; 38:561–7.
6. Hoel AW: Aneurysmal disease: thoracic aorta. Surg Clin N Am 2013; 93:893–910.
7. Hsu C, Kwan G, van Driel M, et al: Distal aortic perfusion during thoracoabdominal aneurysm repair for prevention of paraplegia. Cochrane Database Syst Rev 2012; (3):CD008197.
8. Keenan JE, Benrashid E, Kale E, et al: Neurophysiological intraoperative monitoring during aortic arch surgery. Semin Cardiothorac Vasc Anesth 2016; 20(4):273–82.
9. Milewski RK, Pacini D, Moser GW, et al: Retrograde and antegrade cerebral perfusion: results in short elective arch reconstructive times. Ann Thorac Surg 2010; 89:1448–57.
10. Misfield M, Leontyev S, Borger MA, et al: What is the best strategy for brain protection in patients undergoing aortic arch surgery? A single center experience of 636 patients. Ann Thorac Surg 2012; 93:1502–8.
11. Piazza M, Ricotta JJ: Open surgical repair of thoracoabdominal aortic aneurysms. Ann Vasc Surg 2012; 6:600–5.
12. Shelstad RC, Reeves JG, Yamanaka K, et al: Total aortic arch replacement: advantages of varied techniques. Semin Cardiothorac Vasc Anesth 2016; 20(4):307–13.
13. Wahlgren CM, Blohmé L, Günther A, et al: Outcomes of left heart bypass versus circulatory arrest in elective open surgical descending and thoraco-abdominal aortic repair. Eur J Vasc Endovasc Surg 2017; 53(5):672–8.

ENDOVASCULAR STENT GRAFTING OF AORTIC ANEURYSMS

SURGICAL CONSIDERATIONS

Description

The standard treatment for descending **thoracic aortic aneurysm** is surgical resection of the aneurysm and replacement with a segment of prosthetic graft material. Although resection of aneurysms often can be performed without the need for extracorporeal circulation, the procedure has a reported mortality rate of up to 50% in emergency cases and 12–15% in elective cases. Transluminal endovascular stent grafting (TEVAR) offers an alternative treatment that is less invasive, less hazardous, and potentially less expensive than standard operative repair.

In the initial workup, all patients have contrast-enhanced spiral CT scans of the thorax and thoracic aortography to assess the dimensions of the aneurysms. The most important features to consider in evaluating an aortic aneurysm for endovascular stent graft treatment is the presence of an adequate proximal and distal neck. A minimum neck length of at least 1.5 cm is required to allow secure anchoring of most stent grafts. The distance from the origins of the left subclavian artery and celiac axis to the aneurysm should be at least 1.5–3 cm to ensure that the stent graft does not inadvertently block these arteries. In an effort to reduce the incidence of paraplegia and to limit exclusion of intercostal arteries, the overall length of the stent graft is kept to a minimum. Stent graft technology has made significant advancements in recent years. Debranching grafts (which cover important branch vessels) are now sometimes placed in patients with unfavorable anatomy and lack of traditional landing zones. In these cases, snorkel grafts are placed into branch vessels and run parallel (usually outside the graft) and allow for important branch vessel flow to be maintained.

Another important anatomic consideration is the size of the proposed conduit vessel (e.g., iliac) to ensure that it is adequate for accommodation of the stent introducer system, which usually requires at least an 8-mm diameter vessel. Where the pelvic vessels are <8 mm, either a retroperitoneal iliac or retroperitoneal aortic approach is utilized. The stent is the metallic framework to which the graft material is applied. Various balloon-expandable or self-expanding stents are available. For application in the thoracic aorta, stents of 30–40 mm in diameter (mean 35 mm) are required.

Typically, thoracic aortic aneurysm stent graft procedures are performed in the cath lab (although some may be performed in the OR, depending on equipment availability and local politics), with the patient intubated and under GA. The cath lab is prepared for aortic surgery, with the patient placed on the table in a shallow right decubitus position. For an approach via the common femoral artery, the groin area is prepped for a femoral artery cutdown. In appropriately selected cases, percutaneous suture-mediated closure devices may be deployed in a preclose fashion at the beginning of the procedure. This approach obviates the need for surgical cutdown. When the iliac arteries are of insufficient size, the left lower abdomen is prepped for a retroperitoneal approach to either the aorta or the common iliac artery. High-quality fluoroscopic equipment is essential to ensure accurate placement of the device, and today, most procedures are performed using fixed angiographic units and fusion imaging guidance to take advantage of preprocedure CT imaging for accurate guidance. When the iliac vessels are of sufficient size, a cutdown is performed on a femoral artery, the artery is punctured, and a guide wire is advanced into the thoracic aorta. A pigtail catheter is placed, and an aortogram is performed. The patient is then anticoagulated with iv heparin (100 IU/kg). A long, stiff guide wire is placed, and the endograft delivery system is introduced and advanced to the appropriate location under fluoroscopic guidance.

To reduce the likelihood of inadvertent downstream deployment of the stent graft caused by the force of blood flow during initial delivery, the arterial BP may be lowered to a mean of 50–60 mm Hg using SNP or clevidipine. This is especially valuable the more proximal the intended placement and is standard practice with targets involving the proximal aortic arch and/or ascending aorta. In cases involving the ascending aorta, rapid ventricular pacing with a temporary transvenous pacemaker is commonly performed to decrease the blood pressure at the time of deployment. Once the endograft is in the desired position, it is unsheathed to allow expansion (Fig. 6.3-3). Immediately following deployment, the vasodilator or rapid pacing is discontinued, allowing the BP to normalize. Relative HTN may be implemented to increase spinal perfusion if there is concern for stent graft–related spinal hypoperfusion. Repeat aortogram is performed, and any early leakage of contrast into the aneurysm is treated either with balloon angioplasty of the stent graft or further stent graft placement. Occasionally, a faint, persistent leak of contrast is caused by leakage through the graft material. This typically ceases when the patient's coagulation status returns to normal. Following removal of the delivery sheath, heparin is reversed with protamine sulfate, and the arteriotomy is repaired surgically or closed by the previously placed percutaneous closure systems. For stent placement through the retroperitoneal aorta, the procedure has a more extensive surgical component; however, the technique is similar.

The basic concept of **abdominal aortic aneurysm** (AAA) repair via an endovascular route (EVAR) is similar to the preceding section on thoracic aneurysm repair, with some notable exceptions. AAAs commonly arise inferior to the renal arteries, and 80–90% of cases involve either one or both iliac arteries. For this reason, stent grafts that can accommodate this more complicated anatomy are required (Fig. 6.3-4). The superior aneurysm neck needs to be of sufficient length (1.5–2 cm) inferior to the most inferior renal artery to provide for stable anchoring. Inferiorly, the stent graft must accommodate either or both iliac arteries. The procedure is performed in a two-stage fashion. Initially, an aorta-to-single-iliac-artery device is placed from the infrarenal aortic neck into one of the iliac vessels. A contralateral femoral artery puncture is then performed, and a catheter and guide wire are used to access an open stump of the stent graft from the contralateral limb. At this stage, a modular section of stent graft is placed from the aortic component into the contralateral limb; in this way, an aorta-to-bi-iliac graft is placed.

Usual preop diagnosis

Aortic aneurysm

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options.** **See Appendix K, p. K-1.**

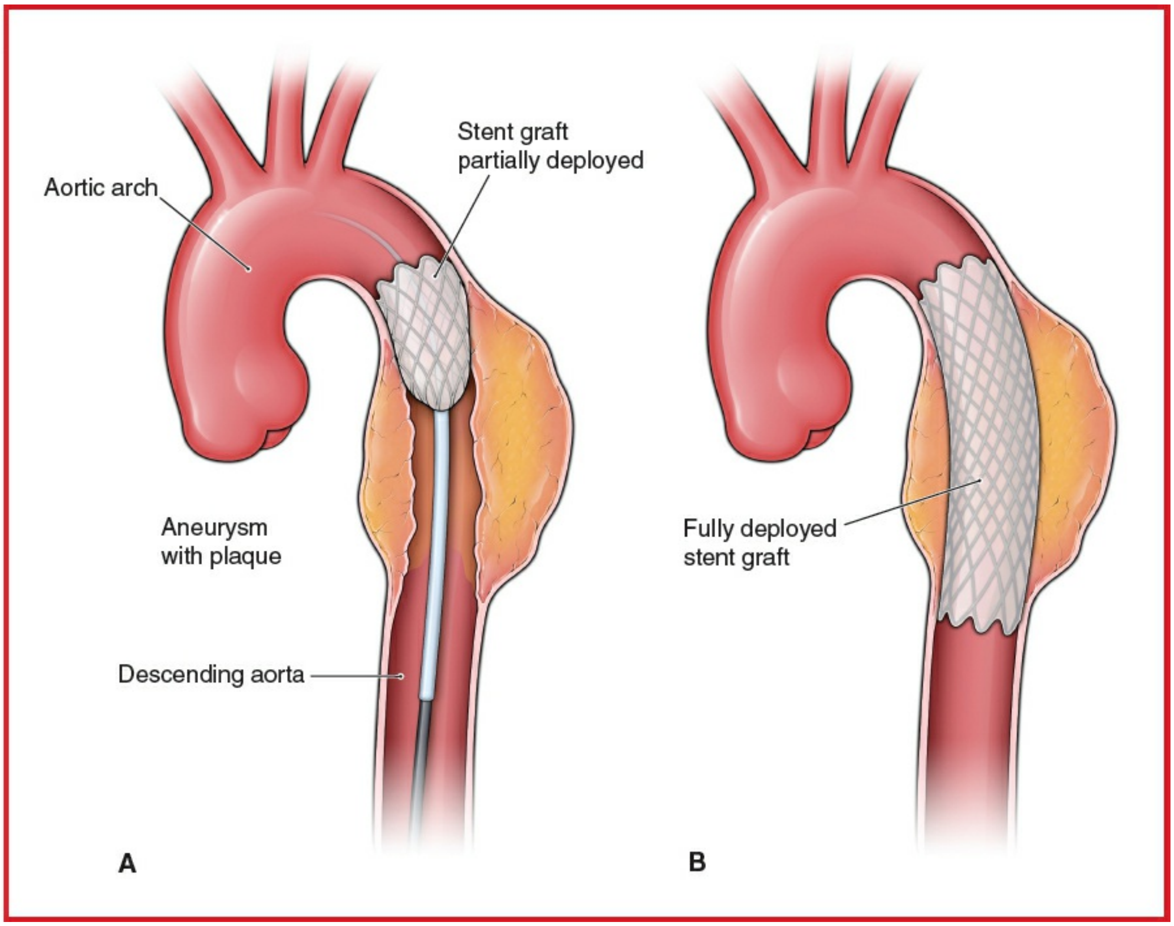

Figure 6.3-3 Deployment of endovascular stent graft in descending aorta.

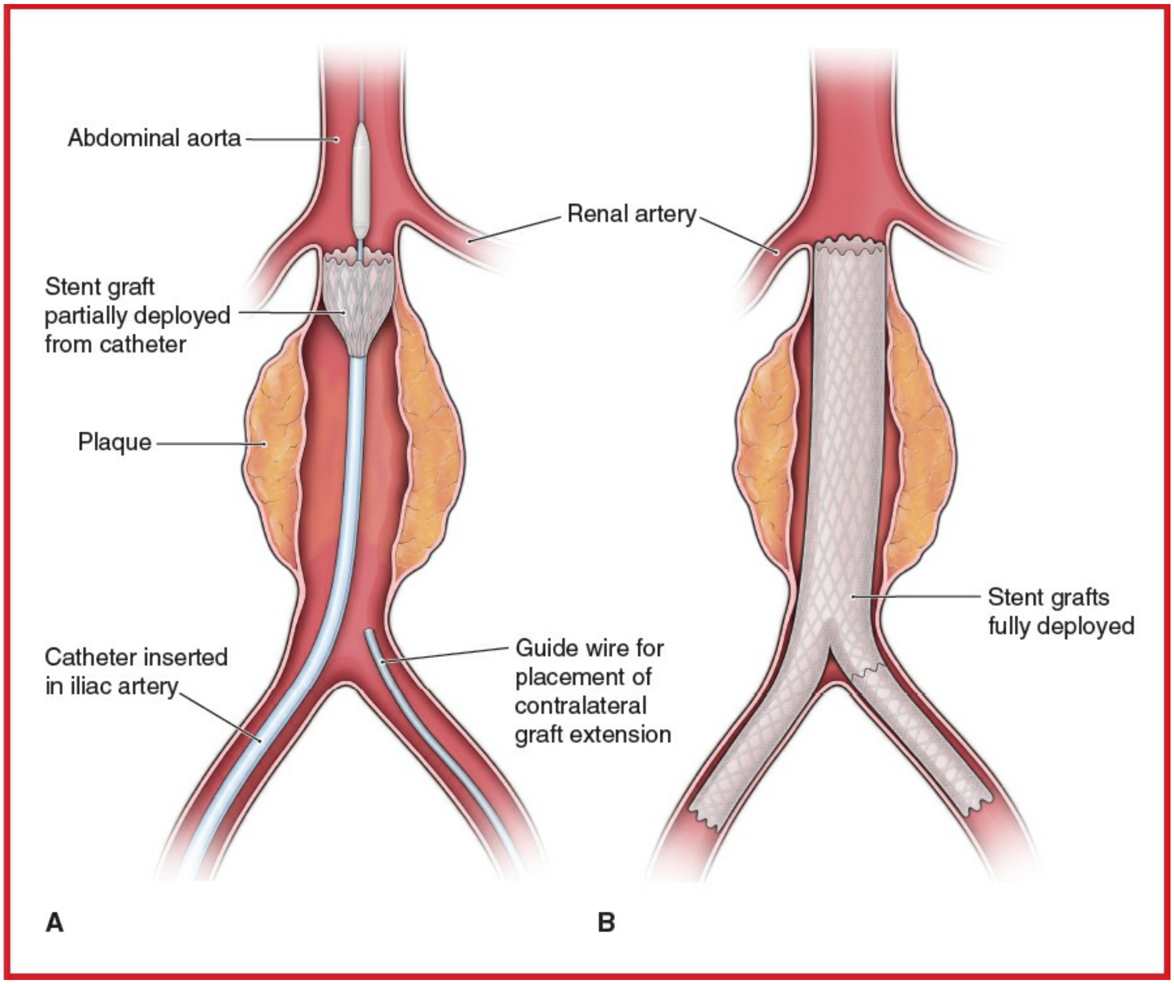

Figure 6.3-4 Deployment of endovascular stent grafts in the infrarenal aorta and iliac arteries.

SUMMARY OF PROCEDURES

	Thoracic Aortic Aneurysms	Abdominal Aortic Aneurysms
Position	Supine/slight right decubitus	Supine
Incision	Femoral artery cutdown	Bilateral femoral artery cutdowns
Special instrumentation	Catheters, guide wires, sheaths, dilators, and stent grafts	⇐
Unique considerations	May require adjunctive procedure (e.g., brachial artery catheterization,	⇐

	balloon angioplasty, possible left subclavian-to-common carotid artery bypass procedure); iv heparin 100–300 IU/kg; preop and intraop hydration may be required given the risk of contrast-induced nephropathy.	
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg)	⇐
Surgical time	1–3 h	⇐
Closing considerations	None	⇐
EBL	Usually minimal; however, can be up to 2–3 U	⇐
Postop care	Routine floor monitoring or ICU × 1 d	⇐
Mortality	3–5%	2–3%
Morbidity	Paraplegia: 2–3% Infection: 1% Surgical conversion: <1%	Acute rupture: <1% ⇐ ⇐
Pain score	3–4	3–4

PATIENT POPULATION CHARACTERISTICS

Age range	30–90 yr	40–90 yr
Incidence	1/10,000 of the population in the United States	1/1,000 of the population in the United States
Etiology	Atherosclerosis; HTN; trauma; aortic dissection; infection	Atherosclerosis; infection
Associated conditions	CAD; aneurysmal disease elsewhere	⇐

ANESTHETIC CONSIDERATIONS

The anesthetic considerations for thoracic and abdominal stent grafting are similar because the surgical techniques, patient comorbid disease, and stent graft deployment techniques are similar. Patients may be asymptomatic; most will have coexisting CAD, peripheral vascular disease (PVD), and/or cerebrovascular disease. A complication notably more likely with thoracic stent grafting is spinal cord ischemia or infarct requiring special consideration.

ERAS

Although not commonly referred to as ERAS, fast-track pathways are used in some centers to facilitate rapid recovery and early discharge after abdominal aortic stent grafting. The risk of spinal cord ischemia in patients coming for stent grafting of thoracic lesions generally precludes fast-track implementation.

At Stanford University, specific criteria for "fast-track EVAR" include:

- Preop: functional independence in all ADLs, social support with a responsible adult to assist patient for first 24 h, absence of significant comorbidities, normal kidney function, and favorable aortic anatomy (e.g., no significant thrombus, no angulated portions)
- Periop: uncomplicated periop course with surgical duration <2 h, uneventful 4 h postop observation period, adequate analgesia with PO analgesics, and tolerating regular diet

For patients meeting these criteria, discharge can be considered within 8–12 h of procedure.

For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See Appendix K, p. K-1.

PREOPERATIVE

Preop considerations for these patients are the same as for any patient undergoing repair of a descending thoracic aneurysm, AAA, or aortobifemoral aneurysm. Many of these patients, however, are not suitable for conventional repair via a thoracotomy because of respiratory disease (e.g., $FEV_1 < 1$ L, severe chronic obstructive pulmonary (COPD) $PaCO_2 > 60$ mm Hg), CAD, renal failure, CHF, or other significant comorbid diseases

As such, they generally are at very high risk for periop morbidity and mortality. Before surgery, the anesthesiologist should consult with the surgical and radiologic teams to decide what will be done should a complication such as penetration or rupture of the aneurysm occur during surgery. Temporary control may be feasible with an occlusive endoaortic balloon; otherwise, an emergency thoracotomy or laparotomy may be necessary in the cath lab.

INTRAOPERATIVE

Anesthetic technique

Usually GETA (OLV may be required for emergency thoracotomy 2° ruptured thoracic aneurysm). Abdominal stent grafts may be placed under neuraxial anesthesia. Rarely, stent grafting may be carried out with local anesthesia or MAC in patients amenable to percutaneous approaches who are otherwise unable to tolerate GA or neuraxial techniques.

Induction	Patients are typically extubated prior to leaving the OR, so the dose of fentanyl (5–10 mcg/kg) is limited to that which will suppress the hypertensive response to intubation while still allowing for early extubation. The same considerations apply to the use of muscle relaxants (e.g., rocuronium [1–1.5 mg/kg]). If a thoracotomy is discussed as the primary plan for unexpected rupture, a DLT should be used or a bronchial blocker (e.g., Uniblocker™) should be available; otherwise, a SLT can be used.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Usually a balanced anesthetic technique of O_2/air/isoflurane or sevoflurane, supplemented with fentanyl (or remifentanil infusion) as needed. Hemodynamic control of BP (esmolol, SNP, NTG, clevidipine) and heart rate to avoid myocardial ischemia is important. **Anesthetic management during stent graft deployment:** During stent graft deployment, the aorta is momentarily occluded.

	Formerly, this resulted in a rapid ↑ BP → stent graft being moved from its intended position. The newer generations of stent grafts, using thermal or mechanical means for rapid deployment, cause minimal hemodynamic change, eliminating the need for aggressive BP control.	
Emergence	Extubation is desirable, although iatrogenic hypothermia (<34°C) may prevent this. If unable to extubate in the cath lab, reversal of NMB and emergence should be expedited to allow for rapid assessment of hip flexion if the spinal cord is at risk for ischemia. Recovery is usually in the ICU for thoracic stent grafts but may be on the floor for abdominal stent grafts. Consider iv acetaminophen if po dose not given. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	IV: 14–16 ga × 2 BSS < 1 L/case PRBC available	Although usually minimal, blood loss can be considerable. See Zero Balance Fluid Management, p. K-2 and p. K-4.
Monitoring	Standard monitors Arterial line ± CVP catheter	Arterial line placement should be on the right as the radiologists may require access to the left brachial artery. Central access for vasoactive drugs may be needed.
	Urinary catheter TEE	TEE is used to aid in the identification of the thoracic aneurysm necks, to monitor deployment of the stent graft, and to identify any continued flow of blood into the

		aneurysmal sac after deployment (endoleak) or aortic dissection.
	CSF pressure/drainage	Lumbar CSF drain → ✓ spinal cord perfusion pressure and ↓ CSF pressure to mitigate risk of spinal cord ischemia (anterior spinal artery syndrome).
Positioning	✓ and pad pressure points ✓ eyes	Supine ± slight right lateral decubitus Underbody Bair Hugger or warming blanket
Complications	Hemorrhage	Hemorrhage at the groin site can be considerable and may be concealed.
	Vessel damage	Damage to the vessels (femoral, iliac, or abdominal aorta) during passage of the insertion system can occur with attendant massive hemorrhage.
	Rupture of the aorta	Rupture or penetration of the thoracic aneurysm also can occur, necessitating rapid conversion to open thoracotomy.
	Deployment failure/incorrect position	The stent graft may deploy in an incorrect position or fail to fully deploy. This may require positioning of further stent grafts, balloon expansion of the stent

		graft, or conversion to thoracotomy.
	Hypothermia	Hypothermia can be a problem as a circulating-water warming blanket cannot be used on the operating table (use of fluoroscopy). Overbody Bair Hugger problematic for surgical prep. Underbody Bair Hugger compatible with fluoroscopy but beware of overzealous heating of back (spinal cord hyperthermia).

POSTOPERATIVE

Complications	Aortic perforation/rupture Migration of grafts Femoral artery dehiscence Distal embolization Paraplegia	Requires rescue with endoaortic occlusive balloon +/− emergency surgery. → loss of distal pulses, mesenteric ischemia, and acute renal insufficiency Requires prompt return to OR ✓ pulses; angiography/surgical intervention ✓ strength of hip flexion. Increase spinal cord perfusion pressure by deliberate hypertension (goal MAP 90–100 mm Hg after stent graft deployment + working

		lumbar CSF drain in patients at risk).
Multimodal pain management	Minimal analgesic requirements	
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	As indicated by patient condition.	CT scan, angiogram prior to discharge and at regular intervals following discharge.

Suggested Readings

1. Daye D, Walker TG: Complications of endovascular aneurysm repair of the thoracic and abdominal aorta: evaluation and management. Cardiovasc Diagn Ther 2018; 8(Suppl 1):S138–56.
2. De Bruin JL, Baas AF, Buth J, et al: Long-term outcome of open or endovascula repair of abdominal aortic aneurysm. N Engl J Med 2010; 362(20):1881–9.
3. Etz CD, Weigang E, Hartert M, et al: Contemporary spinal cord protection durin thoracic and thoracoabdominal aortic surgery and endovascular aortic repair: a position paper of the vascular domain of the European association for cardio-thoracic surgery. Eur J Cardio-thoracic Surg 2014; 47(6):943–57.
4. Findeiss LK, Cody ME: Endovascular repair of thoracic aortic aneurysms.Semin Intervent Radiol 2011; 28(11):107–17.
5. Foley OJ, Criado FJ, Farber MA, et al: VALOR Investigators: long-term results wi the Talent thoracic stent graft in the VALOR trial. J Vasc Surg 2012; 56:1214–21.
6. Miller DC: Through the looking glass: the first 20 years of thoracic aortic stent-grafting. J Thorac Cardiovasc Surg 2013; 145:S142–8.
7. Nicolaou G, Ismail M, Cheng D: Thoracic endovascular aortic repair: update o indications and guidelines. Anesthesiol Clin 2013; 31(2):451–78.
8. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery.J Cardiothorac Vasc Anesth 2018; 32:2760–70. pii: S1053-0770(18)30049-1.

REPAIR OF ACUTE AORTIC DISSECTIONS

SURGICAL CONSIDERATIONS

Description

Repair of **acute aortic dissection** is performed to prevent life-threatening complications such as hemorrhage, tamponade, and heart failure 2° acute aortic valve insufficiency and to redirect flow into the true lumen. Emergent repair of acute ascending dissections is generally accepted therapy to prevent rupture of the aortic root with exsanguination or pericardial tamponade. Mortality for acute ascending dissection is estimated at 1% per hour for the first 48 h. The management of descending thoracic aortic dissections remains controversial, but surgical intervention probably should be recommended only for younger patients, patients with uncontrolled pain or evidence for continued expansion or extravasation, and those with branch vessel compromise.

Ascending aortic dissections typically produce sharp, tearing retrosternal pain that penetrates straight through to the subscapular area. The presentation, however, is so frequently variable that any patient in extremis, especially with migratory pain or vacillating findings, and even asymptomatic patients with aortic valve regurgitation (AR), should be considered for the diagnosis. With a suggestive Hx, a new murmur, or a pulse deficit, an enlarged mediastinal shadow on chest radiography should prompt further diagnostic efforts. CT scanning, MRI, and aortography may all be diagnostic, bu no diagnostic modality has 100% sensitivity. TEE appears to be highly sensitive ir detecting a mobile intimal flap in the ascending or descending aorta. In addition, TEE can provide useful information regarding AR, periaortic hematoma, and flow within a false channel. It is usually available at the bedside or in the emergency ward and does not subject the patient to a contrast load.

After patients are diagnosed, they are transported immediately to the OR. Through a median sternotomy, venous access is gained via the right atrium, and arterial inflow is supplied through the ascending aorta (challenging in cases of aortic dissection as it requires identification of the true lumen and is associated with a significant risk of aortic rupture), a femoral artery or axillary artery. Some surgeons opt for axillary cannulation to avoid direct manipulation of the aorta and to maintain antegrade flow in the aorta. This mode of cannulation may not be possible if the dissection flap involves the innominate artery or if urgent need to initiate CPB is necessary (e.g., tamponade, rupture, or myocardial ischemia causing hemodynamic compromise). Axillary cannulation requires sewing a temporary graft to the right axillary artery and takes time to complete. Femoral cannulation, though occasionally used, has significant risks of dissection (including further propagating the aortic dissection, causing flaps to occlude visceral branches causing organ malperfusion, and embolization of thrombus or debris into the cerebral vasculature). CPB is established, the patient cooled (degree of cooling dependent on need for arch repair and utilization of cerebral perfusion strategies). If arch repair is necessary, circulatory support is discontinued with or without antegrade or RCP. The repair is carried distally into the arch, if the entire dissection can be resected, and the distal aortic layers are reapproximated with a Teflon felt strip supporting the medial and adventitial layers. The distal graft anastomosis is then completed, the graft clamped, the bypass pump restarted, and systemic warming commenced. Proximally, the aortic root may require reconstruction, again using Teflon felt to support the medial and adventitial layers and to resuspend the aortic valve, which can be salvaged in ~85% of cases. The heart is then cleared of air and the cross-clamp removed to allow reperfusion of the coronary circulation. After a sufficient period of resuscitation, the patient is weaned from CPB. Aggressive management of an acquired coagulopathy is not unusual prior to chest closure.

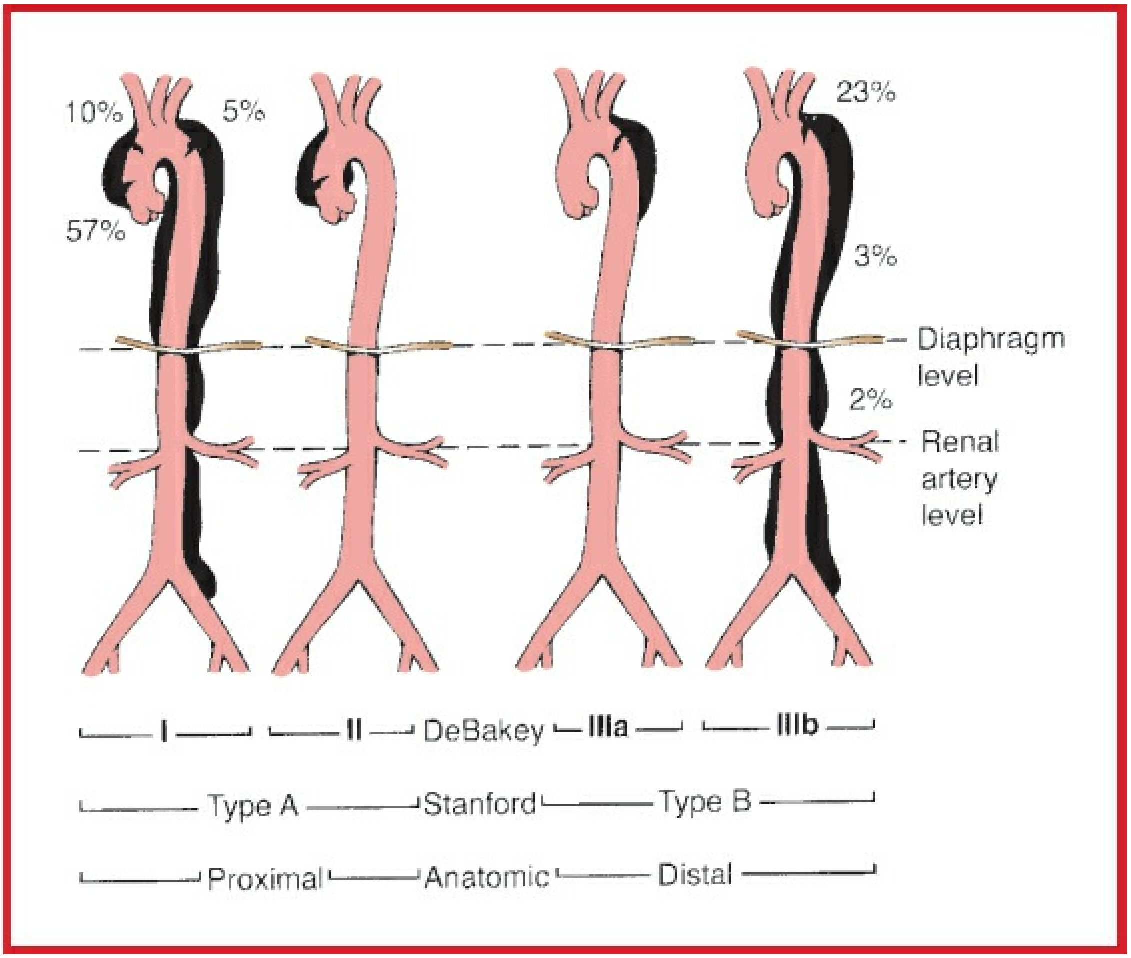

Figure 6.3-5 The three classification systems of aortic dissection and the distribution of the intimal tear.

Open repair of **dissections** involving the **descending thoracic aorta** is accomplished through a left thoracotomy utilizing partial CPB. Venous drainage is usually via the femoral vein, although the pulmonary vein may be used for partial or "left heart" CPB. Arterial access is via the femoral artery. After institution of CPB, the dissected aorta above and below the most damaged area is cross-clamped and the aorta transected. After oversewing patent intercostal arteries, the medial and adventitial layers are buttressed with Teflon felt, and an interposition Dacron graft is sewn into place. After evacuation of air, clamps are removed, and the patient weaned from CPB. Heparin reversal, decannulation, and closure are accomplished in the usual manner.

Usual preop diagnosis

Acute dissection of the aorta (see Fig. 6.3-5 for types of dissection)

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. See Appendix K, p. K-1.**

SUMMARY OF PROCEDURES

	Ascending Aorta	Descending Aorta
Position	Supine	Lateral decubitus, left side up
Incision	Median sternotomy	Left lateral thoracotomy
Unique considerations	Full hemodynamic monitoring, with provisions for circulatory arrest, profound hypothermia, barbiturates, and steroids; TEE	DLT; CPB; TEE; lumbar drain
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg)	⇐
Surgical time	Cross-clamp: 30–50 min Circulatory arrest: 20–30 min CPB: 60–100 min Total: 3–5 h	30–60 min — 35–60 min ⇐

Closing considerations	Aggressively treat coagulopathy (frequently 2° ↓ Plt)	Replace DLT with single-lumen ETT
EBL	400–800 mL	600–800 mL
Postop care	ICU: 1–2 d, intubated	⇐
Mortality	10–25%	⇐
Morbidity	Bleeding: 3–8% Respiratory insufficiency: 2–5% CVA: 2–4%	Paraplegia: 5% CVA: 1–2% MI: 1–2%
Pain score	7–10	9–10

PATIENT POPULATION CHARACTERISTICS	
Age range	40–70 yr
Male:Female	3:2
Incidence	10/100,000
Etiology	Degenerative aortic disease
Associated conditions	HTN; secondary AR; bicuspid aortic valve; Marfan syndrome

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Sx are usually of sudden onset and depend on the site of dissection and specific organ involvement. Aortic dissections are divided into 2 types, depending on the site of intimal tear: Stanford **type A**, ascending and aortic arch, and Stanford **type B**, descending aorta (Fig. 6.3-5). Initial treatment involves the use of antihypertensive medications (e.g., SNP, clevidipine, nicardipine) to control BP and β-blockers (e.g., esmolol) to ↓ contractility. Type A dissections usually require emergent surgery. Patients presenting electively for thoracotomy (descending aorta) may have a thoracic epidural catheter and lumbar CSF drain placed the night before surgery.

Respiratory	✓ for recurrent laryngeal nerve palsy associated with chronic aneurysmal dilation. Tracheal and left main

	bronchus compression → difficult intubation, atelectasis, hemoptysis 2° rupture into lung, hemothorax → compromised oxygenation, and ↑ intrathoracic pressure → ↓ venous return, especially with IPPV **Tests:** CXR: ✓ for widened mediastinum, tracheal or left main bronchus compression and distortion (affects DLT placement), risk of aortic rupture or injury during esophageal manipulation with TEE, atelectasis, and pleural effusion (or hemothorax 2° rupture)
Cardiovascular	Aortic dissections may be associated with chronic HTN, cystic medial necrosis, or other connective tissue disorder (e.g., Marfan syndrome) and trauma. Dissection may result in cardiac tamponade, acute aortic regurgitation, cardiogenic shock, angina, MI, or rupture of the aorta. Dissection of major arteries may result in ↓ or absent peripheral pulses, which may affect the placement sites for intra-arterial monitoring and peripheral cannulation options for CPB. Pain and anxiety may result in HTN, while rupture or leakage may result in ↓ BP and shock. **Tests:** ECG: ✓ for Sx of LVH, ischemia or infarction, or low voltage (tamponade). ECHO: ✓ for dissection confirmation, identification of true/false lumens, thrombus formation (most commonly false lumen), fenestrations, valvular incompetence, LV function, pericardial effusion, or tamponade. CT scan: ✓ site and extent of dissection. Angiography (not warranted in unstable patients): ✓ for site of dissection (type A or B), valvular function, involvement of coronary and other major arteries, sites of rupture, and LV function
Neurologic	Deficits are not uncommon, especially with dissections involving the arch vessels and type B dissections where the blood supply to the spinal cord may be jeopardized. Document preop exam carefully. Neurologic deficits secondary to carotid hypoperfusion in type A dissections infer immense

	surgical risk.
Renal	Renal failure 2° renal artery involvement in dissection, shock, or cardiac failure. UO should be monitored closely during initial medical therapy. **Tests:** UO; BUN; Cr; electrolytes
Gastrointestinal	Compromised blood supply to the bowel or liver may result in ischemia → metabolic acidosis and ↓ liver function. **Tests:** Consider ABG; ✓ persistent metabolic acidosis; LFTs, if indicated by H&P
Hematologic	Coagulopathy may be present 2° massive hemorrhage or liver involvement and can increase risk of the surgery. **Tests:** PT; PTT; Hct/Hb/Plt
Laboratory	Lactic acid if concerned for organ hypoperfusion Other tests as indicated from H&P
Premedication	Because many of these patients present emergently, consider full-stomach precautions with rapid-sequence induction. Cautious use of vasodilators in symptomatic patients prior to induction. Alleviate anxiety and pain (midazolam 0.025–0.05 mg/kg iv), which may worsen HTN, but avoid obtundation.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: Preinduction control of BP and contractility (NTG or SNP to SBP = 105–11
and esmolol to HR = 60–80) is important in preventing extension of the dissection or

rupture of the aneurysm. Fluid resuscitation may be necessary prior to induction. Extreme caution and preemptive hemodynamic support are essential during induction of patients with hemopericardium to avoid hemodynamic collapse due to tamponade.

Induction	Control of hypertensive response to laryngoscopy and intubation is important and may be accomplished with moderate-dose narcotic (fentanyl 10–15 mcg/kg or sufentanil 1–2 mcg/kg) and etomidate (0.1–0.3 mg/kg), esmolol (5–10 mg iv bolus), or SNP (25–50 mcg bolus). Pretreatment with lidocaine (1.5 mg/kg) also may be required to further control the hypertensive response to laryngoscopy. Muscle relaxation may be obtained using vecuronium (0.1 mg/kg) or rocuronium (1–2 mg/kg). Remember the possibility of a full stomach in this patient population. Hence, a modified rapid-sequence induction (cricoid pressure with manual ventilation and NMR) may achieve the two goals of relatively rapid induction and intubation and tight control of BP. Patients with ascending and arch lesions undergoing median sternotomy do not need lung isolation. In patients with descending lesions, lung isolation is necessary to improve surgical access; however, it may be difficult to place a left DLT due to aneurysmal compression of the trachea and left mainstem bronchus, and it is associated with a small risk of aneurysmal rupture. For these reasons, a bronchial blocker or a right DLT may be preferred. FOB is mandatory to verify ET placement.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.
Maintenance	O_2/narcotic/benzodiazepine: low-dose volatile agent

	(isoflurane [0.3–0.5%] or sevoflurane [0.5–1%]) may be used to control BP (MAP = 60–80), although infusion of vasopressors, vasodilators, or inotropes may be necessary. Control HR (<80; anesthesia, esmolol) and contractility (β-blockade or inotropes, depending on circumstances). Benzodiazepines may be used for amnesia (midazolam 100 mcg/kg). If thoracic epidural catheter is in place, fentanyl (20–30 mcg/kg) or sufentanil (4–6 mcg/kg) may be used. Most type A dissections and aneurysms involving the distal ascending or aortic arch require hypothermic CPB (see Anesthetic Considerations for Cardiopulmonary Bypass, p. 386). Intraop anesthetic considerations are governed primarily by the site of the aortic pathology. These considerations are discussed below. Recommend use of antifibrinolytics for all cases requiring CPB.	
Emergence	Transported to ICU, sedated, intubated, and ventilated × 6–48 h. Following repairs of the descending aorta (thoracotomy incision), weaning from ventilator may be aided by epidural narcotics after documentation of normal spinal cord function and coagulation. Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Anticipate large blood loss. IV: 14 ga—7 Fr × 2 BSS @ 6–8 mL/kg/h Warm all fluids. Humidify gases. Cross-match 6–8 U of blood. UO 0.5–1 mL/kg/h	If innominate or axillary artery cannulation is planned, arterial line should be placed in the left arm due to inaccuracies of the right radial arterial line during CPB. Mannitol (0.25–0.5 g/kg iv) should be given if renal perfusion is

		compromised by dissection or before cross-clamping. Consider normovolemic hemodilution if the patient is stable and the Hct > 35. After unclamping: furosemide (1 mg/kg), if hemodynamically stable. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Arterial line CVP, PA catheter (optional) Urinary catheter	Arterial line site is dependent on type of surgery, cannulation strategy, and location of the lesion. Because the right subclavian artery may be compromised in patients with ascending lesions, left upper extremity or femoral arteries may need to be used. Aortic arch lesions may involve the vascular supply to both upper extremities; hence, femoral artery catheterization may be necessary. In select cases, up to three arterial lines may be required, bilateral upper extremity arterial lines to permit estimation of flow with different cerebral perfusion strategies, as well as femoral arterial

		line to demonstrate distal (below clamp) perfusion pressure. Consult with surgeon to plan ideal site.
	TEE	TEE is used to confirm cannula placement and assess ventricular function, regional wall motion abnormalities, valve pathology, and integrity of aortic valve repair/replacement (if required as part of a type A repair). Probe passage may increase compression of the trachea, impeding ventilation. Caution should be used in the presence of aneurysmal compression of the esophagus.
	Temperature	Monitor both core (esophageal/bladder) and either nasopharyngeal (closest to brain T) or bilateral tympanic membranes T (for symmetric cooling)—important in hypothermic circulatory arrest.
	EEG/EP (arch/ascending lesions)	EEG/EP may help in assessing effectiveness of cerebral protection or adequacy of cerebral perfusion.
	SSEPs/MEPs (descending	SSEPs may detect

	lesions)	posterior spinal perfusion problems, while MEPs may detect anterior cord dysfunction. Both EEG and EPs may require expert help to set up, monitor, and interpret.
	Cerebral oximetry	Cerebral oximetry may be helpful in detecting changes in scalp + cerebral blood flow after cannulation of the innominate artery or gross malperfusion issues. Be aware of false-positive and false-negative readings.
Complications	Hemorrhage Coagulopathy	Both are common. Hemorrhage should be treated with crystalloid, colloid, or blood products, as indicated.
Ascending lesions	CPB AR Coronary arteries	Common CPB arterial cannulation sites include the axillary artery, innominate artery, ascending aorta, or femoral artery. Aortic valve replacement may be necessary if at least moderate AR is present. Patients may have myocardial ischemia 2° coronary artery occlusion or dissection; may require CABG or reimplantation of vessels.
	Hypothermic circulatory	Cerebral protective

	arrest (HCA)	measures and selective perfusion of cerebral vessels may be required (see below).
Arch lesions	CPB Hypothermic circulatory arrest	Usual site of cannulation for CPB is femoral or axillary artery. Cerebral protection relies on hypothermia (20–22°C) and drugs (hydrocortisone 100 mg iv or dexamethasone 8 mg iv) to reduce $CMRO_2$ and neuronal injury. The head is surface cooled (protect eyes and ears from cold injury). EEG may be monitored to ensure absence of brain electrical activity. Monitor tympanic membrane T as an indication of brain T. Avoid hyperglycemia and maintain normal pH and $PaCO_2$ when measured at 37°C (alpha stat). Maintain muscle relaxation.
Descending lesions	Pre–cross-clamping	Mannitol (0.5 g/kg) and furosemide (20–40 mg iv) should be given before clamp application to provide renal protection, even if a shunt is placed. Systemic hypothermia may be used to protect the spinal cord.

	Shunt	A heparin-bonded shunt may be used from the aortic arch to the femoral artery to provide distal perfusion.
	Partial bypass	Partial CPB may be used to provide distal perfusion. In this arrangement, the heart perfuses the head and upper extremities, while CPB is used to perfuse and oxygenate the lower body. Venous drainage is from the femoral vein, PA, or left atrium (specifically known as left heart bypass) and returned via the femoral artery. Distal and proximal pressures are altered by controlling cardiac filling, pump flow, and vasodilators. Arterial pressure monitoring pre- and postclamp is essential.
	Cross-clamping	Application of clamp may → acute HTN with ischemia and LV failure. This should be controlled with potent vasodilators (SNP 0.5–2 mcg/kg/min, NTG 0.5–2 mcg/kg/min). Partial CPB or shunting may attenuate hemodynamic response. During cross-clamping,

		monitor UO and ✓ metabolic acidosis (renal or bowel ischemia) with serial ABGs. Cross-clamp time should be <30 min to reduce incidence of paraplegia. A lumbar CSF drain should be inserted to ↓ CSF pressure (to ↓ spinal venous pressure) and, thus, improve spinal cord blood flow.
Descending lesions (cont'd)	Unclamping	Unclamping may result in severe ↓ BP and myocardial depression. Hypovolemia, acidosis, vasoactive factors, and reactive hyperemia have been implicated as the cause. Prior to unclamping, ensure adequate volume status (CVP and/or PCWP 2–5 mm Hg above patient's normal); treat acidosis; have vasopressors available; and the clamp should be released slowly over 1–2 min. Use of partial bypass or a shunt to ensure distal perfusion will mitigate unclamping shock. When hemodynamically stable, give furosemide (1 mg/kg iv). Inotropes (dopamine) may be needed to support circulation.

Positioning	✓eyes ✓ and pad pressure points Arch and ascending: supine, shoulder roll Descending: right lateral decubitus, beanbag, axillary roll, pillow between knees

POSTOPERATIVE

Complications	Myocardial ischemia, CHF Dysrhythmias Hemorrhage Coagulopathy Renal failure Bowel ischemia Respiratory failure	A DLT may be required in lesions of the descending aorta, if hemorrhage continues from the left lung; otherwise, the tube should be exchanged at the end of procedure to a single-lumen ETT. BP should be controlled to MAP 60–80 and HR < 100 to decrease the likelihood of bleeding or graft dehiscence.
	Paraplegia	Anterior spinal artery syndrome
Multimodal pain management	PCA Epidural	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6). Thoracic epidural catheter may be used in patients following thoracotomy if coagulation status is normal.
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be

	preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.
Tests	ECG: ischemia, infarction, dysrhythmias CXR: line and ETT placement; ALI/pulmonary contusion Coagulation profile BUN, Cr: AKI ABG: metabolic acidosis, mesenteric ischemia CT scan: CNS or spinal neurologic deficits Electrolytes, lactic acid: resuscitation goals, malperfusion

Suggested Readings

1. Crumley S, Schraag S: The role of local anaesthetic techniques in ERAS protocol for thoracic surgery. J Thorac Dis 2018; 10(3):1998–2004.
2. Dong CCJ, MacDonald DB, Janusz MT: Intraoperative spinal cord monitoring durin descending thoracic and thoracoabdominal aneurysm surgery. Ann Thorac Surg 2002; 74:S1873–6.
3. Epstein NE: Cerebrospinal fluid drains reduce risk of spinal cord injury fo thoracic/thoracoabdominal aneurysm surgery: a review. Surg Neurol Int 2018; 9:48.
4. Fattori R, Mineo G, Di Eusano M: Acute type B aortic dissection: currer management strategies. Curr Opin Cardiol 2011; 26:488–93.
5. Hughes GC, Andersen ND, McCann RL: Management of acute type B aorti dissection. J Thorac Cardiovasc Surg 2013; 145:S202–7.
6. Kruger T, Conzelmann LO, Bonser RS, et al: Acute aortic dissection type A.Br J Surg 2012; 99:1331–44.
7. Kruger T, Weigang E, Hoffmann I, et al: GERAADA Investigators: cerebra protection during surgery for acute aortic dissection type A: results of the German Registry for Acute Aortic Dissection Type A (GERAADA).Circulation 2011; 124:434–43.
8. Lu S, Sun X, Hong T, et al: Bilateral versus unilateral antegrade cerebral perfusion i arch reconstruction for aortic dissection. Ann Thorac Surg 2012; 93:1917–20.
9. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery.J Cardiothorac Vasc Anesth 2018; 32:2760–70. pii: S1053-0770(18)30049-1.
10. Penco M, Paparoni S, Dagianti A: Usefulness of transesophageal echocardiograph

in the assessment of aortic dissection. Am J Cardiol 2000; 86:53G–56G.
1. Serra R, de Franciscis S, Grande R, et al: Endovascular repair for acute traumatic transection of the descending thoracic aorta: experience of a single centre with a 12-years follow up. J Cardiothorac Surg 2015; 10(1):1–6.

REPAIR OF ANEURYSMS OF THE THORACOABDOMINAL AORTA

SURGICAL CONSIDERATIONS

Description

Aneurysms of the thoracoabdominal aorta (TAAA) may occur because of degenerative aortic disease (atherosclerosis), as a consequence of hereditary disorders of metabolism (Marfan syndrome), or as a sequela of chronic aortic dissections. These aneurysms are classified into four types (Fig. 6.3-6) that occur with equal frequency. Type I consists of aneurysms that involve most of the descending thoracic and upper abdominal aortas. Type II involves most of the descending thoracic aortas and most or all of the abdominal aortas. Type III involves the distal thoracic and varying segments of the abdominal aorta. Type IV involves most or all of the abdominal aortas, including the origins of the visceral vessels.

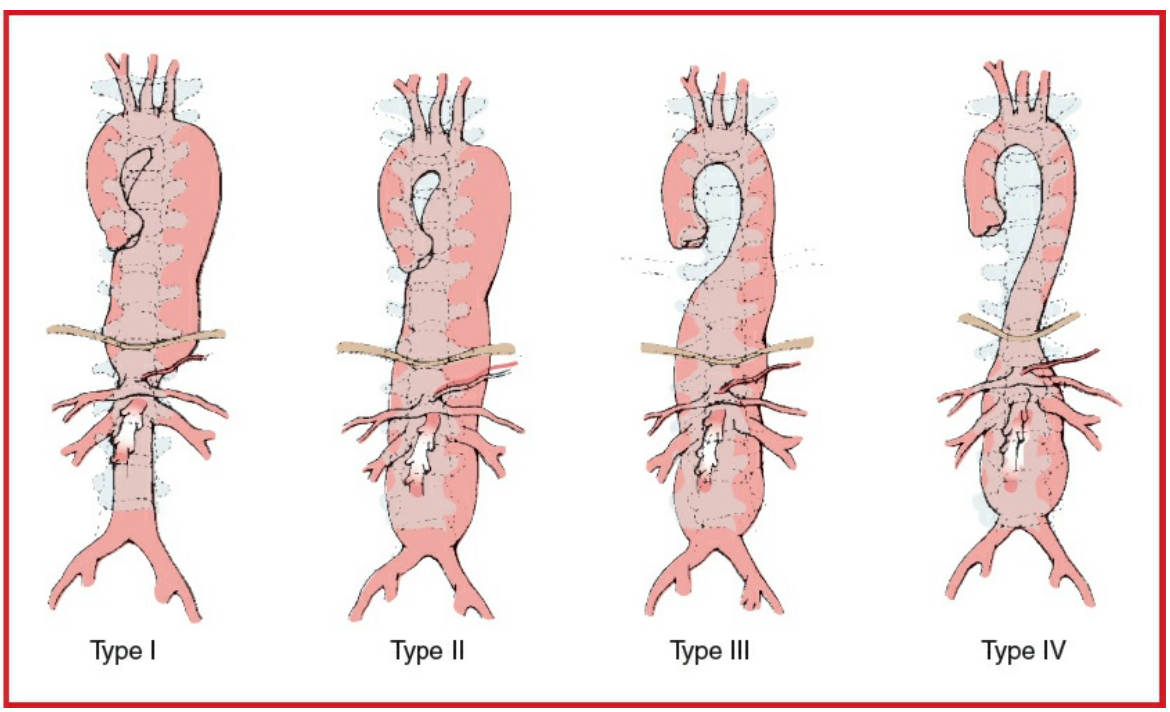

Figure 6.3-6 Crawford classification of thoracoabdominal aortic aneurysms (TAAA). (Reproduced with permission from Baker RJ, Fischer JE, eds Mastery of Surgery, Vol. 2, 4th edition. Lippincott

Williams & Wilkins, Philadelphia: 2001.)

Repair of these aneurysms is an extensive, difficult, and demanding procedure, as blood flow to the entire body below the neck is interrupted, with resultant renal and visceral ischemia. In addition, the blood supply to the spinal cord may arise from lumbar and/or intercostal vessels in the affected aortic segment, producing critical cord ischemia during cross-clamping and postop paraplegia.

Almost all thoracoabdominal aneurysm repairs are performed through a thoracoabdominal incision (Fig. 6.3-7) using the **inclusion technique** (Fig. 6.3-8) as advocated by **Crawford** et al. After opening the chest, the incision is extended across the costal cartilage onto the abdomen. The diaphragm is radially incised to the aortic hiatus, and the retroperitoneal dissection plane established anterior to the psoas musculature. All intra-abdominal contents, as well as the left kidney, are reflected anteriorly (Fig. 6.3-8). Only proximal aortic control is established. After partial heparinization, OLV is established to allow collapse of the left lung, the proximal aorta at the aneurysm neck is cross-clamped, and the aneurysm is incised. Back-bleeding from patent intercostal, mesenteric, and renal vessels can be controlled by balloon catheters, and aggressive blood salvage with autotransfusion devices is mandatory. The repair entails suturing a tube graft proximally to the divided aorta and then sewing islands of aortic tissue containing intercostal visceral vessels onto appropriate-sized holes in the side of the tube graft. This allows reperfusion of important intercostal, celiac axis, superior mesenteric, renal arteries, and finally, the distal aorta or iliac arteries. Because there is obligate visceral ischemia during the period of cross-clamping (which must be limited to <60–75 min), the operation must proceed expeditiously.

Alternatively, in an effort to afford both spinal cord and visceral protection through hypothermia, the operation may be performed on full CPB during a period of profound hypothermic circulatory arrest, which may exacerbate hemorrhagic complications.

After aortic cross-clamping, the aneurysm is opened and the repair performed from within the aneurysm, sewing onlay patches of the intercostal, mesenteric, and renal vessels to openings created in the tube graft. This no-clamp technique allows reasonable management of these very extensive aneurysms but results in an obligatory and ongoing blood loss through back-bleeding of visceral vessels until the anastomoses are complete.

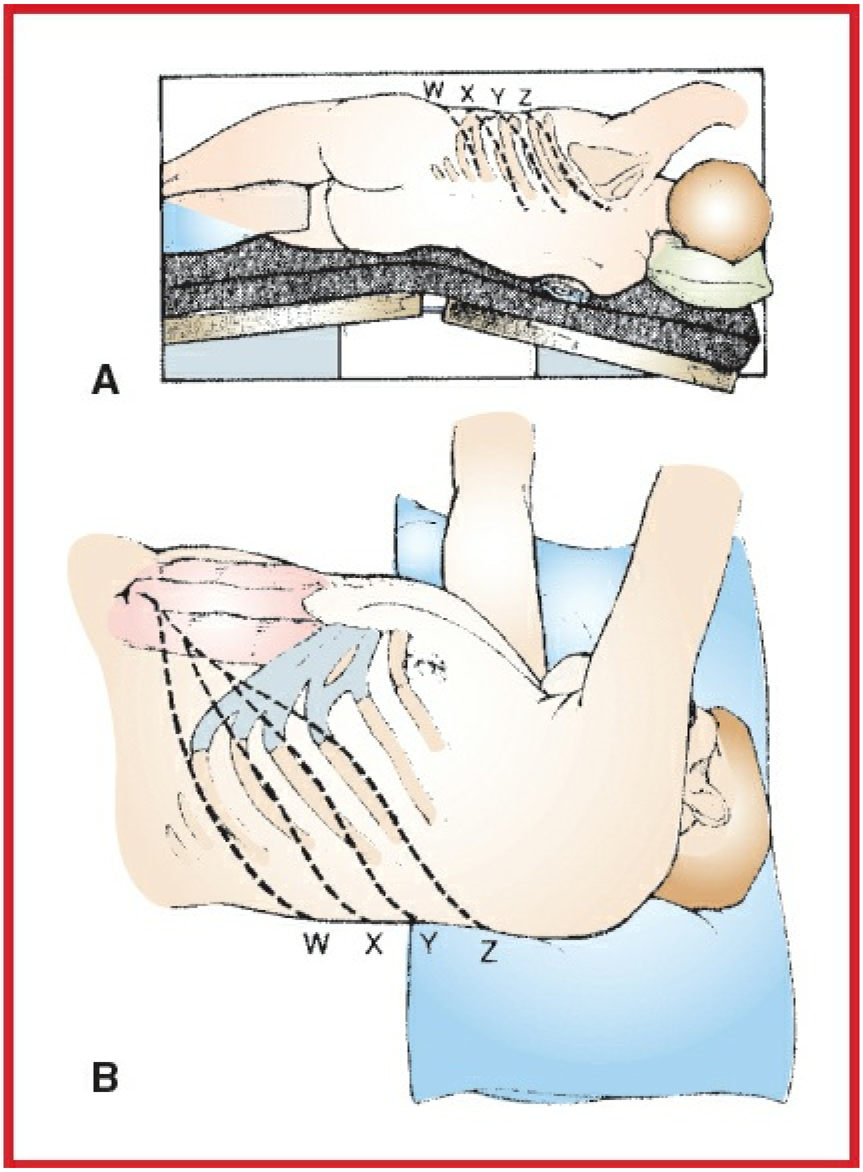

Figure 6.3-7 Lateral **(A)** and frontal **(B)** views of the thoracoabdominal incisions used for repair of type IV (W, X), type III (Y, Z), and types I and II (Z) TAAAs. (Reproduced with permission from Baker RJ, Fischer JE, eds:Mastery of Surgery, Vol. 2, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

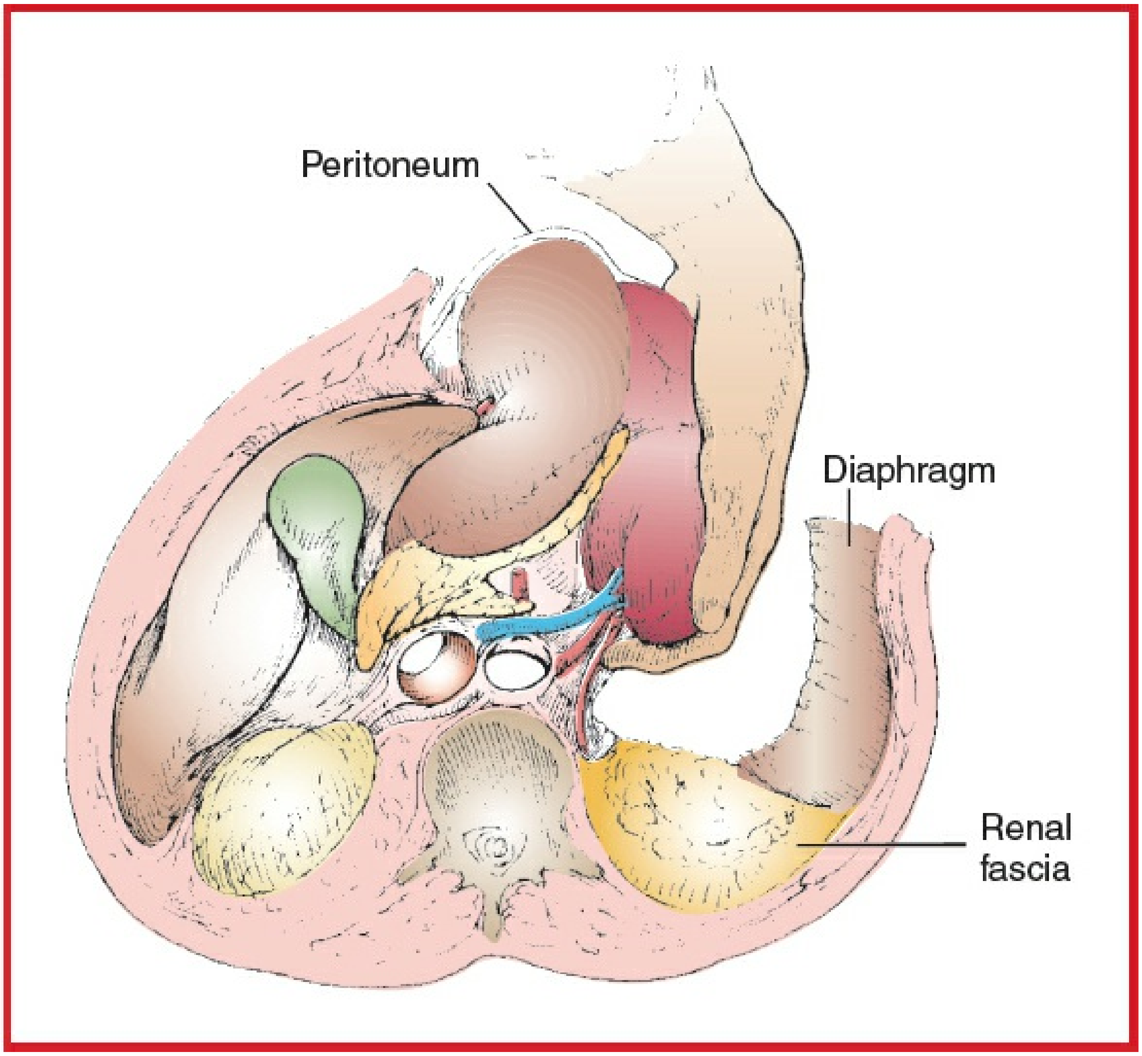

Figure 6.3-8 During the inclusion technique, the anterior renal fascia is opened and the kidney is mobilized, along with the upper abdominal organs (on the left). (Reproduced with permission from Wind GG, Valentine RJ: Anatomic Exposures in Vascular Surgery. Williams & Wilkins, Baltimore: 1991.)

Usual preop diagnosis

Expanding, painful, or large thoracoabdominal aneurysm

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. **See Appendix K, p. K-1.**

SUMMARY OF PROCEDURE

Position	Right lateral decubitus; hips rotated posteriorly to 45° and left arm draped forward over an airplane sling. Axillary roll placed. (See Fig. 6.3-7A.)
Incision	Posterolateral thoracotomy incision, in the appropriate interspace, extended across the costal margin to the midline and then extended inferiorly as a midline abdominal incision (Fig. 6.3-7). The incision is one of the largest incisions in surgery, necessitated by the absolute need for exposure in this difficult area, and is, unfortunately, associated with significant postop pain.
Special instrumentation	DLT; CPB; NG tube; cell saver; rapid infusion device; TEE; lumbar drain
Unique considerations	OLV with collapse of the left lung necessary for most cases. Cold LR may be injected into renal or visceral arteries for organ preservation. Alternatively, operation can be performed under profound hypothermic circulatory arrest for spinal cord protection in patients with chronic dissections. Frequently, operation is performed with only proximal cross-clamping; so there may be an obligate ongoing blood loss.
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg)
Surgical time	6 h. Proximal aortic cross-clamping until completion of visceral revascularization may extend to 60 min. Longer cross-clamp times may be anticipated in patients with aneurysmal dilation from chronic aortic dissection for which profound hypothermic circulatory arrest may be utilized.
Closing considerations	OR → ICU, intubated × 24–72 h
EBL	Ongoing back-bleeding from visceral and iliac vessels results in substantial volume loss during cross-clamping. Most red cell volume is salvaged via cardiotomy suckers and returned through cardiopulmonary bypass circuit or via an RBC salvage system (e.g., cell saver).
Postop care	ICU, intubated and ventilated × 24–72 h. Rewarming, hemodynamic monitoring, neuromonitoring, and volume resuscitation are often required in ICU.

	Aneurysm Group:	**Type I**	**Type II**	**Type III**	**Type IV**
Mortality	Overall: 9%	—	10–25%	—	5%
Morbidity	Paraplegia/spinal cord ischemia	6%	15%	3%	2%
	Other neurologic complications	6%	12%	2%	1%
	Renal insufficiency	—	2–5%	—	—
	Respiratory failure	—	10%	—	—
	Hemorrhage	Common	⇐	⇐	⇐
	Graft infection	—	1–6%	—	—
	MI	Rare	⇐	⇐	⇐
	Graft failure	—	Rare	—	—
	False aneurysm	Rare	⇐	⇐	⇐
	Embolization	Rare	⇐	⇐	⇐
	Bowel ischemia	—	2–10%	—	—
	Impotence	Rare	⇐	⇐	⇐
	Ureteral injury	Rare	⇐	⇐	⇐
Pain score	6–10				

PATIENT POPULATION CHARACTERISTICS

Age range	Non-Marfan, 55–75 yr; Marfan, 35–55 yr
Male:Female	3:1
Incidence	<5 cases/yr in most hospitals, except major referral centers
Etiology	Predominantly atherosclerotic. Patients with Marfan syndrome may present with a progressive dilation of a chronic dissection.
Associated conditions	HTN (75%); CAD (30%); COPD (30%); renal insufficiency (15%)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Sx are usually of sudden onset and depend on the site of dissection and specific organ involvement. Initial treatment involves the use of antihypertensive medications (e.g., SNP) to control BP and β-blockers (e.g., esmolol) to ↓ contractility. Patients presenting electively for thoracotomy (descending aorta) should have a thoracic epidural catheter and a lumbar cerebrospinal fluid (CSF) drain prior to surgical repair, which may be placed the day before or the morning of surgery.

Respiratory	Due to the need for prolonged intraop OLV, chronic pulmonary disease is associated with postop morbidity. Preop preparation with bronchodilators, cessation of smoking, incentive spirometry, and chest physiotherapy may decrease the risk of postop problems. **Tests:** CXR: ✓ for distortion of the left mainstem bronchus, which may affect placement of the DLT. May need PFTs and ABG to determine severity of pulmonary disease.
Cardiovascular	CAD is the most frequent cause of periop and late death in elective thoracoabdominal aortic aneurysm repair. It is commonly associated with HTN. **Tests:** ECG: ✓ for LVH and ischemia
Neurologic	Increased risk of spinal cord ischemia with cross-clamping of the aorta; therefore, any preop

	neurologic deficits should be well documented. The use of CSF drainage and/or systemic hypothermia is protective during thoracic aneurysm repair. Intraop neuromonitoring is frequently employed.
Renal	Preop renal dysfunction increases the potential for postop renal problems. Aneurysmal involvement of the renal arteries also may occur. **Tests:** BUN; Cr. Consider creatinine clearance.
Gastrointestinal	Aneurysmal involvement of the inferior mesenteric and superior mesenteric arteries may cause visceral ischemia. **Tests:** Consider abdominal x-ray (ileus) and ABG (metabolic acidosis).
Hematologic	Preexisting coagulopathy increases risk. Many patients have been on aspirin preop. Rarely, a DIC process may occur within the lumen of the aneurysm. **Tests:** PT; PTT; Plt count; Hct
Laboratory	Electrolytes; radiologic assessment of aneurysm (ultrasonography, CT, and arteriography)
Premedication	Anxiety and pain may contribute to HTN and risk of aneurysmal rupture. Rx: morphine 0.1 mg/kg iv and midazolam 1–5 mg iv. Full-stomach precautions for emergent procedures (e.g., RSI +/− metoclopramide 10 mg iv, famotidine 20 mg iv).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: The goals of anesthesia for this procedure are (a) preserve myocardial, renal, pulmonary, CNS, and visceral organ function; (b) maintain adequate intravascular volume so that CO is not impaired; (c) control BP so that the transmural pressure across the aneurysm does not increase, which would increase the risk of rupture; and (d) provide good perfusion of other organs. CPB is used to accomplish deep hypothermic circulatory arrest. The rationale for this use is controversial. Deep hypothermic cardiac arrest (DHCA) may confer spinal cord protection and is usually reserved for patients with extensive aneurysms or complex chronic dissections. Drainage of CSF has been proposed as another means of protecting the spinal cord against ischemic injury. CPB may be used for distal perfusion of organs and afterload protection of the left ventricle. Partial CPB usually is reserved for suprarenal or supraceliac aneurysms to perfuse bowel and kidneys. (See Anesthetic Considerations for Cardiopulmonary Bypass and Repair of Acute Aortic Dissections, for more discussion on CPB and DHCA.)

Induction	Prevent hypertensive response to laryngoscopy and intubation with moderate-dose narcotic technique (fentanyl 5–10 mcg/kg or sufentanil 0.5–2 mcg/kg) in combination with etomidate (0.2–0.3 mg/kg) or propofol (1–2 mg/kg). Esmolol 100–500 mcg/kg over 1 min, NTG 0.5–3 mcg/kg, or lidocaine administered either by intratracheal spray or an iv dose of 1.5 mg/kg also will decrease the cardiovascular response to intubation. Muscle relaxation for intubation may be achieved with vecuronium (beware of ↓ HR) or rocuronium (1 mg/kg). A modified rapid-sequence induction may be necessary in emergent cases. A DLT is mandatory for this procedure; however, it may be difficult to position due to distorted anatomy. FOB is mandatory to verify position of DLT.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.
Maintenance	O_2/air/narcotic ± low-dose volatile agent.

	Benzodiazepines may be used for amnesia (e.g., midazolam 100 mcg/kg). When epidural catheter is in place, sufentanil (loading dose 25–50 mcg, followed by 4–6 mcg/h) may provide adequate additional analgesia. In hemodynamically unstable patients, scopolamine (400 mcg) provides amnesia. Maintain CO and control of BP at preop levels. These patients may have increased hemodynamic variability on cross-clamping aorta, 2° bleeding, and coexisting disease. Keeping the patient warm may be difficult due to large incision and visceral exposure. Recommend use of antifibrinolytics for all cases requiring CPB.	
Emergence	Deferred to ICU. Postop ventilation 6–72 h. DLT may need to be maintained postop 2° to facial, oral, and airway edema. Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Anticipate large blood loss. IV: 14 ga or 7 Fr × 2; BSS Rapid infuser Cell saver T&C 8–10 U PRBCs Warm fluids and humidify gases. Maintain UO @ 0.5–1 mL/kg/h.	Large incision and visceral exposure may require administration of large volumes of fluid. Consider use of mannitol or furosemide if concerned about renal function and UO. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Arterial lines CVP catheter ± PA catheter ± ST-segment analysis	Consult with surgeon regarding planned location of cross-clamp and cannulas for CPB to ensure ideal arterial line site.

	UO TEE Core temperature ± EEG, SSEP, MEP ± Cerebral oximetry	Consider care when securing central venous lines in the right neck to ensure patency when in RLD position; consider LIJ lines if neck anatomy not favorable. ✓ for ↑ PA pressures and PCWP and ↓ in SvO_2 or CO TEE is a good monitor for ventricular filling, myocardial ischemia and changes in SV. Monitor core (esophageal/bladder) and either nasopharyngeal (closest to brain T) or bilateral tympanic membranes T (for symmetric cooling)—important in hypothermic circulatory arrest. EEG and EPs may be useful in assessment of cerebral protection (SSEP) and spinal cord perfusion problems (MEP ± SSEP). In case of the need of DHCA, cerebral oximetry may be helpful.
Cross-clamping	Clamping	Application of cross-clamp at supraceliac level produces the greatest hemodynamic stress experienced by surgical patients. Application of

	Unclamping	the clamp → HTN and ischemia. Preload, afterload, and HR can be controlled with SNP (0.25–5, mcg/kg/min) and esmolol (100–500 mcg/kg/min) infusions. Bolus dose NTG (100 mcg) may be useful to terminate an acute hypertensive episode. Ensure adequate volume; replace blood loss. Clamp should be removed gradually or stepwise. Just before removal of the cross-clamp, filling volumes are allowed to rise gradually, avoiding the occurrence of myocardial ischemia. Dilators are D/C'd. ↓ BP may occur with removal of cross-clamp 2° hypovolemia, reactive hyperemia, acidosis, ↑ K^+, or myocardial dysfunction. BE > −5 should be corrected before unclamping. If necessary, surgeon can reclamp or occlude the aorta.
Positioning	Right axillary roll ✓ and pad pressure points ✓ eyes	Right lateral decubitus with hips rotated posteriorly. Left arm placed in airplane sling or supported by pillows.

		Ensure neck and spine are aligned, and down ear and eye are free of pressure.
Complications	Myocardial ischemia HTN	
	Coagulopathy Hemorrhage Hemostasis	Coagulopathy due to dilutional and consumptive processes
	Hypothermia Other organ ischemia	Hypothermia may exacerbate coagulopathy, cause dysrhythmias, and depress cardiac contractility.

POSTOPERATIVE

Complications	Myocardial ischemia Neurologic deficits 2° cerebral or spinal cord ischemia Renal failure Respiratory failure	BP should be closely controlled postop to decrease bleeding from graft site and raw surfaces.
Multimodal pain management	Epidural narcotics	In selected patients, a thoracic epidural catheter may be placed the night before surgery; otherwise, an epidural catheter is placed only after normal neurologic and coagulation status is determined. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever

		possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.
Tests	Neuro exam: ASAP at end of surgery, hip flexion bilaterally, should be done even while intubated CXR: line and ETT placement; ALI/pulmonary contusion Coagulation profile BUN, Cr: AKI ABG: metabolic acidosis, mesenteric ischemia CT scan: CNS or spinal neurologic deficits Electrolytes, lactic acid: resuscitation goals, malperfusion	

Suggested Readings

1. Anton JM, Herald KJ: Anesthetic management of open thoracoabdominal aortic aneurysm repair. Int Anesthesiol Clin 2016; 54(2):76–101.
2. Coselli JS, Bozinovski J, LeMaire SA: Open surgical repair of 228 thoracoabdominal aortic aneurysms. Ann Thorac Surg 2007; 83:S862–4.
3. Coselli JS, LeMaire SA, Miller CC, et al: Mortality and paraplegia after thoracoabdominal aortic aneurysm repair: a risk factor analysis. Ann Thorac Surg 2000; 69:408–14.
4. Crumley S, Schraag S: The role of local anaesthetic techniques in ERAS protocols for thoracic surgery. J Thorac Dis 2018; 10(3):1998–2004.
5. Epstein NE: Cerebrospinal fluid drains reduce risk of spinal cord injury for thoracic/thoracoabdominal aneurysm surgery: a review. Surg Neurol Int 2018; 9:48.
6. Etz CD, Weigang E, Hartert M, et al: Contemporary spinal cord protection during thoracic and thoracoabdominal aortic surgery and endovascular aortic repair: a position paper of the vascular domain of the European association for cardio-thoracic surgery. Eur J Cardio-thoracic Surg 2014; 47(6):943–57.

7. Fann JI: Descending thoracic and thoracoabdominal aortic aneurysms.Coron Artery Dis 2002; 13:93–102.
8. Fischbein MP, Miller DC: Long-term durability of open thoracic and thoracoabdominal aneurysm repair. Semin Vasc Surg 2009; 22:74–80.
9. Lima B, Nowicki ER, Blackstone EH, et al: Spinal cord protective strategies during descending and thoracoabdominal aortic aneurysm repair in the modern era: the role of intrathecal papaverine. J Thorac Cardiovasc Surg 2012; 143:945–52.
0. Lips J, de Haan P, de Jager S, et al: The role of transcranial motor evoked potentials in predicting neurologic and histopathologic outcome after experimental spinal cord ischemia. Anesthesiology 2002; 97:183–91.
1. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery.J Cardiothorac Vasc Anesth 2018; 32:2760–70. pii: S1053-0770(18)30049-1.
2. Wong DR, Parenti JL, Green ST, et al: Open repair of thoracoabdominal aortic aneurysm in the modern surgical era: contemporary outcomes in 509 patients. J Am Coll Surg 2011; 212:569–79.

SURGERY OF THE ABDOMINAL AORTA

SURGICAL CONSIDERATIONS

Description

Operations on the abdominal aorta are generally performed for aneurysmal or occlusive diseases. Although **aortic aneurysms** may involve the suprarenal aorta, the majority is infrarenal in origin and may extend into the iliac arteries (Fig. 6.3-9). Most (>95%) are asymptomatic and are discovered incidentally during investigation of another medical problem. Because of the associated increased risk for rupture as the aneurysm increases in size, most vascular surgeons recommend prophylactic repair for aneurysms >5 cm in cross-sectional dimension. Repair also is indicated for painful aneurysms, those that have been associated with atheroembolism, and when there is documented recent increase in size or evidence of leak or rupture. CAD coexists in 30–40% of these patients and should be assessed preop.

Endovascular stent grafting of AAA has emerged as an alternative therapeutic approach. The open surgical repair may be either **transperitoneal** or **retroperitoneal** (Fig. 6.3-10). After exposure of the abdominal aorta from the level of the renal vein distally to the iliac arteries, the aorta is cross-clamped, distally at first to prevent atheroembolism and then proximally. Graft origin is usually from the infrarenal aorta but may arise from the intramesenteric aorta or even the supraceliac aorta. Graft termination may be to the distal aorta above the bifurcation (tube graft), to the common or external iliac arteries (Y graft), or to the femoral arteries. Immediately prior to cross-clamping, vasodilators are increased to reduce afterload, which is significantly increased with application of the aortic cross-clamp. The aorta is then incised, lumbar vessels oversewn, and the aorta transected to allow an interposition graft to be sewn into place. The retroperitoneal approach has many advocates, as it may require less volume intraop, may be associated with less temperature loss, and may result in a shorter period of postop adynamic ileus. In a randomized, prospective study, however, no significant differences could be detected between these two approaches for blood loss or postop recovery time.

Aortoiliac occlusive disease can be a significant cause of lower extremity arterial insufficiency. Although the operative approach may be similar to that for aneurysmal disease, there exists a significant intraop difference in that there are not such profound changes associated with aortic clamping because there is already some element of increased afterload 2° the occlusive disease. Nevertheless, hemodynamic monitoring is mandatory to allow for rapid volume shifts and to ensure adequate preload and sufficient afterload reduction, especially during the period of aortic cross-clamping.

Variant procedure or approaches

In the rare patient with COPD severe enough to preclude weaning from the ventilator postop, extra-anatomic grafts (e.g., axillofemoral, iliofemoral, and femoral-femoral bypass) can be constructed under local anesthesia.

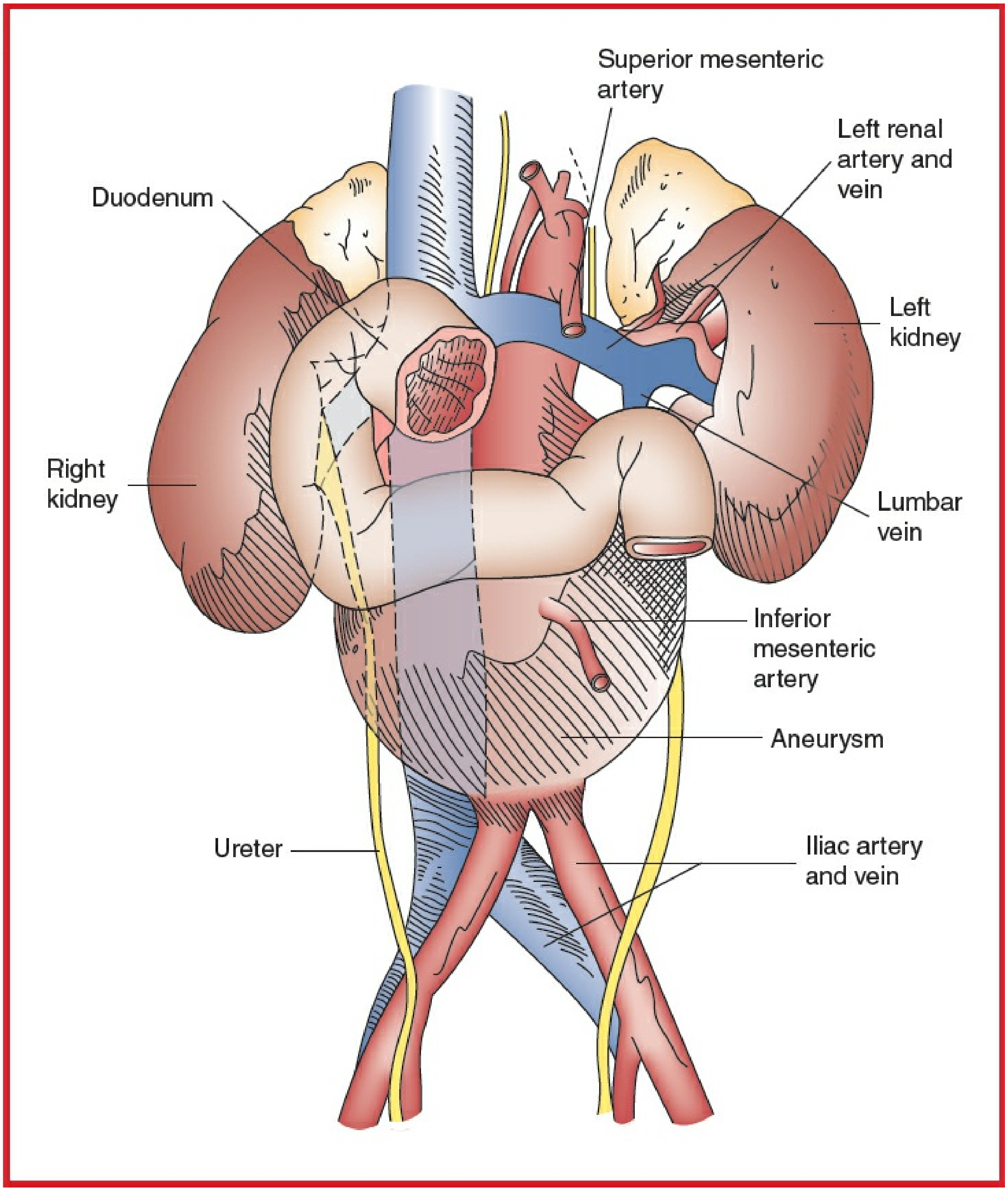

Figure 6.3-9 Aneurysm of the abdominal aorta.

Usual preop diagnosis

Abdominal aortic aneurysm; severe aortoiliac stenosis sufficient to cause debilitating buttock, thigh, or calf claudication; isolated inflow (aortoiliac) disease (rarely the sole cause for ischemic symptoms at rest or for tissue loss, except as a result of embolic complications)

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of**

nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. See Appendix K, p. K-1.

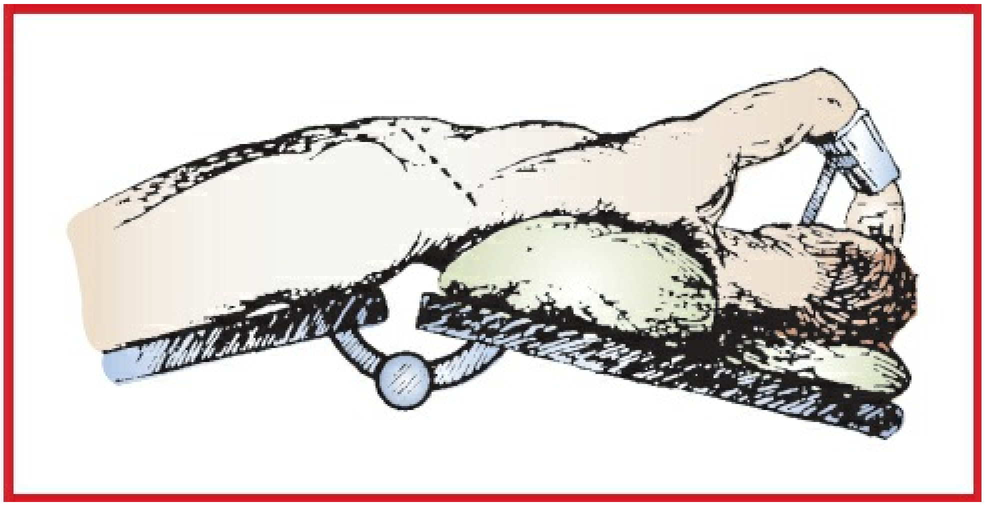

Figure 6.3-10 Retroperitoneal approach to the aorta: the incision is extended from the tip of the 11th rib or 10th intercostal space laterally toward the midhypogastrium. (Reproduced with permission from Scott-Conner CEH, Dawson DL. Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

SUMMARY OF PROCEDURES

	Transperitoneal Approach	Retroperitoneal Approach
Position	Supine	Supine with mild elevation of left flank
Incision	Midline abdominal	Left subcostal, left oblique, along 10th rib toward umbilicus (See Fig. 6.3-10.)
Special instrumentation	Self-retaining retractor; TEE; if suprarenal change, SMA + renal artery perfusion catheters	⇐
Unique considerations	Will need pharmacologic manipulation to ↓ afterload or ↑ preload coincident with clamping or	⇐ (The retroperitoneal approach is not used for emergency cases.)

	unclamping of aorta	
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg)	⇐
Surgical time	3–5 h	⇐
EBL	500 mL	⇐
Postop care	ICU 8–16 h, often extubated. Requires aggressive volume administration to allow for third-space loss in the first 12 h postop. Patient comfort and respiratory management markedly improved by epidural catheter for postop analgesia. Careful cardiac monitoring	⇐
Mortality	Elective: 2–5%; emergent: 50%	⇐
Morbidity	MI: 10–15% (3% fatal) Respiratory insufficiency/pneumonia: 5–10% Lower extremity ischemia: 2–5% Renal insufficiency: 2–5% Bowel complications: 3–4% Hemorrhage: 2–4% CVA: <1% Infection: <1% Paraplegia: <0.4%	⇐
Pain score	8–10	7–10

■ PATIENT POPULATION CHARACTERISTICS	
Age range	55 yr +
Male:Female	4:1
Incidence	3% of males >55 yr; >5% of patients >50 yr in a general cardiology clinic; 30–66/1000
Etiology	Atherosclerosis; Marfan syndrome; Ehlers-Danlos syndrome; dissection; infection, including syphilis
Associated conditions	HTN; CAD; COPD; cerebrovascular disease; renal insufficiency

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for open AAA repair are typically older males with multiple coexisting diseases (e.g., CAD, HTN, PVD, COPD). Most commonly, these patients ar asymptomatic, and the Dx is made as an incidental finding during routine exams or other medical procedures; however, some patients may present with severe ↓ BP following aneurysm rupture. These patients require prompt resuscitation (SBP goal: 90–100 mm Hg) and emergent open or endovascular control (aortic cross-clamp vs. endovascular balloon occlusion). For anesthetic considerations, see Endovascular Stent Grafting of Aortic Aneurysms and Abdominal Trauma: Vascular Injuries.

Respiratory	Many patients have COPD and a long Hx of smoking. Preop preparation—including bronchodilators, cessation of smoking, incentive spirometry, and chest physiotherapy—will ↓ risk of postop complications. **Tests:** CXR. May need PFTs and ABG to determine severity of pulmonary disease.
Cardiovascular	CAD is present in up to 70% of these patients and is the most common cause of morbidity and mortality in this patient population. HTN increases risk of aneurysmal rupture; hence, tight control of BP is essential. ✓ BP in both arms to determine placement of arterial line intraop (arterial line should be placed in the arm with higher BP).

	Tests: Radiologic assessment of aneurysm using ultrasonography, CT, arteriography, and MRI. ECG: ✓ for evidence of ischemia, infarction, and LVH. Stress testing, coronary angiography in case of suspicion of coronary disease
Renal	Chronic renal insufficiency occurs frequently in this patient population 2° HTN, diabetes mellitus (DM), and atherosclerotic renovascular disease. Hypovolemia 2° radiographic dye studies and bowel prep → renal failure. **Tests:** BUN, Cr; consider creatinine clearance; electrolytes
Hematologic	Preop coagulation disorders should be corrected. ASA should be continued whenever feasible in patients with known CAD, but DAPT and other anticoagulants should be stopped or bridged with shorter half-life agents to reduce excess perioperative bleeding. Alcohol abuse can be associated with anemia, thrombocytopenia, and ↓ vitamin K–dependent factors. **Tests:** PT; PTT; Hct; Plt
Laboratory	Other tests as indicated from H&P
Premedication	Anxiety and pain may cause HTN and ↑ risk of aneurysmal rupture. Sedatives and analgesics should be used as indicated. Full-stomach precautions for emergent procedures.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA or combination of epidural and general anesthesia. The goals of anesthesia are to (a) preserve myocardial, renal, pulmonary, and CNS perfusion; (b) maintain adequate intravascular volume and CO; (c) anticipate the surgical maneuvers that will affect BP and blood volume; and (d) control BP to minimize risk of rupture while ensuring perfusion of other organs.

Induction	If it is planned to extubate the patient postop, anesthesia should be induced using fentanyl (4–6 mcg/kg) in conjunction with propofol (1–2 mg/kg) or etomidate (0.2–0.4 mg/kg). If an epidural will be used for postop pain management, decrease intraop iv opioid dose. Patients with an epidural typically receive either morphine (2–4 mg), hydromorphone (0.2–0.8 mg), fentanyl (150–200 mcg), or sufentanil (50 mcg) in 10 mL into the epidural catheter. Local anesthetics may cause hemodynamic instability after unclamping 2° sympathectomy. Muscle relaxants are chosen to minimize tachycardia or ↓ BP (e.g., vecuronium 0.1 mg/kg or rocuronium 1 mg/kg). Anesthesia is deepened before intubation by mask ventilation with 1% sevoflurane. If there is a hemodynamic response to oral airway insertion, tracheal lidocaine or further anesthetic is given before gentle laryngoscopy and intubation. Etomidate is useful for induction in hemodynamically unstable patients. Full-stomach precautions may be necessary.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.
Maintenance	O_2/air/narcotic and volatile agent. N_2O can be used, but it may cause bowel distention. Combining

	epidural and general anesthesia offers good abdominal relaxation. Patients receiving epidural local anesthesia may require phenylephrine to treat ↓ BP 2° sympathetic blockade. Epidural catheter placement before systemic anticoagulation is a safe technique. Narcotic epidural analgesia should be given within 1 h of skin incision if morphine 3–8 mg is used. If hydromorphone (0.5–0.8 mg) is given epidurally, it is given within 1 h of abdominal closure. Patient may become hypothermic 2° large incision with visceral exposure and prolonged operation. Recommend use of antifibrinolytics for all cases requiring CPB.	
Emergence	Normothermic (>35.5°C) patients who have had uneventful surgery, especially utilizing a retroperitoneal approach, are frequently extubated in the OR or shortly after arrival in the ICU. Patients with significant cardiac or pulmonary disease generally are left intubated and weaned from the ventilator in the ICU. Prevention of HTN and tachycardia during emergence may require titration of β-adrenergic–blocking drugs (e.g., esmolol 50–300 mcg/kg/min) and/or vasodilators (e.g., SNP 0.25–2.0 mcg/kg/min) or NTG (0.25–2.0 mcg/kg/min). Epidural morphine (2–4 mg) or hydromorphone (0.5–0.8 mg) is given ~60 min before the end of surgery. It is important to ensure full reversal from neuromuscular blockade. Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Major blood loss IV: 14 ga (or 7 Fr) × 2; BSS 4 U PRBC UO 0.5–1 mL/kg/h	Use of rapid infusion and blood-salvaging devices is helpful. Consider acute normovolemic hemodilution.

	Warm all fluids. Humidify all gases.	Hemorrhage should be treated with crystalloid, colloid, or blood products as appropriate. Upperbody or underbody Bair Hugger or equivalent. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors	Monitor for cardiac ischemia. Automated ST-segment analysis is useful.
	CVP line	The measurement of CVP may be sufficient in patients with good ventricular function and exercise tolerance.
	PA catheter	In patients with recent MI, Hx of CHF, Hx of unstable angina, or multisystem disease, or in those presenting emergently (without the benefit of a complete workup), a PA catheter is appropriate.
	TEE	TEE is invaluable for assessing cardiovascular changes 2° cross-clamping and unclamping the aorta.
	Arterial line Urinary catheter	The arterial line is generally placed in the radial artery of the arm having the higher BP (if difference exists).

Cross-clamping	↑↑ afterload ↑ preload ↑ filling pressures ↓ CO ± ↓ spinal cord perfusion ± ↓ renal perfusion ↓ perfusion to viscera below the clamp ✓ ABG/lytes and UO regularly	Application of aortic cross-clamp causes sudden increase in SVR, LVEDV, wall tension, and preload with a decrease in cardiac output. In a healthy heart, this should be well tolerated, with minimal increase in filling pressure. In case of poor left ventricular function, filling pressures generally rise. Control of preload, afterload, and HR can be accomplished with vasodilators and β-blockers. Negative inotropic agents (e.g., β-blockers, inhalational anesthetics) are used cautiously. Ensure adequate intravascular volume and CO and consider mannitol (0.25–0.5 g/kg) and furosemide (20–80 mg iv) before clamping to maintain UO.
Aortic unclamping	↓↓ afterload ↓↓ preload Lactate washout ± phenylephrine	Immediately before unclamping, D/C dilators and negative inotropic agents. Gradually increase filling pressures (volume loading) to avoid myocardial ischemia, start vasoconstrictors (phenylephrine or norepinephrine) to target

		MAP ~20% above baseline, and consider inotropic support if there is preexisting ventricular dysfunction. ↓↓ BP may occur with removal of cross-clamp 2° to hypovolemia, reperfusion syndrome, or myocardial dysfunction. Reperfusion of lower limbs and washout of lactate do not usually require use of HCO_3. If ↓↓ BP persists, ask surgeon to reclamp or partially clamp the aorta. If hemodynamically stable with ↓ UO, consider furosemide 20–80 mg.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Myocardial ischemia HTN Hemorrhage Coagulopathy Hypothermia Organ ischemia	Hypothermia may cause dysrhythmias, depress contractility, and exacerbate coagulopathy.

POSTOPERATIVE

Complications	Myocardial ischemia Renal failure Respiratory failure	
Multimodal pain	Epidural narcotics	Emphasize multimodal

management		approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.
Tests	CXR to ✓ line and ETT placement ABG, lactic acid Coagulation profile BUN; Cr	BP should be closely controlled postop to decrease bleeding from graft site and raw surfaces.

Suggested Readings

1. Boccara G, Jaber S, Eliet J, et al: Monitoring of end-tidal carbon dioxide partia pressure changes during infrarenal aortic cross-clamping: a non-invasive method to predict unclamping hypotension. Acta Anesthesiol Scand 2001; 45:188–93.
2. Conrad MF, Crawford RS, Pedraza JD, et al: Long-term durability of ope abdominal aortic aneurysm repair. J Vasc Surg 2007; 46:669–75.
3. Crumley S, Schraag S: The role of local anaesthetic techniques in ERAS protocol for thoracic surgery. J Thorac Dis 2018; 10(3):1998–2004.
4. Filinger M: Who should we operate on and how do we decide: predicting rupture and survival in patients with aortic aneurysm. Semin Vasc Surg 2007; 20:121–7.
5. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery.J Cardiothorac Vasc Anesth 2018; 32:2760–70. pii: S1053-0770(18)30049-1.
6. Panaretou V, Toufekzian L, Siafaka I, et al: Postoperative pulmonary function after open abdominal aortic aneurysm repair in patients with chronic obstructive pulmonary disease: epidural versus intravenous analgesia. Ann Vasc Surg 2012; 26:149–55.
7. Posner M, Gelman S. Pathophysiology of aortic cross-clamping and unclamping.Best

Pract Res Clin Anaesthesiol 2000; 14(1):143–60.
8. Takayama T, Yamanouchi D: Aneurysm disease: the abdominal aorta. Surg Clin N Am 2013; 93:877–91.
9. Tang TY, Walsh SR, Fanshawe TR, et al: Comparison of risk-scoring methods in predicting the immediate outcome after elective open abdominal aortic aneurysm surgery. Eur J Vasc Endovasc Surg 2007; 34:505–13.
0. Tsai S, Conrad MF, Patel VI, et al: Durability of open repair of juxtarenal abdomina aortic aneurysms. J Vasc Surg 2012; 56:2–7.

INFRAINGUINAL ARTERIAL BYPASS

SURGICAL CONSIDERATIONS

Description

Due to the limited durability of distal bypass procedures, surgical bypass to the infrainguinal arteries is indicated only for salvage of the severely ischemic lower extremity, as manifest by gangrene, ischemic ulceration, or ischemic rest pain. Less frequently, it is used to alleviate functional ischemia of claudication, or leg discomfort with exercise. Its use is predicated on the existence of adequate inflow to the level of the groin or femoral artery. Other necessary components include an adequate target vessel, preferably in continuity with runoff into the plantar arch of the foot, and an adequate conduit, preferably autologous saphenous vein. Long-term patency rates of nonautologous conduits to the below-knee arteries are distinctly inferior to that of saphenous vein and should be avoided whenever possible. This population, almost by definition, includes patients with diabetes mellitus (DM), CAD, and cerebrovascular disease, all of which must be assessed preop.

The operative repair usually involves incisions at the groin and distal bypass sites to expose the donor and recipient arteries and to harvest leg or arm venous conduit. The operative approaches are rather similar. An unobstructed inflow source—usually the common femoral, superficial femoral, or deep femoral artery—is exposed in the groin. The target distal artery, usually at the level of the knee or below, can be approached through a medial incision. More distally, the peroneal and anterior tibial arteries can be approached laterally at the midtibial level. At the level of the malleolus, both the dorsalis pedis and posterior tibial arteries can be revascularized. After control of donor and recipient vessels, an anatomic tunnel is created, and the bypass conduit (saphenous vein, prosthetic graft) is passed through its length. After administration of 10,000 U of heparin, the distal anastomosis is constructed first, followed by the proximal anastomosis. A completion arteriogram confirms unobstructed flow. Heparin is then partially reversed and meticulous hemostasis obtained, and the wounds are closed.

Although conventional wisdom has held that the anesthetic and operative risks for patients undergoing distal reconstructions are low, there may be little difference in postop morbidity and mortality when compared with patients undergoing a major inflow procedure within the abdomen.

Usual preop diagnosis

Severe PVD; CAD; DM; HTN; obstructive PA disease (These are almost ubiquitou comorbidities.)

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. **See Appendix K, p. K-1.**

SUMMARY OF PROCEDURE	
Position	Supine
Incision	Groin, medial knee, ± distal leg incisions, + incision to harvest venous conduit
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg) after induction of anesthesia
Surgical time	2–5 h
Closing considerations	Optimization of coagulation status
EBL	200–300 mL
Postop care	Possible ICU × 24 h; patients are at risk for myocardial ischemia
Mortality	2–4%
Morbidity	MI: 5–12% Respiratory insufficiency: 5% Infection: 2–5% Amputation: 2–4% CVA: <1%

Pain score	4–6

■ PATIENT POPULATION CHARACTERISTICS	
Age range	>55 yr (mean age ≥ 70 yr)
Male:Female	4:1
Incidence	Claudication fairly common in elderly; however, 75% will remain untreated, but stable, over 2–5 yr, with <5% requiring amputation.
Etiology	Arteriosclerosis (primarily); chronic embolic disease (rare); vasculitis; popliteal artery entrapment; cystic adventitial disease of popliteal artery
Associated conditions	CAD; cerebrovascular disease; DM; HTN; COPD

ANESTHETIC CONSIDERATIONS

(Procedures covered: infrainguinal arterial bypass; arterial embolectomy; lumbar sympathectomy; venous thrombectomy or vein excision)

PREOPERATIVE

Patients presenting for peripheral vascular surgery may suffer from major systemic diseases, including CAD, HTN, and DM. Three types of occlusive vascular diseas have been described. Type 1, isolated to the aortic and iliac bifurcations, is not associated with CAD. Type 2, diffuse pattern involving coronary and cerebral circulations, is associated with a higher incidence of DM and HTN. Type 3, involving small vessels, especially of the lower limbs, is associated with higher postop morbidity and mortality.

Respiratory	Vascular patients frequently have Hx of smoking and COPD. Preop evaluation of pulmonary function helps to determine whether regional anesthesia vs. GA is appropriate while providing baseline values for postop comparison. **Tests:** CXR; consider PFTs; ABG
Cardiovascular	Vascular surgery of the lower extremities is often

	associated with ↑ morbidity and mortality 2° ↑ incidence of CAD and HTN in this patient population. **Tests:** ECG: ✓ for LVH and ischemia; other tests as indicated from H&P
Neurologic	↑ incidence of cerebrovascular disease. Careful neurologic assessment is necessary to document existing deficits.
Endocrine	↑ incidence of DM, which may be associated with peripheral and autonomic neuropathies (silent MI, labile BP) and delayed gastric emptying. Insulin requirements for most diabetic patients can be managed by administering one-half the usual a.m. dose of insulin, once a dextrose-containing iv (e.g., D5LR) has been established. Frequent blood glucose measurements should be made periop.
Renal	There is a higher incidence of renal artery disease and renal insufficiency in this patient population. **Tests:** BUN; Cr; consider creatinine clearance; electrolytes; UA
Hematologic	Many patients presenting for this surgery are taking anticoagulant/anti-Plt medications. Inquire as to bleeding or bruising tendency. **Tests:** Hct; Plt; consider PT, PTT
Laboratory	As indicated from H&P
Premedication	Continue usual medications up to time of surgery. Anxiety can contribute to HTN and tachycardia. Premedicate conservatively (midazolam 0.5–2 mg iv) for geriatric patients. If ischemic pain is present, a narcotic such as fentanyl (25–50 mcg iv) can be given. Avoid im administration in patients on anticoagulant therapy.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Either regional anesthesia or GETA may be used. For **infrainguinal arterial bypass,** there is evidence that regional anesthesia is superior for promoting graft survival. To date, no study has shown a difference in patient mortality between these anesthetic techniques. **Lumbar sympathectomy** may be accomplished with regional or general anesthesia. For **thrombectomy,** GETA with IPPV may reduce the risk of pulmonary emboli.

General anesthesia

Induction	Hemodynamic stability is important; therefore, a slow, gradual induction is carried out. Preoxygenation is followed by smooth iv induction, using small, incremental doses of fentanyl (1–3 mcg/kg); then propofol is given in divided doses until the patient is asleep. Alternatively, if the LV function is poor, etomidate (0.2 mg/kg iv) in combination with fentanyl (100–200 mcg) is appropriate. A full dose of muscle relaxant (vecuronium or rocuronium) is given, and ventilation is controlled. Muscle relaxation is not necessary during the procedure, but muscle relaxant is given to facilitate intubation. The muscle relaxant is chosen to avoid undesirable effects on HR.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance with fentanyl (1–2 mcg/kg/h) and inhaled anesthetic. Continued muscle relaxation is usually unnecessary. These surgical procedures are associated with minimal hemodynamic instability.
Emergence	Tracheal extubation should be based on standard criteria, such as adequacy of ventilation, return of airway reflexes, and reversal of muscle relaxation. Control of BP is accomplished with vasodilators (e.g., NTG 0.1–4.0 mcg/kg/min or SNP 0.25–5.0

	mcg/kg/min). Esmolol can be given in incremental doses of 10 mg or by infusion (50–200 mcg/kg/min). Consider iv acetaminophen if po dose not given. PONV prophylaxis should include ondansetron.

Regional anesthesia

Patients presenting for regional anesthesia must have a normal coagulation profile; also, they cannot be on heparin (including low molecular weight), urokinase, or streptokinase. The important goals in regional anesthesia are to achieve hemodynamic stability while establishing an adequate block. An epidural or spinal catheter (multi orifice) is frequently used to infuse local anesthetic for a gradual onset and to prevent an excessively high block. Hemodynamic stability can be improved by infusing 500–1000 mL of crystalloid before performing the block. The patient may be placed in the lateral decubitus position (with operative side down), which may provide a denser and more prolonged block on that side. Achieving a T8-T10 level is optimal. Overzealous hydration may → CHF in this patient population, when the vasodilation 2° regional sympathectomy dissipates. The American Society of Regional Anesthesia provides guidelines on perioperative management of neuraxial catheters and anticoagulation; generally, heparin dosing should be delayed until 1 h after manipulation of neuraxial space, and coags should be checked and corrected prior to catheter removal. If blood is aspirated after placement of an epidural, we remove the catheter and then replace it at a different interspace. Patients on minidose heparin should have a normal partial thromboplastin time (PTT) before use of regional anesthesia.

Spinal	**One-shot spinal:** 1% hyperbaric tetracaine (6–10 mg) or 0.75% bupivacaine (10–15 mg). Add epinephrine 0.2 mg (generally increases duration of block about 50%) or phenylephrine 5 mg (generally increases duration of block about 100%). Do not add vasoconstrictor if patient is diabetic. Large doses of hyperbaric local anesthetic should be avoided as they may cause postop cauda equina syndrome. **Continuous spinal:** A 20-ga catheter is placed via an 18-ga Tuohy needle. 0.5% hyperbaric tetracaine or 0.75% bupivacaine is titrated to desired anesthetic level (T8-T10). Catheters should be redosed every 60–80 min or as soon as BP trends upward. The risk of spinal headache is very low with

	continuous spinals.	
Epidural	Lidocaine 2% with 1:200,000 epinephrine or 0.5% bupivacaine is used. Titrate to desired anesthetic level (T8-T10).	
Blood and fluid requirements	IV: 14 or 16 ga × 1 BSS @ 3–5 mL/kg/h Warm fluids. Humidify gases. Maintain UO 0.5–1 mL/kg/h.	Minimal-to-moderate blood loss. Use forced-air warmer (e.g., Bair Hugger). See Zero Balance Fluid Management, p. K-2 and p. K-4.
Monitoring	Standard monitors	
	Arterial line ST-segment analysis	An arterial line is most often used in patients with severe cardiopulmonary disease or brittle diabetes. Blood sampling for ABGs, electrolytes, glucose, and Hct is facilitated by the use of an arterial line.
	± CVP/PA catheters ± TEE	CVP and PA catheters are not used routinely because large changes in intravascular volumes are uncommon. Patients with poor LV function, recent MI, or severe valvular disease may need PA and/or TEE monitoring.
Positioning	✓ and pad pressure points ✓ eyes	Diabetic patients may be at risk of skin ischemia due to poor positioning or inadequate padding of limbs.
Complications	HTN	

	Ischemic reperfusion syndrome (postembolectomy) Hypothermia Hemorrhage	Reperfusion of an ischemic limb → ↓ pH, ↑ K^+, and release of myoglobin from injured muscle → ATN.

POSTOPERATIVE

Complications	CHF	Infrainguinal bypass: avoid overhydration; when epidural sympathectomy fades, these patients are at increased risk for CHF. Maintain normovolemia so that peripheral vasoconstriction (which may limit outflow to the graft) does not occur.
	Hypothermia Graft occlusion	Hypothermia causes vasoconstriction and also may limit outflow to the graft.
Multimodal pain management	Epidural/spinal opiates PCA	Epidural (or spinal) catheter may be used for postop analgesia. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Tests	CXR if central line placed Hct

Suggested Readings

1. Clair D, Shah S, Weber J: Current state of diagnosis and management of critical limb ischemia. Curr Cardiol Rep 2012; 14:160–70.
2. Conte MA: Critical appraisal of surgical revascularization for critical limb ischemia. J Vasc Surg 2013; 57:8s–13s.
3. Hull DJ, Kizilgul M, Aung W, et al: Frequency of coronary artery disease in patients undergoing peripheral artery disease surgery. Am J Cardiol 2012; 110:736–40.
4. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery.J Cardiothorac Vasc Anesth 2018; 32:2760–70. pii: S1053-0770(18)30049-1.
5. Singh N, Sidawy AN, Dezee K, et al: The effects of the type of anesthesia in outcomes of lower extremity infrainguinal bypass. J Vasc Surg 2006; 44:964–8.

ARTERIAL EMBOLECTOMY

SURGICAL CONSIDERATIONS

Description

The etiology of acute arterial insufficiency is often the result of thromboembolism, which is usually of cardiac origin. Patients at high risk for embolization are those with MI, mitral stenosis, or atrial fibrillation (AF), all of which increase the risk of intracardiac thrombus formation. Noncardiac causes include thoracic and abdominal aortic pathology, such as aneurysms, severe atherosclerotic disease, and paradoxical embolus via a patent foramen ovale (PFO). The emboli usually become lodged at tapering regions and branch sites and most commonly involve the extremities; the cerebrovascular and mesenteric vasculature also may be involved. In the lower extremities, emboli frequently lodge at the iliac, femoral, or popliteal arteries; the common femoral artery is involved in up to 50% of all embolic events. Emboli to the upper extremities typically affect the brachial artery. Because the collateral circulation may not be well developed in patients without underlying PVD, muscle necrosis can appear within 4–6 h after onset, although this time frame is highly variable. Patients are heparinized at time of diagnosis. Therapy directed at the specific etiology is critical to achieve a successful outcome. Revascularization after 8–12 h of ischemia usually is less effective. If the underlying source of the emboli is not adequately treated, recurrence is likely and is associated with poor prognosis. The surgical approach to femoral embolectomy is a groin incision and isolation of the common, superficial, and deep femoral arteries. Both the superficial and deep femoral systems are explored, and small embolectomy catheters are passed into the distal lower extremity. In addition to passage of embolectomy catheters, angioscopy may be helpful. Residual thrombus on intraop arteriogram requires distal surgical exposure and exploration.

Usual preop diagnosis

Peripheral artery embolism

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. **See Appendix K, p. K-1.**

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Limited groin, medial knee, possible distal leg incisions
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg)
Surgical time	Variable, 1–4 h
Closing considerations	Ischemia-reperfusion syndrome with acidosis and hyperkalemia, renal tubular necrosis from myoglobinuria
EBL	100–300 mL
Postop care	May need ICU for cardiac monitoring; monitor metabolic disturbances, neurovascular status, and for compartment syndrome
Mortality	10–40% (~80% cardiac); older age; acute onset →↑ risk
Morbidity	Arterial reocclusion or thrombosis: 20% Wound hematoma: 10–20% Recurrent emboli: 6–45% (less with anticoagulation) Amputation: 5–10% Metabolic complications: Frequent Compartment syndrome: Occasional

Pain score	4–6

PATIENT POPULATION CHARACTERISTICS	
Age range	30–90 yr
Male:Female	3:2
Incidence	50/100,000 admissions
Etiology	Cardiac origin (AF accounts for 50–75% of patients; less frequent, LV thrombus from MI) or noncardiac origin (mural thrombus from thoracic or abdominal aortic pathology)
Associated conditions	Rheumatic heart disease and mitral stenosis, mitral or aortic valve prosthesis, CAD and MI, CHF, endocarditis, peripheral vascular and cerebrovascular disease

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Infrainguinal Arterial Bypass.

Suggested Readings

1. Bandyk DF, Johnson BL, Kirkpatrick AF, et al: Surgical sympathectomy for reflex sympathetic dystrophic syndromes. J Vasc Surg 2002; 35:269–77.
2. Davis FM, Albright J, Gallagher KA, et al: Early outcomes following endovascular open surgical, and hybrid revascularization for lower extremity acute limb ischemia. Ann Vasc Surg 2018; 51:106–12.
3. Jaffery Z, Thornton SN, White CJ: Acute limb ischemia.Am J Med Sci 2011; 342:226–34.
4. Setacci C, de Donato G, Setacci F, et al: Hybrid procedures for acute limb ischemia J Cardiovasc Surg 2012; 53:133–43.

LUMBAR SYMPATHECTOMY

SURGICAL CONSIDERATIONS

Description

The role of open surgical or endoscopic **lumbar sympathectomy** is not well defined.

The procedure is used selectively in patients with causalgia, inoperable lower limb ischemia with rest pain or toe gangrene, symptomatic vasospastic disorders (e.g., Raynaud phenomenon, frostbite), and sometimes as an adjunct to distal revascularization procedures. Causalgia responds well to lumbar sympathectomy, especially if performed early in the clinical course. There may be some benefit of the procedure in 50–60% of patients with rest pain or ischemic ulceration. Sympathectomy may increase collateral blood flow and local skin blood flow. Scleroderma with impending "minor" amputation may benefit from sympathectomy with improved wound healing. Sympathetic denervation involves the division of preganglionic fibers along their segmental origins and resection of corresponding relay ganglia. For most clinical indications, L2 and L3 **ganglionectomy** sufficiently sympathectomizes the lower extremity (Fig. 6.3-11).

The **anterolateral retroperitoneal approach** is the most commonly performed open surgical technique because of the adequate exposure and a relatively well-tolerated incision. For this approach, an oblique incision is made through the abdominal musculature, extending from the lateral border of the rectus abdominis to the anterior axillary line. Cephalad and caudad blunt dissection is performed between the transversalis fascia and peritoneum. The dissection is continued in a retroperitoneal fashion. The psoas muscle is identified, with care being taken to leave the ureter and gonadal vessels attached to overlying peritoneum. The sympathetic chain is identified between the psoas muscle and the vertebra (medial to psoas and overlying the transverse process of lumbar vertebra). On the left side, the sympathetic chain is lateral to the abdominal aorta; on the right, the sympathetic chain is beneath the inferior vena cava (IVC). The sympathetic chain is dissected free from surrounding tissue, clipped proximally and distally, and resected. Hemostasis is achieved, and the abdominal wall is closed in layers.

A **posterior approach** is used less often because of significant postop paraspinous muscle spasms. In this approach, a transverse lumbar incision is made, and the paraspinous muscles are partially divided and retracted to expose the vertebra. The sympathetic chain is identified and resected as described. The **anterior/transperitoneal approach** is performed in conjunction with an abdominal aortic or intraperitoneal procedure. Dissection is carried to the psoas muscle; the retroperitoneum is entered, and the sympathetic chain is isolated in the groove between the psoas and the vertebra. The sympathetic chain is clipped proximally and distally and resected.

Usual preop diagnosis

Causalgia; inoperable arterial occlusive disease with limb-threatening ischemia causing rest pain, ulceration, or superficial digital gangrene; symptomatic vasospastic disorders (e.g., Raynaud phenomenon or frostbite)

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. **See Appendix K, p. K-1.**

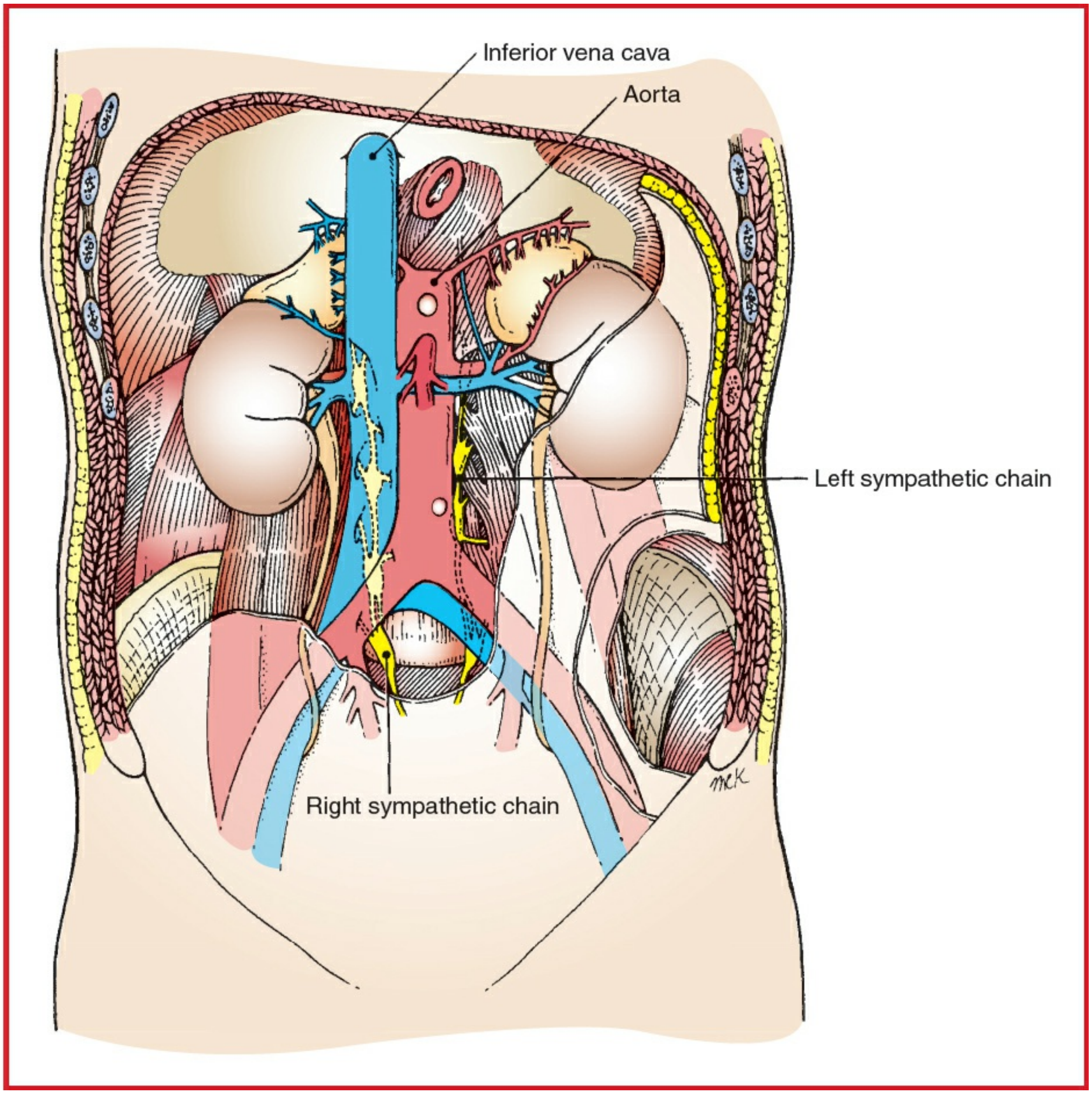

Figure 6.3-11 Surgical anatomy for lumbar sympathectomy. (Reproduced with permission from Scott-Conner CEH, Dawson DLOperative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

SUMMARY OF PROCEDURES

	Anterolateral	Posterior	Anterior

	(Flowthow)	(Royle)	(Adson)
Position	Supine (flank slightly raised); widen distance between costal margin and iliac crest	Prone	Supine
Incision	Oblique (lateral edge of rectus to ribs at anterior axillary line)	Posterior transverse over midlumbar region	Transverse or midline abdominal
Special instrumentation	Self-retaining retractor	⇐	⇐
Unique considerations	May use frozen section to confirm specimen	⇐	⇐
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg)	⇐	⇐
Surgical time	2–3 h	3 h	4 h
Closing considerations	Flex table to facilitate abdominal wall closure; occasionally drain is placed	⇐	⇐
EBL	50–100 mL (unless complicated)	50–100 mL	⇐
Postop care	PACU → ward (may require cardiac and hemodynamic monitoring in high-risk patients, or if combined with distal revascularization)	⇐	⇐
Mortality	Minimal	⇐	⇐

Morbidity	Postsympathectomy neuralgia: 50% Sexual derangement—retrograde ejaculation (usually bilateral L1 sympathectomy): 25–50% Wound hematoma: 10% Wound infection: 1–3%	~50% ~25–50% ~10% ~1–3% Paraspinous muscle spasms	⇐ ⇐ ⇐ ⇐
Pain score	5	5	5 (related to primary procedure)

PATIENT POPULATION CHARACTERISTICS

Age range	38–91 yr; younger patients with vasospastic disease; older patients with PVD
Male:Female	2:1
Incidence	Unknown. ~30 cases/yr at Stanford University Medical Center, mainly for inoperable lower extremity ischemia or as an adjunct to revascularization procedures
Etiology	PVD (rest pain and tissue loss); vasospastic disorders; causalgia
Associated conditions	PVD (rare); vasospastic disorders (rare)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Infrainguinal Arterial Bypass.

Suggested Readings

1. Nemes R, Surlin V, Chiutu L, et al: Retroperitoneoscopic lumbar sympathectomy:

prospective study upon a series of 50 consecutive patients. Surg Endosc 2011; 25(9):3066–70.
2. Sen I, Agarwal S, Tharyan P, et al: Lumbar sympathectomy versus prostanoids for critical limb ischaemia due to non-reconstructable peripheral arterial disease. Cochrane Database Syst Rev 2018; 4:CD009366.
3. Setaccia C, de Donatoa G, Teraa M, et al: Chapter IV: treatment of critical limb ischemia. Eur J Vasc Endovasc Surg 2011; 42(S2):S43–59.

VENOUS SURGERY—THROMBECTOMY OR VEIN EXCISION

SURGICAL CONSIDERATIONS

Description

Standard therapy for acute deep venous thrombosis (DVT) consists of anticoagulation, bed rest, and elevation of the extremity. Endovascular or catheter-directed thrombolysis and thrombectomy systems have been shown to extract large venous thrombus burden. Open surgical thrombectomy for acute iliofemoral DVT remains controversial. **Venous thrombectomy** is recommended for patients with threatened limb loss or venous gangrene caused by massive DVT associated with high compartment pressures and arterial insufficiency (phlegmasia cerulea dolens). Surgical venous thrombectomy requires exposure of the femoral vein via a groin cutdown. The common femoral vein is isolated (located medial or posteromedial to the femoral artery) and controlled proximally and distally. The patient is given iv heparin at this stage, if not already heparinized. A transverse venotomy is followed by extraction of the thrombus, using forceps and Fogarty embolectomy catheters (Fig. 6.3-12). Distal thrombi are expressed through the same incision with the aid of an Esmarch bandage placed on the extremity. After complete removal of the thrombus, the venotomy is closed with nonabsorbable sutures, and flow through the femoral vein is reestablished. The femoral incision is closed in layers. A plastic/Silastic drain may or may not be used. The best results of thrombectomy are obtained in young patients with the first episode of proximal (iliofemoral) thrombosis.

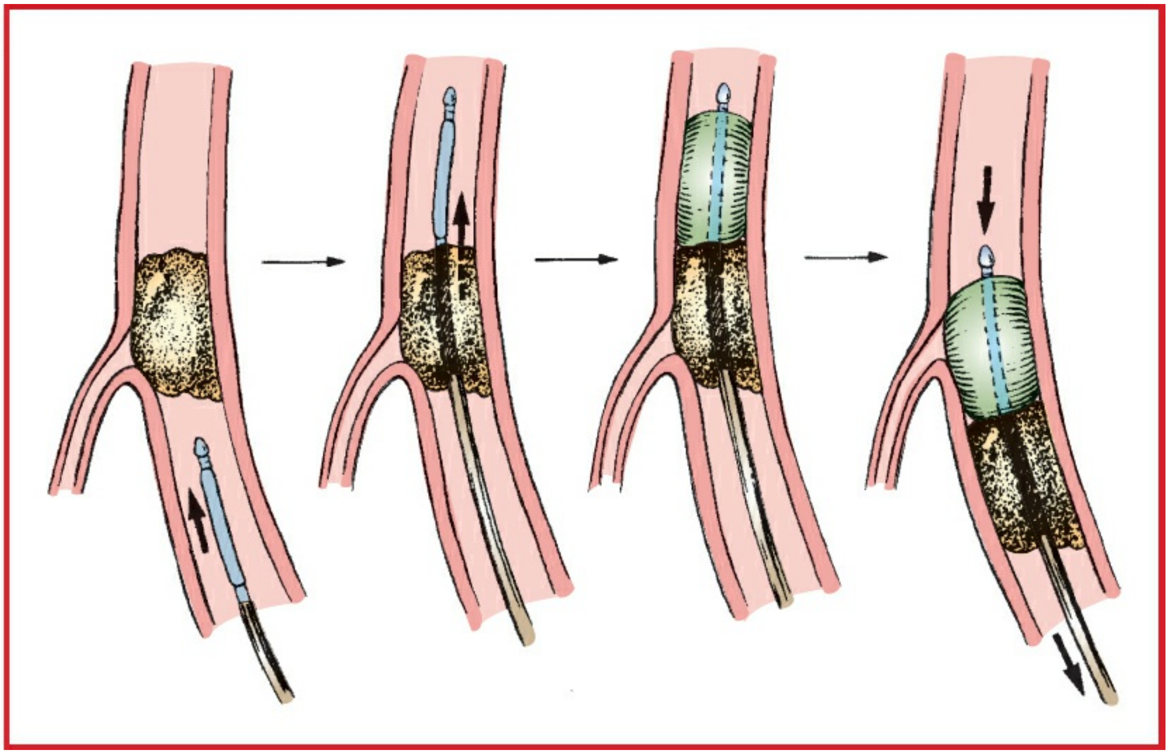

Figure 6.3-12 Fogarty catheter embolectomy with catheter insertion in the distal vessel. (Reproduced with permission from Baker RJ, Fischer JEMastery of Surgery, Vol. 2, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Nonsuppurative thrombophlebitis of the superficial veins may develop due to local trauma, prolonged inactivity, fungal infection, or the use of oral contraceptives. Suppurative thrombophlebitis may occur as a complication of iv line placement or iv drug abuse. Underlying varicose veins may predispose to thrombophlebitis. Migratory superficial thrombophlebitis may be associated with chronic ischemia of the extremities in Buerger disease, or it may develop in patients with a malignancy. Conservative management with hot compresses, nonsteroidal anti-inflammatory medications, and elevation of the extremity is effective in most cases. Rarely, **excision** of the acutely thrombosed greater saphenous vein is indicated to prevent progression of thrombosis to the saphenofemoral junction and into the deep venous system. Vein excision is simply approached by a longitudinal incision directly over the affected vein. The phlebitic vein is dissected from the surrounding tissue, ligated proximally and distally, and removed. The surrounding fibrotic tissue is debrided gently. The wound is irrigated and packed open with moist gauze. Suppurative phlebitis is treated with iv antibiotics and complete excision of the involved vein segment through multiple small incisions.

Usual preop diagnosis

Lower limb venous thrombosis threatening viability; femoral thrombosis <10 d; iliac thrombosis <3 wk; floating thrombi at hip level; acute deep or superficial venous thrombosis; suppurative thrombophlebitis

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. See** **Appendix K, p. K-1****.**

SUMMARY OF PROCEDURES

	Thrombectomy	Vein Excision
Position	Supine	⇐
Incision	Ipsilateral longitudinal or oblique groin incision	Multiple small incisions along the course of vein to be excised
Special instrumentation	Fogarty embolectomy catheters; RBC salvage system	None
Unique considerations	IPPV during thrombectomy may decrease chance of PE 2° ↓ venous return → more complete extraction of the thrombus.	None
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg). If the patient has septic thrombus, antibiotic is dependent on blood culture.	Dependent on culture of aspirate. Usually, a septic patient is on broad-spectrum antibiotics (guided by culture).
Surgical time	2–3 h	1–3 h
Closing considerations	None	Wound packed open for drainage
EBL	50–250 mL	⇐
Postop care	PACU → ward; ICU in high-risk patients; heparin administered before and	PACU → ward; support stockings

	after procedure, followed by Coumadin for 1–6 mo	
Mortality	Minimal (depending on underlying illness)	Minimal
Morbidity	Postthrombotic syndrome: <10–44% Venous stasis Nonpitting edema Brawny induration Aching pain	Minimal
Pain score	3	2

PATIENT POPULATION CHARACTERISTICS	
Age range	Young adult—elderly
Male:Female	1:2
Incidence	6–7 million in the United States have adverse effects from chronic venous stasis; 500,000 have complications of leg ulceration.
Etiology	Chronic primary varicose veins; defective venous valves; impaired pumping action of muscles in the leg; previous iliofemoral thrombophlebitis; obstruction of venous return
Associated conditions	Multifactorial: varicose veins; underlying malignancy; altered coagulation status (hypercoagulability, acquired or congenital); Hx of DVT/thrombophlebitis

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Infrainguinal Arterial Bypass.

Suggested Readings

1. Comerota AJ: The current role of operative venous thrombectomy in deep vein thrombosis. Semin Vasc Surg 2012; 25:2–12.
2. Davenport DL, Xenos ES: Early outcomes and risk factors in venous thrombectomy: an analysis of the American College of Surgeons NSQIP dataset.Vasc Endovascular Surg 2011; 45:325–8.

3. Kim HS, Patra A, Paxton BE, et al: Catheter-directed thrombolysis with percutaneous rheolytic thrombectomy versus thrombolysis along an upper and lower extremity deep vein thrombosis. Cardiovasc Intervent Radiol 2006; 29:1003–7.
4. Wissgott C, Kamusella P, Andresen R: Percutaneous mechanical thrombectomy: advantages and limitations. J Cardiovasc Surg 2011; 52(4):477–84.

SURGERY FOR PORTAL HYPERTENSION

SURGICAL CONSIDERATIONS

Description

Alcoholic liver disease is the major cause of portal HTN, and end-stage liver disease (ESLD) with cirrhosis is the 10th leading cause of death in the United States (exceeding 23,000/yr). Of patients with portal HTN, 15–20% have variceal hemorrhage during the 1st yr of diagnosis (additional 5–10% incidence of bleeding per yr); and the initial episode of variceal hemorrhage is associated with 50% mortality. Portal HTN (>15 mm Hg) develops when splanchnic venous flow to the right heart becomes impeded. Medical and surgical therapeutic interventions are directed not at portal HTN per se but at its complications—notably bleeding esophageal varices. Intractable ascites and hypersplenism are less common indications for operative therapy. Presinusoidal portal HTN, unlike sinusoidal or postsinusoidal obstruction, is not associated with severe hepatocellular disease; thus, the prognosis for patients with presinusoidal block is better than for those with sinusoidal or postsinusoidal disease. Surgical approaches can be divided into shunt and nonshunt procedures. **Shunt procedures** can be classified as either **total** (decompression of the portal venous system) or **selective** (decompression of only the varix-bearing area). It should be noted, however, that surgery for portal HTN has been replaced largely by the TIPS procedure.

Shunt procedures

There are two general types of total shunt (**portosystemic shunt**) procedures. The **end-to-side portacaval shunt** (Fig. 6.3-13A) is technically simpler and may be more appropriate in emergency situations. It is associated with immediate control of hemorrhage in the majority of cases. The portacaval shunt, however, does eliminate portal perfusion of the liver and does not decompress the hepatic sinusoids. Alternatively, the **functional side-to-side shunt** (Fig. 6.3-13B) allows decompression of hepatic sinusoids and may preserve some degree of portal perfusion of the liver. Also, it is more effective in controlling ascites. Variations of the side-to-side shunt include **portacaval, splenorenal, mesocaval** (**Clatworthy**), and **portarenal** (rarely used).

For the **end-to-side** and **side-to-side portacaval shunts,** the approach is via an extended right subcostal incision. **Cholecystectomy** usually is not performed because of the likelihood of profuse bleeding from the liver bed. The hepatoduodenal ligament is identified, and the portal vein is exposed from the hilum of the liver to the pancreas. The gastroduodenal and right gastric branches may be divided to provide additional exposure of the portal vein. The IVC is exposed by incising the peritoneum just beneath the hepatic triad. Proximal and distal control of the portal vein is achieved, and a side-biting clamp is placed on the IVC. To perform an end-to-side portacaval shunt, the portal vein is divided and oversewn proximally; the end-to-side anastomosis is performed from the portal vein to the IVC. The alternative is to perform a side-to-side anastomosis of the portal vein to the IVC without division of the portal vein.

The **proximal splenorenal shunt** is approached through a left thoracoabdominal or transabdominal incision. The spleen is isolated and removed, and the distal splenic vein is mobilized from the distal pancreatic bed. The left renal vein is exposed and controlled. The distal splenic vein is then anastomosed in an end-to-side fashion to the midrenal vein. The **mesocaval shunt** (Fig. 6.3-13C) is indicated in cases of ascites, periportal fibrosis, portal vein thrombosis, and Budd-Chiari syndrome. The mesocaval shunt is approached through a vertical midline incision. The colon is retracted cephalad, and the superior mesenteric vein (SMV) is identified at the root of the mesentery. A length of the SMV is isolated and encircled. A **Kocher maneuver** (mobilization of the duodenum) is performed, and the IVC is exposed anteriorly and laterally. A side-biting clamp partially occludes the IVC, and a 14- to 20-mm Dacron graft is anastomosed in an end-to-side fashion to the IVC. The graft is clamped and the side-biting clamp removed from the IVC, thereby restoring flow via the IVC. The SMV is clamped, and an end-to-side anastomosis is created from the graft to the SMV. Flow is, thus, reestablished from the SMV through the graft to the IVC.

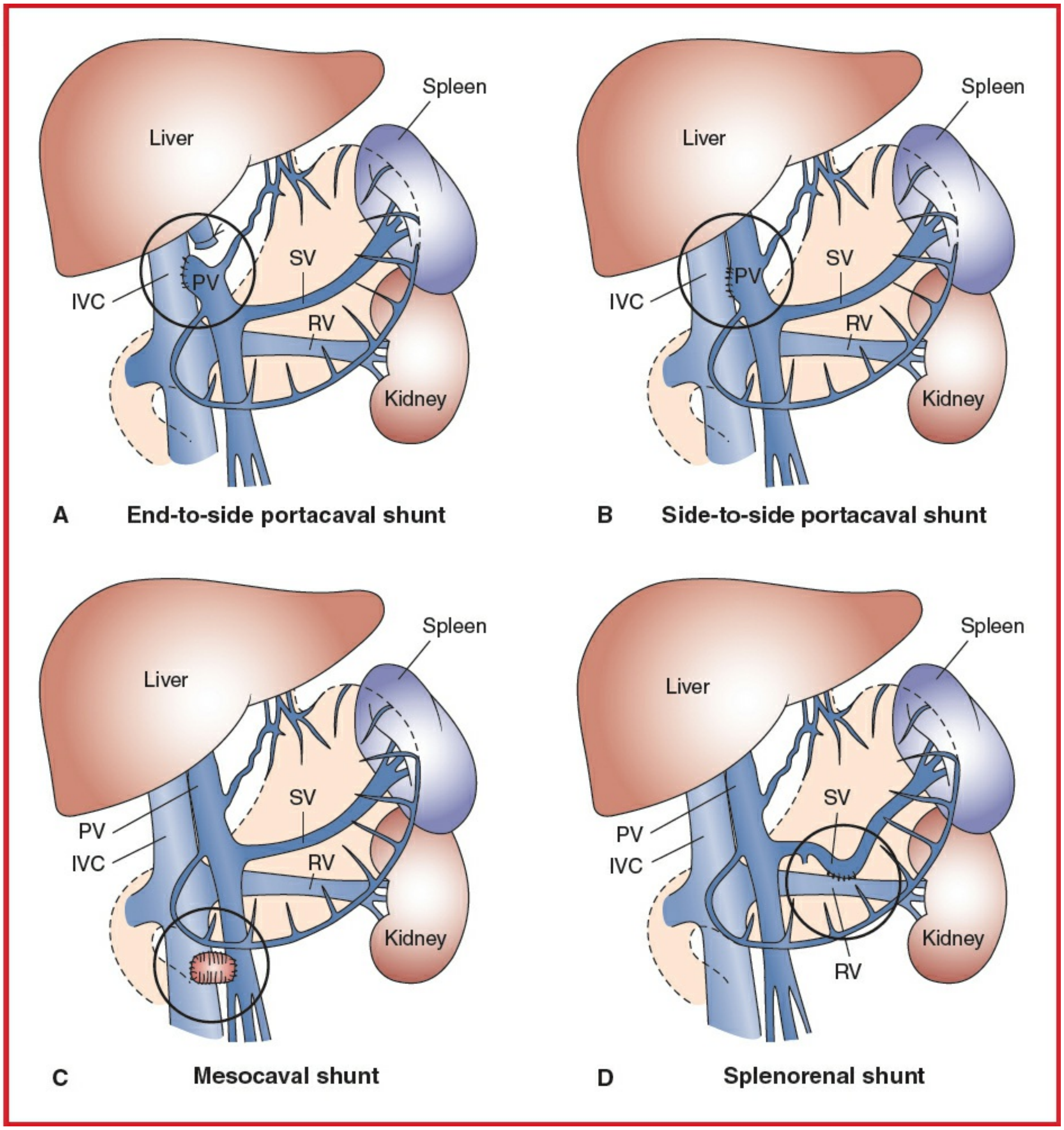

Figure 6.3-13 Types of shunts. **A:** End-to-side portacaval. **B:** Side-to-side portacaval. **C:** Mesocaval shunt. **D:** Distal splenorenal. (Reproduced with permission from Hardy JD:Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)

Selective shunts are designed to decompress esophageal varices, while some portal perfusion of the liver is maintained. The hallmark example of this approach is the **distal splenorenal shunt** (**Warren**; Fig. 6.3-13D), which is seldom used in emergency situations. The principal feature of this shunt is disconnection of the splenic and superior mesenteric venous drainage systems. The distal splenorenal shunt is approached through a left chevron or extended left subcostal incision. The lesser sac is entered after division of the gastroepiploic vessels and mobilization of the splenic flexure of the colon. The stomach is retracted cephalad, and the peritoneum overlying the inferior aspect of the pancreas is incised. The splenic vein is identified and controlled proximally and distally. The inferior mesenteric vein is divided. The splenic vein is divided proximally, and the proximal stump is oversewn. Then the splenic vein is mobilized from the pancreatic bed. The left renal vein is identified, and 5–7 cm of the vein is isolated. The splenic vein is anastomosed in an end-to-side fashion to the renal vein. The coronary vein is ligated close to its origin. The distal splenorenal shunt decompresses the stomach, distal esophagus, and spleen and controls variceal hemorrhage in 85% of patients.

Variant procedure or approaches

Total shunt (e.g., portacaval, proximal splenorenal, mesocaval); **selective shunt** (e.g., **Warren**); **nonshunt procedures** (e.g., **Sugiura, Hassab,** and **esophageal transection with stapling**)

Usual preop diagnosis

Bleeding esophageal varices (as a result of portal HTN); ascites; hypersplenism

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options.** **See Appendix K, p. K-1.**

SUMMARY OF PROCEDURES (TOTAL AND SELECTIVE SHUNTS)

	Portacaval	Proximal Splenorenal	Mesocaval (Clatworthy)	Distal Splenorenal (Warren)
Position	Supine	Supine ± left flank elevated	Supine	Supine, left side elevated slightly
Incision	Extended right subcostal—may be lengthened or converted to left thoracoabdominal	Left thoracoabdominal or left subcostal or vertical midline	Vertical midline abdominal	Left chevron or left subcostal with midline extension
Special instrumentation	Self-retaining retractor	⇐	⇐	⇐
Unique considerations	May need FFP; consider cell saver	⇐	⇐	⇐
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg)	⇐	⇐	⇐
Surgical time	4–6 h	⇐	⇐	⇐
Closing considerations	None	⇐	⇐	⇐
EBL	1000–2000 mL	⇐	⇐	⇐
Postop care	Patient → ICU; careful fluid management; consider Na^+ restriction; may require PA catheter	⇐	⇐	⇐

Mortality	Emergency: 38% Elective: 3% Child's A: 0–15% Child's B: 1–43% Child's C: 6–58%	5–10%	⇐	1–16%
Morbidity	Late liver failure: 50% Encephalopathy: 32% Child's A: 8% Child's B: 20% Child's C: 30% Early rebleeding: 0–19%	~50% ~32% ~0–19% 20%	⇐ ⇐ ⇐ 9–30%	— 5–47% (overall) 5% at 2 yr 12% at 3–6 yr 27% at 10 yr — —
	Shunt thrombosis: 1–10%		Duodenal obstruction; erosion through bowel wall	Loss of selectivity: 60% (@ 2 yr) Recurrent variceal hemorrhage (shunt occlusion): 3–19% Portal vein thrombosis: 4–10%
Pain score	6	6	6	6

Note: In addition to the shunt procedure itself, other factors that determine postop morbidity and mortality include severity of hepatocellular disease, degree of hepatic reserve, and urgency of the procedure. Although the specific numbers may vary, several comparative series have shown no difference in operative mortality rate or long-term survival rate among the various open shunt procedures.

Nonshunt procedures

Nonshunt procedures are designed to devascularize the esophagogastric region, thus eliminating acute variceal hemorrhage. Procedures of this type include **portoazygos disconnection**, **splenectomy**, **coronary vein ligation**, and **transesophageal** or **transgastric varix ligation**.

The **Sugiura operation,** approached via abdominal and thoracic incisions, includes esophageal transection and devascularization, splenectomy, and pyloromyotomy. This procedure is usually performed in two operative stages, sometimes with a delayed second stage. The **Hassab procedure** involves devascularization of the upper half of the stomach and splenectomy, thus effectively disconnecting the esophageal varices. (This operation has been reserved in some centers for the failures of sclerotherapy.) Finally, **esophageal transection,** using a stapling device, disconnects the varices in the lower esophagus. This is accomplished by placing a row of staples at the esophagus just above the esophagogastric junction. Because portal perfusion of the liver is maintained after the nonshunt procedures, hepatic function is preserved.

■ SUMMARY OF NONSHUNT PROCEDURES

	Sugiura (Two Stages)	Hassab	Esophageal Transection
Position	Supine, left side elevated	Supine	⇐
Incision	Abdominal stage: midline; thoracic stage: left thoracotomy	Vertical midline	⇐
Special instrumentation	Self-retaining retractor	⇐	⇐
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg)	⇐	⇐
Surgical time	4–6 h for both stages	4 h	⇐
EBL	1000–2000 mL	⇐	⇐
Postop care	ICU; careful hemodynamic monitoring; PA catheterization	⇐	⇐
Mortality	Overall: 5–14% Emergent: 14% (≤60%) Elective: 3%	12–38% 10%	10–83%
Morbidity	Recurrent hemorrhage: 2–50% Esophagopleural leak: 6% Encephalopathy: 3% Ascites: 2% Wound infection: 2%	7% — 1–25% 4% —	0–50% 3% 33% 11% 11%

Pain score	6	5	5

■ PATIENT POPULATION CHARACTERISTICS FOR SHUNT A NONSHUNT PROCEDURES	
Age range	11–72 yr (mean age = 48 yr); depending on etiology: pediatric (e.g., congenital hepatic fibrosis) to adult (e.g., alcoholic liver disease)
Male:Female	2–8:1
Incidence	Rare
Etiology	Alcoholic liver disease (alcoholic hepatitis, chronic alcoholism, cirrhosis); postnecrotic cirrhosis; portal vein thrombosis; splenic vein occlusion; hematologic diseases; hepatic vein occlusion; schistosomiasis; congenital hepatic fibrosis; sarcoidosis; sinusoidal occlusion (vitamin A toxicity, Gaucher disease); veno-occlusive disease
Associated conditions	Alcohol dependency; cirrhosis/liver failure; poor nutritional status; coagulopathy; encephalopathy

ANESTHETIC CONSIDERATIONS

(Procedures covered: shunt and nonshunt procedures for portal HTN surgery)

PREOPERATIVE

Respiratory	Hypoxemia may be present 2° ascites, V/Q mismatch, ↑ R→L pulmonary shunting, atelectasis, pulmonary infections, and ↓ pulmonary diffusing capacity. Note whether patient most comfortable sitting, semirecumbent, or supine, and maintain that position prior to intubation and periextubation to optimize respiratory mechanics. **Tests:** Consider ABG; PFTs, as indicated; CXR
Cardiovascular	Patients presenting with portal HTN often have a

	hyperdynamic circulatory state with ↑ plasma volume, ↑ CO, and ↓ SVR, with a decreased ability to ↑ SVR or ↑ HR in response to stimuli. Ventricular performance may be abnormal (CHF), especially in patients with alcoholic liver disease. Ascites →↑ intrathoracic pressure, ↓ FRC, ↓ venous return, and ↓ CO. Older patients in this population usually have CAD. **Tests:** ECG. If LV function in question, ECHO or angiography
Hepatic	The physical manifestations of hepatic disease include palmar erythema, caput medusae, spider angiomas, and gynecomastia. Albumin and other products of liver synthesis (e.g., coagulation factors) may be decreased. Encephalopathy may be present 2° impaired ammonia metabolism. **Tests:** Bilirubin; albumin; PT; SGOT; SGPT; ammonia; alkaline phosphatase
Gastrointestinal	Portal HTN eventually → esophageal and gastric varices. Patients may present emergently with profuse GI bleeding. Ascites occurs in ~80% of patients with portal HTN, and splenomegaly is invariably present. As a result of elevated intra-abdominal pressure from ascites and slow gastric emptying, a rapid-sequence induction with full-stomach precautions will be necessary.
Renal	Portal HTN →↓ GFR and ↓ renal blood flow → renal failure (hepatorenal syndrome). **Tests:** Consider UA; creatinine clearance as indicated from H&P
Hematologic	These patients are often anemic as a result of poor nutrition, malabsorption, and intestinal tract blood loss. Hypersplenism may be present (Plt count < 50,000 and WBC < 2000). Synthesis of all coagulation factors is decreased except factor VIII and fibrinogen. A low-grade DIC may be present. T&C for 8–10 U PRBC. **Tests:** CBC; Plt count; PT; PTT; consider DIC screen

Pharmacologic	The liver is the major site of drug biotransformation; however, the effects of hepatic dysfunction on drug elimination and disposition are inconsistent.
Laboratory	These patients may have significant electrolyte disturbances (e.g., ↓↓ Na^+, ↓ K^+↓ albumin). **Tests:** Electrolytes and others as indicated from H&P
Premedication	If premedication is appropriate, small doses of anxiolytic, such as midazolam (0.5–1 mg iv), are preferable. Avoid im medications in patients with possible coagulopathy. Full-stomach precautions are necessary. Metoclopramide (10 mg iv) and famotidine (20 mg iv) may be given 60 min before surgery.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: Preservation of intravascular volume and myocardial stability can be a challenge in these patients.

Induction	In stable patients rapid-sequence induction with propofol (1–2 mg/kg) and succinylcholine (1–2 mg/kg) should be used. Replace blood loss and ensure normovolemia before induction (if possible). Etomidate (0.2 mg/kg) or ketamine (1 mg/kg) may be preferable for induction in hemodynamically unstable patients.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Combination of narcotics (fentanyl) and inhalation agents (isoflurane/sevoflurane) titrated to

	hemodynamic condition. Midazolam (0.1–0.2 mg/kg) may be given in conjunction with the narcotic to ensure amnesia during times of hemodynamic instability when volatiles cannot be tolerated but beware hepatic elimination. N_2O is avoided to prevent bowel distention. Muscle relaxation is needed, titrate using a nerve stimulator. **NB:** After drainage of ascitic fluid, there may be a precipitous drop in BP requiring rapid volume replacement ± vasopressor.	
Emergence	Generally deferred to ICU due to large fluid shifts and transfusion requirements. Patients who have undergone uneventful and nonemergent surgery may be candidates for extubation. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Anticipate large blood loss. IV: 14 ga × 2 or 7 Fr × 2; BSS Rapid infuser RBC salvage device 8–10 U PRBC Warm all fluids. Humidify all gases.	FFP, Plt, and cryoprecipitate should be available to treat coagulopathy. See Zero Balance Fluid Management, Appendix K, p. K-2 and p. K-4.
	Warming blanket	e. g., Bair-Hugger warmer
Monitoring	Standard monitors	
	Arterial line CVP or PA catheter TEE	Arterial and central pressure monitoring are essential. A PA catheter is useful in this setting because most patients are cirrhotic and may have excessive blood loss and large fluid shifts. TEE is reserved for patients with severe cardiopulmonary disease

		or instability intraop due to risk of variceal bleeding.
	UO	UO is measured and is helpful as a monitor of renal perfusion. Mannitol (0.25–1 g/kg iv) may be needed to maintain UO.
	Temperature	In these procedures, prevention of hypothermia is important. In patients with large varices, avoid esophageal placement of T probes or stethoscopes.
	ABGs	Serial ABGs to determine adequacy of ventilation and normal acid-base status should be done.
	Hct, coags, Ca^{++}	Hct, coagulation, and Ca^{++} should be measured following replacement of large blood volumes.
	Electrolytes	Electrolytes and glucose also should be monitored.
	Blood glucose	Glucose metabolism in liver disease may be impaired → ↓ glucose.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Coagulopathy Hemorrhage Hypothermia	

POSTOPERATIVE

Complications	Coagulopathy Hypothermia Encephalopathy Renal failure	
Multimodal pain management	PCA Parenteral opiates	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	CXR: line placement Hct Electrolytes Glucose	DIC screen, if continued bleeding

Suggested Readings

1. Costa G, Cruz RJ Jr, Abu-Elmagd KM: Surgical shunt versus TIPS for treatment of variceal hemorrhage in the current era of liver and multivisceral transplantation. Surg Clin N Am 2010; 90:891–905.
2. Henderson JM: The distal splenoral shunt. Surg Clin North Am 1990; 70(2):405–23.
3. Henderson JM: Surgery versus transjugular intrahepatic portal systemic shunt in the treatment of severe variceal bleeding. Clin Liver Dis 2006; 10:599–612.
4. Jenkins RL, Gedaly R, Pomposelli JJ, et al: Distal splenorenal shunt: role, indications, and utility in the era of liver transplantation. Arch Surg 1999; 134:416–20.
5. Liu Y, Li Y, Ma J, et al: A modified Hassab's operation for portal hypertension: experience with 562 cases. J Surg Res 2013; 185:e1–e6.
6. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery.J Cardiothorac Vasc Anesth 2018; 32:2760–70. pii: S1053-0770(18)30049-1.
7. Voros D, Polydorou A, Polymeneas G, et al: Long-term results with the modified Sugiura procedure or the management of variceal bleeding: standing the test of time in the treatment of bleeding esophageal varices. World J Surg 2012; 36:659–66.

ARTERIOVENOUS ACCESS FOR HEMODIALYSIS

SURGICAL CONSIDERATIONS

Description

Peripheral subcutaneous arteriovenous (AV) fistula, or **prosthetic graft,** is the current procedure of choice for patients requiring permanent hemodialysis access. The blood flow in the autogenous AV fistula increases with time, and the resulting vein wall thickening prevents venous tears and infiltration during dialysis.

The standard AV fistula is usually constructed by anastomosing the cephalic vein to the radial artery at the wrist level (**Brescia-Cimino fistula**; Fig. 6.3-14). Other locations include the "snuff box," or antebrachium. Vascular access using vascular substitutes or prosthetic grafts is performed when there is a lack of suitable veins in patients who have had failed-access procedures, peripheral vein sclerosis, or severe arterial disease involving the upper extremity. Forearm grafts are constructed as a direct communication between the radial or ulnar artery and the antecubital or brachial vein or as a "loop" between the brachial artery and these veins (Fig. 6.3-15). Similarly, an access can be constructed in the upper arm as a communication between the brachial artery above the elbow and the basilic or axillary vein in a straight fashion. The polytetrafluoroethylene (Teflon) graft has become the mainstay for hemodialysis access in patients who are not candidates for Brescia-Cimino fistula placement. These grafts are associated with a primary patency rate of 50–60% at 2–3 yr.

Variant procedure or approaches

Loop or straight graft, using vascular substitute in the forearm; **upper arm straight graft**

Usual preop diagnosis

End-stage renal failure requiring graft hemodialysis

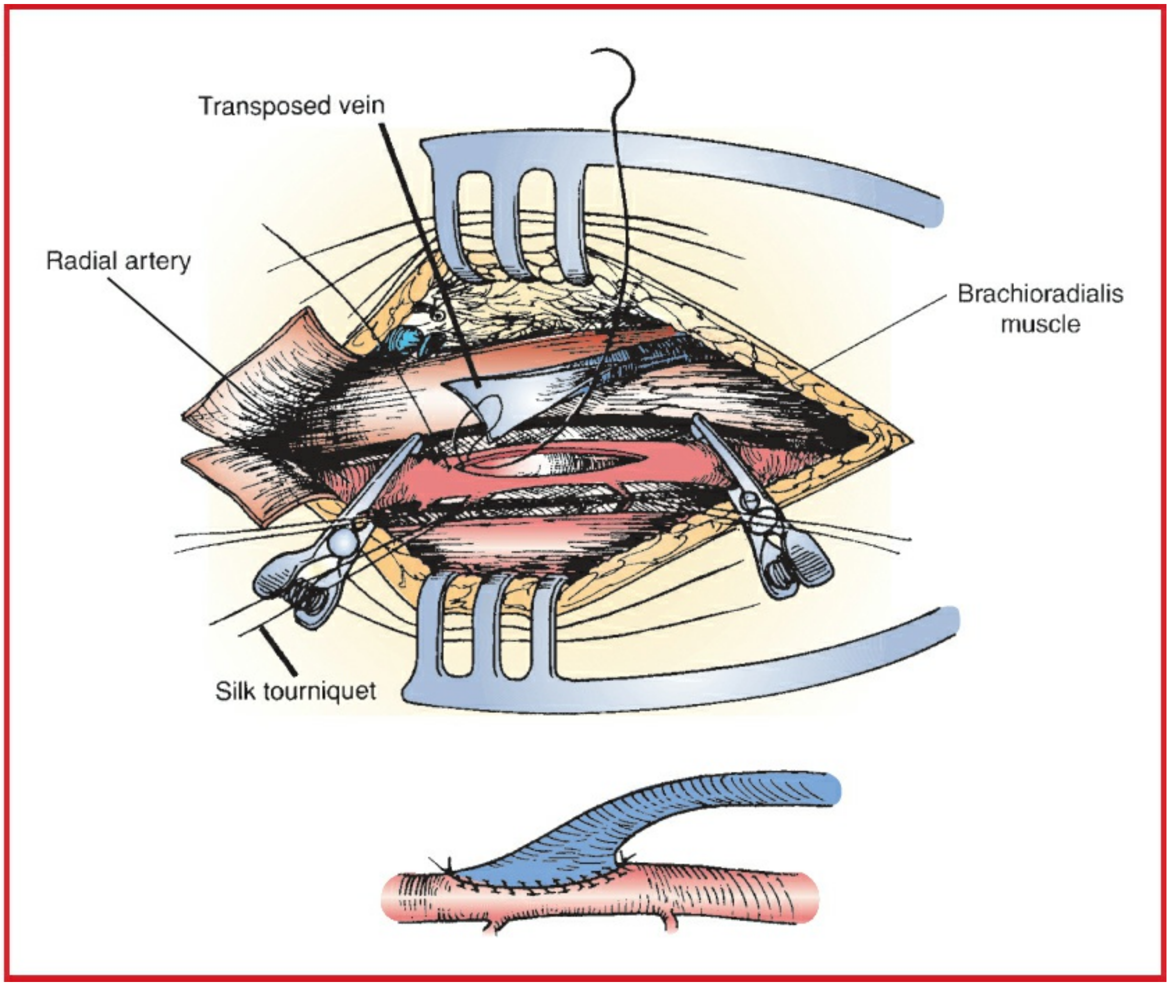

Figure 6.3-14 Brescia-Cimino fistula (side-to-end anastomosis). (Reproduced with permission from Scott-Conner CEH, Dawson DL. Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

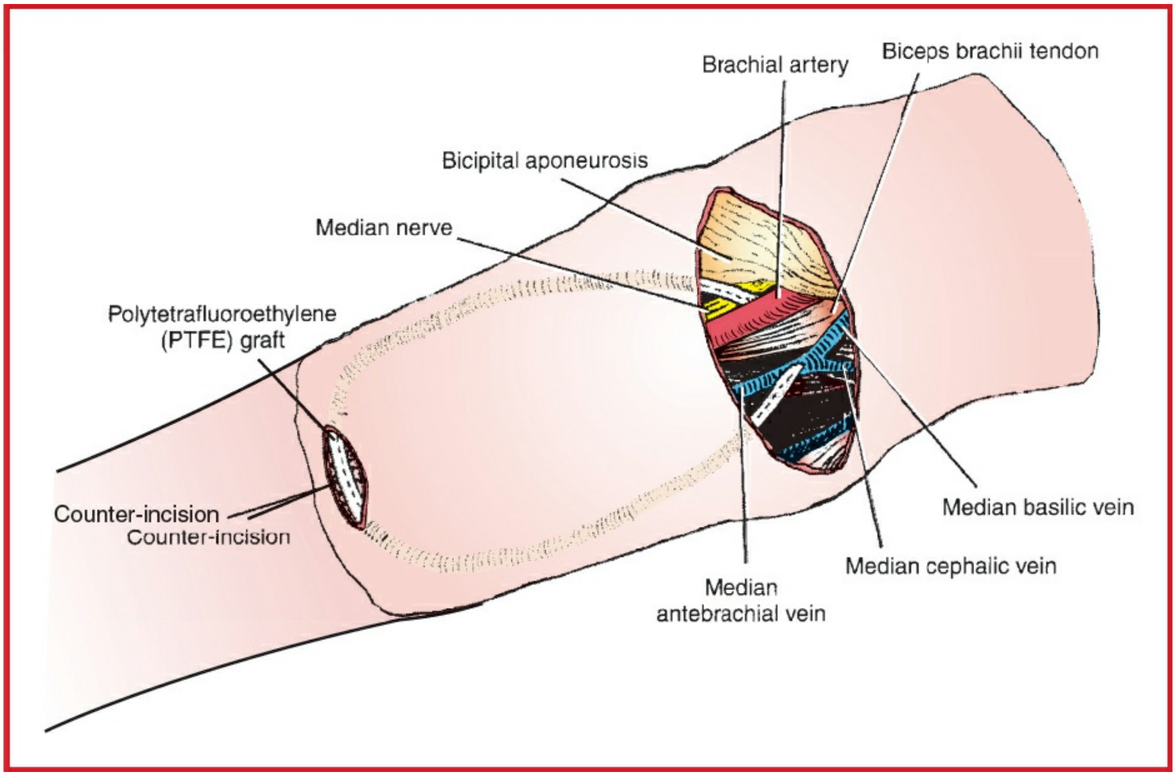

Figure 6.3-15 In arteriovenous hemodialysis access with prosthetic graft, the brachial artery and the median antebrachial, median basilic, and median cephalic veins are exposed via a horizontal incision below the antecubital joint crease. (Reproduced with permission from Scott-Conner CEH, Dawson DL: Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options. **See Appendix K, p. K-1.**

SUMMARY OF PROCEDURES

	Forearm AV Fistula	Forearm Loop or Straight Graft	Upper Arm Straight Graft
Position	Supine, arm abducted	⇐	⇐
Incision	Longitudinal or transverse at wrist, or “snuff box”	Transverse at antecubital fossa and/or at wrist; counterincision in the forearm	Transverse or longitudinal in the upper arm
Special	Arm/hand table	⇐	⇐

instrumentation	(for arm abduction); Doppler flow probe may be used		
Unique considerations	Local anesthesia; consider brachial plexus block (increased incidence of hematoma); heparinization	⇐	⇐
Antibiotics	Cefazolin 2 g iv (3 g if >120 kg)	⇐	⇐
Surgical time	1–2 h	⇐	⇐
Closing considerations	Use of Doppler to 3 shunt patency	⇐	⇐
EBL	25–50 mL	25–100 mL	⇐
Postop care	Hemodynamic monitoring for poor-risk patients; can be done as outpatient	⇐	⇐
Mortality	Minimal, depending on associated risk factors	⇐	⇐
Morbidity	Thrombosis: 20% Technical failure: 10–15% Arterial steal: Rare Cardiac failure: Rare Infection: Rare	8–32% ⇐ ⇐ ⇐ 10% 6% ⇐ ⇐	~8–32% ⇐ ⇐ ⇐ ⇐ ~6% ⇐ ⇐

	No venous outflow: Rare Seroma: Rare Venous aneurysm: Rare Venous HTN: Rare	⇐	⇐
Pain score	1	2	2

PATIENT POPULATION CHARACTERISTICS	
Age range	Pediatric and adult population
Male:Female	1:1
Incidence	Hemodialysis access is one of the most commonly performed procedures by vascular surgeons.
Etiology	Glomerulonephritis; diabetes; HTN; pyelonephritis
Associated conditions	Diabetes mellitus; PVD/CAD

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Permanent Vascular Access.

Suggested Readings

1. Bylsma LC, Gage SM, Reichert H, et al: Arteriovenous fistulae for haemodialysis: a systematic review and meta-analysis of efficacy and safety outcomes. Eur J Vasc Endovasc Surg 2017; 54(4):513–22.
2. Rose DA, Sonaike E, Hughes K: Hemodialysis access. Surg Clin N Am 2013; 93:997–1012.
3. Smith GE, Gohil R, Chetter IC: Factors affecting the patency of arteriovenous fistulas for dialysis access. J Vasc Surg 2012; 55:849–55.
4. Spergel LM, Ravani P, Asif A, et al: Autogenous arteriovenous fistula options. J Nephrol 2007; 20(3):288–98.

PERMANENT VASCULAR ACCESS

SURGICAL CONSIDERATIONS

Description

Silastic or plastic catheters are placed in patients who require venous access for chronic antibiotic therapy, TPN, chemotherapy, or hemodialysis. **Hickman, Broviac,** and **Groshong catheters** are made of silicone rubber or plastic with a cuff near the skin exit site, which (in theory) serves as a barrier to infection. Access is generally achieved via subclavian, IJ, or femoral vein puncture. These catheters are available in various sizes and in single- or double-lumen configurations. Larger diameter (13 Fr) Hickman or **PermCath DL** catheters have been introduced for hemodialysis. **Mediport** and **Portacath devices** have a metallic or plastic reservoir connected to the catheters and are intended for complete subcutaneous implantation. These catheters are used in chronically ill patients, particularly those requiring chemotherapy. The implantable access ports have been associated with improved patient comfort and reduced infection rates. Long-term catheter survival is limited by infection. Removal and replacement of the catheter is the only way to eradicate the infection.

Variant procedure or approaches

Two major distinctions: Hickman/Broviac catheters (no reservoir) versus Mediport/Portacath catheters (subcutaneous with reservoir). Subclavian, IJ, or femoral vein puncture is selected, depending on vein status, previous operations, and patient comfort.

Usual preop diagnosis

Chronic antibiotic therapy; TPN; chemotherapy; end-stage renal failure

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, and postop pain management options.** **See Appendix K, p. K-1.**

SUMMARY OF PROCEDURES

	Hickman/Broviac/Groshong	Mediport/Portacath
Position	Supine, slight Trendelenburg	⇐
Incision	Puncture site (subclavian, IJ, or femoral vein); subcutaneous tunnel for passage of catheter.	Puncture site (subclavian, IJ, or femoral vein); subcutaneous pocket for

	Alternative: cephalic or IJ vein cutdown to achieve access	port. Alternative: cephalic or IJ vein cutdown to achieve access
Special instrumentation	I.I. or fluoroscope	⇐
Unique considerations	Local anesthesia; may need iv sedation; monitor for ectopy during placement	⇐
Antibiotics	Cefazolin 2 g iv	⇐
Surgical time	30–90 min	45–90 min
EBL	10–25 mL	25–50 mL
Postop care	CXR in recovery room; may be outpatient	⇐
Mortality	Minimal	⇐
Morbidity	Catheter thrombosis: 25% Skin exit site infection: 13% Poor flow: 10–13% Catheter sepsis: 5–8% Arterial puncture: 6% Local bleeding: 5% SVC thrombosis: 5% Catheter displacement: 3% Subclavian thrombosis: 2% Pneumothorax: 1–2% Failed attempt: 1% Infection: 3–5/1000 cath days	4% — ~10–13% 2% ~6% — ~5% ~3% ~2% ~1–2% ~1% — Pocket hematoma: 2% Pocket infection: 3% Catheter leakage: 1%
Pain score	1	2

PATIENT POPULATION CHARACTERISTICS

Age range	Pediatrics and adults, 8–80 yr

Male:Female	1:1
Incidence	Depending on underlying disease
Etiology	Access for chemotherapy; infections; hemodialysis; chronic TPN
Associated conditions	Malignancy; chronic illness/infection; renal failure

ANESTHETIC CONSIDERATIONS FOR VASCULAR ACCESS

(Procedures covered: arteriovenous access for hemodialysis; permanent vascular access)

PREOPERATIVE

The patient populations presenting for vascular access surgery are extremely diverse. Patients requiring vascular access for chemotherapy, TPN, and chronic antibiotic therapy frequently can be done with MAC. Also presenting for these procedures are end-stage renal failure patients who need AV access for hemodialysis (generally involving the upper extremity). These patients return to the OR frequently for revising or replacing of fistulas. They are often ASA III and IV patients who may require GA or an upper extremity block. Anesthetic considerations for the **chronic renal failure patient** are discussed below.

Respiratory	Pulmonary edema may be present from fluid overload. CHF and uremic pleuritis can occur. Pneumonia occurs more frequently in these patients due to depressed immune systems. Hemodialysis contributes to hypoxemia due to V/Q mismatch and hypoventilation. **Tests:** CXR; consider ABG
Cardiovascular	Often have HTN related to hypervolemia and a disorder of the renin-angiotensin system. May have LVH 2° to HTN. Cardiomyopathy, pericarditis, and pericardial effusion occur with uremia. Hypervolemia and hypoalbuminemia can contribute to CHF. Uremic patients often have defective aortic and carotid body reflex arcs. Ejection murmurs are common.

	Tests: ECG; echo; others as indicated from H&P
Renal	Full details of recent dialysis therapy; patient optimal and current weight. Dialysis is usually advisable before anesthesia and surgery (<24 h if possible) as complications of uremia (Plt dysfunction, electrolyte/fluid abnormalities, CNS, and GI disturbances) improve with dialysis. If a transfusion is needed, consider giving during dialysis so intravascular volume can be controlled. **Tests:** Serum BUN; Cr. If patient produces urine, consider creatinine clearance and UA.
Hematologic	Chronic anemia with Hct ranging from 15 to 21 g/dL. Normochromic, normocytic anemia presents 2° to bone marrow depression, lack of erythropoietin, nutritional deficiency, and diminished red cell survival time. Patients adjust to chronic anemia through ↑ CO and ↑ 2,3-DPG levels. Accumulation of waste products inhibits Plt function. Defects do occur in the coagulation cascade, but PT and PTT are usually normal. **Tests:** CBC; Plt; PT; PTT
Gastrointestinal	Uremic patients commonly have hiccups, anorexia, N/V, and diarrhea. They are very prone to developing GI bleeds. Renal failure causes ↓ gastric emptying. Premedication with metoclopramide (10 mg iv) and famotidine (20 mg iv) will ↓ gastric volume and pH.
Nervous system	CNS Sx of uremia range from malaise to Sz to coma. Fatigue and intellectual impairment commonly occur. Peripheral and autonomic neuropathies exist. Peripheral neuropathy presents as itching and paresthesia of the lower extremities. Autonomic dysfunction can cause postural ↓ BP. Document deficits carefully.
Endocrine	Diabetes frequently may be the cause of renal failure, with its attendant problems. **Tests:** Glucose
Immune system	Often impaired. Patients prone to sepsis. Hepatitis

	and HIV infections from blood products may exist.
Metabolic and biochemical	Accumulation of K^+ urea, parathyroid hormone (hypercalcemia), Mg, aluminum (neurotoxicity), acid metabolites, and phosphate occurs. Knowing hyperkalemia exists is of importance because of the potential for fatal cardiac dysrhythmias. Shift of the oxyhemoglobin curve to the right occurs due to the metabolic acidosis and ↑ 2,3-DPG (improves tissue oxygenation). Hyponatremia is a common electrolyte disturbance in chronic renal failure.
Premedication	If warranted, it is best to use low-dose premedication with benzodiazepines or opioids due to the possibility of exaggerated effects.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status.

INTRAOPERATIVE

Anesthetic technique

GA, upper extremity block, or MAC

Regional anesthesia

May be advantageous due to decreased number of drug effects. See section on upper extremity blocks (Anesthetic Considerations for Wrist Procedures). If the patient was very recently dialyzed, there may be a residual heparin effect. Regional anesthesia is contraindicated if coagulopathy is present.

General anesthesia

The duration of action and elimination of many anesthetic drugs is altered in the patient with renal failure.

Induction	If general anesthesia is indicated, induction with propofol and low-dose fentanyl (25–50 mcg) is usually well tolerated. Succinylcholine is appropriate if $K^+ < 5.5$ mEq/L. If succinylcholine is contraindicated, cisatracurium (0.1–0.2 mg/kg) is the alternative muscle relaxant of choice because its elimination is not affected by renal failure; if

	sugammadex is available, then rocuronium may also be considered. Etomidate (0.2–0.3 mg/kg) may be used in case of preop hemodynamic instability.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Inhalation anesthetics (isoflurane, sevoflurane) offer the advantage of not requiring renal elimination. Biotransformation may produce some inorganic fluoride (nephrotoxin); however, this is not an issue in dialysis patients. Opioids can produce an increased magnitude and duration of effect. Increased accumulation of morphine glucuronides → prolonged respiratory depression. Fentanyl and remifentanil are good choices, 2° rapid tissue redistribution (fentanyl) or rapid metabolism (remifentanil). Benzodiazepine dosage should be adjusted because protein binding in renal insufficiency/failure is reduced.	
Emergence	Prolonged effect of anticholinesterases (e.g., neostigmine and edrophonium) effectively offsets prolongation of blockade. Other factors affecting reversal of nondepolarizers should be taken into account. These include acid-base status, depth of blockade, temperature, and use of drugs such as diuretics or antibiotics, which can potentiate blockade.	
Blood and fluid requirements	IV: 18–20 ga × 1 BSS @ 1–2 mL/kg/h	IV access may be difficult; avoid iv placement in the operated arm. Minimize fluids in renal failure patients. See Zero Balance Fluid Management, Appendix K, p. K-2.
Monitoring	Standard monitors	Avoid BP cuff placement on the operated arm.

Positioning	✓ and pad pressure points ✓ eyes
Complications	Local anesthetic toxicity

POSTOPERATIVE

Complications	Nerve damage Hematoma	These are rare complications of brachial plexus blocks.
Multimodal pain management	PO analgesics	
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	

Suggested Readings

1. Bour ES, Weaver AS, Yang HC, et al: Experience with the double lumen Silastic catheter for hemoaccess. Surg Gynecol Obstet 1990; 171(1):33–9.
2. Monk J: Hemodialysis catheters and ports. Semin Nephrol 2002; 22:211–20.
3. Vasquez MA: Vascular access for dialysis: recent lessons and new insights. Curr Opin Nephrol Hypertens 2009; 18:116–21.

VENOUS SURGERY—VEIN STRIPPING AND PERFORATOR LIGATION

SURGICAL CONSIDERATIONS

Description

Chronic venous insufficiency results from static blood flow in the deep, superficial, and perforating veins of the lower extremities. Clinical manifestations include pathologic changes in the skin and subcutaneous tissues, such as pigmentation, dermatitis, induration, and ulceration around the lower portion of the leg. The condition is most commonly caused by defective venous valves and less often by obstruction to the venous return or impaired pumping action of the muscles in the leg. The disorder is sometimes the residual of previous iliofemoral thrombophlebitis. Varicose veins of the primary type, particularly those of long duration, are a common cause of chronic venous insufficiency of milder degrees. Most symptoms respond well to conservative management, which includes compression stockings, elevation of the extremity, and topical treatment of ulcerations. Failure of medical management is an indication for surgical intervention. Split-thickness skin grafting is indicated for large ulcers to accelerate healing and shorten hospitalization time. **Ligation of perforators** is best performed when the ulcer has completely healed. The classic approach of **Linton** is rarely used today. If the quality of the skin overlying the perforators prevents a direct approach, **subfascial ligation** of the perforators may be performed through a short, posterior midline incision. The incompetent greater or lesser saphenous veins are resected only if patency of the deep system is confirmed. Venous ulcers recur in 30% of patients after surgical therapy, and ulcerations persist for prolonged period in 15% of patients. Adjunctive procedures include **valvuloplasty, vein transposition,** and **venous valve transplant.**

Alternative procedures

Minimally invasive radiofrequency techniques have been used successfully for ablation of varicose veins.

Usual preop diagnosis

Chronic deep venous insufficiency

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, carb loading, and postop pain management options. See Appendix K, p. K-1.**

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	**Vein stripping:** longitudinal or oblique groin incision and transverse incision at medial malleolus; transverse incision over posterior lower leg for lesser saphenous vein stripping. **Perforator ligation:** longitudinal incision along medial aspect of tibia to posterior medial malleolus
Special instrumentation	Vein stripper
Antibiotics	If patient has an associated venous ulcer, preop antibiotics should be based on culture results; generally, cefazolin 2 g iv (3 g if >120 kg), if culture

	results are not available.
Surgical time	3 h
EBL	50–250 mL
Postop care	PACU → ward; antiembolism stockings and SCDs
Mortality	Minimal
Morbidity	Persistence of nonhealing ulcer: 20–53%
Pain score	3

■ PATIENT POPULATION CHARACTERISTICS	
Age range	Young adult–elderly (generally older adults, although present in younger patients as well)
Male:Female	1:2
Incidence	6–7 million people in the United States have adverse effects from chronic venous stasis; 500,000 have complications of leg ulceration.
Etiology	Chronic primary varicose veins; defective venous valves; impaired pumping action of muscles in the leg; previous iliofemoral thrombophlebitis; obstruction of venous return
Associated conditions	Varicose veins; Hx of DVT/thrombophlebitis

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Varicose Vein Stripping.

Suggested Readings

1. Geersen DF, Shortell CEK: Phlebectomy techniques for varicose veins.Surg Clin North Am 2018; 98(2):401–14.
2. Roth SM: Endovenous radiofrequency ablation of superficial and perforator veins. Surg Clin North Am 2007; 87(5):1267–84.
3. See Suggested Readings for Venous Surgery—Thrombectomy or Vein Excision, p. 493.

VARICOSE VEIN STRIPPING AND ABLATION

SURGICAL CONSIDERATIONS

Description

In patients with primary varicose veins, no definite cause has been identified, although age, female sex, pregnancy, obesity, and positive family Hx are predisposing factors. The causes of secondary varicosity include incompetence or obstruction of the deep veins as a result of previous DVT, tumor, trauma, or congenital or acquired AV fistulas. Usual indications for operative therapy include aching, swelling, heaviness, cramps, itching, cosmesis, stasis dermatitis, pigmentation, burning, and ulcers. Surgical treatment is contraindicated in pregnant patients, elderly patients who are considered high risk, and patients with arterial insufficiency of the lower extremities, lymphedema, skin infection, or coagulopathy. Endovenous thermal (laser/RF) ablation techniques have largely replaced the surgical treatment of varicose veins in the lower leg. These endovenous procedures are usually performed under local anesthesia.

There are two principal approaches: the **stab avulsion technique** and **high ligation and stripping**. With **stab avulsion,** the varicosities are marked preop. Small transverse or longitudinal incisions are made directly over these varicosities, which are dissected from the surrounding subcutaneous tissue (with undermining of the skin) and bluntly removed or avulsed. Firm pressure over the region being operated on will achieve hemostasis. After removal of all marked varicosities, sterile dressings are placed and a compression bandage wrapped around the affected leg. The patient is instructed to keep the leg elevated as much as possible while convalescing at home. The chief advantage of the stab avulsion technique is preservation of the saphenous vein when it is not directly involved with varicosities.

If there is valvular incompetence of the saphenous vein, the treatment of choice is **stripping** (**avulsion**) of the incompetent portion of the greater and lesser saphenous veins, together with avulsion of the superficial varicose veins of the thigh and calf. **High ligation and stripping** refers to the removal of the greater saphenous vein from the level of medial malleolus to the saphenofemoral junction. A small transverse incision is made at the level of the ankle and the saphenous vein is dissected free. A longitudinal or oblique incision at the groin permits isolation of the saphenous vein at the saphenofemoral junction. The greater saphenous vein is ligated proximally and distally. After a **venotomy,** a plastic or metallic vein stripper is passed and the vein is removed or stripped in a distal-to-proximal fashion. Sterile dressings are applied, followed by a compressive dressing.

Although high ligation and stripping is the gold standard in the treatment of varicose veins, it has largely been replaced by thermal ablation in the United States. The use of either RF or laser ablation, which results in closure and often reabsorption of the vein, is highly successful and causes less postop discomfort. However, surgical ligation and stripping still has a role in the management of varicose veins. If all varicose veins are removed and the incompetent segment of the saphenous vein is stripped, 85% of the patients will have good-to-excellent results at late follow-up. These procedures can be performed with regional or general anesthesia.

Usual preop diagnosis

Varicose veins; symptoms of venous insufficiency; cosmetic considerations

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, carb loading, and postop pain management options.** **See Appendix K, p. K-1.**

SUMMARY OF PROCEDURES

	Stab Avulsion Technique	High Ligation and Stripping
Position	Supine	⇐
Incision	Varicosities marked preop; short stab incisions made and veins avulsed with small forceps	Varicosities marked preop; small transverse incision over saphenous vein proximal to medial malleolus; proximal saphenous vein exposed via groin incisions and stripper passed
Special instrumentation	None	Vein stripper
Unique considerations	May be facilitated by tourniquet	⇐
Antibiotics	None	⇐
Surgical time	2–3 h	⇐
Closing	Leg compressed with	⇐

considerations	elastic wrap	
EBL	50–250 mL	50–150 mL
Postop care	PACU → ward; support stockings	⇐ + elevate foot of bed 10°; short periods of ambulation
Mortality	Minimal	⇐
Morbidity	Recurrence: <10% Hematoma: Rare Infection: Rare Lymph fistula: Rare Nerve injury: Rare Postop DVT: Rare Femoral artery injury: Nil	⇐ ⇐ ⇐ ⇐ ⇐ 5% Very rare
Pain score	2	2

PATIENT POPULATION CHARACTERISTICS

Age range	Wide range, young adult–elderly (average = 48 yr)
Male:Female	1:3
Incidence	24,000 in the United States
Etiology	Primary varicose veins: no definite cause (Predisposing factors include age, female sex, pregnancy, obesity, and positive family Hx.) Secondary varicose veins: incompetence or obstruction of the deep veins from DVT, tumor, trauma, or high venous pressures due to congenital or acquired arteriovenous fistulas
Associated conditions	Older age; pregnancy; obesity; DVT; tumor; trauma

ANESTHETIC CONSIDERATIONS

(Procedures covered: vein stripping and perforator ligation; varicose vein stripping)

PREOPERATIVE

Patients presenting for varicose vein surgery are a generally healthy patient population (ASA I & II). Preop considerations and tests, therefore, should be guided by the H&P.

Hematologic	If regional anesthesia planned, check patient's coagulation status. **Tests:** Plt count; Hct
Laboratory	Tests as indicated from H&P
Premedication	If necessary, standard premedication.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status.

INTRAOPERATIVE

Anesthetic technique

General, regional, and local (tumescent) anesthesia, ± sedation, are all appropriate anesthetic techniques. Choice depends on factors such as extent of surgery, patient physical status, and patient and surgeon preference.

Regional anesthesia

Spinal	**Single shot vs. continuous:** Patient in sitting or lateral decubitus position (operative site down) for placement of hyperbaric subarachnoid block. Doses of local anesthetic for T10-T12 level: 0.75% bupivacaine in 8% dextrose (7–10 mg) or 0.5% tetracaine in 5% dextrose (10–12 mg). For continuous spinal (multiorifice catheter), titrate local anesthetic to desired surgical level (T12). Large doses of hyperbaric local anesthetic should be avoided as they may cause postop cauda equina syndrome.
Epidural	Patient in sitting or lateral decubitus position for placement of epidural catheter. After locating the epidural space, administer a test dose (e.g., 3 mL of 1.5% lidocaine with 1:200,000 epinephrine) to elucidate whether the catheter is subarachnoid

	(rapid onset outer block) or intravascular (↑ HR, tinnitus, etc.). Titrate local anesthetic until desired surgical level is obtained (3–5 mL at a time) usually <15 mL.
Local/tumescent	Requires gentle surgical technique. Surgical field block, plus ilioinguinal and iliohypogastric nerve blocks using 0.5% bupivacaine with 1:200,000 epinephrine. Usually done by the surgeon, tumescent anesthesia consists of injecting a dilute buffered solution of lidocaine (0.05%) with epinephrine around the vein with ultrasound guidance. Total lidocaine dose may be up to 10 mg/kg.

General anesthesia

Induction	LMA/mask vs. ETT: Standard induction. LMA or mask GA may be suitable for many patients.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance	
Emergence	No special considerations. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 18 ga × 1 BSS @ 2–3 mL/kg/h	See Zero Balance Fluid Management, Appendix K, p. K-2 and p. K-4.
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	

POSTOPERATIVE

Complications	Cauda equina syndrome	The diagnosis of cauda equina syndrome (urinary and fecal incontinence, paresis of lower

		extremities, perineal hypoesthesias) should be sought in the postop period in patients who have received large doses of intrathecal local anesthetic during continuous spinal techniques. ✓ patients for bowel or bladder dysfunction and perineal sensory deficits. If present, consider a neurology consultation and continue follow-up of the patient's neurologic dysfunction.
	Urinary retention common with regional anesthesia	Patients with urinary retention may require intermittent catheterization until urinary function resumes.
Multimodal pain management	PO analgesics: Acetaminophen and codeine Tylenol # 3 1–2 tab q4–6h Oxycodone and acetaminophen Percocet 1 tab q6h	Regional anesthesia should provide sufficient analgesia postop. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition and mobilization.

Suggested Readings

1. Biemans AA, Kockaert M, Akkersdijk GP, et al: Comparing endovenous laser ablation, foam sclerotherapy, and conventional surgery for great saphenous varicose veins. J Vasc Surg 2013; 58(3):727–34.
2. Brown KR, Rossi PJ: Superficial venous disease. Surg Clin N Am 2013; 93:963–82.
3. Gibson K, Gunderson K. Liquid and foam sclerotherapy for spider and varicose veins. Surg Clin North Am 2018; 98(2):415–29.
4. Holt NF: Tumescent anaesthesia: its applications and well tolerated use in the out-of-operating room setting. Curr Opin Anaesthesiol 2017; 30(4):518–24.
5. Kheirelseid EAH, Crowe G, Sehgal R, et al. Systematic review and meta-analysis of randomized controlled trials evaluating long-term outcomes of endovenous management of lower extremity varicose veins. J Vasc Surg Venous Lymphat Disord 2018; 6(2):256–70.
6. Mowatt-Larssen E, Shortell CK: Treatment of primary varicose veins has changed with the introduction of new techniques. Semin Vasc Surg 2012; 25:18–24.
7. Rasmussen L, Lawaetz M, Bjoern L, et al: Randomized clinical trial comparing endovenous laser ablation and stripping of the great saphenous vein with clinical and duplex outcome after 5 years. J Vasc Surg 2013; 58:421–6.

CHAPTER 6.4

Heart/Lung Transplantation

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SURGERY FOR HEART TRANSPLANTATION

SURGICAL CONSIDERATIONS

Description

Although heart transplantation has been practiced since 1967, it has had its greatest expansion since the early 1980s with the introduction of cyclosporine. Currently, there are ~150 transplant centers and 2200 heart transplant procedures performed yearly in the United States. Indications for heart transplantation range from hypoplastic left heart syndrome (HLHS) in the neonate to cardiomyopathy and ischemic heart disease in the adult. Recipients usually have end-stage heart disease manifested by congestive heart failure (CHF) and a prognosis of <1-yr survival. Many patients are on inotropic drugs or on some type of additional mechanical assist, such as the use of an intra-aortic balloon pump (IABP) or an implanted LV assist device. Current immunosuppressive protocols consist of a combination of a calcineurin inhibitor with prednisone and mycophenolate mofetil. Immunosuppression begins either immediately preop or perioperatively and will continue throughout the life of the patient. Induction immunosuppression is commenced in the operating room after protamine administration with methylprednisolone 500 mg iv and basiliximab or rATG in the ICU after transplantation. Current 1-yr survival averages 85% in most centers with 3-yr survival of ~80% and a median survival approaching 10 yr.

In adult heart transplantation, following median sternotomy, the pericardium is opened with care being taken to preserve the phrenic nerve. The aorta and vena cava are cannulated, the aorta is cross-clamped, and caval tapes (tourniquets to prevent venous air embolism [VAE]) are applied. The aorta and PA are then transected. This is followed by an incision through the atria, and the recipient heart is removed. The donor heart is prepared by opening the left atrium through the pulmonary veins, separating the aorta and PA. The donor heart is attached by a long, continuous suture line around the left atrium (Fig. 6.4-1), followed by separate anastomoses to the inferior and superior venae cavae. Alternatively, the donor right atrium is anastomosed to the recipient right atrium with a single long continuous suture. Next, the PA and aorta are anastomosed to their respective recipient vessels. Multiple deairing maneuvers are followed by aortic unclamping and rewarming and resuscitation of the heart. Normal sinus rhythm (NSR) is established and CPB d/c'd. Heparin is reversed, hemostasis is secured, and the chest is closed in a routine manner. Following chest closure, these patients will often have implanted defibrillators that will be removed. An incision is made to access the pacemaker pocket, and the device is explanted. After hemostasis is established, the pocket is closed (see p. 384 for discussion of cardiopulmonary bypass [CPB]).

Neonatal heart transplantation differs in that the PA is cannulated if the ductus arteriosus is patent. Reconstruction of the aortic arch in the patient with HLHS requires CPB with deep hypothermia (<18°C) and circulatory arrest. The heart is then excised, and the transverse aortic arch is opened beyond the ductus arteriosus to minimize risk of late coarctation. The donor heart is prepared, with special attention given to trimming the transverse aortic tissue for subsequent reconstruction. The left and right atrium, PA, and aorta are sutured in place. The new ascending aorta and right atrium are cannulated, and CPB with rewarming is reinstituted. Chest closure is routine (see Pediatric Transplantation).

Pediatric heart transplantation has become commonplace in the past 10 yr. Patients often have had previous cardiac surgery, and reentry and excision of the native heart are complicated by the presence of adhesions and graft material from previous attempts at palliative/corrective surgery. Patients are often highly sensitized and may require intraoperative plasmapheresis while on CPB. Preparation for CPB is often similar to the adult heart transplant patient, and the implantation procedure is also similar. However, provisions should be made for

prolonged cross-clamp times necessary for implantation in the setting of abnormal systemic-cardiac or pulmonary-cardiac connections (see Pediatric Transplantation).

Usual preop diagnosis

Cardiomyopathy; coronary artery disease (CAD) with ischemic cardiomyopathy congestive heart disease (CHD) (e.g., HLHS or anomalous left coronary artery); end stage valvular heart disease

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. **See Appendix K: Enhanced Recovery After Surgery, p. K-1.**

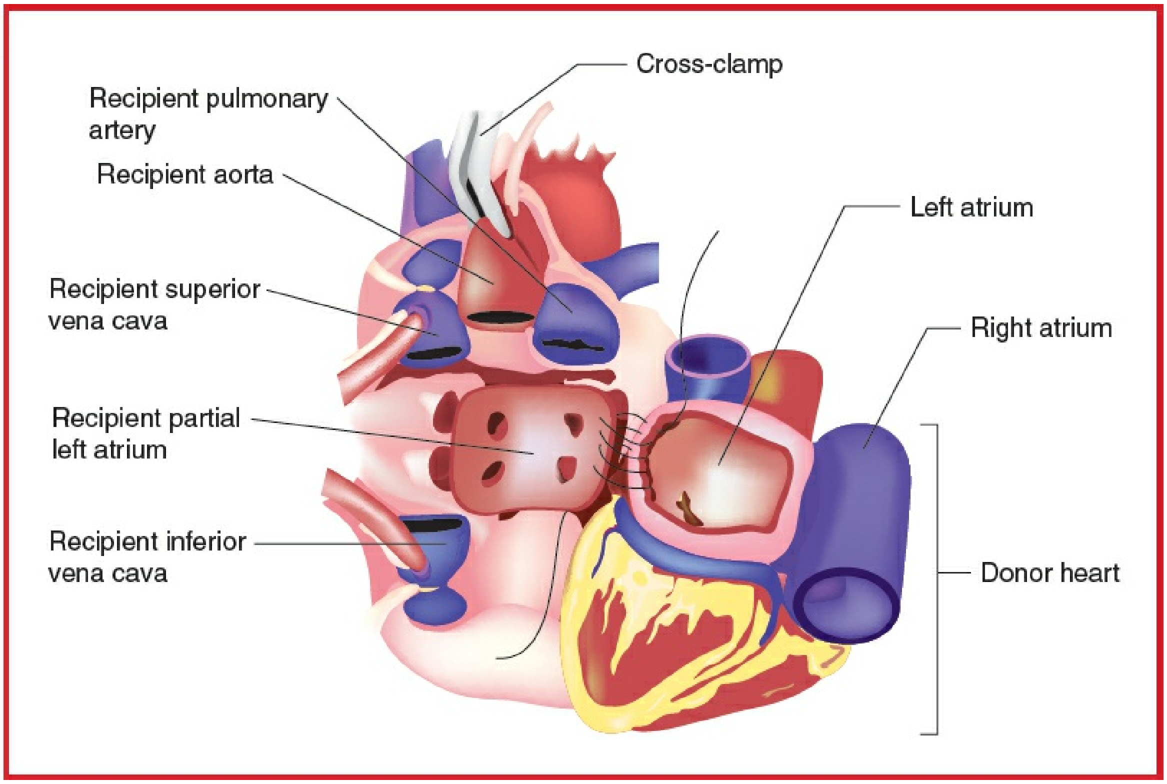

Figure 6.4-1 The donor left atrium is sutured to the recipient left atrial remnant and the graft vessels are anastomosed.

SUMMARY OF PROCEDURES

	Adult Heart Transplantation	Neonatal/Pediatric Heart Transplantation

Position	Supine	⇐
Incision	Median sternotomy	⇐
Special instrumentation	Ascending aortic, SVC, and IVC cannulae	Ascending aortic and right atrial cannulae
Unique considerations	Due to complete excision of the heart, the use of a PA catheter is usually not feasible	Deep hypothermia with circulatory arrest
Antibiotics	Cefazolin 2 g iv	Cefazolin 25–50 mg/kg iv
Surgical time	Cross-clamp: 45–60 min Surgery: 3–4 h	Circulatory arrest: 45–60 min Cross-clamp: 80–100 min Surgery: 4–6 h
Closing considerations	Temporary AV pacing wires are occasionally placed. A PA catheter may be advanced, especially with concerns about residual pulmonary HTN and donor right heart function. An isoproterenol infusion is started intraop to keep HR = 100–110 and help improve right heart function and ↓ PVR. Inhaled NO may be used to further ↓ PVR	Temporary ventricular pacing wire is usually placed. Temporary transthoracic left atrial line may be placed. Extensive aortic suture line requires avoidance of postop hypertensive episodes.
EBL	500–1500 mL	50–100 mL
Postop care	Cardiac ICU × 1–2 d of assisted ventilation, 2- to 3-d stay	Pediatric ICU × 1–2 d of assisted ventilation; 4- to 5-d stay, with attention to pulmonary care
Mortality	<5%	⇐
Morbidity	Early acute rejection	⇐

	episodes from 10–21 d: 50%	
	Infection, particularly pulmonary: 10% Pulmonary HTN with right heart dysfunction: <10%	Infection: 10% Respiratory problems: 20%
	Dysrhythmias with nodal rhythms: 5% Bleeding: 2–4%	Pulmonary vasospasm with right heart dysfunction: <10% Bleeding: 2–4%
	Hyperacute rejection: rare (<1%)	
	Intracoronary air emboli	
Pain score	8–10	8–10

PATIENT POPULATION CHARACTERISTICS		
Age range	18–75+ yr (average 50–55 yr)	1 d to 2 mo (neonatal), 2 mo to 18 yr (pediatric)
Male:Female	7:3	1:1
Incidence	2200/yr (United States)	Rare
Etiology	Cardiomyopathy (50%), CAD with multiple previous infarcts (48%), others (2%)	No apparent correlation with any specific genetic disorder. Cardiomyopathy (49%), CHD (42%), others (7%)
Associated conditions	CHF	Other congenital anomalies

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients scheduled for heart transplantation are functionally compromised, typically with CHF, which is associated with a mortality of >50% in 2 yr. (Studies have shown that patients with severe CHF have a mortality of 50% in 6 mo.) The progression o cardiovascular disease is usually well documented in these patients. A Hx of recent exacerbation of cardiac dysfunction should be sought, and all data should be interpreted in light of interval changes. Patients may arrive in the operating room from home or in hospital with inotropic infusions (dobutamine, milrinone), VADs, or extracorporeal membrane oxygenation (ECMO) in place.

Respiratory	The presence of pulmonary HTN and ↑ PVR may be disclosed by catheterization. The severity of the abnormality and the responsiveness to specific vasodilators must be determined. **Test:** Right heart catheterization
Cardiovascular	Indicators to consider include hemodynamic status; LV EF (mortality is rapid in patients with EF <10% and is worse for patients with EF of 10–20%, as compared with those with EF >20%); myocardial structure and morphology, symptoms, and functional capacity; neuroendocrine status; serum sodium; and dysrhythmia. Unfortunately, while these measures show trends with mortality, they are not individually strong enough to predict a particular patient's course. Low maximum O_2 consumption (<10 mL/kg/min) is associated with poor survival. Normal O_2 consumption is 40 mL/kg/min. In practice, however, this measure is too severe because many patients awaiting heart transplantation have maximum O_2 consumption of 20 mL/kg/min. Dysrhythmia is a major cause of death; unfortunately electrophysiology studies of these patients may not be helpful because dysrhythmia tends to be noninducible. This phenomenon frustrates efforts to select and test antidysrhythmic drug therapy. The effectiveness of past antidysrhythmic therapy should be reviewed. **Tests:** ECG; cardiac catheterization; ECHO
Hematologic	Patients with dilated cardiomyopathy, previous cardiac surgery including VADs and ECMO, are frequently treated with anticoagulants to reduce the

	risk of thrombus formation. Hepatic dysfunction may result from RV failure and may reduce synthetic function. Mild hepatic dysfunction and chronic anticoagulation may contribute to postop bleeding. The anticoagulant effect of warfarin should be reversed with FFP or prothrombin complex concentrates (PCCs) such as Kcentra™ and vitamin K (iv or subcutaneous). **Tests:** Hct; PT; PTT; fibrinogen; Plts.
Endocrine	Neuroendocrine abnormalities are often present in severe CHF cases. The cardiomyopathy produces low CO → compensatory sympathetic activation and renin-angiotensin activity. The result is excessive vasoconstriction with salt and H_2O retention, which further impair myocardial performance. Markedly worse survival is seen in CHF patients with serum sodium <130. This may indicate the importance of neuroendocrine pathophysiology or may simply be evidence of the severity of the CHF. It may also simply indicate that patients with more severe CHF are treated with more diuretics. When patients are treated with an angiotensin-converting enzyme inhibitor, such as enalapril, the serum sodium is normalized, and survival chances are improved because of the slowing of the progression of CHF, not from alteration in the incidence of sudden death. **Tests:** Electrolytes; Cr
Laboratory	Evidence of renal and hepatic dysfunction should be sought by H&P and lab studies. Mild hypokalemia is generally not treated in view of the K^+ in the graft (see CPB).
Premedication	Although anxious, these patients are usually well informed and psychologically prepared to undergo heart transplantation. They respond well to the reassurance of the preop visit, and pharmacologic premedication usually is not necessary. O_2 therapy should commence prior to transport of the patient to the OR. Reassuring the family of a patient who

	suffers from rapidly progressive cardiac dysfunction also is valuable. The patient may be at increased risk for pulmonary aspiration of gastric contents because of the unscheduled nature of the surgery and use of oral cyclosporine immediately preop. Famotidine (20 mg) and metoclopramide (10 mg) may be administered iv, most efficiently accomplished in the OR.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: After the patient is placed on the operating table, O_2 and noninvasive monitors are applied. Dyspnea (a complication of the supine position) can be treated by raising the back of the table. As infection is a serious complication in the immunosuppressed transplant patient, aseptic technique is extremely important. Aseptic technique is used in inserting and securing all vascular catheters. The anesthesia machine should be equipped with a supply of air to control the FiO_2.

Induction	An arterial line for BP and blood gas monitoring should be inserted, using liberal amounts of local anesthetics before induction. There are rare exceptions to this rule, but the presence of real-time BP monitoring is critical during induction. If infusion drugs (e.g., epinephrine) are necessary before insertion of the CVP catheter, they can be infused temporarily through a separate peripheral iv. In patients who have a ↓ EF, it is often helpful to infuse epinephrine 0.05–0.10 mcg/kg/min during induction to avoid ↓ HR and ↓ CO. Anesthesia is not induced

	until the team harvesting the graft reports that the donor heart appears to be normal. The patient is preoxygenated (FiO_2 = 1.0), and cricoid pressure is applied just before induction. Induction agents include fentanyl (2–4 mcg/kg) or sufentanil (0.5–2 mcg/kg). Propofol (0.5–1 mg/kg, titrated) or etomidate (0.1–0.2 mg/kg) is useful in permitting rapid control of the airway. Midazolam also may be used. Rocuronium 1–1.5 mg/kg should be administered immediately to permit airway control. Care must be taken to avoid bradycardia, which often results in low CO in these patients. Immediate control of the airway is crucial, as hypercarbia and hypoxia must be avoided. The patient can be expected to have a low CO, resulting in a delayed induction of anesthesia, which must be anticipated to avoid anesthetic overdosage. Hypotension should be treated promptly with inotropes (bolus and/or infusion). Fluid boluses may be poorly tolerated in patients with diminished contractility and should be administered cautiously. The usual aids for managing the unexpectedly difficult airway should be readily available. Antibiotics are administered, and additional monitors (urinary catheter with thermistor, TEE) are set up. Nasopharyngeal temperature probes are avoided because full heparinization is needed for CPB. If there is a delay in the anticipated arrival of the graft, the recipient should be covered and kept warm, and skin prep should be delayed. Additional narcotics should be administered only in immediate anticipation of the commencement of surgery.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after

	CPB.
Maintenance	Typical cumulative anesthetic doses for the entire intraop course are fentanyl 10–20 mcg/kg or sufentanil 1.5–5 mcg/kg, midazolam 0.15 mg/kg, and rocuronium 3–4 mg/kg. Recommend the use of antifibrinolytics for all cases requiring CPB.
Termination of CPB	Junctional rhythm is common in the denervated transplanted heart. Isoproterenol 20–50 ng/kg/min or epinephrine 50–100 ng/kg/min may be used to achieve a HR of 100–120 bpm. Isoproterenol is also useful in providing inotropic support and pulmonary vasodilation (see below). Atropine and neostigmine do not affect HR. HTN does not produce reflex bradycardia. The graft atrium produces normally conducted P waves if the right atrial cuff anastomosis is used. The graft-conductive tissue contains adrenergic receptors and responds normally to norepinephrine, epinephrine, and isoproterenol. Inotropic support with epinephrine (20–100 ng/kg/min) +/− milrinone 0.2–0.5 mcg/kg/min may be necessary, especially if pulmonary HTN promotes RV failure. Inhaled NO (20 ppm) or epoprostenol (0.05 mcg/kg/min) may provide selective pulmonary vasodilation. Vasoplegia may occur particularly in patients with an LVAD and may require iv vasopressin (0.01–0.06 U/h) or norepinephrine 0.05–0.2 mcg/kg/min. A PA catheter may be helpful in guiding the use of inotropes and vasodilators and may be placed after CPB. After termination of CPB, TEE may be of particular value in assessing RV dysfunction, estimating PA pressures, and guiding appropriate fluid therapy, pharmacologic support, and mechanical support as necessary. RV failure may be produced by the presence of air in the RCA. Visual inspection may demonstrate this problem, and one should wait for the passage of the air and the resolution of ischemia

	before terminating CPB. SNP (0.2–2.0 mcg/kg/min) or clevidipine (1–6 mg/h) is used for afterload reduction. Inhaled NO, epoprostenol or iv prostaglandin E_1 (20–100 ng/kg/min), and NTG (0.2–2.0 mcg/kg/min) may be used for pulmonary vasodilation, especially if a preop catheterization study demonstrates responsiveness of the pulmonary circulation. Isoproterenol infusion (10–100 ng/kg/min) may provide appropriate pulmonary vasodilation, chronotropy, and inotropy. IV fluid and vasodilators must be given with particular care, as the flow produced by the denervated heart is quite sensitive to preload.	
Postbypass hemorrhage	Postbypass bleeding is a common problem brought on by the preop use of anticoagulants, the depressed synthetic function of the liver in chronic heart failure, and the effects of CPB. Following administration of protamine, infusion of Plts, FFP, and RBCs may be necessary. Cryoprecipitate is often indicated, especially for patients with previous chest surgery. Also consider PCCs for patients on preop warfarin. Epsilon-aminocaproic acid (EACA) or tranexamic acid may be appropriate in some cases. If bleeding persists, consider recombinant factor VIIa or FEIBA.	
Immunosuppression	Methylprednisolone 500 mg is given on bypass by the perfusionist at the time of graft reperfusion.	
Diuresis	There may be little urine production, especially if patient received high-dose diuretics preop. Mannitol and furosemide may be needed to induce diuresis.	
Transport	A Mapleson D or Jackson-Rees circuit is used in transporting the patient to the ICU. Sedation with iv infusion of propofol (20–50 mcg/kg/min) and/or dexmedetomidine (0.4–0.8 mcg/kg/min) may be used.	
Blood and fluid requirements	Possible severe bleeding IV: 14–16 ga × 1–2 BSS @ 4–6 mL/kg/h	Bleeding is often a problem after termination of CPB. A second iv

		catheter is inserted in patients with previous chest surgery. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Arterial line CVP/PA catheter UO	Although it may be helpful to have a triple-lumen CVP line before induction for preload monitoring and infusion of potent infusion drugs, it is not essential. This line is ideally inserted after the patient is intubated to avoid patient dyspnea and discomfort. A PA catheter is usually not advanced before bypass because it must be removed during surgery. An 8.5-Fr introducer is used in anticipation that a PA catheter may be necessary to manage right heart failure following the transplantation. In some institutions (not Stanford), the left IJ vein is the preferred site of cannulation, which leaves the right IJ unscarred for repeated postop endomyocardial biopsy of the transplanted heart.
	Cerebral oximetry	Cerebral oximetry readings may be

		confounded by changes in scalp perfusion. Be aware that false-positive and false-negative readings are not uncommon.
	TEE	TEE is used to assess air in the heart while weaning from CPB and optimize fluid therapy, inotropic agents, vasodilators, and chronotropic agents. ✓ for any LV regional wall motion abnormalities.
Positioning	✓ and pad pressure points ✓ eyes Arms padded at sides Chest roll	

POSTOPERATIVE

Complications	RV dysfunction	RV failure may occur in patients with pulmonary HTN and high RV afterload (see pulmonary HTN, below). Coagulopathy requiring high-volume resuscitation may also result in RV dysfunction (TACO).
Complications (cont'd)	Pulmonary HTN	Maneuvers that exacerbate pulmonary HTN should be avoided. These include hypoxia, hypercarbia, acidosis, and low minute ventilation. Inhaled NO (20 ppm

	Oliguria Drug side effects: • Corticosteroids: glucose intolerance, HTN, obesity, hyperlipidemia, aseptic necrosis of the hip, bowel perforation, infection	inspired concentration) or epoprostenol (50 mcg/kg/min) can be used to treat pulmonary HTN (20–40 ppm inspired concentration). Efforts to treat pulmonary HTN with iv vasodilator therapy may be complicated by impaired V/Q matching with hypoxemia and by systemic ↓ BP producing poor coronary perfusion and RV ischemia. Inotropic support of the RV may be necessary (epinephrine 0.02–0.1 mcg/kg/min, milrinone 0.2–0.4 mg/kg/min). Isoproterenol infusion (10–150 ng/kg/min) may be attractive as it combines inotropy, pulmonary vasodilation, and chronotropy. Preexisting impairment may → chronic ↓ UO state. Other renal problems may be related to cyclosporine toxicity, diuretic toxicity, or CPB. Rx by optimizing hemodynamics. Consider reduction or elimination of nephrotoxins and continuing use of diuretics. Cyclosporine nephrotoxicity occurs in

		most patients. A functional toxicity with ↓ GFR occurs at low dose and is reversible. Tubular toxicity with morphologic changes occurs at high doses and is generally clinically unimportant and reversible. The most serious damage is vascular interstitial toxicity, which occurs over months at high doses and is not reversible.
Multimodal pain management	PCA after weaning from mechanical ventilation	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.
Tests	Cr Hct	

Suggested Reading

1. See Suggested Readings on p. 532.

SURGERY FOR THE LUNG AND HEART/LUNG TRANSPLANTATION

SURGICAL CONSIDERATIONS

Description

With the availability of cyclosporine, the ability to successfully transplant the heart and both lungs was proven in monkeys and then successfully applied in a patient in March 1981. Subsequently, single-lung transplantation was successfully performed in 1984 and an en bloc, double-lung transplant in 1986. Clinical lung transplantation of these various types has increased markedly in the last few years, and currently, ~600 single-lung transplants, 1000 bilateral-lung transplants, and 30 heart/lung transplants are performed in the United States each year.

Current indications for heart/lung transplantation are primarily those of combined heart and lung disease, including Eisenmenger syndrome due to a congenital heart defect with irreversible pulmonary hypertension (HTN). Certain types of diffuse lung disease, such as primary pulmonary HTN with significant right heart failure, are also treated in some centers by heart/lung transplantation. Recipients for single-lung transplant usually have end-stage pulmonary disease without significant sepsis. This includes patients with interstitial fibrosis, emphysema, and lymphangioleiomyomatosis. Some patients with pulmonary vascular disease, such as primary pulmonary HTN or pulmonary HTN associated with an ASD, have undergone single-lung transplantation with or without cardiac repair. Bilateral-lung transplantation is now performed usually as a sequential single-lung transplant with the major indications being septic lung disease, such as cystic fibrosis, chronic bronchiectasis, severe bullous emphysema, or pulmonary vascular disease with or without cardiac repair. Current immunosuppressive protocols consist of a combination of tacrolimus with prednisone and mycophenolate mofetil, with or without early induction therapy, using a cytolytic agent (e.g., antithymocyte globulin) or an IL-2 receptor or blocker (e.g., basiliximab). Immunosuppression may begin preop and continue throughout the life of the patient. Current 1-yr survival rates average 70% for heart/lung transplants and 80% for lung transplants.

Heart/lung transplants usually are performed through a median sternotomy, although occasionally bilateral, transsternal thoracotomy has been used. **Single-lung transplants** (usually left side) and **bilateral sequential lung transplants** use a lateral thoracotomy, transsternal bilateral thoracotomy, or median sternotomy. Single- and double-lung transplants are greatly facilitated with one-lung ventilation (OLV), which is essential for these procedures. If this type of ventilation is not feasible, a bronchial blocker must be inserted through the operative field during pneumonectomy and reimplantation. CPB is routinely used for heart/lung transplantation and is used for either single- or bilateral-lung transplantation, depending on the stability of the patient during OLV and/or clamping of the PA. Patients with severe pulmonary HTN undergoing single- or double-lung transplantation will almost always require CPB to reduce PA pressure during clamping. For combined transplants, the recipient heart is removed as for standard heart transplantation (see Surgery for Heart Transplantation). A portion of the PA near the ligamentum arteriosum, however, is left intact to preserve the recurrent laryngeal nerve. Next, each recipient lung is excised, and the trachea is transected above the carina. For single-lung transplant, usually the recipient lung is excised, leaving a bronchial stump and vascular pedicles for the PA and veins (left atrium). For bilateral-lung transplants, both recipient lungs are removed, both bronchi are resected just distal to the carina, and the main PA and left atrium are prepared for subsequent anastomosis.

Implantation of the grafts involves a tracheal anastomosis, aortic and separate superior vena cava (SVC) and inferior vena cava (IVC) anastomoses for heart/lung transplants, and a bronchial anastomosis with PA and pulmonary venous anastomosis for single-lung transplantation. **Bilateral sequential lung transplants (BSLT)** are performed as if they were single-lung transplants. CPB requires heparinization and protamine reversal. Prior to closure, extensive exploration for potential bleeding sites within the posterior mediastinum is carried out with placement of right and left pleural and mediastinal drainage. Thoracotomies are closed in standard fashion with routine chest tube drainage.

Usual preop diagnosis

End-stage heart and lung disease, such as Eisenmenger syndrome; cystic fibrosis; primary pulmonary HTN; emphysema; bronchiectasis; lymphangioleiomyomatosis; interstitial pulmonary fibrosis; sarcoidosis; other unusual forms of lung disease

ERAS: **For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, possible need for a chest tube and postop mechanical ventilation, and physiotherapy plan. See Appendix K, p. K-1.**

■ SUMMARY OF PROCEDURES

	Heart/Lung	Single-Lung	Bilateral-Lung

	Transplantation	Transplantation	Transplantation
Position	Supine	Lateral thoracotomy	Supine
Incision	Median sternotomy, usual; bilateral anterior thoracotomy, occasionally	Posterolateral thoracotomy	Transsternal bilateral thoracotomy, median sternotomy
Special instrumentation	Ascending aortic, SVC, and IVC cannulae	Occasional need for BB to be inserted through the operative field	± Aortic, SVC, and IVC cannulation; occasional need for BB
Unique considerations	CPB. If recipient has had previous thoracotomies, mediastinal collaterals may cause troublesome bleeding. Some patients with cystic fibrosis have severe bilateral scarring, requiring extensive dissection to remove the recipient lung.	± CPB. Patient may become severely hypoxic or hypercarbic during OLV, requiring CPB. During right thoracotomy, cannulation can be performed through the thorax, but left thoracotomy may require femoral artery and vein cannulation.	CPB. Thoracotomy usually is performed on the left side first, with implantation of the lung on this side, followed by completion of right-side thoracotomy and right-lung transplantation. If patient becomes unstable, cannulation in the thorax is usually possible to facilitate transplantation.
Antibiotics	Continue specific antibiotic regimen. Coverage for	Cefazolin 2 g iv and then 2 g q4h	⇐ With appropriate coverage for

	pseudomonas is suggested in patients with cystic fibrosis.		pseudomonas in patients with cystic fibrosis
Surgical time	4–5 h	2–3 h	5–6 h
Closing considerations	Temporary pacing wire is applied and isoproterenol infusion is usually started intraop to keep HR between 100 and 110, as with a cardiac transplant.	Ventilation with as low an FiO_2 as possible to maintain a PO_2 >90. Minimize iv fluids.	⇐
EBL	500–2000 mL	<500 mL (more if CPB is used)	500–2000 mL
Postop care	Cardiac ICU: 3–7 d, 1- to 2-d assisted ventilation	⇐	⇐
Mortality	10–15%	5%	5%
Morbidity	Early acute rejection episodes from 10 to 21 d: 75%	—	—
	Infection, particularly pulmonary: 30–40%	⇐	⇐
	Pulmonary interstitial edema: 20%	Bleeding: 2–4%	⇐
		Bronchial leak or stenosis: 2–4%	⇐
	Return for bleeding: 4–6%		
	Hyperacute		

	rejection: rare		
Pain score	8–10	8–10	8–10

■ PATIENT POPULATION CHARACTERISTICS

	Heart/Lung Transplantation	Single-Lung Transplantation	Bilateral-Lung Transplantation
Age range	3 mo to 55 yr (average 30–40 yr)	1–65 yr	⇐
Male:Female	1:1	⇐	⇐
Incidence	~30/yr (United States) 200/yr (worldwide)	600/yr (United States) 1000/yr (worldwide)	1000/yr (United States) 2000/yr (worldwide)
Etiology	Eisenmenger syndrome, CHD, cystic fibrosis, pulmonary HTN, other lung diseases	Acquired chronic lung disease, pulmonary HTN	Cystic fibrosis, interstitial fibrosis, emphysema
Associated conditions	Severe cyanosis and polycythemia, diabetes in patients with cystic fibrosis, sinus infections in patients with cystic fibrosis	Right heart dysfunction, pulmonary valve insufficiency, and tricuspid valve insufficiency	Diabetes mellitus, sinus infections in patients with cystic fibrosis

ANESTHETIC CONSIDERATIONS FOR HEART/LUNG TRANSPLANTATION

PREOPERATIVE

Patients scheduled for heart/lung transplantation are terminally ill, although they may still be able to maintain limited activity. Indications include primary pulmonary HTN with irreversible heart failure, Eisenmenger syndrome, and combined cardiac and pulmonary disease. The standard preanesthetic evaluation is supplemented with considerations particular to these patients. The progression of disease is usually well documented.

Respiratory	Patients with severe pulmonary HTN (80/50 mm Hg) have enlarged PAs. Vocal cord dysfunction (Sx: hoarseness, inability to phonate “e”) may occur when the left recurrent laryngeal nerve is stretched by an enlarged PA, making these patients at increased risk for pulmonary aspiration. Appropriate precautions to avoid aspiration should be taken (see Induction, below). **Tests:** ABG; cardiac catheterization
Cardiovascular	Hx of recent exacerbation of symptoms should be sought and cardiac catheterization data interpreted in light of interval changes. The severity of pulmonary HTN and the responsiveness to specific vasodilators during catheterization should be reviewed. **Tests:** ECG; cardiac catheterization; ECHO
Neurologic	R → L intracardiac shunting may be present in patients with pulmonary HTN, and Hx of embolic episodes should be sought. Extra care should be used to avoid injection of even small quantities of intravenous air.
Hematologic	The medication schedule should be verified with particular attention to the recent use of anticoagulants. **Tests:** Hct; PTT; PT (special tubes required if severe polycythemia is present 2° ↓ plasma volume); Plt count; fibrinogen
Laboratory	Evidence of renal and hepatic dysfunction should be sought by H&P and lab studies. Mild hypokalemia is generally not treated because the heart/lung graft is preserved with K^+ and implantation will reverse hypokalemia.
Premedication	Although anxious, these patients are usually well informed and psychologically prepared. They respond well to the reassurance of the preop visit, and

	pharmacologic premedication usually is not necessary. O_2 therapy should commence prior to transport of patient to the OR. Patient may be at increased risk for pulmonary aspiration of gastric contents because of the unscheduled nature of the surgery and the presence of recurrent laryngeal nerve damage. Famotidine (20 mg) and metoclopramide (10–20 mg) may be administered iv preop.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural, continuous paravertebral, or chest wall nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: Infection is a serious potential complication in the immunosuppressed transplant patient; thus, aseptic technique is important. Aseptic technique is used in inserting and securing all vascular catheters. The anesthesia machine should be equipped with a supply of air to permit control of the FiO_2, and anesthesia is not induced until the team harvesting the graft reports that it appears to be normal on direct inspection.

Induction	In the OR, the patient should be placed on the operating table and O_2 and noninvasive monitors applied. An arterial line for BP and blood gas monitoring should be inserted, using liberal amounts of local anesthetics before induction. The presence of real-time BP monitoring can be critical during induction. A patient who is dyspneic in the supine position may be treated by raising the back of the table. Cricoid pressure may be used when the patient

	is at risk for aspiration of gastric contents. A major goal of anesthetic induction is the avoidance of further increases in PVR by guarding against respiratory acidosis, hypoxia, N_2O, light anesthesia, and extremes of lung volume. When hemodynamically tolerated, anesthetic is induced with titrated propofol (0.5–1 mg/kg) and fentanyl (2–4 mcg/kg). Etomidate (0.1–0.2 mg/kg) may be used when hypotension limits administration of propofol and narcotics. Rocuronium (1–1.5 mg/kg) should be administered early to permit rapid airway control. Midazolam and scopolamine produce amnesia. N_2O is not used because it may exacerbate pulmonary HTN, reduces FiO_2, and expands intravascular air bubbles. Patient should be ventilated by mask and cricoid pressure released only after the airway has been secured with a cuffed ETT. An ETT of internal diameter of 8.0 mm or greater will facilitate FOB at the end of the procedure and postop.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.
Maintenance	Typical total anesthetic doses for the entire intraop course are fentanyl 10–20 mcg/kg or sufentanil 10–15 mcg/kg, midazolam 0.05–0.15 mg/kg, vecuronium 0.3–0.5 mg/kg, or rocuronium 2–3 mg/kg. Recommend the use of antifibrinolytics for all cases requiring CPB.
Termination of CPB	After the tracheal anastomosis is complete, the lungs are suctioned with a sterile catheter and ventilated manually with air (FiO_2 = 0.21). Tidal volumes should be sufficient to reinflate areas of atelectasis

under direct vision. Steroids (methylprednisolone 250–500 mg) may be given by the perfusionist as the lung is reperfused. Once rewarmed, full ventilation is begun with the lowest FiO_2 to achieve $SpO_2 \geq 90\%$, usually $FiO_2 = 0.4–0.5$. Peak inspiratory pressures are typically 20–30 cm H_2O in the open chest. Optimal tidal volumes 6–8 mL/kg can be in part determined by direct visualization and lung protection strategy. PEEP may be added to optimize SpO_2 as well as inhaled NO.

Junctional rhythm is common in the denervated transplanted heart. Epinephrine (50–100 ng/kg/min) or isoproterenol (10–75 ng/kg/min) is used to achieve a HR of 100–120. When sinus rhythm is achieved, it is common to see two P waves if a biatrial anastomosis has been used. The residual atrial tissue produces nonconducting P waves. Responses mediated by vagal tone will be seen in the rate of the original atrial tissue and have no clinical importance beyond the ease with which the ECG is interpreted.

In the denervated heart, atropine and neostigmine will not affect HR, and HTN will not produce reflex bradycardia. The graft atrium produces normally conducted P waves. The transplanted heart contains adrenergic receptors and responds normally to norepinephrine, epinephrine, and isoproterenol.

The CO of the denervated heart is quite sensitive to preload; thus, iv fluid and vasodilators must be given with particular care. SNP or clevidipine is used for afterload reduction. TEE may be of particular value in assessing RV dysfunction and guiding appropriate fluid therapy, pharmacologic support, and mechanical support as necessary. Inhaled NO or epoprostenol, iv PGE_1 (20–100 ng/kg/min), isoproterenol, and NTG (0.2–2.0 mcg/kg/min) also may be used for pulmonary vasodilation. Inotropic support with epinephrine (0.05–0.10 mcg/kg/min), milrinone (0.2–

	0.4 mcg/kg/min), or isoproterenol may be necessary, especially if pulmonary HTN and RV failure occur. TEE is useful in evaluating the PA anastomosis (diameter >1 cm) and pulmonary veins (diameter >0.5 cm). Flow rates by Doppler should be ≤100 cm/sec.	
Immunosuppression	Methylprednisolone 500–1000 mg iv is given at the time of graft reperfusion while on CPB by the perfusionist.	
Diuresis	There may be little urine production, especially if patient received high-dose diuretics preop. Cyclosporine may exacerbate renal dysfunction. Mannitol and furosemide may be needed to induce diuresis.	
Transport	A sterile, disposable Jackson-Rees system is used in transporting the patient to the ICU.	
Blood and fluid requirements	Anticipate large blood loss IV: 14–16 ga × 2 BSS	Bleeding is often a major problem after termination of CPB. Patients with intracardiac defects are at increased risk for cerebral embolic events. Care must be taken to remove all air bubbles from intravascular lines. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Control of blood loss	Postbypass bleeding is a common problem	Postbypass bleeding is exacerbated by preop use of anticoagulants, depressed synthetic liver function, trauma of CPB, and/or previous chest surgery.
	Coagulation therapy	Coagulation therapy may

	necessary	include protamine (30 mg/kg), Plts, cryoprecipitate, FFP, RBCs, EACA (300 mcg/kg), and DDAVP (0.3 mcg/kg).
	Possible severe bleeding	Severe bleeding prompts further therapy: PCCs, FEIBA® (anti-inhibitor coagulant complex), and/or recombinant factor 7a.
Monitoring	Standard monitors Arterial line	
	CVP/PA catheter UO Cerebral oximetry	The CVP catheter is ideally placed following anesthetic induction. An introducer permits the rapid insertion of a PA catheter when necessary. A "double-stick" technique, placing both a triple-lumen CVL and a 9-Fr introducer in the right IJ, is sometimes used. In some centers (not Stanford), the left IJ vein is the preferred site of cannulation, leaving the right IJ unscarred for repeated endomyocardial biopsies of the transplanted heart. Cerebral oximetry readings may be confounded by changes in scalp perfusion. Be aware that false-

		positive and false-negative readings are not uncommon.
	TEE	TEE is used to optimize volume status (LV + RV size), inotropic agents (LV + RV contractility), vasodilators, and chronotropic agents.

POSTOPERATIVE

Complications	Oliguria	Diuresis may be induced with mannitol and furosemide.
	Pulmonary edema	Given the lack of lymphatic drainage in the transplanted lung, pulmonary edema may occur. Diuresis and restriction of iv fluid may be required.
	RV dysfunction	RV failure may occur in patients with pulmonary HTN and high RV afterload (see Pulmonary HTN Rx, below).
	Hypoxia	If oxygenation does not improve with increasing FiO_2, PEEP, tidal volumes, and/or inhaled NO, ECMO may be considered.
	Pulmonary HTN	Maneuvers that exacerbate pulmonary HTN should be avoided. These include hypoxia, hypercarbia, acidosis, and hypoventilation. Inhaled

		NO can be used to treat pulmonary HTN (20 parts per million inspired concentration). Efforts to treat pulmonary HTN with vasodilator therapy may be complicated by impaired V/Q matching with hypoxemia and by systemic hypotension, producing poor right coronary perfusion and RV ischemia. Inotropic support of the RV may be necessary. Isoproterenol is attractive because it combines inotropy, pulmonary vasodilation, and chronotropy.
	Rejection Infection	Monitor rejection Sx with transvenous endomyocardial biopsy and transbronchial biopsy.
	Drug side effects: • Cyclosporine: HTN, nephrotoxicity, hepatotoxicity • Corticosteroids: glucose intolerance, HTN, obesity, hyperlipidemia, aseptic necrosis of the hip, bowel perforation, infection • Azathioprine: anemia, thrombocytopenia, leukopenia, hepatotoxicity	Cyclosporine nephrotoxicity occurs in most patients. A functional toxicity with reduced GFR occurs at low dose and is reversible. Tubular toxicity with morphologic changes occurs at high doses and generally is clinically unimportant and reversible. The most serious damage is

		vascular interstitial toxicity, which occurs over months at high doses and is not reversible.
Multimodal pain management	PCA	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Reading

1. See Suggested Readings on p. 532.

ANESTHETIC CONSIDERATIONS FOR LUNG TRANSPLANTATION

PREOPERATIVE

The patient presenting for lung transplantation typically has end-stage pulmonary fibrosis, emphysema, cystic fibrosis, or bronchiectasis. Pulmonary HTN also may be treated by lung transplantation. The progression of the disease is usually well documented; however, Hx of recent exacerbation of symptoms should be sought.

Respiratory	Assess the patient’s ability to undergo OLV by review of the V/Q scan. If little perfusion of the nonoperative lung is present, anticipate the need for

	CPB. The extent of the restrictive lung disease and diffusion abnormality must be assessed preop. For example, room-air PaO_2 <45 mm Hg predicts the need for CPB. **Tests:** PFT; V/Q scan; ABG
Airway	Patients with severe pulmonary HTN (80/50 mm Hg) have enlarged pulmonary arteries. Vocal cord dysfunction (Sx: hoarseness, inability to phonate "e") may occur when the left recurrent laryngeal nerve is stretched by an enlarged PA, making these patients at increased risk for pulmonary aspiration; therefore, appropriate precautions to avoid aspiration should be taken (see Induction, below).
Cardiovascular	Evidence of RV dysfunction with tricuspid regurgitation should be sought by physical exam, ECHO, and cardiac catheterization. RV ejection fraction (EF) may be estimated with radionuclide ventriculography (normal EF ≥50%). Pulmonary HTN is considered to be severe and may produce RV failure when pressure is >2/3 of systemic arterial pressure. Note response to specific vasodilators recorded during catheterization. Patients are often aggressively diuresed. Some patients (cystic fibrosis, bronchiectasis) may develop vasoplegia especially on CPB. BP support with iv vasopressin and norepinephrine (infusion and/or boluses) may be necessary. **Tests:** Preview cardiac catheterization data; ECG; mean PAP >40 mm Hg and PVR >5 mm Hg/min/L may predict the need for partial CPB.
Neurologic	R → L intracardiac shunting may be present in patients with pulmonary HTN, and Hx of embolic episodes should be sought. Extra care should be used to avoid injection of even small quantities of intravenous air.
Musculoskeletal	Chronic cachexia precludes the procedure.

Hematologic	Polycythemia 2° chronic hypoxemia is common. Autologous blood is collected as CPB is initiated. **Tests:** Hct; coagulation studies require special blood tubes to correct for low plasma volume in patients with severe polycythemia.
Laboratory	Other tests as indicated from H&P
Premedication	Patients awaiting lung transplantation are generally well informed about the planned perioperative course. These patients respond well to the reassurance of the preop visit, and pharmacologic premedication usually is not necessary. O_2 therapy, with the usual home O_2 regimen, should commence prior to transport to the OR. Patient may be at ↑ risk for pulmonary aspiration of gastric contents because of the unscheduled nature of the surgery, the use of oral cyclosporine immediately before surgery, and the presence of recurrent laryngeal nerve damage. Ranitidine (50 mg) and metoclopramide (10–20 mg) may be administered iv before surgery.

INTRAOPERATIVE

Anesthetic technique

GETA: Typically, OLV through a DLT is required for single-lung transplants and may be requested for BSLT to facilitate pre-CPB dissection and reperfusion. Also, some institutions (not Stanford) perform BSLT without CPB (standby only), thus requiring OLV. Consider ETT/BB in cystic fibrosis patients with tenacious sputum. Infection is a serious complication in the immunosuppressed transplant patient; thus, the aseptic technique is important. Aseptic technique is used in inserting and securing all vascular catheters.

Induction	An arterial line for BP and blood gas monitoring should be inserted, using liberal amounts of local anesthetics before induction. The presence of real-time BP monitoring can be critical during induction. After receiving confirmation from the procurement team regarding donor lung suitability and estimated arrival time, induction is begun. After

	preoxygenation, midazolam 0.05–0.1 mg/kg, fentanyl 2–4 mcg/kg, propofol (0.5–1 mg/kg) or etomidate 0.2–0.3 mg/kg, and rocuronium 1–1.5 mg/kg or vecuronium (0.1–0.2 mg/kg) are typically used. Cricoid pressure must be used when the patient is at risk for aspiration because of the unscheduled nature of the surgery and possible vocal cord dysfunction associated with stretch injury of the recurrent laryngeal nerve. Avoid further increases in PVR by guarding against hypoxemia, acidosis, hypercarbia, light anesthesia, and inadequate minute ventilation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider placement of parasternal or chest wall nerve blocks if preop blocks have not been placed. Ketamine (0.5 mg/kg iv) at induction may improve pain management and reduce delirium after CPB.
Maintenance	Typically, narcotic/O_2/air/± isoflurane (in the absence of hypoxemia and right heart failure). Typical total anesthetic doses for the entire intraop course: fentanyl 10–20 mcg/kg or sufentanil 10–15 mcg/kg, midazolam 0.2 mg/kg, vecuronium 0.5–1 mg/kg, or rocuronium 0.2–0.3 mg/kg. Inhaled NO or epoprostenol may be initiated after induction to help improve right heart function and/or oxygenation. Patients with end-stage cystic fibrosis may require frequent endotracheal suctioning. Recommend the use of antifibrinolytics for all cases requiring CPB.
Emergence	Before closure of the chest, lungs are inflated under direct vision to reinflate atelectatic areas and check adequacy of bronchial closure. At the conclusion of surgery, if a DLT is used, both lumens should be suctioned and the tube replaced with a single-lumen 8.0-mm or larger ETT to facilitate bronchoscopy. Some prefer to change the DLT for a SLT while on

	CPB with FOB guidance. The patient is transported to the ICU intubated and ventilated. Discuss with surgeon local wound infiltration or chest wall nerve block (e.g., via prefascial plane catheter) placement. Consider iv acetaminophen if po dose is given >4–6 hr ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 14 or 16 ga × 1–2 BSS @ 4 mL/kg/h	Patients with intracardiac defects are at increased risk for cerebral embolic events; take care to remove all air bubbles from intravascular lines and syringes. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Arterial line PA catheter Urinary catheter with thermistor	ECG leads should be covered with tape to ensure that electrical contact is not degraded by prep solution or blood. Be careful of air embolization during catheter insertion. Patients who are profoundly dyspneic (→ high negative intrathoracic pressure) and have low central heart pressures (2° diuresis) are at high risk for VAE; consider inserting the catheter after GA and IPPV have been instituted. Oxygenation must be watched closely.

	Cerebral oximetry	Cerebral oximetry readings may be confounded by changes in scalp perfusion. Be aware that false-positive and false-negative readings are not uncommon.
	TEE	Assessment of RV size and function (normal EF = 0.5–0.7) may be helpful.
OLV	DLT: typically 39–41 Fr (men), 35–37 Fr (women) OLV: necessary for SLT. May not be feasible for BSLT in the most severely compromised patients	A DLT is inserted in the left mainstem bronchus to permit surgical access. FOB is used to verify proper tube placement. Positioning of the bronchial cuff in the proximal left mainstem bronchus does not interfere with surgical access to the bronchus. The position of the DLT should be verified after the patient is moved to the lateral position for SLT. Finally, verify ventilation and proper functioning of the tube. Apply O_2 with CPAP at 5 cm H_2O to the nondependent lung. Further adjustment of CPAP may enhance oxygenation. Also consider using inhaled NO or epoprostenol. The

		nondependent lung may be reinflated with O_2, if necessary, to achieve adequate oxygenation. If adequate oxygenation cannot be achieved, CPB should be initiated.
	Frequent suctioning	Frequent suctioning is necessary in patients with tenacious secretions.
PA clamping (for CPB S/B cases)	Improve V/Q mismatch Improve oxygenation PAP ↑↑→ RV failure	Clamping of the PA will improve V/Q mismatch and oxygenation; however, severe pulmonary HTN and RV failure may develop. Vasodilators, such as NTG (0.2–2 mcg/kg/min), SNP (0.2–10 mcg/kg/min), PGE_1 (20–100 ng/kg/min), or inhaled NO (20–40 ppm), should be used to treat pulmonary HTN and ↓ RV afterload; however, care must be taken to avoid systemic hypotension. Inotropic support for the RV may be necessary (epinephrine [20–100 ng/kg/min]). TEE and right atrial pressure should be monitored for evidence of tricuspid regurgitation associated with RV dilation.

PA unclamping (for CPB S/B cases)	PIP: 20–25 cm H_2O O_2 sat: 90–100% PEEP: 5–10 mm Hg	Temporary unclamping of the PA may be necessary to allow further pharmacologic therapy. If RV failure cannot be controlled pharmacologically, CPB should be initiated. The PA should not be unclamped until ventilation is possible to the transplanted lung. Perfusion without oxygenation of the transplanted lung would produce profound shunt and hypoxemia. TV should be adjusted to eliminate atelectasis and to achieve a PIP of 20–25 cm H_2O with the chest open. The FiO_2 (0.3–0.5) is limited in the hope of curtailing free radical injury. PEEP and inhaled NO may be used to enhance oxygenation.
Positioning	For single lung: • Supine to lateral decubitus • Axillary roll • Airplane splint • Avoid arm hyperextension (>90°) For double lung: • Supine with arms above head for bilateral subcostal incision ✓ and pad pressure	Verify correct position of DLT or BB after moving patient to the lateral position. Difficult access to airway after patient positioned. **NB:** potential for kinking of iv and arterial lines

	points ✓ eyes

POSTOPERATIVE

Complications	Pulmonary edema Infection: bacterial, viral, fungal, or protozoan	Given the lack of lymphatic drainage in the transplanted lung, pulmonary edema may occur. Diuresis and restriction of iv fluid may be required. Mannitol and furosemide can be used to induce diuresis.
	Side effects of immunosuppressive agents: • Corticosteroids: HTN, osteoporosis, glucose intolerance, hyperlipidemia • Azathioprine: anemia, thrombocytopenia, leukopenia	Immunosuppression drugs typically include tacrolimus, mycophenolate mofetil, and corticosteroids. Early induction therapy with a cytolytic agent (e.g., antithymocyte globulin) may be used.
Multimodal pain management	Epidural narcotics Parenteral narcotics	Postop analgesia may be provided by infusion of narcotics through an epidural catheter. If CPB is used, the insertion of the epidural catheter should be delayed until normal coagulation function is documented in the ICU. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid

		analgesics with regional anesthetics whenever possible (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	If postop sedation is required, dexmedetomidine may be preferable to propofol. Encourage fast-track extubation, early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Arora RC, Brown CH IV, Sanjanwala RM, et al: "NEW" rehabilitation: a 3-wa approach to improve postoperative survival and health-related quality of life in cardiac surgery patients. Can J Cardiol 2018; 34(7):839–49.
2. Crumley S, Schraag S: The role of local anaesthetic techniques in ERAS protocol for thoracic surgery. J Thorac Dis 2018; 10(3):1998–2004.
3. Kwak J, Majewski M, LeVan PT: Heart transplantation in an era of mechanica circulatory support. J Cardiothorac Vasc Anesth 2018; 32:19–31.
4. Murray AM, Quinlan JJ: Anesthesia for heart, lung, and heart-lung transplantation. In Kaplan J, Cronin B, Maus T, eds. Chapter 19: Essentials of Cardiac Anesthesia fo Cardiac Surgery, 2nd edition. Elsevier, Philadelphia: 2018, 473–503.
5. Nicoara A, Anderson-Dam J: Anesthesia for lung transplantation. Anesthesiology Clin 2017; 35:473–89.
6. Noss C, Prusinkiewicz C, Nelson G, et al: Enhanced recovery for cardiac surgery. J Cardiothorac Vasc Anesth 2018; 32:2760–70. pii: S1053-0770(18)30049-1.
7. Rao V, Ghadimi K, et al: Inhaled nitric oxide and inhaled epoprostenol use in cardiothoracic surgical patients. J Cardiothorac Vasc Anesth 2018; 32:1452–7.
8. Sellers D, Cassar-Demajo W, et al: The evolution of anesthesia for lung transplantation. J Thorac Cardiovasc Anesth 2017; 31:1071–9.
9. Semenkovich TR, Hudson JL, Subramanian M, et al: Enhanced Recovery Afte Surgery (ERAS) in thoracic surgery. Semin Thorac Cardiovasc Surg 2018; 30:342–9. pii: S1043-0679(18)30164-3.